MIL-T-62050B(AT)
31 October 1985
SUPERSEDING
MIL-T-62050A(AT)
19 March 1973
M60PD-T-62226(AT)
27 February 1975

MILITARY SPECIFICATION

TANK, COMBAT, FULL-TRACKED: 105MM GUN, M48A5; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the processing of the M48A5, 105MM, Gun, Full-tracked, Combat, Tank, for storage and shipment.

1.2 Classification. Processing shall be of the following levels as specified (see 6.1):

Level A	- Processing for domestic or oversea shipment and any storage outside of buildings in excess of 90 days from date of processing (periodic care and preservation during storage required).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standard.zation Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AREA PACK

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Level B	- Limited processing for immediate use shipment and for domestic or oversea shipment (excluding open deck loading) and any storage not to exceed 90 days from date of processing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

A-A-883	- Tape, Pressure Sensitive Adhesive, Masking.
A-A-1685	- Tape, Pressure Sensitive Adhesive, (High Tensile Strength, Black, Weather-Resistant, Glass Filament Reinforced).
O-E-760	- Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents.
O-S-576	- Sodium Bicarbonate, Technical.
O-S-801	- Sulfuric Acid, Electrolyte; for Storage Batteries.
P-D-220	- Detergent, General Purpose.
T-R-650	- Rope, Yarn and Twine, Bast Fiber.
V-F-106	- Fasteners, Slide, Interlocking.
V-T-295	- Thread Nylon.
FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
NN-P-530	- Plywood, Flat Panel.
QQ-A-200	- Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for.
QQ-A-200/5	- Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/8	- Aluminum Alloy 5086, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-A-1876	- Aluminum Foil.
QQ-B-650	- Brazing Alloys, Copper, Copper-Zinc and Copper-Phosphorus.
QQ-E-450	- Electrodes, Welding, Covered: Mild Steel.
QQ-S-635	- Steel Plate, Carbon.
QQ-S-698	- Steel, Sheet and Strip, Low Carbon.
QQ-S-741	- Steel, Carbon: Structural Shapes; Plates; and Bars.

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QQ-S-781	- Strapping, Steel, and Seals.
QQ-W-461	- Wire, Steel, Carbon (Round, Bare, and Coated).
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-485	- Enamel, Semi-Gloss, Rust-Inhibiting.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-C-282	- Chipboard.
UU-P-268	- Paper, Kraft, Wrapping.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative.
WW-T-700/6	- Tube, Aluminum Alloy, Drawn Seamless, 6061.
ZZ-T-351	- Tip, Cane and Crutch, Rubber.
ZZ-T-416	- Tire, Pneumatic - Retread and Repair Materials.
MMM-A-178	- Adhesive, Paper Label, Water-Resistant.
MMM-A-1617	- Adhesive, Rubber Base, General Purpose.
NNN-P-40	- Paper, Lens.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive, Packaging/Paper (for Carton Sealing).

MILITARY

MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packaging.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-450	- Coating Compound, Bituminous Solvent Type, Black (for Ammunition).
MIL-W-530	- Webbing Textile, Cotton, General Purpose, Natural or in Colors.
MIL-P-3420	- Packaging Materials, Volatile Corrosion Inhibitor Treated, Opaque.
MIL-W-3912	- Wood Parts and Wood Substitutes, Fabricated: For Transport Vehicle Bodies and Similar Applications.

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MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-I-8574 - Inhibitors, Corrosion, Volatile, Utilization of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-B-12841 - Basic Issue Items for Military Vehicles, Carriages and Equipment; Preparation for Shipment and Storage of.
MIL-W-13518 - Wood Preservative: Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound.
MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline, Removable.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-C-16555 - Coating Compound, Strippable, Sprayable.
MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic).
MIL-R-17343 - Rope, Nylon.
MIL-C-20696 - Cloth, Coated, Nylon, Waterproof.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085 - Tapes, Pressure - Sensitive, Adhesive, Preservation and Sealing.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-R-24049 - Rope Polypropylene.
MIL-W-45205 - Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys Readily Weldable for Structures Excluding Armor.
MIL-P-46002 - Preservative Oil, Contact and Volatile Corrosion-Inhibited.
MIL-T-50036 - Talc, Technical, T1 and T3.
MIL-D-81298 - Dye, Liquid, for the Detection of Leaks in Aircraft Fuel Systems.
MIL-W-83420 - Wire Rope, Flexible, for Aircraft Control.

STANDARDS
FEDERAL

FED-STD-595 - Colors.
FED-STD-751 - Stitches, Seams and Stitchings.

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MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MIL-STD-2073-1A	- DoD Materiel Procedures for Development and Application of Packaging Requirements.
MS16842	- Clamp, Wire, Rope, Saddled, Single Grip, Steel.
MS16997	- Screw, Cap, Socket Head and Screw, Cap, Socket Head, Self-Locking; Alloy Steel Cadmium Plated, UNC-3A.
MS27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (in/mm).
MS35206	- Screw, Machine - Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNC-2A (in/mm).
MS35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series.
MS35355	- Bolt, Machine, Square Head, Steel, Cadmium or Zinc Plated, UNC-2A.
MS35691	- Nut, Plain, Hexagon (Jam) UNC-2B and UNF-2B.
MS51943	- Nut, Self-Locking, Hexagon-Prevailing Torque, for Critical Installations, 250°F, UNC-3B and UNF-3B.
MS51967	- Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNC-2B (in/mm).
MS51968	- Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNF-2B (in/mm).
MS90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

7407417	- Lockwasher.
8711310	- Clamp Assembly.
10870140	- Hose Assembly, Rubber.
10870861	- Packing, Preformed Rubber.
BCYX6(TAXI Dwg.)	- Screw, Lockwasher.
QACX1(TAXI Dwg.)	- Chape, Web.

PUBLICATIONS
TECHNICAL MANUALS

TM9-2350-258-34-2	- Turret Maintenance Manual.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108	- Steel Bars, Carbon, Cold-Finished, Standard Quality, Specification for.
ASTM A512	- Cold-Drawn Buttweld Carbon Steel Mechanical Tubing, Specification for.
ASTM A513	- Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing, Specification for.
ASTM A519	- Seamless Carbon and Alloy Steel Mechanical Tubing, Specification for.
ASTM A576	- Steel Bars, Carbon, Hot-Wrought Special Quality, Specification for.
ASTM A675	- Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties, Specification for.
ASTM A682	- Steel, Strip, High Carbon, Cold-Rolled, Spring Quality, General Requirements, Specification for.
ASTM D4066	- Nylon Injection and Extrusion Materials (PA), Specification for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.10	- Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes, Specification for.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Northwest LeJuene Road, P.O. Box 351040, Miami, FL 33135.)

ASSOCIATION OF AMERICAN RAILROADS PUBLICATIONS

Section No. 1	- General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, IL 60605.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.1), one of the first 10 production processed vehicles shall be subjected to the inspection specified in 4.3. Approval of this vehicle shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Unless otherwise specified by the Government, any change to materials or design after approval shall require an additional vehicle to be inspected as specified in 4.3. The Government representative shall select the vehicle to be inspected.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards. Materials shall be free of defects which adversely affect performance or serviceability (see 4.5.1).

3.3 Processing. Processing for levels A and B shall be the same except when differences are specified.

3.3.1 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, the vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that the vehicle may be operated, shipped, and placed into service with a minimum of delay (see 4.5.2).

3.3.1.1 Record forms. Two copies of DD Form 1397 shall be provided. Information on the forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 shall be placed in a bag conforming to type 1, style 2 of MIL-B-117. The bag shall be closed by heat sealing and securely attached inside of the vehicle. The other copy of DD Form 1397 shall be waterproofed with adhesive conforming to MMM-A-178 and securely attached in a conspicuous location on the exterior of the vehicle. A copy of DA Form 2408-4, "Weapons Record Data", shall be provided (see 4.5.2).

3.3.1.2 Processing records. Records of vehicle processing shall be maintained and shall be readily available for review by the Government representative (see 4.5.2).

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3.3.2 Cleaning and drying.

3.3.2.1 Vehicle interior. Unless otherwise specified herein, the interior surfaces of the vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791 and warm water. Water or other liquids under pressure, or steam cleaning shall not be used. Cleaned surfaces shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing to assure that no liquids enter instruments, connections, or other components susceptible to water damage and that water does not accumulate in areas where it cannot drain or be dried (see 4.5.2.1).

3.3.2.2 Vehicle exterior. The vehicle exterior shall be cleaned using a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in water or steam. Cleaning shall remove all foreign matter. Cleaned surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into the engine compartment, turret ring, commander's cupola, range finder and boxes, or armament (see 4.5.2.1).

3.3.2.3 Gun. When inspection indicates the need for reprocessing of the gun, the gun shall be cleaned in accordance with process C-3 of MIL-P-116 and shall be thoroughly dried in accordance with procedure D-4 of MIL-P-116 (see 4.5.2.3).

3.3.3 Disassembly. Projecting parts whose removal will accomplish desired cube reduction, and parts susceptible to damage and pilferage, shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Removed parts shall be preserved, packaged, and packed in accordance with the special packaging instructions for the part. Packed parts shall be identified and stowed securely within the vehicle (see 4.5.2).

3.3.3.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to type A of UU-T-81 and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178 (see 4.5.2).

3.3.4 Relubrication. When the vehicle has been operated in excess of 50 miles since the previous lubrication, or after the vehicle has been cleaned in accordance with 3.3.2, the vehicle shall be lubricated using materials conforming to drawings, specifications, or the lubrication order applicable to the vehicle. All exposed oil can points such as, but not limited to, levers, locking bars, strikers, hinges, hinge pins, locking pins, pintle pins, locking levers, wing nuts, latches, door locks, hand-operated locking knobs, linkage, and threaded ends of yokes and related clevis pins shall be lubricated with oil conforming to VV-L-800. Excess lubricant shall be removed after lubrication (see 4.5.2).

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3.3.5 Battery supports and retainers. Battery supports and retainers shall be cleaned with a solution of 0.5 pound of sodium bicarbonate conforming to O-S-576 per gallon of water. Cleaned surfaces shall be flushed with clean water, then thoroughly dried. Battery supports and retainers shall be preserved with compound conforming to MIL-C-450 (see 4.5.2).

3.3.6 Transmission and final drives.

3.3.6.1 Level A. The transmission shall contain lubricating oil conforming to grade 10W of MIL-L-21260 filled to the operating level. The final drives shall contain lubricating oil conforming to grade 50 of MIL-L-21260, filled to the operating level. DD Form 1397 shall be annotated with the grade of lubricant used (see 3.3.1.1 and 4.5.2).

3.3.6.2 Level B. Transmission and final drives shall contain normal seasonal operational lubricant as specified on applicable drawings, specifications, or lubrication orders and shall be filled to the operating level. DD Form 1397 shall be annotated to indicate the grade of lubricant or preservative oil used (see 3.3.1.1 and 4.5.2).

3.3.7 Engine crankcase.

3.3.7.1 Level A. The engine crankcase shall be filled to the operating level with lubricating oil conforming to MIL-L-21260 of the seasonable grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the grade of lubricant used (see 3.3.1.1 and 4.5.2).

CAUTION: For shipment to, storage in, or prior to placing the vehicle in operation in areas where the ambient temperature is below -20 degrees Fahrenheit (°F), the operating oil must be drained from the crankcase and the crankcase filled with the appropriate seasonable grade of oil as specified on the applicable specification, drawing, or lubrication order. For shipment made to these areas, a red warning tag, bearing the information "DRAIN AND FILL WITH APPLICABLE OPERATING OIL", shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be completed to indicate the grade of lubricating or operating oil used (see 3.3.1.1).

3.3.7.2 Level B. Engine crankcase shall contain the normal seasonal operational lubricant as specified on the lubrication order, filled to the operational level. DD Form 1397 shall be annotated to indicate grade of lubricant used (see 3.3.1.1 and 4.5.2).

3.3.8 Engine preservation.

3.3.8.1 Level A. The engine shall be preserved in accordance with 3.3.8.1.1 through 3.3.8.1.5 (see 4.5.2.5).

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3.3.8.1.1 Fuel system and combustion chamber. Prior to processing, the engine shall be cooled to assure that the cylinder head temperature, measured at the injector nozzle flange surface of all cylinders, is not more than 100°F. Cooling shall be accomplished by induced air currents or by waiting the period of time required to attain the above specified temperature. When the ambient temperature exceeds 100°F, the engine shall be cooled to a temperature equivalent to the ambient. After the engine has been cooled, the fuel supply system from the fuel tank shall be shut off. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with preservative oil conforming to grade 1 of MIL-P-46002 to which has been added an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with the appropriate grade of diesel fuel conforming to VV-F-800. Uncouple the quick-disconnect on the inlet fuel line to the primary fuel filter. Remove the filter cans and elements from both the primary and secondary filters. Drain the diesel fuel from the filter cans and fill with preservative oil conforming to grade 1 of MIL-P-46002 and reinstall the filter cans without the filter elements. Uncouple the engine fuel return line quick-disconnect and fasten a transparent plastic recovery line to the engine line. Provide a recovery container for the transparent line. Deliver MIL-P-46002 oil from a portable tank while operating the engine at idle speed [700 revolutions per minute (rpm)] for not more than 60 seconds to consume fuel in the injection system and to pump the MIL-P-46002 oil into the engine fuel system. Following the 60 seconds of engine operation, the hand purge pump shall be operated, without depressing the solenoid switch until the preservative oil is flowing through the transparent line into the recovery container. (The change should occur after approximately 2 gallons of fuel flow-out.) De-energize the intake manifold heaters and operate the hand purge pump 20 strokes, with the solenoid switch depressed, to pump preservative oil conforming to MIL-P-46002 into the intake manifold heaters. Unclamp the hose from each turbocharger inlet. Use removed clamp (PN 8711310) for securing an air restrictor cover, fabricated in accordance with figures 1 and 2 to each turbocharge inlet. While the portable tank furnishes MIL-P-46002 oil, depress the accelerator to full throttle and crank the engine for three 30-second periods as follows: Cranking shall be for a period of 30 seconds, the starter shall be allowed to cool for not less than 3 minutes; the engine shall be cranked for an additional 30 seconds, the starter shall be allowed to cool for not less than 3 minutes; the engine then shall be cranked for a final period of 30 seconds (see 4.5.2.5).

CAUTION: The engine shall be cranked for a period of not less than 25 seconds and not more than 30 seconds. The starter shall be allowed to cool down for not less than 3 minutes after each 30-second period of cranking. Special precautions shall be taken to assure that the time limits specified shall be adhered to as the engine, the starter, or the starter solenoid may be damaged.

NOTE: The engine may fire for 5 seconds while being cranked with air restrictors installed.

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Reconnect the electrical leads to the intake manifold heater igniter plugs. Remove the air restrictor covers, but do not reconnect the turbocharger hoses (see 3.3.8.1.3).

3.3.8.1.2 Engine purging. The portable container shall be adjusted to provide diesel fuel for purging the return fuel line, fuel filters, engine fuel pump, and injector pump. Remove the filter cans, drain the preservative oil, and wipe clean with lint free wiping material. Replace the filter elements in the cans, fill with diesel fuel, and reinstall the filter cans. Hold the engine fuel cut off switch in the OFF position while operating the hand purge pump, without depressing the solenoid switch, until diesel fuel is flowing through the transparent line into the recovery container. (This should occur after approximately 2 gallons of oil flow-out.) Disconnect the recovery line and reconnect the engine fuel return line to the fuel tanks. Turn on the vehicle fuel supply system and reinstall any parts disassembled during engine processing (see 4.5.2.5).

3.3.8.1.3 Preservation of turbochargers. After preservation in accordance with 3.3.8.1.1 and 3.3.8.1.2, 2 ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into each turbocharger inlet. The removed clamps shall be reinstalled on each turbocharger air inlet horn. The turbocharger inlets and the air cleaner hoses shall be sealed with tape conforming to type II of MIL-T-22085. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL HOSE", shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be annotated to show the engine is preserved with a volatile corrosion-inhibitor (VCI) oil (see 3.3.1.1 and 4.5.2.5).

3.3.8.1.4 Preservation through exhaust system. After preservation in accordance with 3.3.8.1.1 through 3.3.8.1.3, 2 ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into each exhaust opening. The openings shall then be sealed with tape conforming to type II of MIL-T-22085. The engine crankcase breathers shall then be sealed with plastic plugs conforming to MIL-C-5501 or with tape conforming to type II of MIL-T-22085. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL HOSE", shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be annotated to show the engine is preserved with a VCI oil (see 3.3.1.1 and 4.5.2.5).

3.3.8.1.5 Preservation through dipstick shroud opening and oil filler tube. After preservation in accordance with 3.3.8.1.1 through 3.3.8.1.4, 6 ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the crankcase through the oil filler cap opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. The spray nozzle shall not be submerged in the crankcase oil. (In case of an inability to use the oil tube filler system to atomize spray due to a change in the location of the filler tube, the dipstick shroud opening shall be used.) After spraying has been accomplished, the dipstick

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shall be reinstalled, the oil filler cap shall be closed, and all the openings to the engine interior, including the dipstick shroud opening and the oil filler cap, shall be sealed with tape conforming to type II of MIL-T-22085. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL AIR CLEANER INLET HOSES", shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be annotated to show that the engine is preserved with a VCI oil (see 3.3.1.1 and 4.5.2.5).

3.3.8.2 Level B. The engine shall not require preservation for level B shipment and storage.

3.3.9 Personnel heater and heater fuel pump.

3.3.9.1 Level A. After processing the engine as specified in 3.3.8, uncouple the quick disconnect from the personnel heater fuel pump line and drain the fuel line. Seal the ends of the disconnected fuel line with plastic caps or plugs conforming to MIL-C-5501. The external heater exhaust opening shall be sealed with tape conforming to type II of MIL-T-22085. Warning tags, bearing the information "HEATER FUEL LINE DISCONNECTED AND SEALED - REMOVE SEALS, OPERATE HEATER FUEL PUMP TO DRAIN MINIMUM OF ONE QUART FUEL - RECONNECT FUEL LINE TO HEATER PRIOR TO STARTING", shall be secured to the heater unit, to the heater fuel pump and to the heater operating switch on the driver's control panel (see 4.5.2).

3.3.9.2 Level B. Personnel heaters and fuel pumps shall be in a ready-to-use condition (see 4.5.2).

3.3.10 Air cleaners.

3.3.10.1 Level A. After completion of acceptance tests and prior to tank shipment, the air cleaner supply shall be adjusted to assure that air will be taken from the engine compartment and not the crew compartment. Exterior air cleaner discharge elbows shall be sealed with tape conforming to type II of MIL-T-22085 (see 4.5.2).

3.3.10.2 Level B. The air cleaners shall not be disassembled or sealed for level B shipment and storage (see 4.5.2).

3.3.11 Fuel tanks.

3.3.11.1 Level A. Each fuel tank shall be drained to the maximum extent possible. The fuel tank cap and the filler screen shall be removed and coated with lubricating oil conforming to grade 30 of MIL-L-21260. One quart of lubricating oil conforming to grade 10W of MIL-L-21260 shall be added to each 5 gallons or portion thereof of residual fuel. The tank cap and filler screen shall be reinstalled (see 4.5.2.2).

3.3.11.2 Level B. Unless otherwise specified (see 6.1), vehicles shall be shipped without draining residual fuel from the fuel tanks (see 4.5.2).

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3.3.12 Turret ring bearing. Lubrication of the turret ring bearing for new production vehicles shall be in accordance with specified manufacturing vehicles. For other than new production vehicles, lubricate the turret ring bearing in accordance with the recommended semi-annual maintenance procedures (see 4.5.2).

3.3.13 105MM gun tube installed in mount. Immediately after cleaning (see 3.3.2.3), the bore and chamber of the gun shall be coated with oil conforming to VV-L-800. Excess oil shall be allowed to drain from coated surfaces. A strip of VCI treated carrier material conforming to type I, class 3, style A of MIL-P-3420 shall be cut and rolled into a tube with the VCI treated surface on the outside. The carrier material shall be of a size that will provide a continuous cover for the bore and chamber surfaces. The rolled carrier material tube shall be inserted into the gun extending the entire length of the bore and chamber. The tube shall not be forced or kinked in a manner that would obstruct the chamber. The gun shall remain in battery with the turret travel lock secured in the lock position. The VCI material shall be applied in accordance with MIL-I-8574 (see 4.5.2.3).

3.3.13.1 Muzzle plug. A plug shall be provided for the muzzle end of the gun cannon (see figure 3). The muzzle plug shall be completely overwrapped with aluminum foil conforming to grade B of QQ-A-1876, positioned in the muzzle end and secured in place with tape conforming to type II of MIL-T-22085. The joint around the muzzle plug and gun shall be completely sealed with tape conforming to type II of MIL-T-22085. A polyethylene bag conforming to type I, class B, style 2 of MIL-B-117, 6 mils, shall be provided. The bag shall be 7 inches long and of applicable width. The bag shall be installed over the muzzle end and secured in place with four strips of tape conforming to type II of MIL-T-22085. Tape shall be applied lengthwise at the top, bottom, and each side. The tape shall be 12 inches long and at least 1 inch wide. The tape shall be applied equally, 6 inches onto the bag and 6 inches onto the painted surface. The bag shall be sealed to the gun with tape conforming to type II of MIL-T-22085. The tape shall be of applicable length and 6 inches wide to provide a continuous seal around the circumference of the gun. The tape shall be applied equally, 3 inches onto the bag and 3 inches onto the painted surface. An additional 1-inch strip of tape shall be applied midway between the muzzle end and the area where the bag is sealed to the gun. This piece of tape shall be applied completely around the circumference of the bag to provide additional securement of the bag to the gun (see 4.5.2.3).

3.3.13.2 Breech mechanism. All unpainted surfaces, including the phosphated surfaces of the breech block, breech operating mechanism, and firing mechanism, shall be coated with grease conforming to MIL-G-10924. A plug shall be provided for the breech (see figure 3). The plug shall be completely overwrapped with aluminum foil conforming to grade B of QQ-A-1876. The breech shall be opened, the breech plug positioned in the gun chamber and the breech shall be closed (see 4.5.2).

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3.3.13.3 Exercising of recoil mechanism. When the recoil mechanism has not been exercised, proof fired, overhauled, or manufactured within 4 months prior to preparation for storage or shipment, the recoil mechanism shall be exercised at least three extensions of the recoil piston. Extension shall be at least 6 inches. DA Form 2408-4 "Weapons Record Data" shall be annotated to show that the recoil has been exercised (see 4.5.2).

3.3.13.4 Recoil mechanism (after exercising). Accessible machined surfaces of the gun immediately forward of the recoil mechanism shall be coated with grease conforming to MIL-G-10924. Inaccessible machined surfaces shall be fogged with lubricating oil conforming to grade 10W of MIL-L-21260. Processing shall be accomplished by removing the cover of the gun shield. The surface of the recoil mechanism immediately forward of the breech ring collar shall be coated with grease conforming to MIL-G-10924. The grease shall be applied while the gun cannon is out of battery during exercising and upon the last extension prior to return to battery (see 4.5.2).

3.3.13.5 Exercising of replenisher. The replenisher assemblies shall be exercised coincidently with the recoil mechanism (see 3.3.13.3 and 4.5.2).

3.3.13.6 Replenisher (after exercising). The replenisher shall be filled to the bleed position with hydraulic fluid conforming to MIL-H-6083, then drained to the operating level (see 4.5.2).

3.3.13.7 Gun mount. The exposed unpainted surfaces of the elevating cylinder, trunnions, trunnion caps and bearings shall be coated with grease conforming to MIL-G-10924 (see 4.5.2).

3.3.14 105MM gun tube removed from mount. For all oversea shipments, the gun tube shall be removed from the turret (see 6.4) and stowed on the vehicle fender in accordance with figures 4 through 21. Extreme care shall be exercised to prevent damage to the threaded portion of the tube during the following operations (see 4.5.2).

3.3.14.1 Gun tube. The gun tube shall be processed in accordance with 3.3.2.3, 3.3.13, and as specified herein. Bare metal surfaces of the breech end of the tube, including the threaded area, shall be wrapped with treated carrier material conforming to type I, class 1, style C of MIL-P-3420. The VCI material shall be applied in accordance with MIL-I-8574. The treated side of the carrier shall be applied against the bare metal surface and shall provide a continuous cover starting from 2 inches on the painted surface and extending to approximately 6 inches beyond the breech end of the tube. The carrier material shall overlap approximately 3 inches over the entire length and shall completely cover the breech plug when folded toward the tube center. The carrier material shall be secured in place with tape conforming to type II of MIL-T-22085. The tape and carrier material shall then be oversprayed with a coating compound conforming to type II, class 1 of MIL-C-16555. The coating shall be at least 0.040 inch thick, when measured after 4 hours of drying (see 4.5.2.4). The coating shall extend at least 2 inches onto the painted surface of the tube. The threaded portion of the tube and area of the tube recoil surface which will contact the fender mount shall be protected by applications of two layers of the carrier material.

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The threaded portion of the tube shall be secured in place with tape conforming to type II of MIL-T-22085. The removed gun tube locking pin and set screw shall be lubricated with oil conforming to VV-L-800, wrapped in barrier material conforming to type II, grade A, class 2 of MIL-B-121 and placed in a 4- by 8-inch bag conforming to type I, class B, style 2 of MIL-B-117, 4 mils. The bag shall be identified, heat sealed and secured in the oddment tray with tape conforming to type II of MIL-T-22085 (see 4.5.2.3 and 4.5.2.4).

3.3.14.2 Gun mount (after removal of the gun tube). After the gun tube has been removed, the gun mount shall be placed in full elevation. All bare metal surfaces exposed by removal of the gun tube shall be cleaned in accordance with process C-3 of MIL-P-116, dried in accordance with process D-4 of MIL-P-116, and sealed with barrier material conforming to type I, grade C of MIL-B-121. A wood disc (see figure 19) shall be inserted into the opening of the mantlet cover and secured in position using the mantlet cover clamp. The bellows section of the mantlet cover shall be compressed and secured against the gun shield in accordance with figure 5 (see 4.5.2).

3.3.14.3 Breech mechanism after removal of the gun tube. The breech mechanism shall be processed in accordance with 3.3.13.2 (see 4.5.2).

3.3.14.4 Reinstallation of gun tube in mount. Prior to reinstallation of the gun tube, the bare metal surfaces of the gun mount exposed by previous removal of the gun tube, and the exterior bare metal surfaces of the gun tube shall be cleaned in accordance with process C-3 of MIL-P-116, dried in accordance with process D-4 of MIL-P-116, and lubricated with grease conforming to MIL-G-10924. The gun tube shall then be reinstalled (see 4.5.2 and 6.4).

3.3.15 Fire control items.

3.3.15.1 Level A. Removed fire control items shall be cleaned, dried, and immediately packaged and packed in accordance with level A requirements of MIL-STD-2073-1A and MIL-P-116, then stowed with the Basic Issue Items (BII), or as otherwise specified by the procuring agency (see 4.5.2).

3.3.15.1.1 Periscopes. All periscopes except the gunner's periscope shall be removed from the vehicle. The commander's periscope link shall not be removed. The end of the link assembly shall be secured to the support provided for this purpose. To prevent damage in this condition, the handcrank shall be wired to the elevation screwjack body. The periscope guard on the cupola roof and attaching hardware shall be removed. The exposed metal surfaces of the periscope in the cupola roof shall be preserved in accordance with 3.3.24. The opening in the cupola roof shall be sealed with tape conforming to type II of MIL-T-22085 (see 4.5.2).

3.3.15.1.2 Gunner's periscope and telescope. The M32 periscope and the M105 telescope shall not be removed from installed positions in the vehicle. Exposed optical components shall be cleaned by blowing on exposed optical glass surfaces with air from a hand syringe, or by use of a clean camel's hair brush, followed by the use of ethyl alcohol conforming to type III of O-E-760. In cases of contamination not removable by alcohol, cleaning shall be accomplished by use of a solution consisting of 2 ounces of detergent

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conforming to MIL-D-16791, 0.5 gallons of ethyl alcohol conforming to type III of O-E-760 and 1 gallon of distilled water. The optical glass surfaces shall be washed with the cleaning solution, using a swab made of lens tissue conforming to type I of NNN-P-40. Washing shall be repeated using a clean swab each time, until no dirt or other foreign matter remains on the surfaces. Cleaning shall be accomplished with a minimum of pressure and rubbing, without the use of cloth or rubbing materials, to prevent damage to lens coatings. Immediately after cleaning, the optics shall be covered or wrapped with lens tissue conforming to type II of NNN-P-40 and secured with tape conforming to type II of MIL-T-22085. All exposed, unpainted, unplated, metal surfaces shall be coated with grease conforming to MIL-G-10924 (see 4.5.2).

3.3.15.1.3 Instrument lights. If installed, instrument lights shall be removed from the vehicle. Instrument lights shall be packaged without batteries (see 4.5.2).

3.3.15.1.4 Range finder. The optical surfaces, including end box windows, shall be cleaned as prescribed in 3.3.15.1.2, covered with four thicknesses of lens tissue conforming to type II of NNN-P-40. The wrapping shall be secured with tape conforming to type II of MIL-T-22085. All exposed, unpainted, unplated metal surfaces shall be coated with grease conforming to MIL-G-10924 (see 4.5.2).

3.3.15.1.4.1 Purging of range finder. If there is evidence of internal moisture in the range finder at the time of preparation for shipment or storage, the range finder shall be processed as follows: Exercise all controls to full limits in all normal directions of movement and return to normal position. Remove the caps from the right and left valves of the range finder. Remove the valve core from one valve of the range finder. Connect the other valve to a cylinder of dry nitrogen and allow nitrogen to flow at 5 pounds per square inch (psi) through the range finder for 20 to 30 minutes. Reduce pressure to 1 to 2 psi and install the valve core.

CAUTION: Check to insure that the cylinder pressure does not exceed capacity of the pressure regulator.

Disconnect the nitrogen cylinder and place caps on the right and left valves. After purging, process the range finder in accordance with 3.3.15.1.4. The end box windows shall be purged only when internal moisture is found during inspection (see 4.5.2).

3.3.15.1.5 Mounts, ballistic computer and drive. The ballistic drive coupling and the wedge of the gunner's periscope shall be coated with lubricating oil conforming to grade 10W of MIL-L-21260. All other exposed, unpainted, unplated, metal surfaces of the periscope and telescope mounts, computer and drive, shall be coated with grease conforming to MIL-G-10924 (see 4.5.2).

3.3.15.1.6 Level vial covers. All level vial covers shall be positioned over the vials (see 4.5.2).

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3.3.15.1.7 Exposed optical glass. Any exposed optical glass, not otherwise provided for herein, shall be cleaned, wrapped and taped as specified in MIL-STD-2073-1A (see 4.5.2).

3.3.15.2 Level B. All fire control items shall remain installed. Cleaning and preservation shall be in accordance with 3.3.15.1.2. All ballistic shields shall be secured in the down position (see 4.5.2).

3.3.16 Ventilation.

3.3.16.1 Level A. All access plates and gaskets on the underside of the vehicle and the lower bulkhead door shall be removed. The drain valve in the driver's compartment shall be secured in the open position. Unpainted metal surfaces exposed by disassembly shall be coated with a compound conforming to grade 1 of MIL-C-16173. All access plates, gaskets, and preserved drain plugs shall be packaged in a box conforming to type CF, class weather-resistant of PPP-B-636. The box shall be closed with tape conforming to PPP-T-76, identified as to contents, and securely stowed within the vehicle. Threaded portions exposed by removal of these items shall be coated with a compound conforming to grade 4 of MIL-C-16173. Screens conforming to figures 22 through 28 shall be constructed and installed in the access cover openings and the driver's periscope opening as specified. The driver's center daylight periscope cover shall be blocked in an open position in accordance with figure 26. The following note shall be stenciled on the exterior of the vehicle: "REMOVE SCREENS, INSTALL ACCESS PLATES, COVERS AND GASKETS, REMOVE BLOCKING IN DRIVER'S CENTER PERISCOPE OPENING, AND CLOSE DRAINS BEFORE OPERATING VEHICLE." Stenciling shall be applied using white or yellow paint conforming to MIL-P-13983. Characters shall be at least 0.75 inch in height (see 4.5.2).

3.3.16.2 Level B. All access plates and gaskets on the underside of the vehicle, and the lower bulkhead door, shall be removed, and exposed surfaces shall be preserved in accordance with 3.3.16.1. Removed items shall be processed in accordance with 3.3.16.1. Both drain valves shall be secured in the open position. The driver's center daylight periscope opening cover shall not be blocked open and screened as described in 3.3.16.1. The following shall be stenciled on the exterior of the vehicle: "INSTALL ACCESS PLATES, COVERS, GASKETS, AND CLOSE DRAIN VALVES BEFORE OPERATING VEHICLE" (see 4.5.2).

3.3.17 Fire extinguishers. Fire extinguisher cylinders shall be filled at least 90 percent of the rated full charge. All seals shall be intact. DA Form 253 shall be completed and securely attached to each cylinder. Exterior fire extinguisher handles and protective shields shall be completely sealed with tape conforming to type II of MIL-T-22085. A red warning tag, bearing the information "EXTERIOR FIRE EXTINGUISHER HANDLE SEALED WITH TAPE - REMOVE TAPE BEFORE STARTING ENGINE OR PLACING VEHICLE IN SERVICE" shall be placed in a conspicuous location within the driver's compartment (see 4.5.2 and 6.2).

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3.3.18 Hatches. Rubber seals around hatches shall be coated with powdered talc conforming to type VI, class C of ZZ-T-416 or type TI of MIL-T-50036. During shipment, the loader's hatch shall be closed and secured from the outside with a bolt having a nut drawn up tight and tack welded to the bolt, or with an approved Government padlock. All other hatches shall be closed and locked from the inside. The brakes shall not be set on vehicles having hatches sealed and secured (see 4.5.2).

3.3.19 Backrests, seats, headrests and crash pads. Cushion components shall be cleaned with a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in warm water. Cushions shall be wiped with cloths and saturated with clean water to remove the detergent solution. Care shall be taken not to saturate the cushions with detergent solution or water. After rinsing, the cushions shall be dried. Cushioned components shall be covered with paper conforming to grade B of UU-P-268 with a basis weight of at least 60 pounds. Paper shall be secured with tape conforming to type I of A-A-883 (see 4.5.2).

3.3.20 Dry charged batteries and cables. Dry charged batteries shall be installed in the vehicle battery carrier. Filler cap openings shall be sealed by placing a 2-inch wide by 3-mil thick strip of film conforming to type II of MIL-B-22191 over all filler cap openings with caps removed. The film shall be of sufficient length to allow the film to be depressed into the filler cap opening to the same depth as the filler cap. Filler caps shall be screwed into the filler openings to form a complete seal without damaging the plastic film. The battery cables shall be secured to the battery carrier with a 0.75-inch wide tape conforming to A-A-1685 (see 4.5.2).

3.3.20.1 Electrolyte. Electrolyte shall be packaged, packed, and marked as specified in O-S-801, except that exterior containers shall conform to PPP-B-601 or PPP-B-621. The packed electrolyte shall be stowed with the BII and secured independently to permit separate removal (see 4.5.2).

3.3.21 Tow hooks. Tow hooks and related hardware shall be removed for shipment and packaged in a box conforming to type CF, class weather-resistant of PPP-B-636. The box shall be closed with tape conforming to PPP-T-76, identified as to contents, and securely stowed within the vehicle (see 4.5.2).

3.3.22 BII. The BII shall be packaged, packed, and stowed in accordance with MIL-B-12841 or other documents designated by the responsible agency. The vehicle shall be provided with BII racks assembled and installed in accordance with figure 29. Engine grille bolts which were removed and replaced by longer bolts shall be coated with a compound conforming to grade 4 of MIL-C-16173, wrapped with barrier material conforming to type II, grade A, class 1 of MIL-B-121, placed in a box conforming to PPP-B-636 sealed with tape conforming to PPP-T-76, identified and securely stowed within the vehicle. Sensitive cargo, such as machine guns, shall be stowed and shipped as specified in 3.3.26 (see 4.5.2).

3.3.22.1 BII containers. BII containers shall be identified by vehicle serial number and stored inside buildings when vehicles are in outside storage (see 4.5.2).

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3.3.23 Telephone box. The external telephone box door shall be secured in the closed position with 0.5-inch wide metal strapping conforming to class 1, type I of QQ-S-781, located between the signal light and the door latch handle. The strap tension shall be sufficient to secure the door but shall not damage or distort the box. Tape conforming to type II of MIL-T-22085 or equivalent material, shall be applied to box edges under the strapping to prevent damage to painted surfaces (see 4.5.2).

3.3.24 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted, metal surfaces on the exterior of the vehicle, except the track, shall be coated with a compound conforming to grade 1 of MIL-C-16173. All exposed, unpainted, unplated ferrous metal surfaces on the interior of the vehicle shall be coated with a compound conforming to grade 4 of MIL-C-16173 (see 4.5.2).

3.3.25 Vehicle closure.

3.3.25.1 Level A.

3.3.25.1.1 Closure kit. Unless otherwise specified (see 6.1), each vehicle shall be provided with a vehicle protective closure. The closure shall be fabricated, assembled and installed in accordance with figures 30 through 46. All sharp corners of the framework shall be cushioned with at least a 0.75-inch thickness of cushioning material conforming to PPP-C-1797, secured in place with tape conforming to type II of MIL-T-22085. To prepare the vehicle for loading, the front and rear stiffener rods shall be removed from the cover. The cover shall be rolled away from the front and rear bows to expose the vehicle lifting eye. The double zipper fastener located at the front of the closure is provided for use when loading two vehicles on a railroad car when the gun tube is installed in the gun mount. When two vehicles are loaded on one flat car, the gun tube of one of the vehicles shall be removed from the travel lock, depressed and secured in accordance with figure 47. The gun tube opening in the closure shall be sealed in accordance with the applicable requirements of figure 47, sheet 2. When the gun tube is removed, the gun tube opening in the closure cover shall be sealed in accordance with figures 15 and 16 (see 4.5.2).

3.3.25.1.2 Closure marking. The information "TO PREPARE VEHICLE FOR LOADING, REMOVE FRONT AND REAR STIFFENER RODS FROM COVER, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES - AFTER LOADING, SECURE COVER IN THE ORIGINAL POSITION" shall be stenciled on the front and rear of the closure in characters at least 0.75 inch in height using white enamel conforming to TT-E-529.

3.3.25.1.3 Closure disposition marking. The following information shall be stenciled on the outside, front and rear of the cover: "REUSABLE CLOSURE (COVER AND FRAMEWORK) - DO NOT DESTROY - WHEN REMOVED AND NO LONGER REQUIRED FOR PROTECTION, PACKAGE AND SHIP PER INSTRUCTIONS ON INSIDE OF COVER". The following information shall be stenciled on the inside, front and rear of the cover:

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"PACKAGING AND SHIPPING INSTRUCTIONS:

1. SECURELY BUNDLE LIKE ITEMS.
2. PACKAGE HARDWARE IN CLOTH BAGS.
3. GROUP LARGEST, HEAVIEST ITEMS IN BOTTOM OF BOX.
4. PLACE OTHER ITEMS IN VOID BETWEEN ABOVE ITEMS.
5. FOLD COVER - PLACE ON TOP OF ABOVE ITEMS.
6. IMMOBILIZE ALL ITEMS AND SECURE COVER.
7. SHIP TO (address to be furnished by contracting officer)."

All stenciled characters shall be at least 0.75 inch high using white enamel conforming to TT-E-529 (see 4.5.2).

3.3.25.2 Level B. Vehicle closures shall not be provided on vehicles processed for level B shipment and storage (see 4.5.2).

3.3.26 Security of sensitive cargo. Sensitive cargo may be stowed and shipped within the locked vehicle. When this is not possible, sensitive cargo shall be stored separately under proper security or shipped by a separate mode of transportation under proper security. All hatches shall be secured in accordance with 3.3.18 and the parking brake shall be set in the released position for shipment of the vehicle (see 4.5.2).

3.3.27 Exposed vehicle openings.

3.3.27.1 Level B only. Vehicle openings, such as the cupola machine gun cover opening, cupola shell ejection port, exposed portions of the cupola gun shield, exterior race ring spaces between the turret and hull and between the cupola and turret, and the range finder blister ports, shall be sealed with 4-inch wide tape conforming to type II of MIL-T-22085. Exterior surfaces of the cupola vision blocks shall be gently cleaned with a solution of detergent conforming to P-D-220 and warm water, rinsed with clean water, and dried. Cleaned optical surfaces shall be covered with pieces of lightweight chipboard, approximate size 7-1/2 by 2-1/4 inches, conforming to UU-C-282, or other suitable filler, and sealed to the cupola with type II of MIL-T-22085 tape. All surfaces to which tape is applied shall be clean and dry to assure effective tape adhesion (see 4.5.2).

3.3.28 Cargo straps and fender boxes.

3.3.28.1 Level B only. Fabric retaining straps on exterior cargo racks and in the fender boxes shall be removed, identified with their Army part numbers, and placed in a plastic bag conforming to type II, class B, style 2 of MIL-B-117. The bag shall be closed and stowed inside the vehicle. Fender box covers shall be closed, and handles shall be locked and secured with a suitable gage wire conforming to QQ-W-461 (see 4.5.2).

3.3.29 Marking. In addition to any special marking required in the contract or purchase order, the vehicle shall be marked as specified herein and in accordance with MIL-STD-129 (see 4.5.2).

3.3.29.1 Lifting points. The information "LIFT HERE" with an arrow pointing to the lifting eye shall be stenciled adjacent to each lifting eye using black ink conforming to type I of TT-I-1795, color number 37038 (see 4.5.2).

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3.4 Loading.

3.4.1 Loading flat cars. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of Section 1, General Rules Governing the Loading of Commodities on Open Top Cars, and figures 80 and 81 of Section 6, Rules Governing the Loading of Department of Defense Material on Open Top Cars, publication of the Association of American Railroads (see 4.5.2).

3.4.2 Reprocessing engine after loading.

3.4.2.1 Level A. If the engine is operated while loading or being moved to the loading area, the engine shall be reprocessed as specified in 3.3.8.1. The vehicle cover shall be rolled clear of the engine intake and exhaust to provide air circulation and to prevent damage to the cover. After reprocessing, the closure shall be placed in its original position (see 4.5.2).

3.4.2.2 Level B. If the engine is operated in connection with the movement of the vehicle during loading or unloading, additional processing of the engine will not be required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. The contractor shall maintain records of all inspections performed, and such records shall be readily available for review by the Government representative.

4.2 Classification of inspection:

a. First article inspection (see 4.3).
b. Quality conformance inspections (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.1), the Government shall select one of the first 10 production processed vehicles produced under the production contract for first article inspection. First article samples shall be inspected in accordance with 4.5 for conformance to the requirements of table I. Approval of the first article sample by the

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Government shall not relieve the contractor of his obligation to supply processed vehicles that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

4.4 Quality conformance inspection. Each vehicle shall be inspected in accordance with 4.5.2, 4.5.2.1.1, 4.5.2.2, 4.5.2.3 and 4.5.2.5 for conformance to the requirements of table I. One vehicle each day shall be inspected in accordance with 4.5.2.1.2 and 4.5.2.4.

4.4.1 Failure. Failure of any production vehicle to pass any quality conformance inspection shall be cause for the Government to refuse acceptance of the vehicles, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5 Methods of inspection.

4.5.1 Materials. Except for materials that have been inspected by the Government at source, all materials to be used in processing of vehicles shall be inspected in accordance with the applicable material specifications to determine conformance to 3.2; or certified inspection and laboratory test reports shall be provided which show that furnished materials conform to the applicable material specification. When materials are listed on a Qualified Products List, they shall be obtained from one of the approved sources.

4.5.2 Processing. Conformance to 3.3.1 through 3.3.1.2, 3.3.3 through 3.3.7.2, 3.3.9.1 through 3.3.10.2, 3.3.11.2 through 3.3.12, 3.3.13.2 through 3.3.14, 3.3.14.2 through 3.4.1 and 3.4.2.1 shall be by qualitative observation that the specified processing and material application have been satisfactorily applied.

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TABLE I. Classification of requirements for Levels A and B.

Requirement	Cleaning	Preservation	Packaging/Stowage
Preparation prior to processing			3.3.1
Record forms			3.3.1.1
Processing records			3.3.1.2
Cleaning and drying	3.3.2		
Vehicle interior	3.3.2.1		
Vehicle exterior	3.3.2.2		
Gun	3.3.2.3		
Disassembly			3.3.3
Matchmarking			3.3.3.1
Relubrication		3.3.4	
Battery supports and retainers	3.3.5	3.3.5	
Transmission and final drives 1/		3.3.6	
Engine crankcase 1/		3.3.7	
Engine preservation		3.3.8	
Fuel system and combustion chamber		3.3.8.1.1	
Engine purging		3.3.8.1.2	
Preservation of turbochargers 1/		3.3.8.1.3	
Preservation through exhaust system 1/		3.3.8.1.4	
Preservation through dipstick shroud opening and oil filler tube 1/		3.3.8.1.5	
Personnel heater and heater fuel pump		3.3.9	
Air cleaners		3.3.10	
Fuel tanks		3.3.11	
Turret ring bearing		3.3.12	
105MM gun tube installed	3.3.2.3	3.3.13	3.3.13
Muzzle plug		3.3.13.1	3.3.13.1
Breech mechanism		3.3.13.2	3.3.13.2
Exercising recoil mechanism		3.3.13.3	
Recoil mechanism (after exercising)		3.3.13.4	
Exercising replenisher		3.3.13.5	
Replensiher (after exercising)		3.3.13.6	
Gun mount		3.3.13.7	
105MM gun tube removed from mount			3.3.14
Gun tube	3.3.2.5	3.3.13 and 3.3.14.1	3.3.14.1
Gun mount (after removal of the gun tube)	3.3.14.2	3.3.14.2	3.3.14.2
Breech mechanism (after removal of the gun tube)		3.3.14.3	3.3.14.3
Reinstallation of gun tube in mount	3.3.14.4	3.3.14.4	
Fire control items	3.3.15	3.3.15	
Periscopes	3.3.15.1	3.3.15.1.1	3.3.15.1

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Check the source to verify that this is the current version before use.

TABLE I. Classification of requirements for Levels A and B - Continued.

Requirement	Cleaning	Preservation	Packaging/Stowage
Gunner's periscope and telescope	3.3.15.1.2	3.3.15.1.2	3.3.15.1
Instrument lights		3.3.15.1.3	3.3.15.1
Range finder	3.3.15.1.2	3.3.15.1.4	3.3.15.1.4
Purging of range finder		3.3.15.1.4.1	
Mounts, ballistic computer and drive		3.3.15.1.5	
Level vial covers		3.3.15.1.6	
Exposed optical glass	3.3.15.1.7	3.3.15.1.7	
Ventilation		3.3.16	3.3.16
Fire extinguishers		3.3.17	3.3.17
Hatches		3.3.18	
Backrests, seats, headrests and crash pads	3.3.19		3.3.19
Dry charged batteries and cables			3.3.20
Electrolyte			3.3.20.1
Tow hooks			3.3.21
BII		3.3.22	3.3.22
BII containers			3.3.22.1
Telephone box			3.3.23
Miscellaneous preservation		3.3.24	
Vehicle closure			3.3.25
Security of sensitive cargo			3.3.26
Exposed vehicle openings			3.3.27
Cargo straps and fender boxes			3.3.28
Marking			3.3.29
Lifting points			3.3.29.1
Loading flat cars			3.4.1
Reprocessing engine after loading		3.4.2	

1/ Note: Inspect DD Form 1397.

4.5.2.1 Cleaning.

4.5.2.1.1 Examination. To determine conformance to 3.3.2.1 and 3.3.2.2, the vehicle interior and exterior shall be examined for cleanliness.

4.5.2.1.2 Test. Additionally, conformance to 3.3.2.1 shall be verified by testing in accordance with the applicable provisions of MIL-P-116.

4.5.2.2 Fuel tanks. To determine conformance to 3.3.11.1, the fuel tank interior shall be visually inspected to verify that the specified processing has been accomplished.

4.5.2.3 Gun. To determine conformance to 3.3.2.3, 3.3.13, 3.3.13.1, and 3.3.14.1, the gun shall be visually inspected to determine the condition and effectiveness of processing. If necessary the gun shall be reprocessed and inspected.

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4.5.2.4 Coating compound thickness. To determine conformance to 3.3.14.1, at least one vehicle per day shall be inspected for thickness of coating compound. After 4 hours of drying, four 1-inch square specimens of carrier material shall be cut from areas (flat and contour) selected at random by the Government inspector, and measured for specified thickness.

4.5.2.5 Engine. To determine conformance to 3.3.8.1, the interior of the engine shall be examined for surface coverage. Engines shall be disassembled to the extent necessary to permit visual examination of all surfaces within the combustion chamber. The combustion chamber is all surfaces within the cylinder from, and including, the crown of the piston, to and including the surface of the head within the cylinder. All surfaces within the combustion chamber shall have a "wet" coating of preservative oil such as obtained when an item is dipped or flushed with the oil. The processing method of the approved preserved engine shall be applied to subsequent production vehicles.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Ordering data. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Applicable level of processing (see 1.2).
c. If first article sample is not required (see 3.1).
d. If draining of residual fuel from fuel tanks is required for level B processing (see 3.3.11.2).
e. If vehicle closure kit is not required (see 3.3.25.1).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If first article inspection is not required (see 4.3).

6.2 Safety precautions. Caution should be exercised in handling carbon dioxide fire extinguisher cylinders. Cylinders shall not be dropped, be permitted to strike each other, or handled roughly. Extreme care shall be exercised during the reinstallation operation to avoid tripping the fire extinguisher control system (see 3.3.17).

6.3 Forms. A copy of "The Equipment Log Book" and all required forms will be furnished to the contractor by the Government at least 30 days before shipment of the equipment as required by the contract delivery schedule (see 3.3.1.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
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6.4 Installation and removal of 105MM gun tube. Installation and removal instructions for the 105MM gun tube are specified in Technical Manual TM9-2350-258-34-2.

6.5 Dimensions. Unless otherwise specified herein, dimensions are in inches.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
Army - SM

(Project No. PACK-A317)

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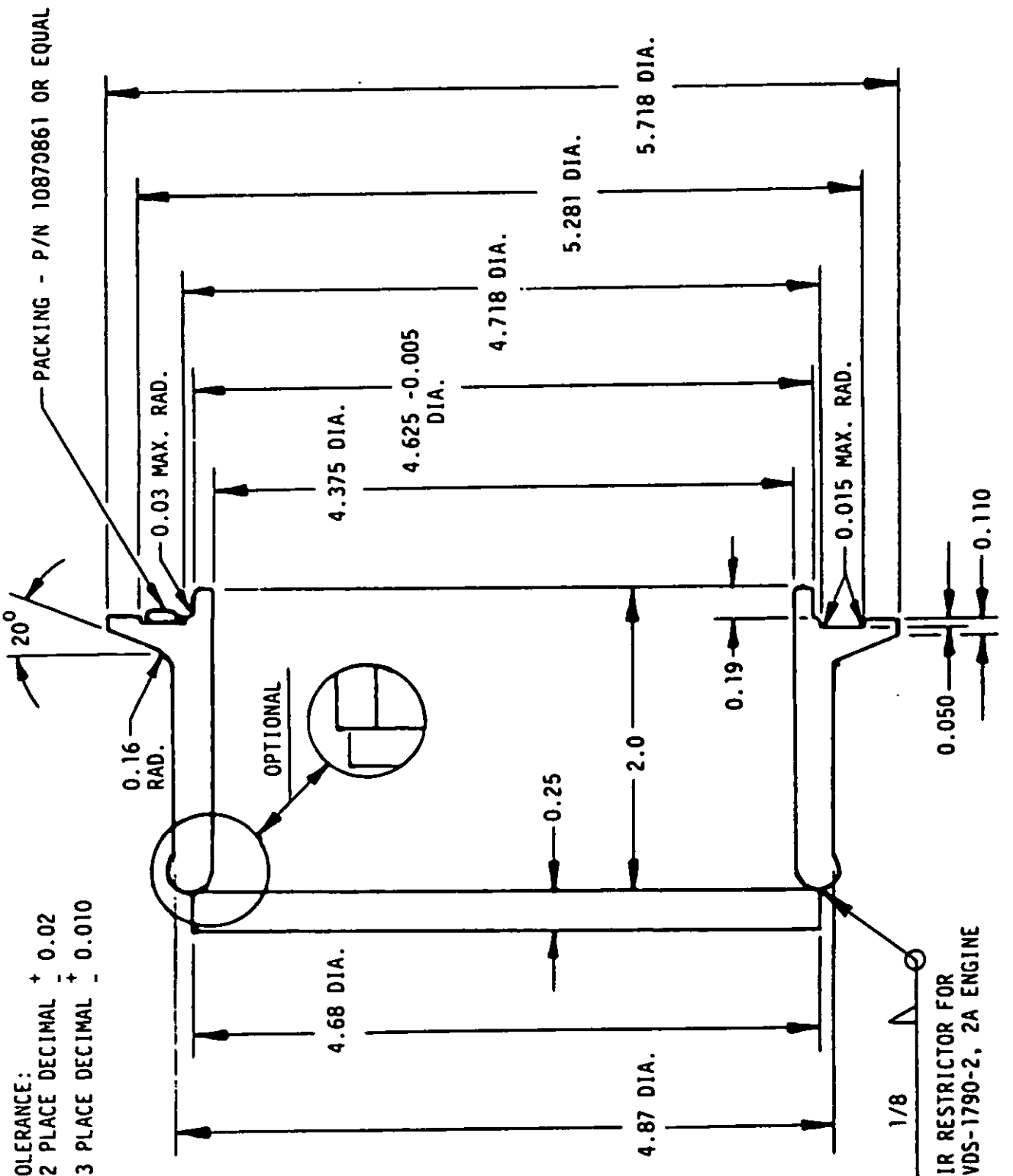


FIGURE 1. Air restrictor cover.

SHEET 1 OF 2

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AIR RESTRICTOR

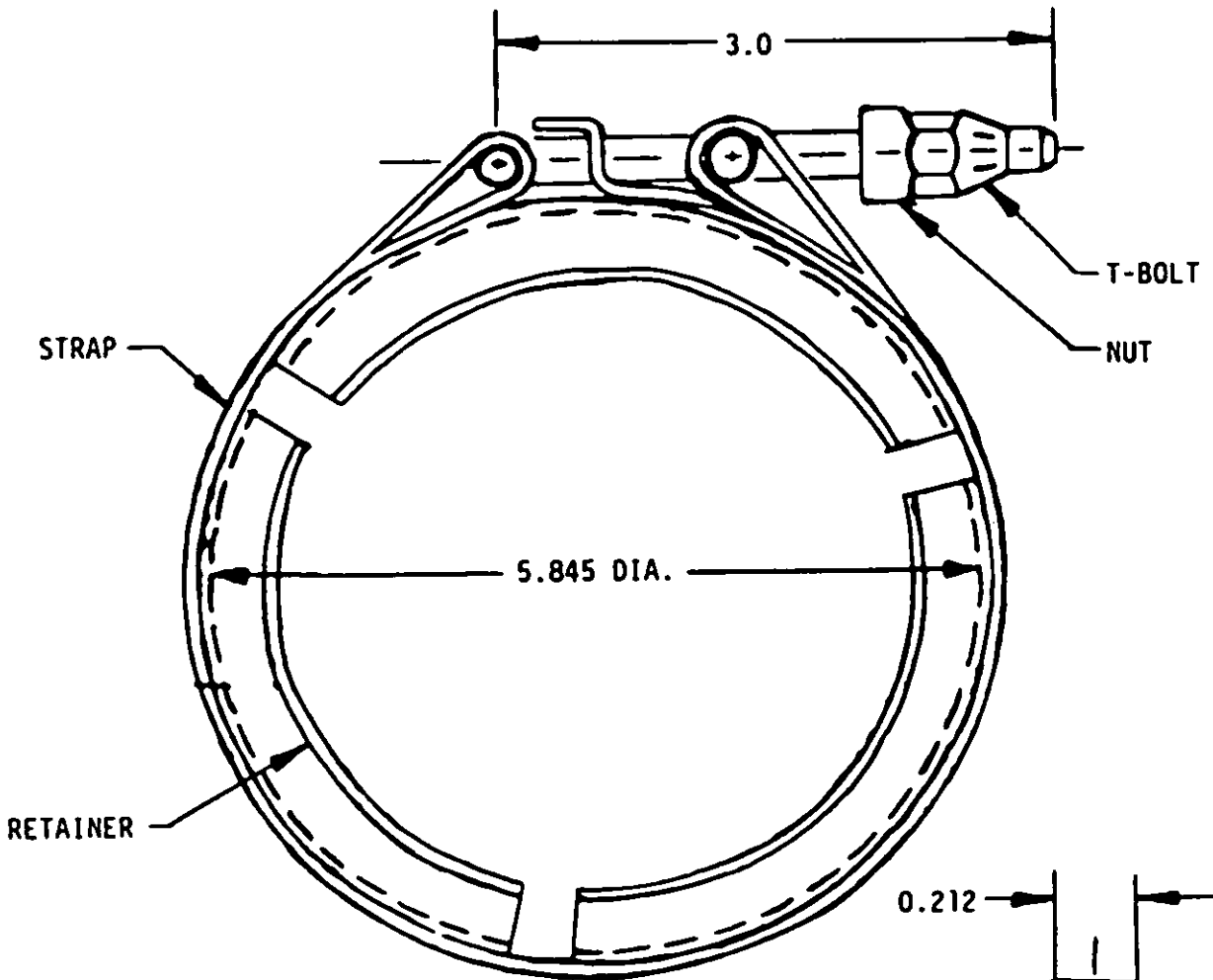
MATERIAL FLANGE ASSEMBLY	ALUMINUM ALLOY CASTING 355, 356 OR ALLCAST CONDITION T6, PER QQ-A-601. OPTIONAL: TUBE, ALUMINUM 5086, PER QQ-A-200/5 OR TUBE, ALUMINUM, COMP 6061, TEMPER T6, TYPE I, PER WW-T-700/6. STEEL, CARBON, TUBING, MT 1010-MTX 1020, TYPE AND CONDITIONAL OPTIONAL PER ASTM A512, A513, OR A519.
PLATE	ALUMINUM PLATE, 6061 TEMPER T6, 5083 OR 5456, PER QQ-A-200. OPTIONAL: STEEL, CARBON PER QQ-S-741 PLATE SHALL BE WELDED TO FLANGE ASSEMBLY. WELDING OF STEEL COMPONENTS SHALL CONFORM TO MIL-STD-1261. WELDING OF ALUMINUM COMPONENTS SHALL CONFORM TO MIL-W-45205, CLASS A AND USE ALUMINUM FILLER METAL CONFORMING TO AWS A5.10. WELDING SHALL ASSURE AN AIR TIGHT JOINT.
PACKING	PACKING SHALL CONFORM TO U.S. ARMY DRAWING NO. 10870861. BOND PACKING TO FLANGE WITH ADHESIVE, CONFORMING TO TYPE II, OF MMM-A-1617. BONDED RUBBER MUST WITHSTAND A 5-POUND PULL PER INCH OF WIDTH PERPENDICULAR TO THE SURFACE.
NOTE:	U.S. ARMY P/N 10870140 HOSE ASSEMBLY MAY BE USED BY REMOVING THE METAL FLANGE ASSEMBLIES FROM THE ENDS AND WELDING A PLATE TO EACH FLANGE.

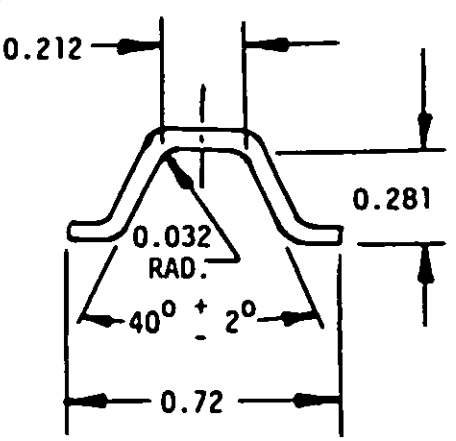
SHEET 2 OF 2

FIGURE 1. Air restrictor cover - Continued.

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Check the source to verify that this is the current version before use.

U.S. ARMY PART NO. 8711310
CLAMP ASSY FOR USE WITH
AIR RESTRICTOR ON
MODEL AV-1790-2
AV-1790-2A ENGINES





TYPICAL RETAINER
CROSS-SECTION

APPROVED SOURCES:

AEROQUIP CORP. (MARMAN PLANT)
11214 EXPOSITION BLVD.
LOS ANGELES, CALIF. 90064

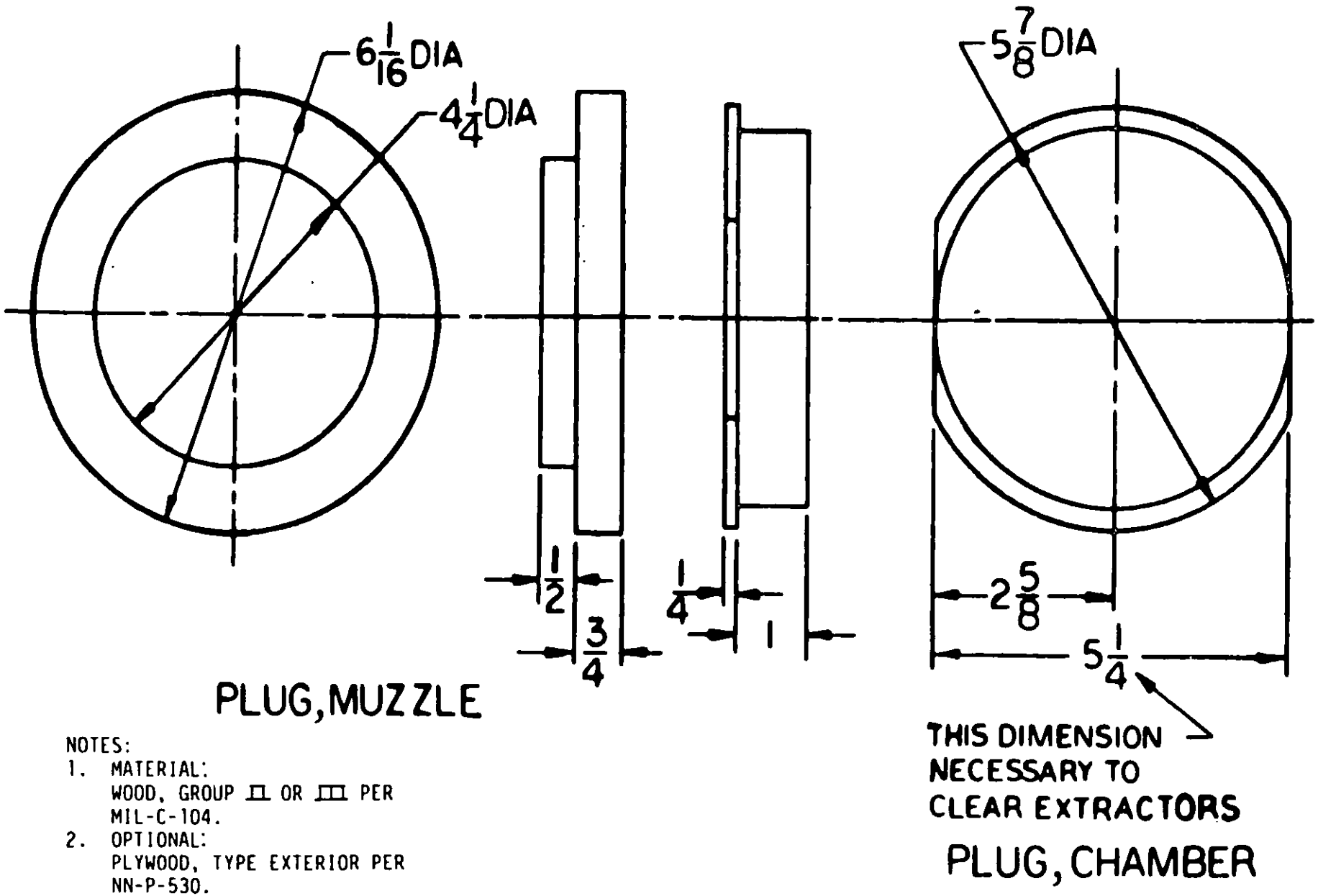
PART NO. 50097-584

NATIONAL UTILITIES CORP.
135 E. RAILROAD AVE.
MONROVIA, CALIF. 91016

PART NO. U4301S3-584

FIGURE 2. Clamp assembly.

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NOTES:
1. MATERIAL:
   WOOD, GROUP II OR III PER MIL-C-104.
2. OPTIONAL:
   PLYWOOD, TYPE EXTERIOR PER NN-P-530.

FIGURE 3. Muzzle and chamber plugs.

MIL-T-62050B(AT)
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

GUN TUBE SHIPPING KEY E1		
KEY NO.	FIG. NO.	DESCRIPTION
E2	5	INSTALLATION, SHIPPING 105 MM GUN TUBE
E3	6	SUPPORT ASSEMBLY, FRONT
E4	7	BRACE, FRONT
E5	8	ANGLE, FRONT
E6	9	BLOCK, FRONT (LOWER)
E7	10	WASHER, FRONT
E8	10	WASHER, REAR
E9	11	SUPPORT ASSEMBLY, REAR
E10	12	BRACE, LEFT REAR
E11	12	BRACE, RIGHT REAR
E12	13	PLATE
E13	14	BLOCK, REAR (LOWER)
E14	15	WIRE
E15	15	INSTALLATION, GUN TUBE OPENING COVER IN CLOSURE COVER
E16	16	COVER
E17	17	PLATE
E18	18	WASHER (8 REQ'D MIN) (SHOWN ON FIGURE 5, SHEETS 3 AND 4)
E19	5	CLIP (SHOWN ON SHEET 5)
E20	5	SCREW (SHOWN ON SHEETS 3 AND 4)
E21	5	BOLT (SHOWN ON SHEET 3)
E22	5	BOLT (SHOWN ON SHEET 4)
E23	5	LOCKWASHER (SHOWN ON SHEETS 3 AND 4)
E24	6 & 11	NUT
E25	5	LOCKWASHER (SHOWN ON SHEETS 3 AND 4)
E26	5	WIRE, CABLE (SHOWN ON FIGURE 5, SHEET 5)
E27	5	WIRE (SHOWN ON FIGURE 5, SHEET 2)
E28	19	DISC
E29	20	BRACE FRONT
E30	21	PLATE (2 REQ'D)
E31	9	BLOCK FRONT (UPPER)
E32	14	BLOCK REAR (UPPER)

FIGURE 4. List of key numbers for gun tube shipping.

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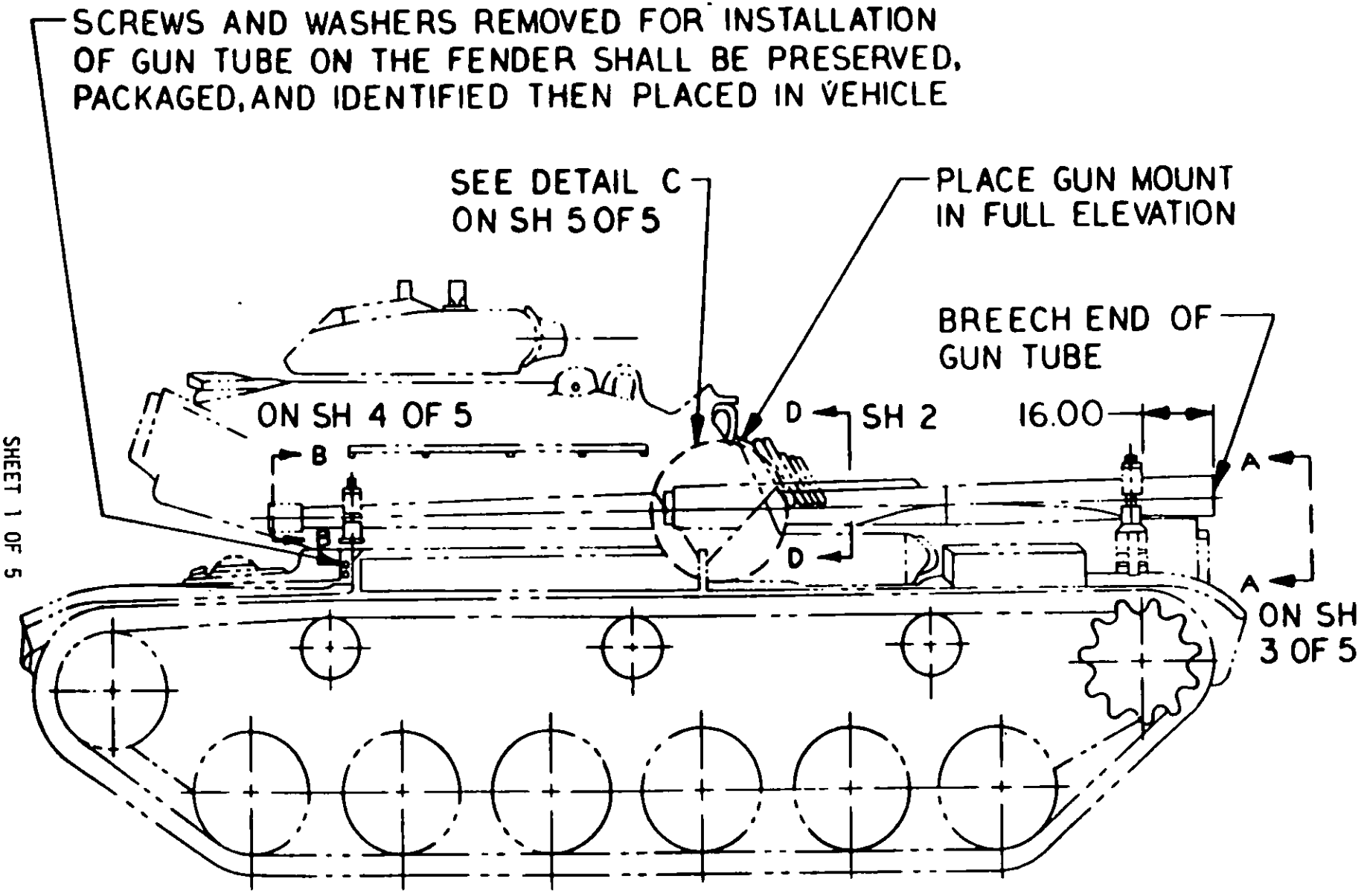
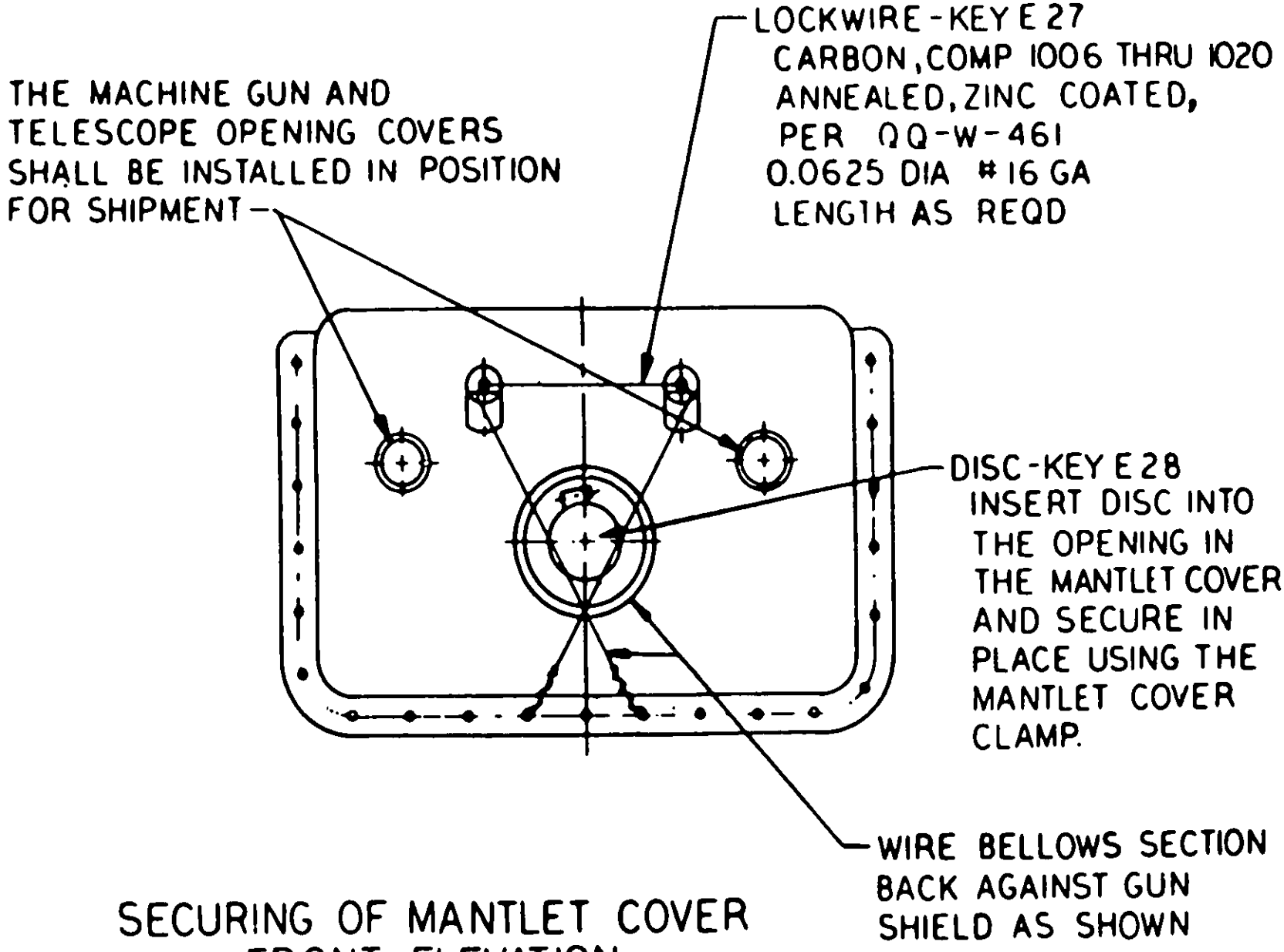


FIGURE 5. Shipping installation for the 105 mm gun tube.

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Check the source to verify that this is the current version before use.



SECURING OF MANTLET COVER
FRONT ELEVATION
KEY E 2

FIGURE 5. Shipping installation for the 105 mm gun tube - Continued.

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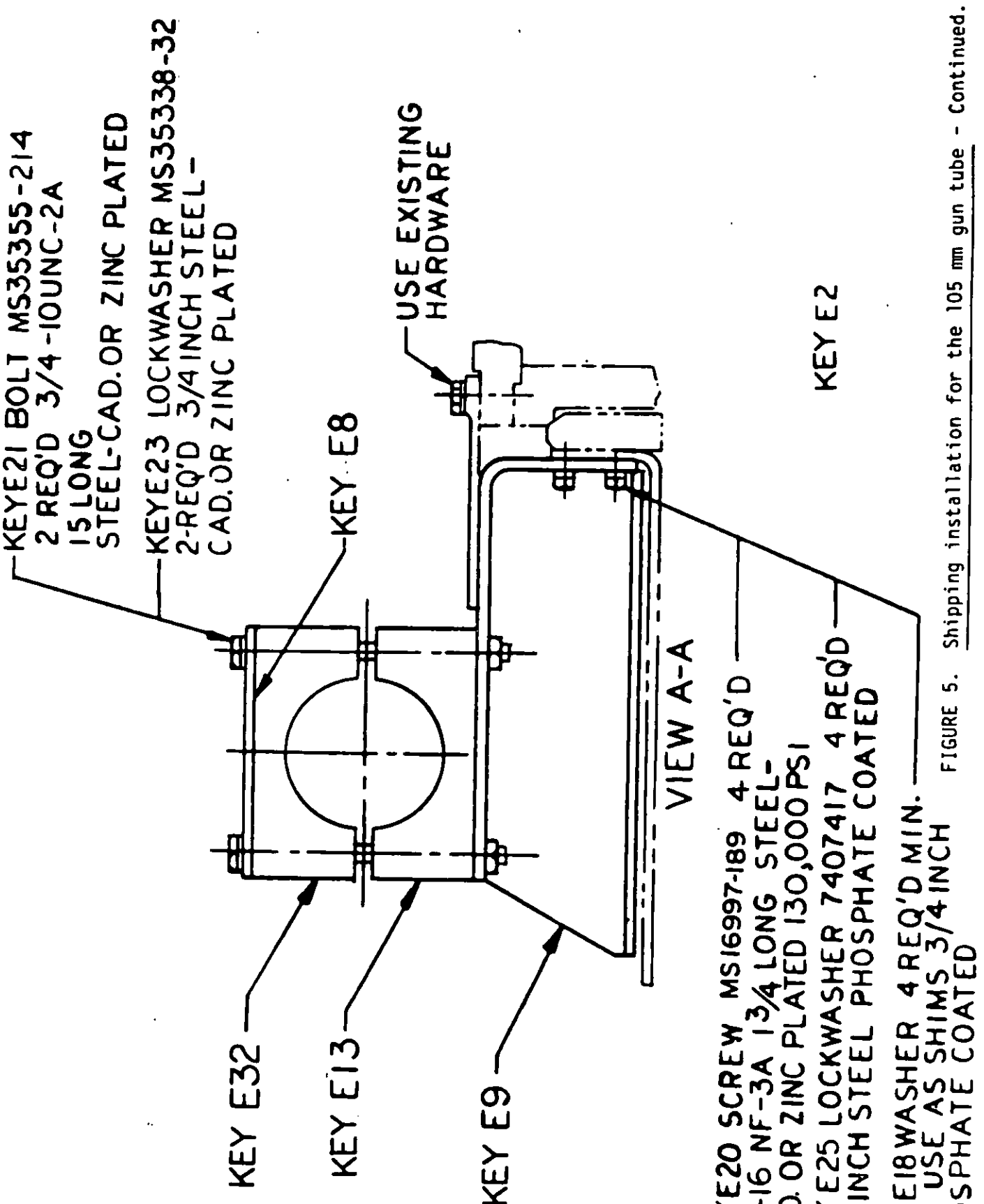


FIGURE 5. Shipping installation for the 105 mm gun tube - Continued.

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Check the source to verify that this is the current version before use.

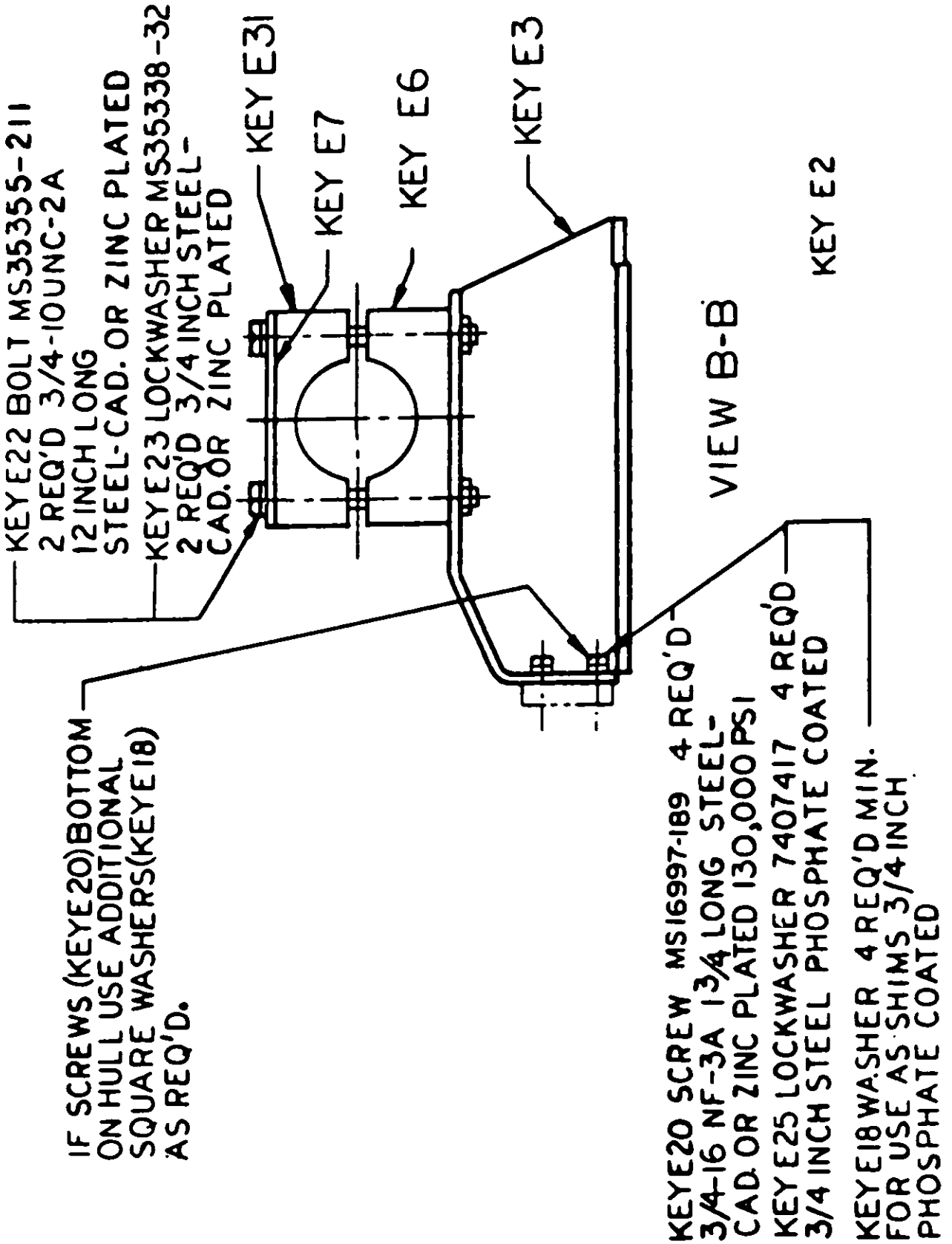


FIGURE 5. Shipping installation for the 105 mm gun tube - Continued.

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Check the source to verify that this is the current version before use.

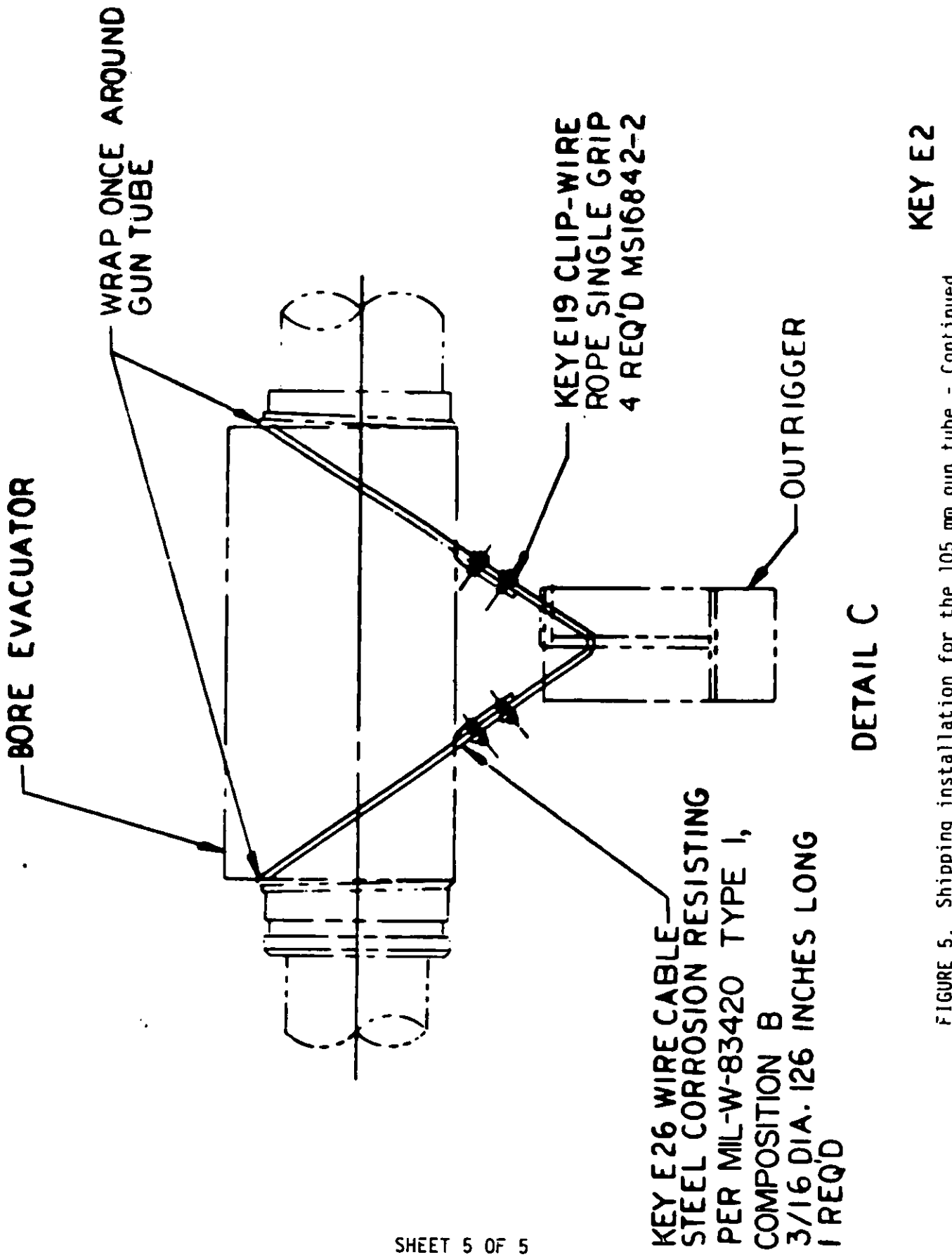
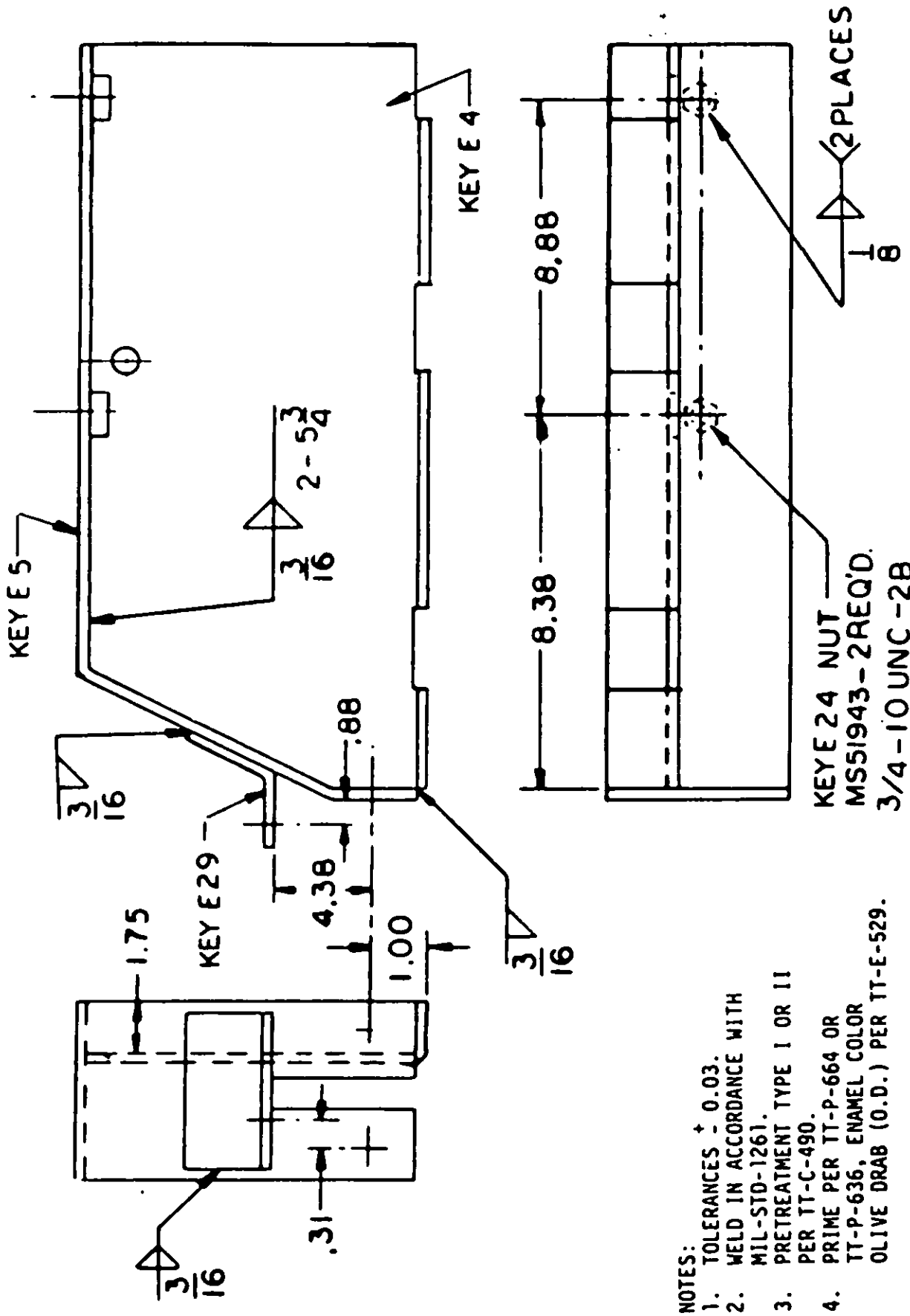


FIGURE 5. Shipping installation for the 105 mm gun tube - Continued.

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NOTES:
1. TOLERANCES ± 0.03.
2. WELD IN ACCORDANCE WITH MIL-STD-1261.
3. PRETREATMENT TYPE I OR II PER TT-C-490.
4. PRIME PER TT-P-664 OR TT-P-636, ENAMEL COLOR OLIVE DRAB (O.D.) PER TT-E-529.

FIGURE 6. Front support assembly.

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Check the source to verify that this is the current version before use.

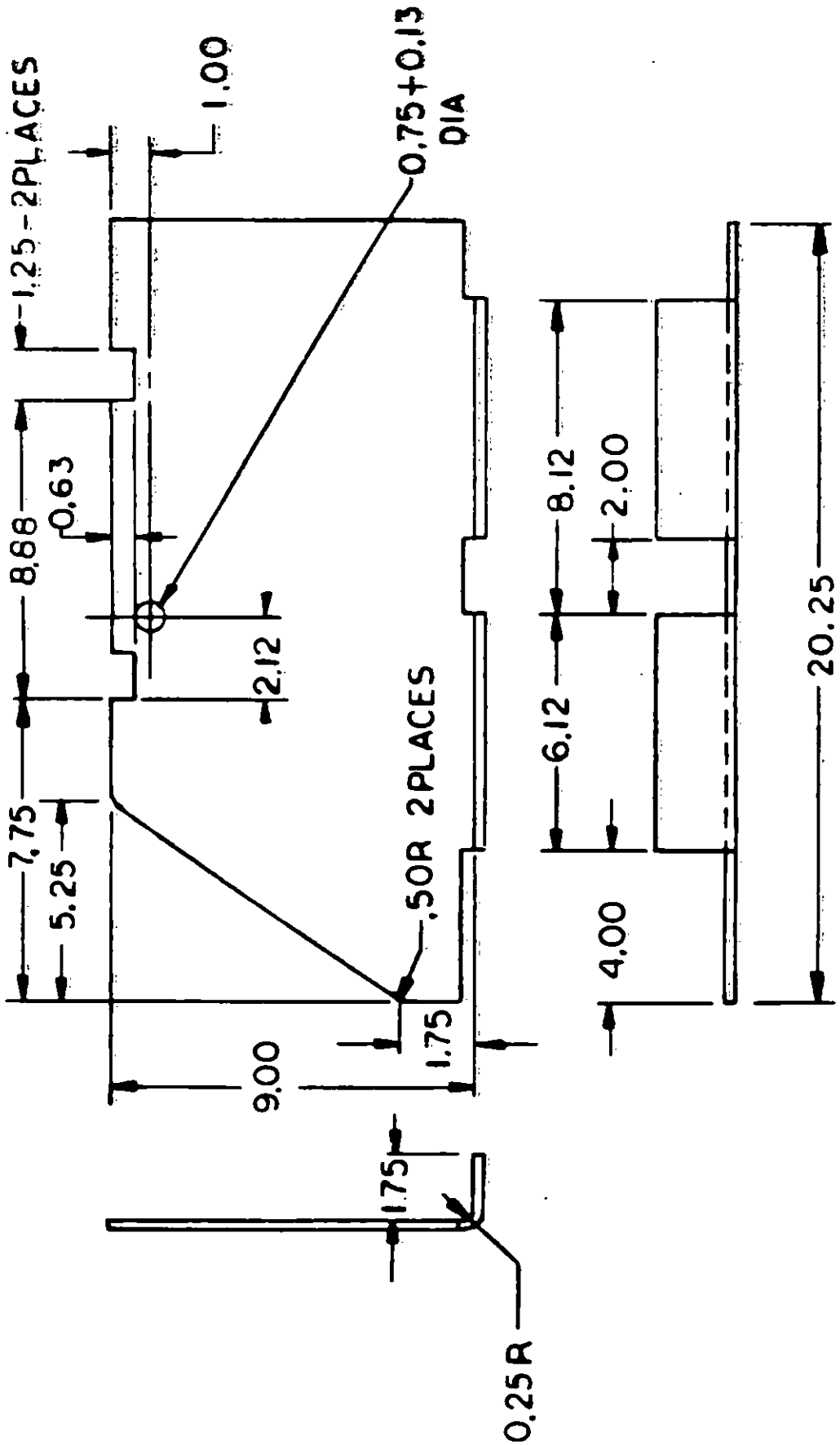
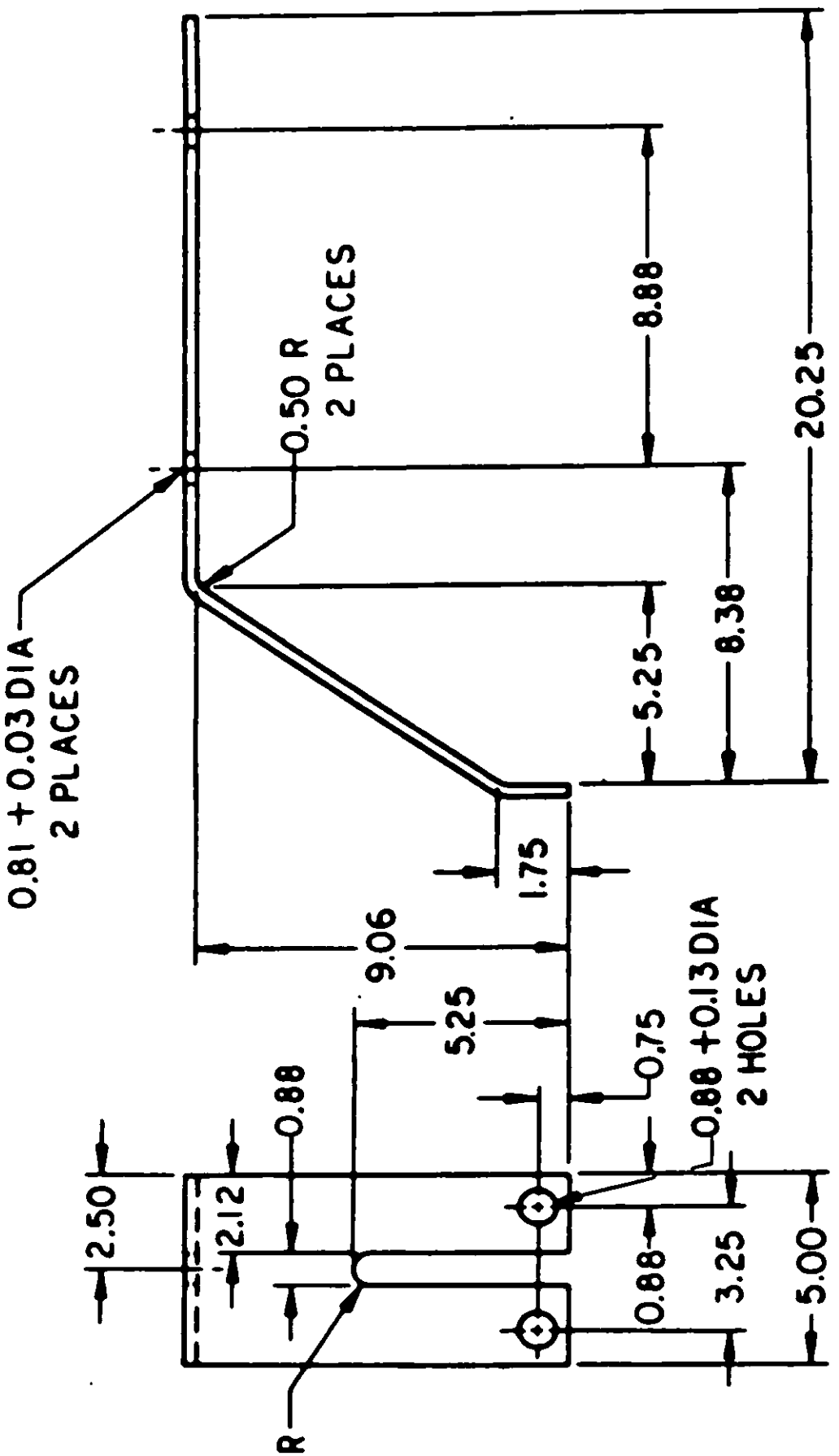


FIGURE 7. Front brace.

NOTES:
1. MATERIAL: STEEL, STRUCTURAL, B OR C, PER QQ-S-741, 1/4 THICK.
2. TOLERANCES $^{+}_{-}$ 0.03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

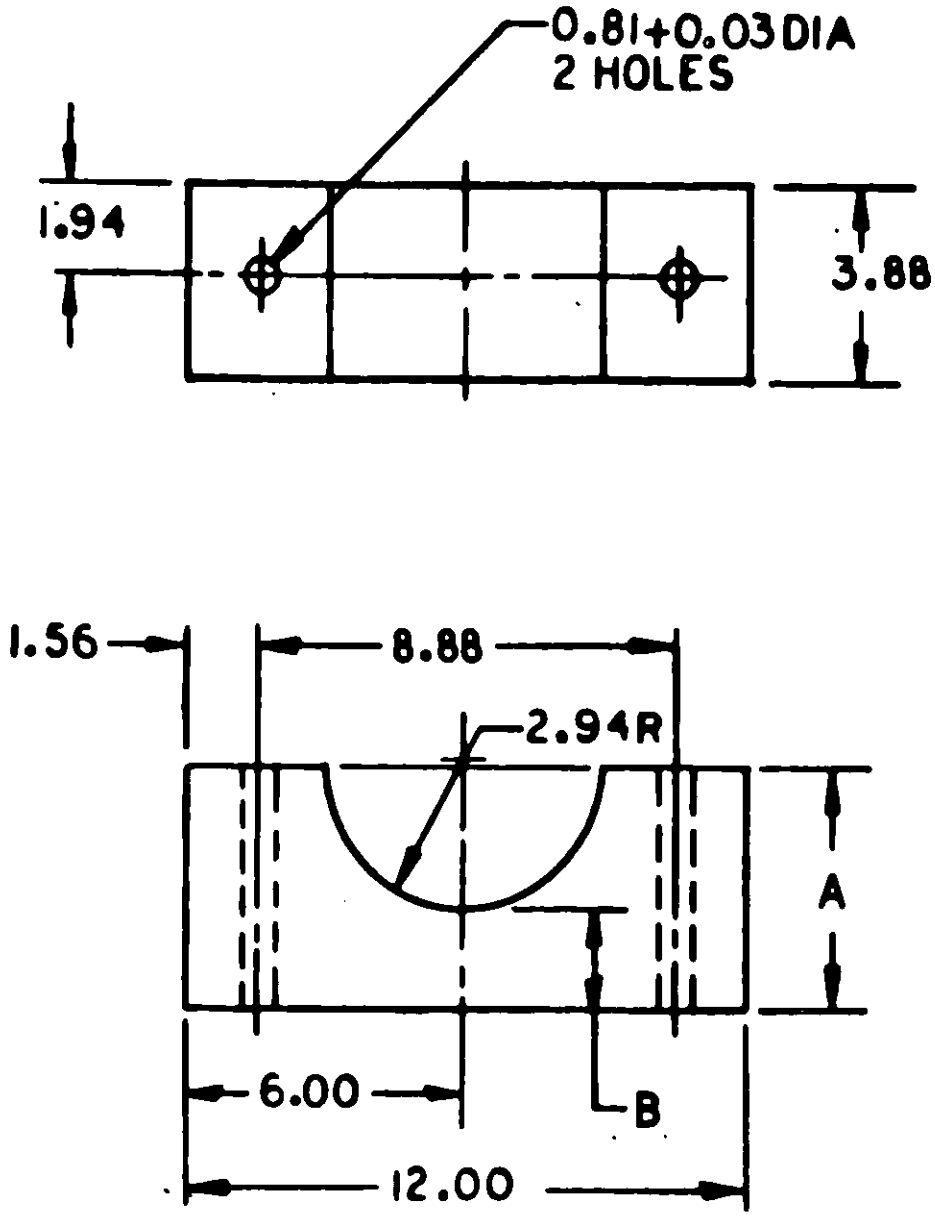


KEY E5

NOTES:
1. MATERIAL: STEEL, STRUCTURAL, GRADE B OR C, PER QQ-S-741, $\frac{5}{16}$ THICK.
2. TOLERANCES $^{+}_{-}$ 0.03.

FIGURE 8. Front angle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



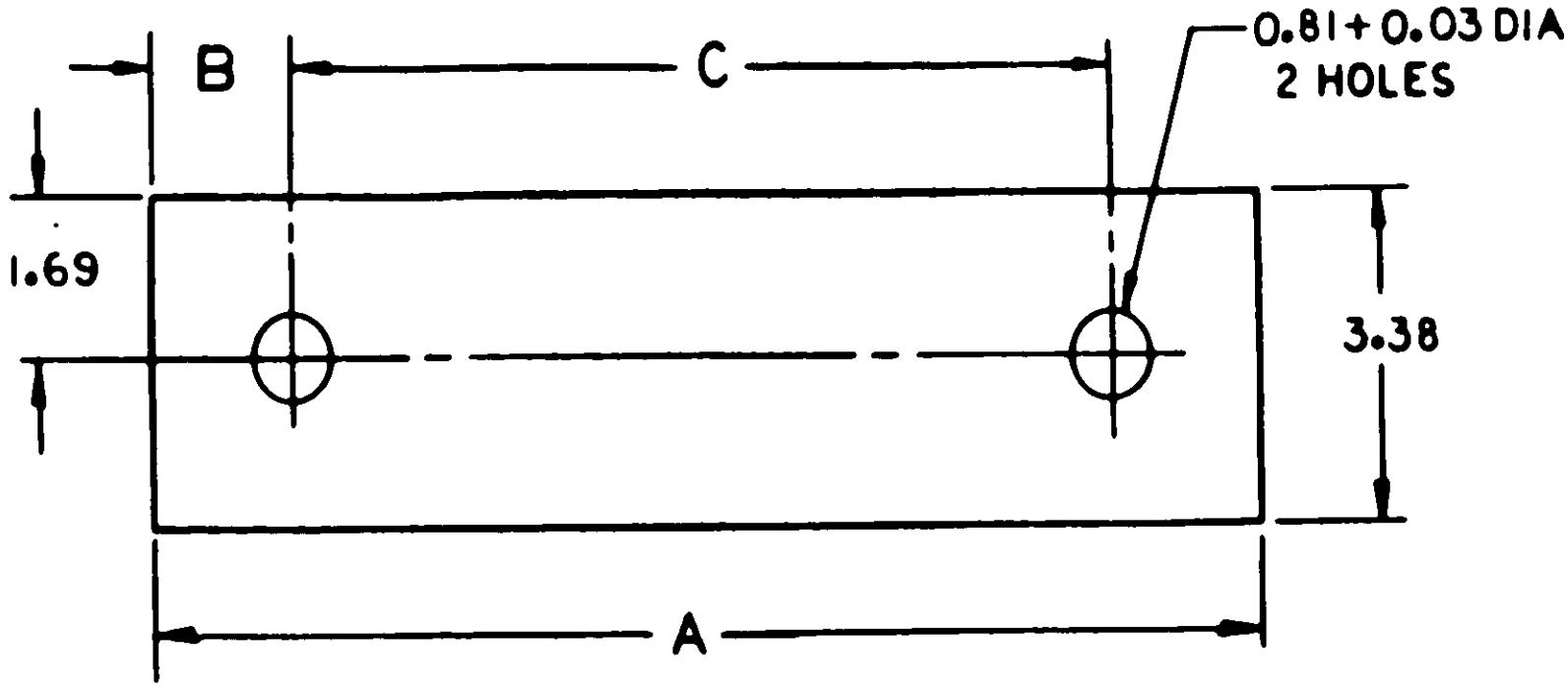
BLOCK	A	B
KEY E6 FRONT LOWER	5.83	2.69
KEY E31 FRONT UPPER	3.75	1.06

NOTES:
1. MATERIAL: WOOD, CLASS A, PER MIL-W-3912.
2. ENAMEL COLOR O.D. PER TT-E-529.
3. TOLERANCES ± 0.03

FIGURE 9. Front block.

KEY E6 & E31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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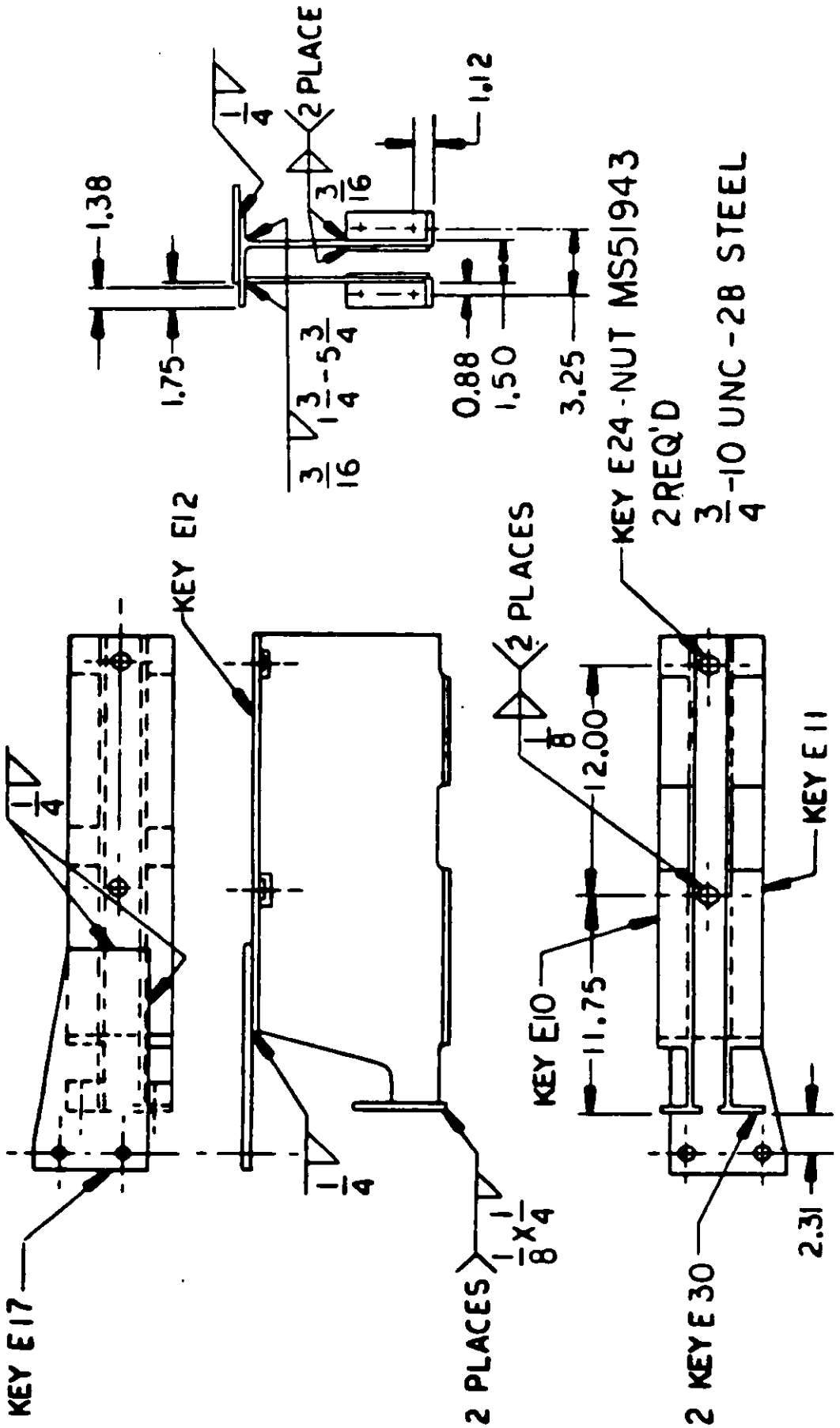
NOTES:
1. MATERIAL:
   STEEL, COMMERCIAL QUALITY PER QQ-S-698, 0.164 THICK.
2. TOLERANCES ± 0.03.
3. PRETREATMENT TYPE I OR III PER TT-C-490.
4. PRIME PER TT-P-664 OR TT-P-636. ENAMEL COLOR O.D. PER TT-E-529.

KEY NO.	A	B	C
KEY E7 FRONT	12.00	1.56	8.88
KEY E7 REAR	15.00	1.50	12.00

KEY E7 & E8

FIGURE 10. Washer.

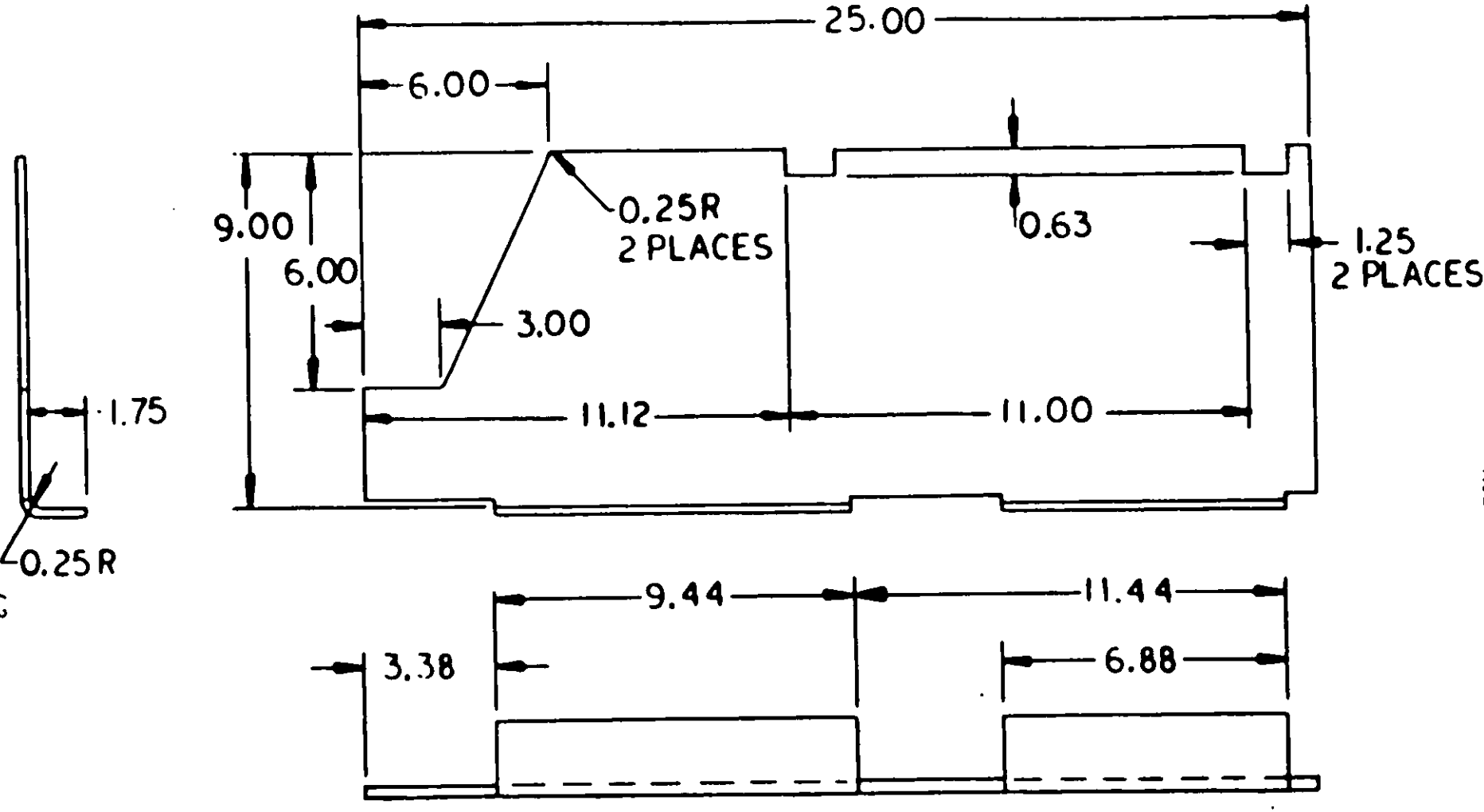
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Check the source to verify that this is the current version before use.



NOTES:
1. PRETREATMENT TYPE I OR II PER TT-C-490.
2. PRIME PER TT-P-664 OR TT-P-636
   ENAMEL COLOR O.D. PER TT-E-529.
3. WELD IN ACCORDANCE WITH MIL-STD-1261, CLASS 1.
4. TOLERANCES ± 0.03.

FIGURE 11. Rear support assembly.

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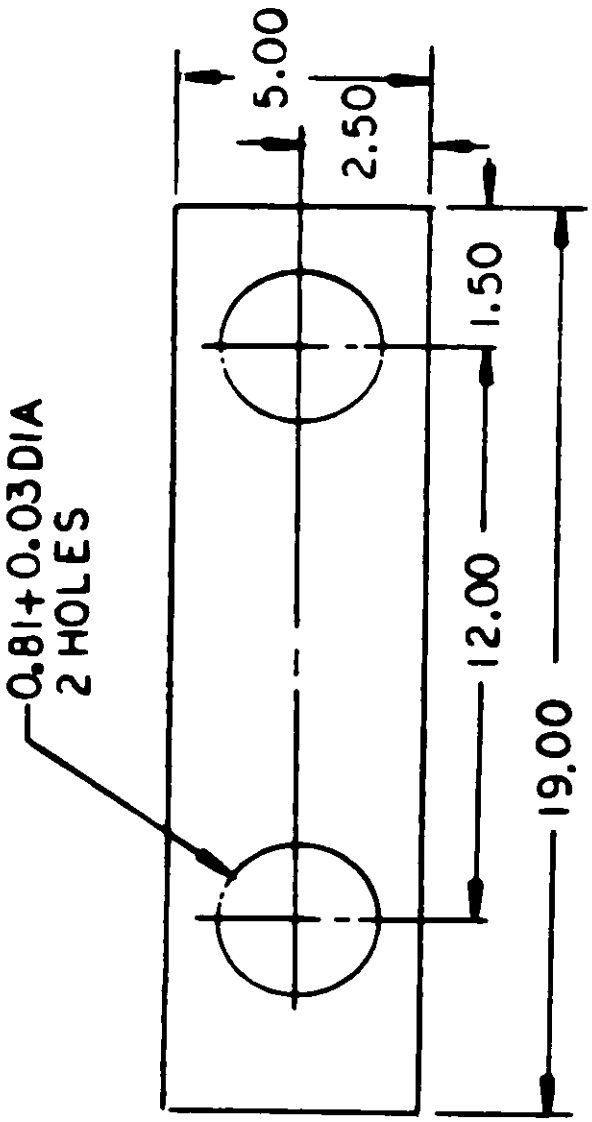


BRACE REAR
KEY E10 (SHOWN)
KEY E11 (OPPOSITE)

NOTES:
1. MATERIAL: STEEL, STRUCTURAL, TYPE II, PER QQ-S-741, 1/4 THICK.
2. TOLERANCES ± 0.03.

FIGURE 12. Rear brace.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

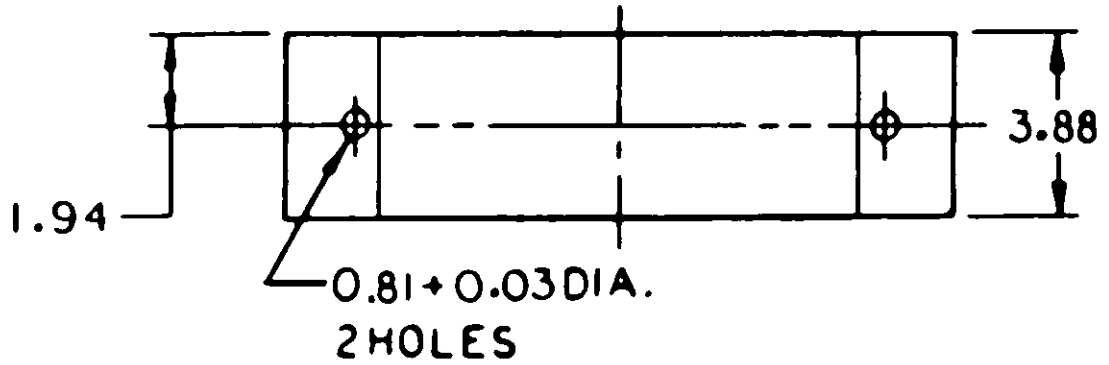


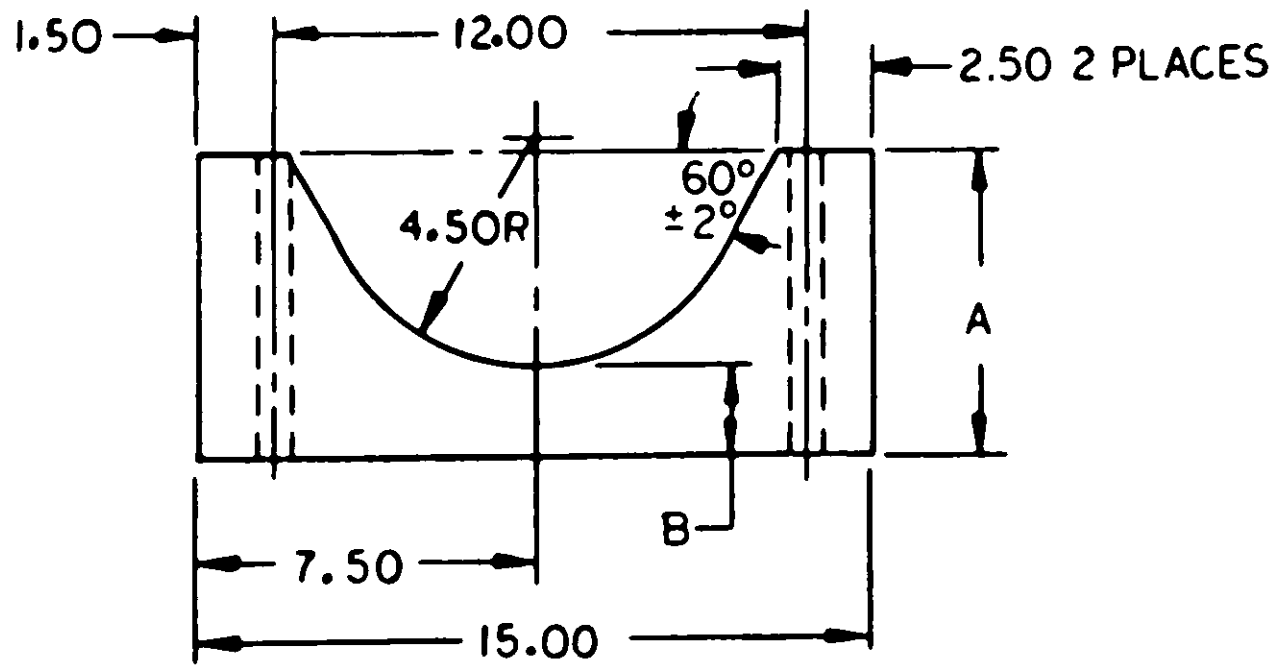
KEY E12

NOTES:
1. MATERIAL:
   STEEL, STRUCTURAL, TYPE II,
   PER QQ-S-741, 5/16 THICK.
2. TOLERANCES ± 0.03.

FIGURE 13. Plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.





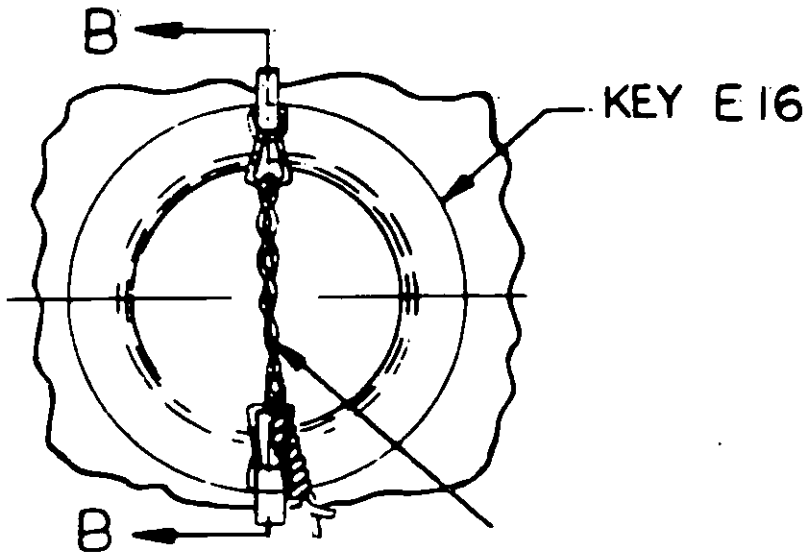
NOTES:
1. MATERIAL: WOOD, CLASS A, PER MIL-W-3912.
2. ENAMEL COLOR O.D., PER TT-E-529.
3. TOLERANCES ± 0.03.

BLOCK	A	B
KEY E 13 REAR LOWER	6.94	2.69
KEY E 32 REAR LOWER	5.31	1.06

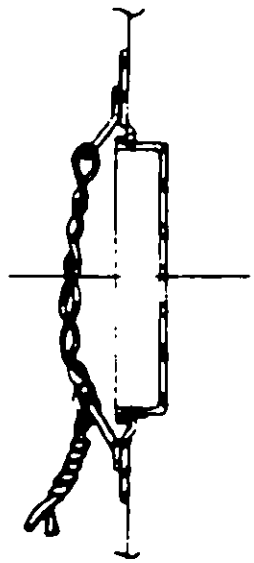
KEY E 13 & E 32

FIGURE 14. Rear block.

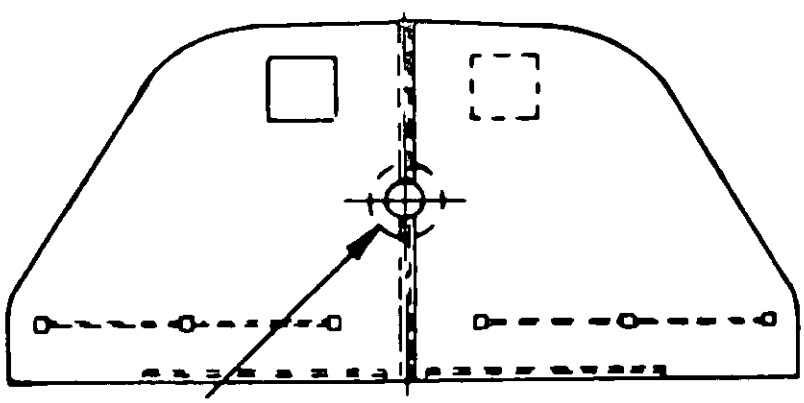
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



DETAIL A



SECTION B-B



SEE DETAIL A

LOCK WIRE - KEY E14
CARBON, COMP 1006
THRU 1020 ANNEALED,
ZINC COATED PER
QQ-W-461, 0.0625 DIA.
#16 GA 20 IN. LENGTH

KEY E15
INSTALLATION, GUN TUBE
OPENING COVER IN
CLOSURE COVER.

KEY E14 & E15

FIGURE 15. Lockwire and opening cover in closure cover.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

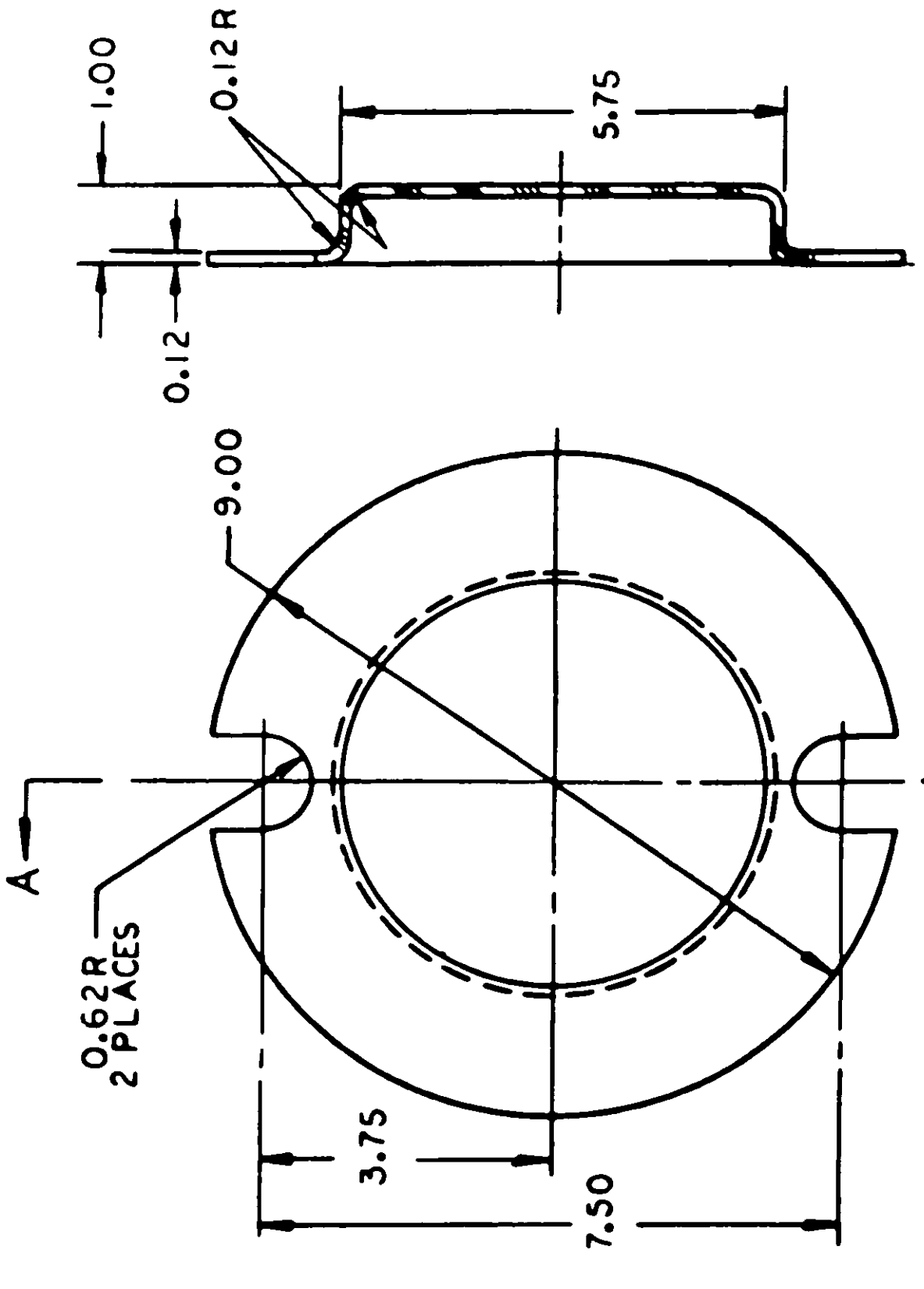
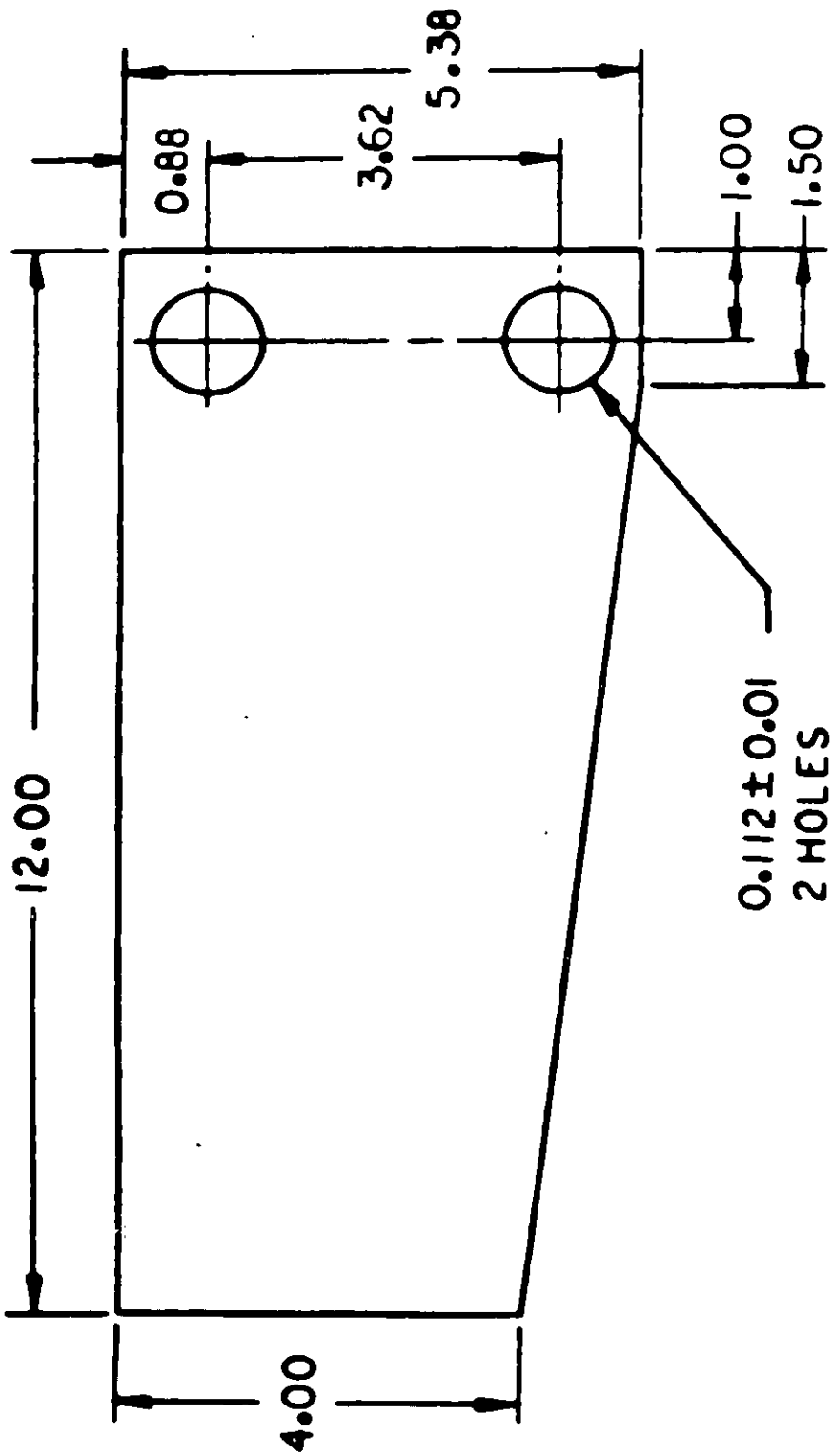


FIGURE 16. Cover.

NOTES:
1. MATERIAL: COVER, NYLON, GROUP 1, CLASS 1, GRADE 1 OR 2 OF ASTM D4066, COLOR O.D.
2. TOLERANCES ± 0.06.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

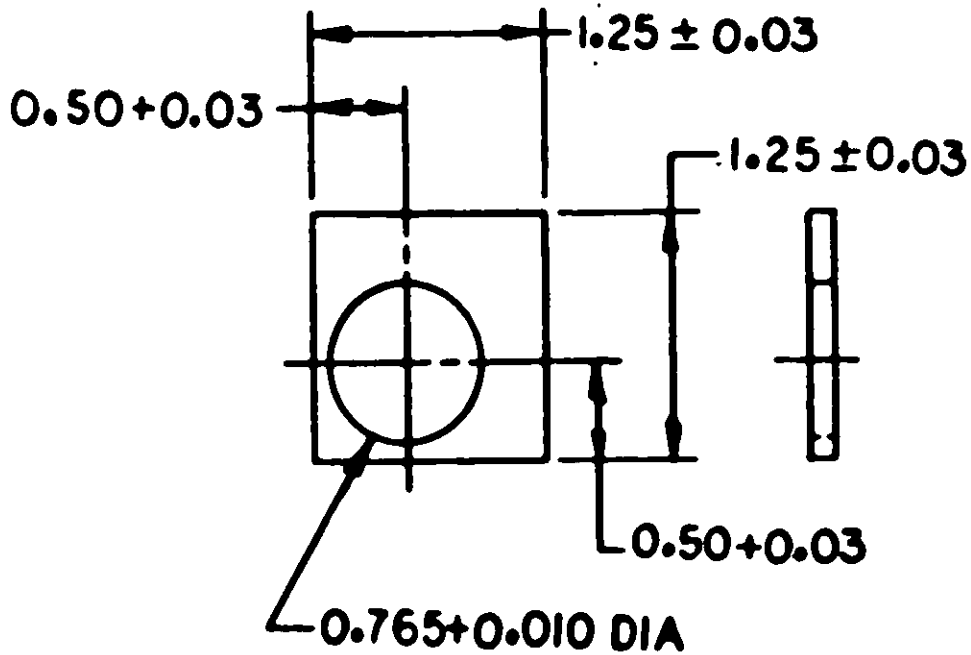


KEY E17

NOTES:
1. MATERIAL:
   STEEL, STRUCTURAL, TYPE II,
   PER QQ-S-741, 5/16 THICK.
2. TOLERANCE ± 0.03.

FIGURE 17. Plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

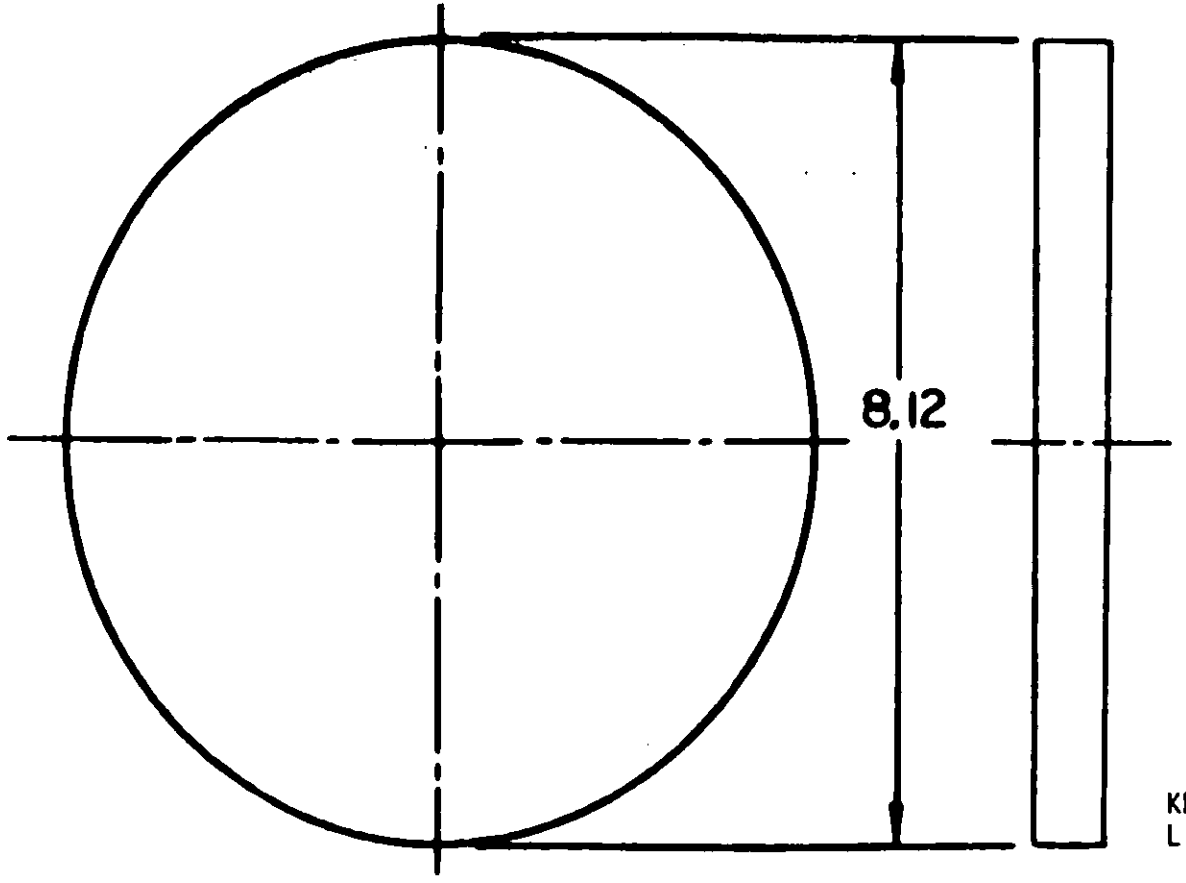


KEY E18

NOTES:
1. MATERIAL:
   STEEL, SPRING UNTEMPERED PER
   ASTM A682
   0.118 THICK.
2. HEAT TREATMENT:
   QUENCH AND TEMPER HARDNESS 42/47
   ROCKWELL C.
3. PRETREATMENT TYPE I OR ⊔⊔ PER
   TT-C-490.
4. PRIME PER TT-P-664 OR TT-P-636,
   ENAMEL COLOR O.D. PER TT-E-529.

FIGURE 18. Washer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



KEYS E19 THRU E27 ARE LISTED ON FIG. 4

KEY E28

NOTES:
1. MATERIAL: PLYWOOD, GROUP B, TYPE C, PER NN-P-530, 3/4 THICK.
2. FINISH: SEAL PER MIL-W-13518, ENAMEL COLOR O.D. PER TT-E-485.

FIGURE 19. Disc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

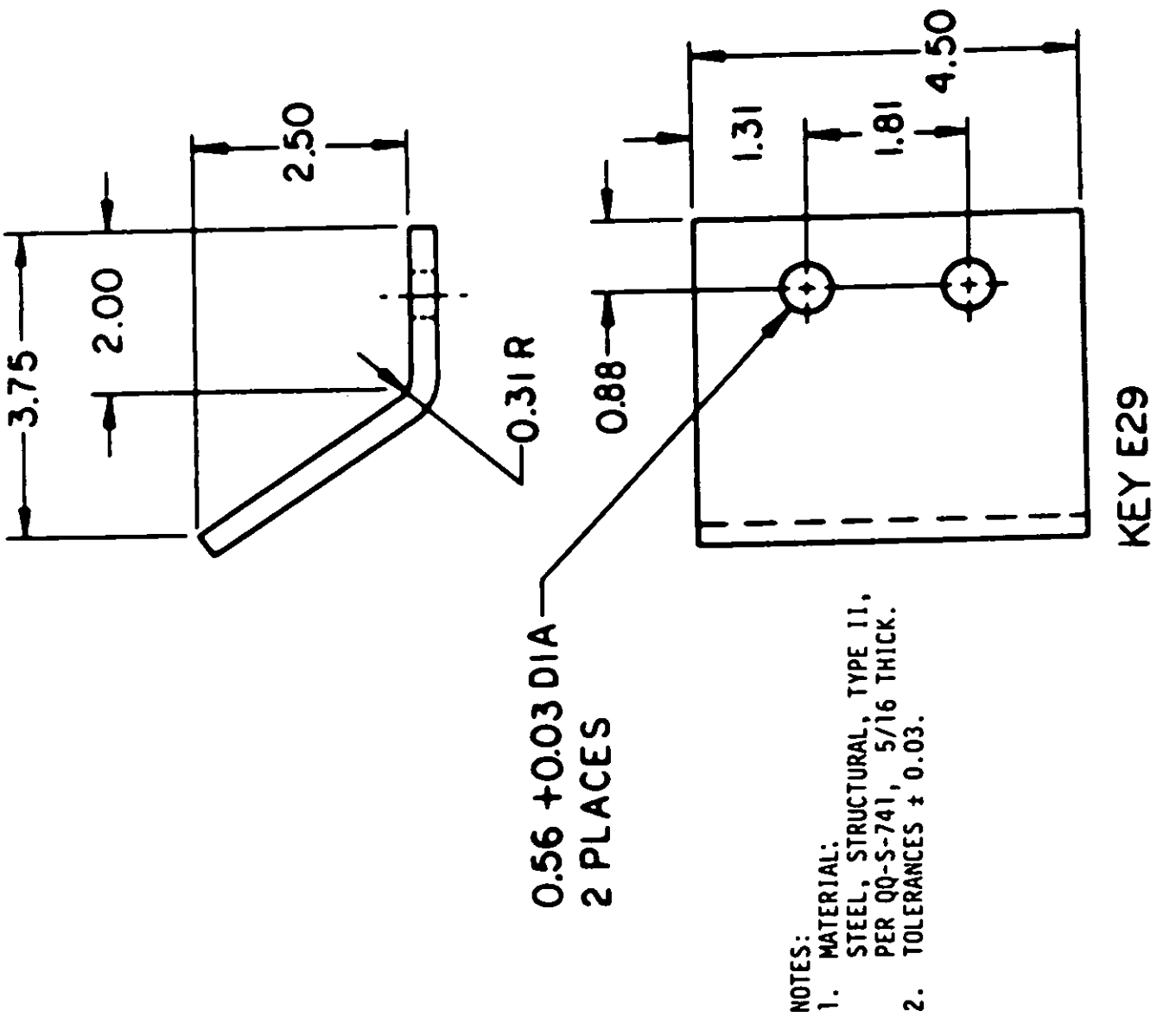
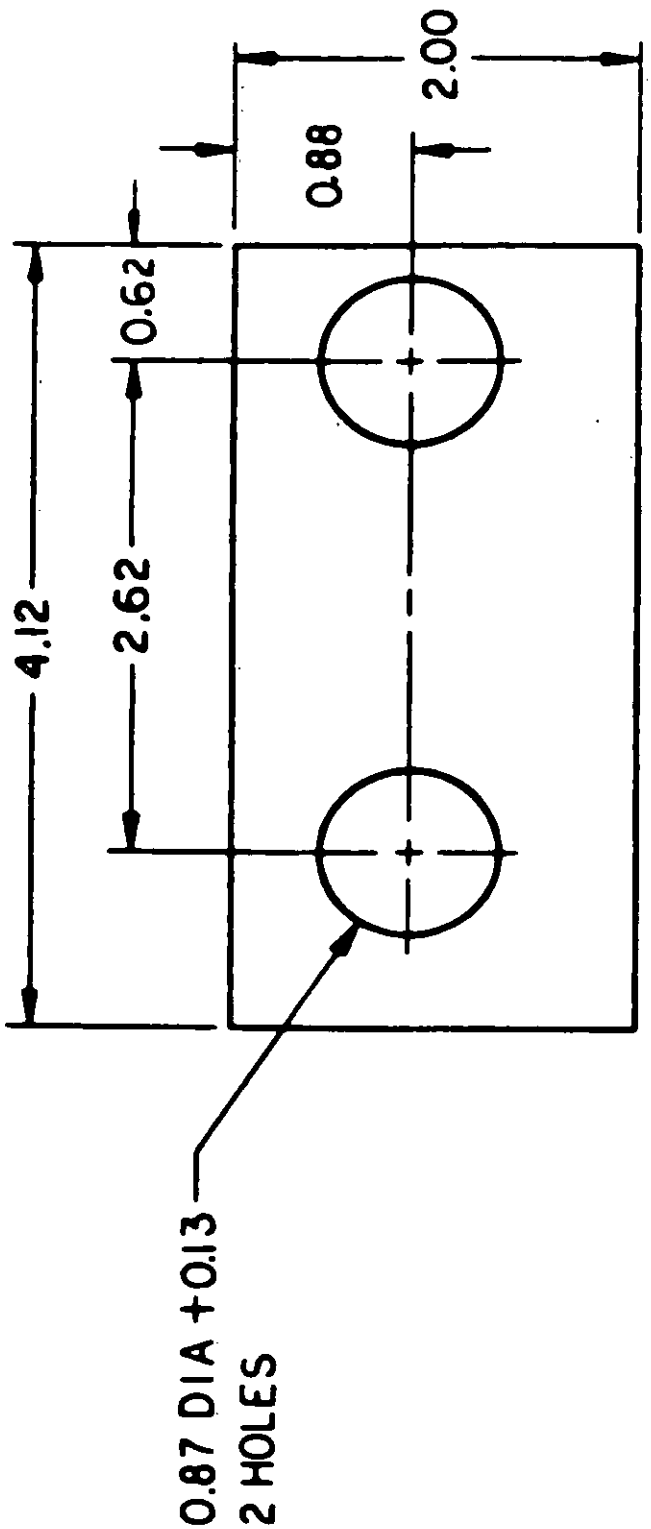


FIGURE 20. Front brace.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



KEY E30
2 REQ'D

NOTES:
1. MATERIAL:
   STEEL, STRUCTURAL, TYPE II, PER
   QQ-S-741, 1/4 THICK.
2. TOLERANCES ± 0.03.

FIGURE 21. Plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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INSTALLATION SCREENS IN BRAKE ACCESS OPENING

ITEM NO.	KEY NO.	NO. REQ'D	NAME	MAT'L	STOCK SIZE
1	B3	2	SCREEN	STEEL	GALVANIZED WIRE 0.047 DIA. 4X4 MESH
2	--	24	WASHER	STEEL	3/8 I.D. PLAIN
3	--	12	SPACER	STEEL TUBING	9/16 O.D. X 0.083 WALL X 7/8 LG.

INSTALLATION SCREENS IN FUEL TANK ACCESS OPENING

ITEM NO.	KEY NO.	NO. REQ'D	NAME	MAT'L	STOCK SIZE
4	B4	2	SCREEN	STEEL	GALVANIZED WIRE 0.047 DIA. 4X4 MESH
5	--	8	SPACER	STEEL TUBING	11/16 O.D. X 0.065 WALL X 9/16 LG.
6	--	16	WASHER	STEEL	1/2 I.D. PLAIN

KEY B2
SHEET 1 OF 2

FIGURE 22. Installation of screens in brake and fuel tank access openings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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INSTALLATION OF SCREEN IN DRIVERS PERISCOPE OPENING

ITEM NO.	KEY NO.	NO. REQ'D	NAME	MAT'L	STOCK SIZE
7	B7	1	SCREEN	STEEL	GALVANIZED WIRE 0.047 DIA. 4X4 MESH
8	B7	1	BLOCK	PLYWOOD	1/2 THICK
9	—	4	STAPLES	STEEL	1/2 LONG
10	B6	AS REQ'D	LOCKWIRE	STEEL	0.0625 DIA (#16 GA)

INSTALLATION OF SCREEN AROUND DRIVERS COMPARTMENT DRAIN VALVE

ITEM NO.	KEY NO.	NO. REQ'D	NAME	MAT'L	STOCK SIZE
11	B8	1	SCREEN	STEEL	GALVANIZED WIRE 0.047 DIA. 4X4 MESH

KEY B2
SHEET 2 OF 2

FIGURE 22. Installation of screws in brake and fuel tank access openings - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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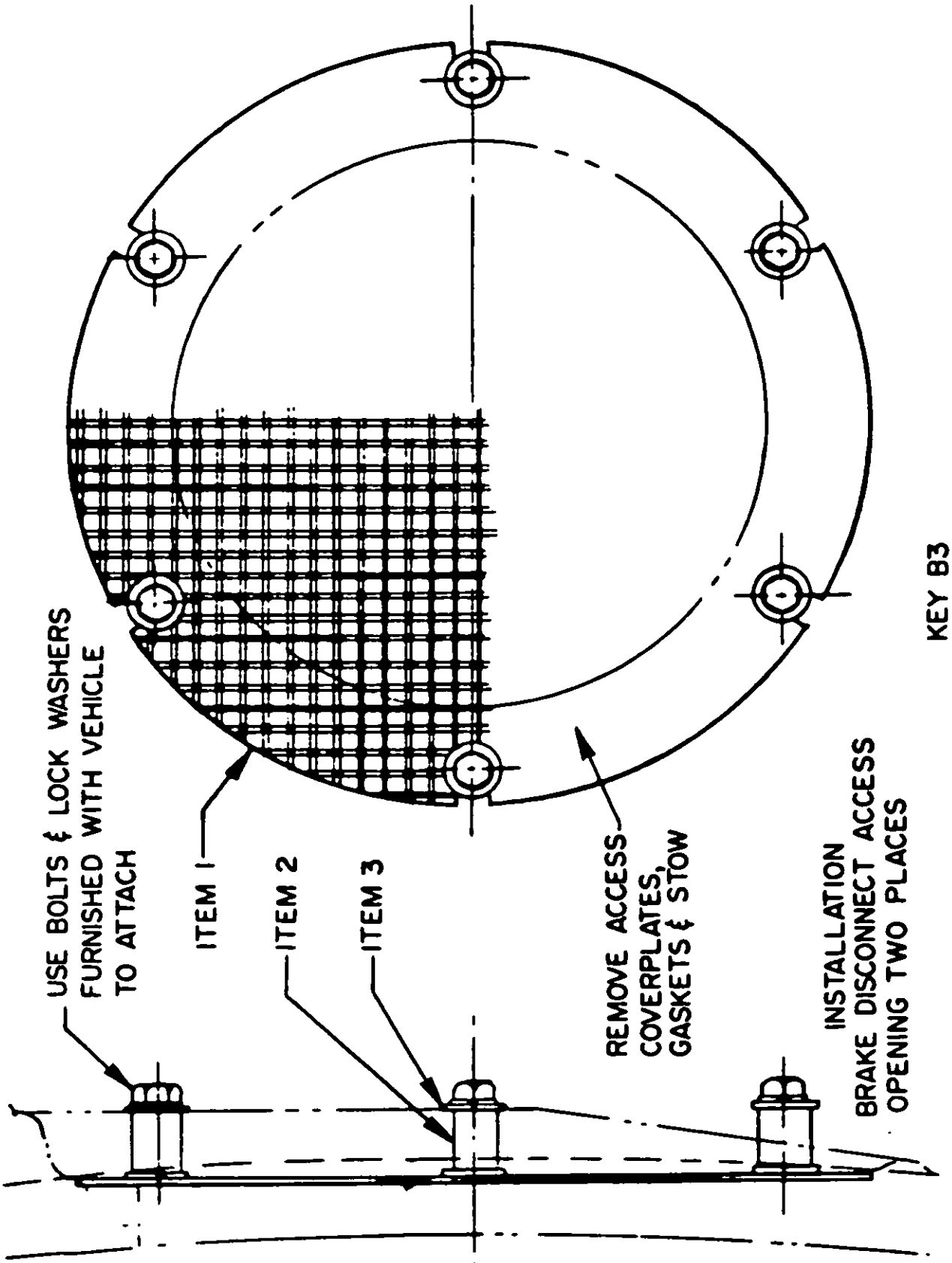


FIGURE 23. Brake disconnect access.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

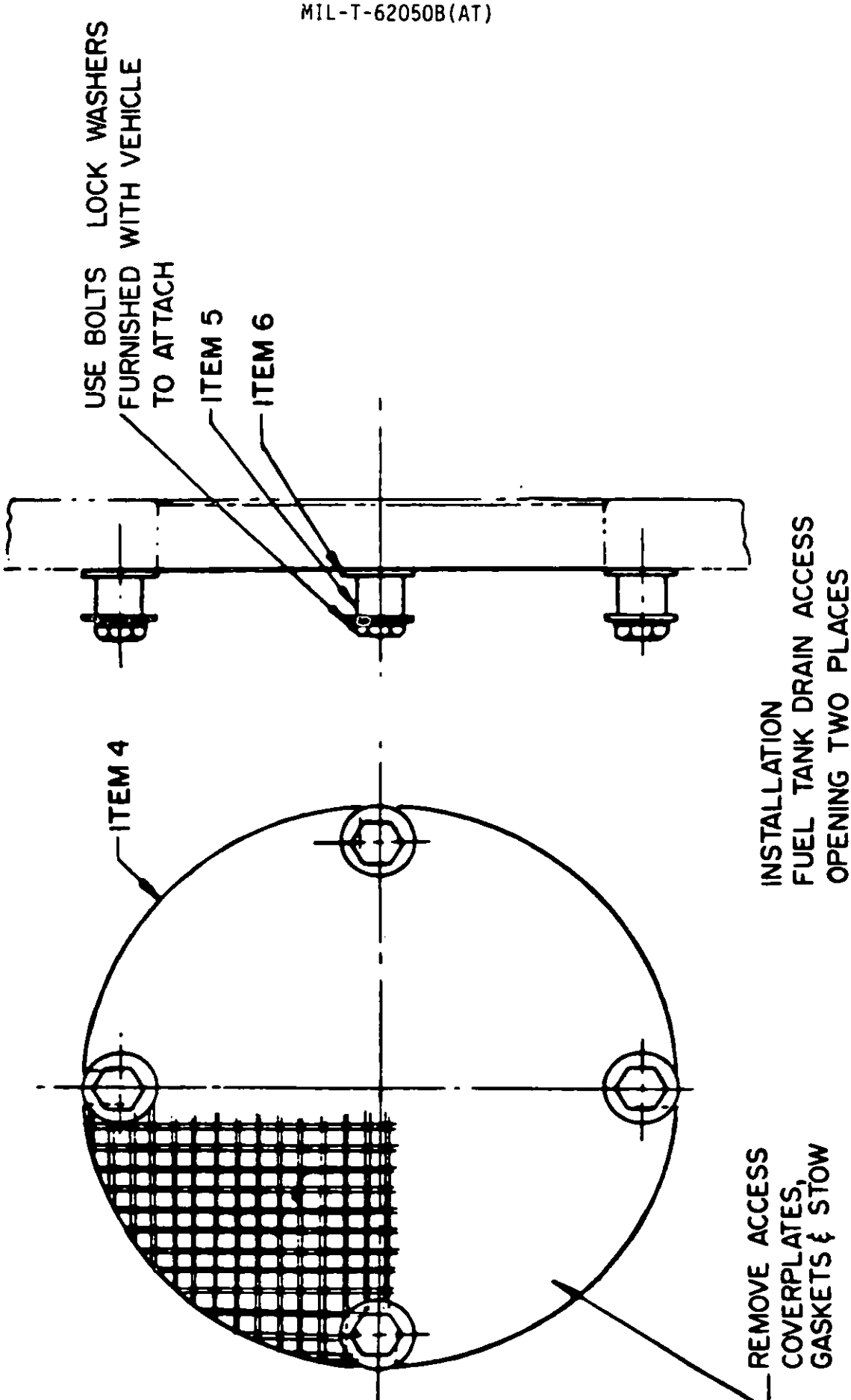
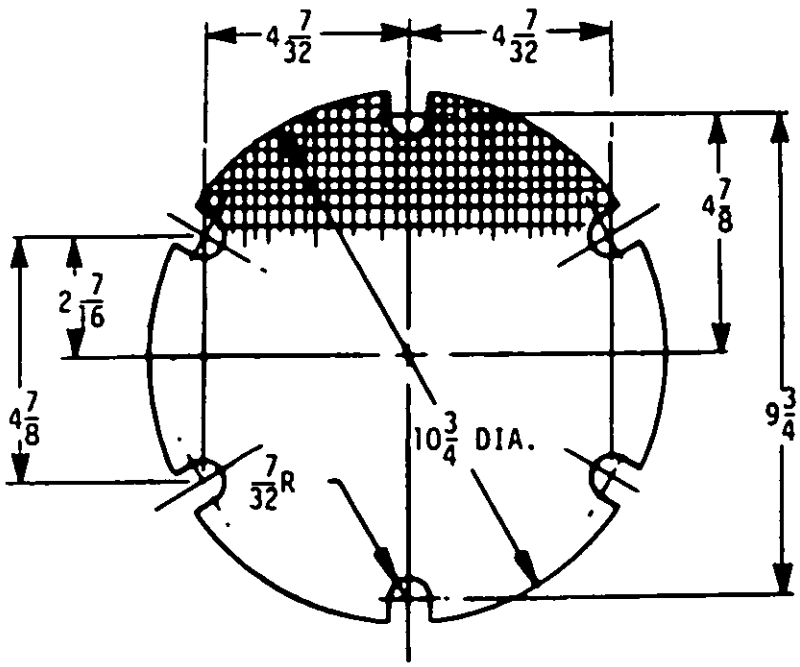
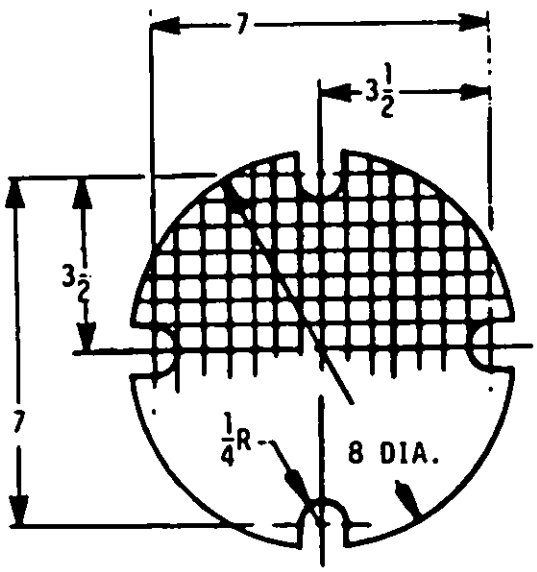


FIGURE 24. Fuel tank drain access.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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SCREEN
ITEM 1

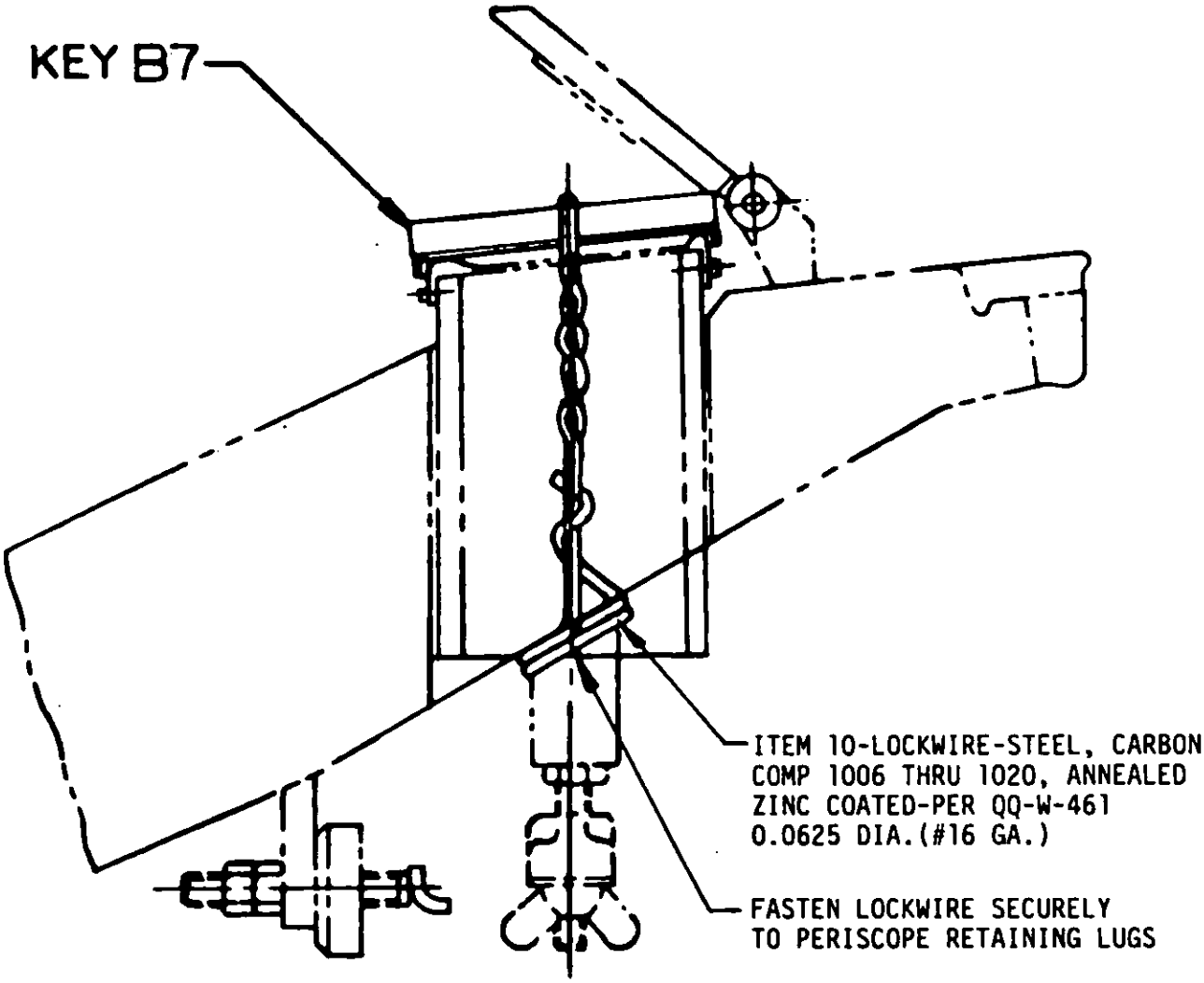


SCREEN
ITEM 4

KEY B5

FIGURE 25. Screens.

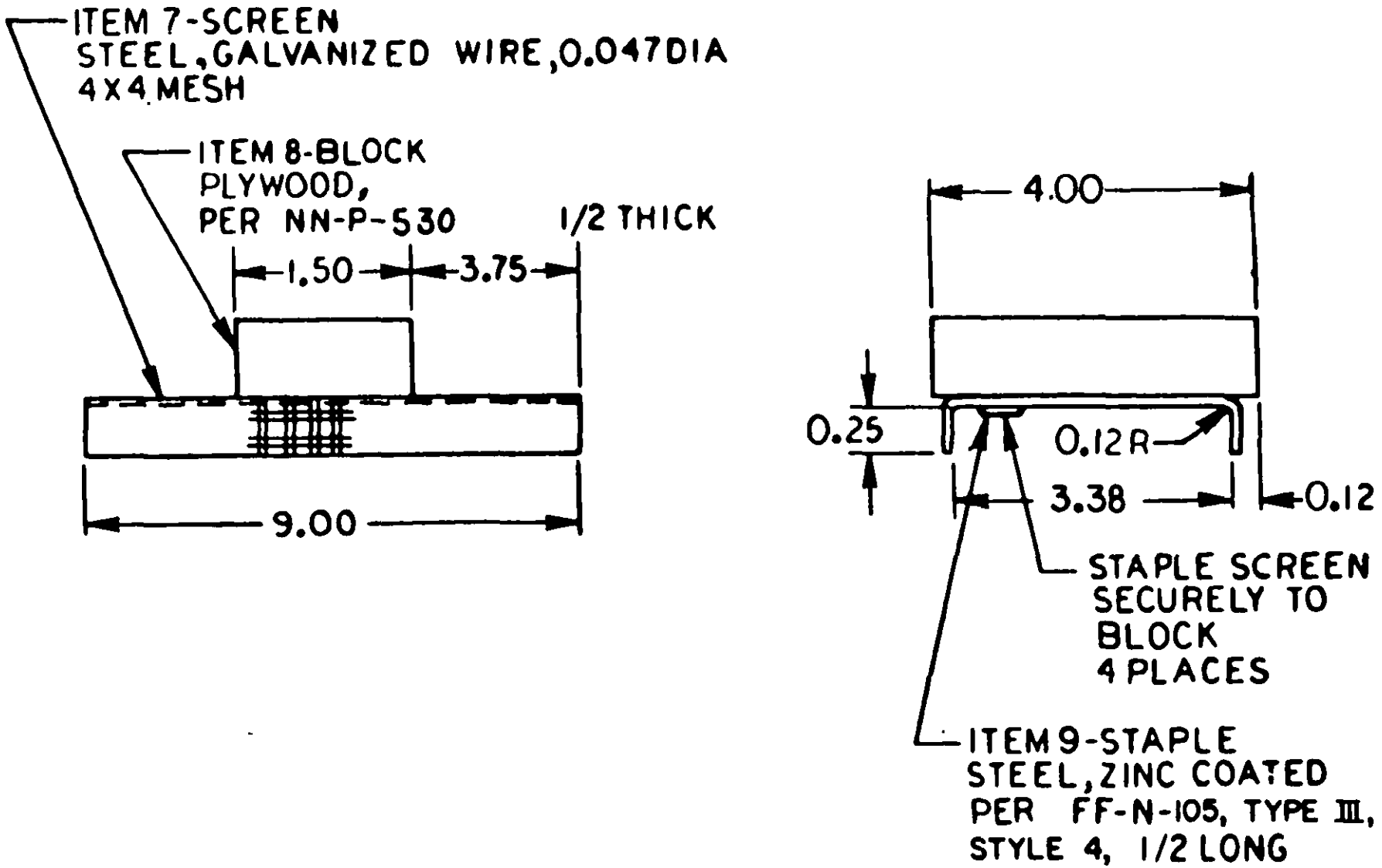
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



INSTALLATION
DRIVERS PERISCOPE OPENING
(CENTER OPENING ONLY)
KEY B6

FIGURE 26. Installation of screens in drivers periscope opening.

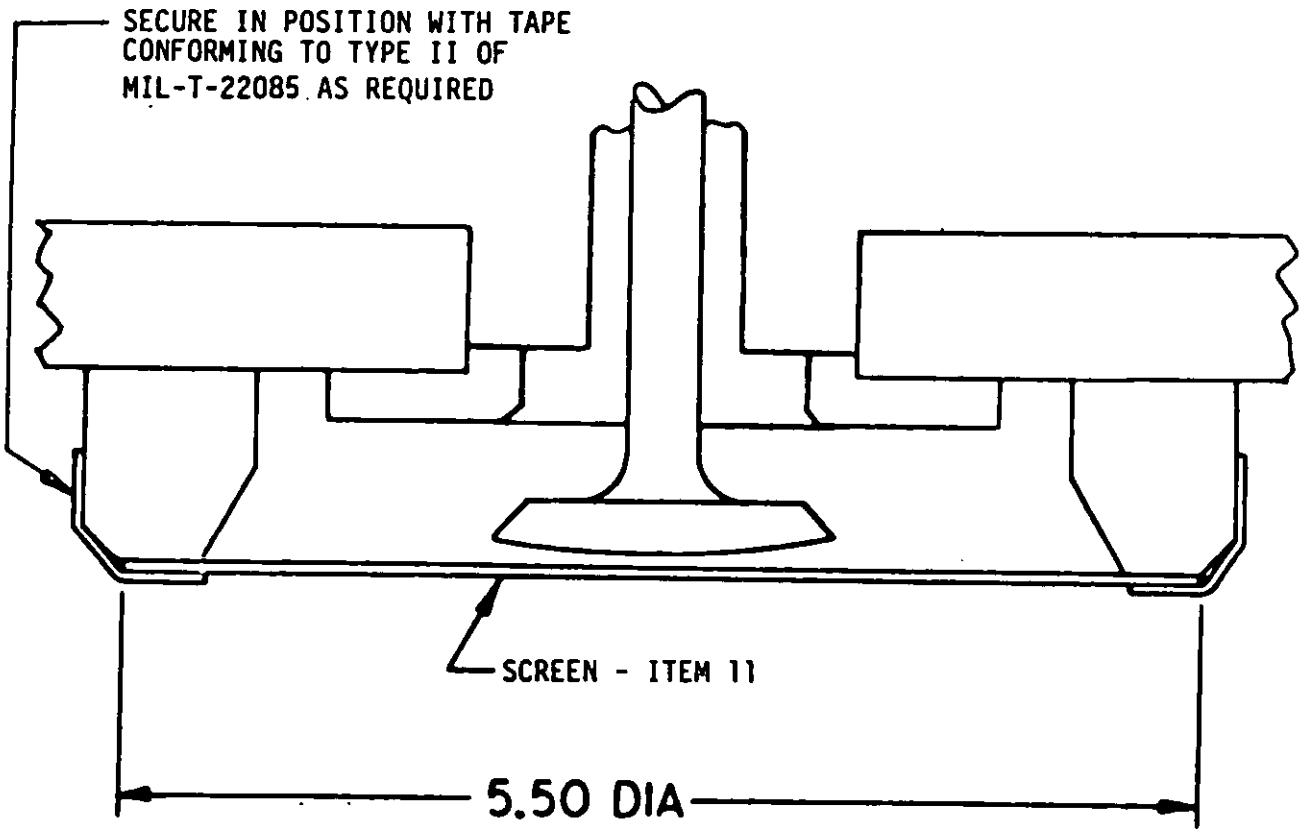
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



KEY B7

FIGURE 27. Driver's periscope screen assembly.

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KEY B8

FIGURE 28. Drain valve screen.

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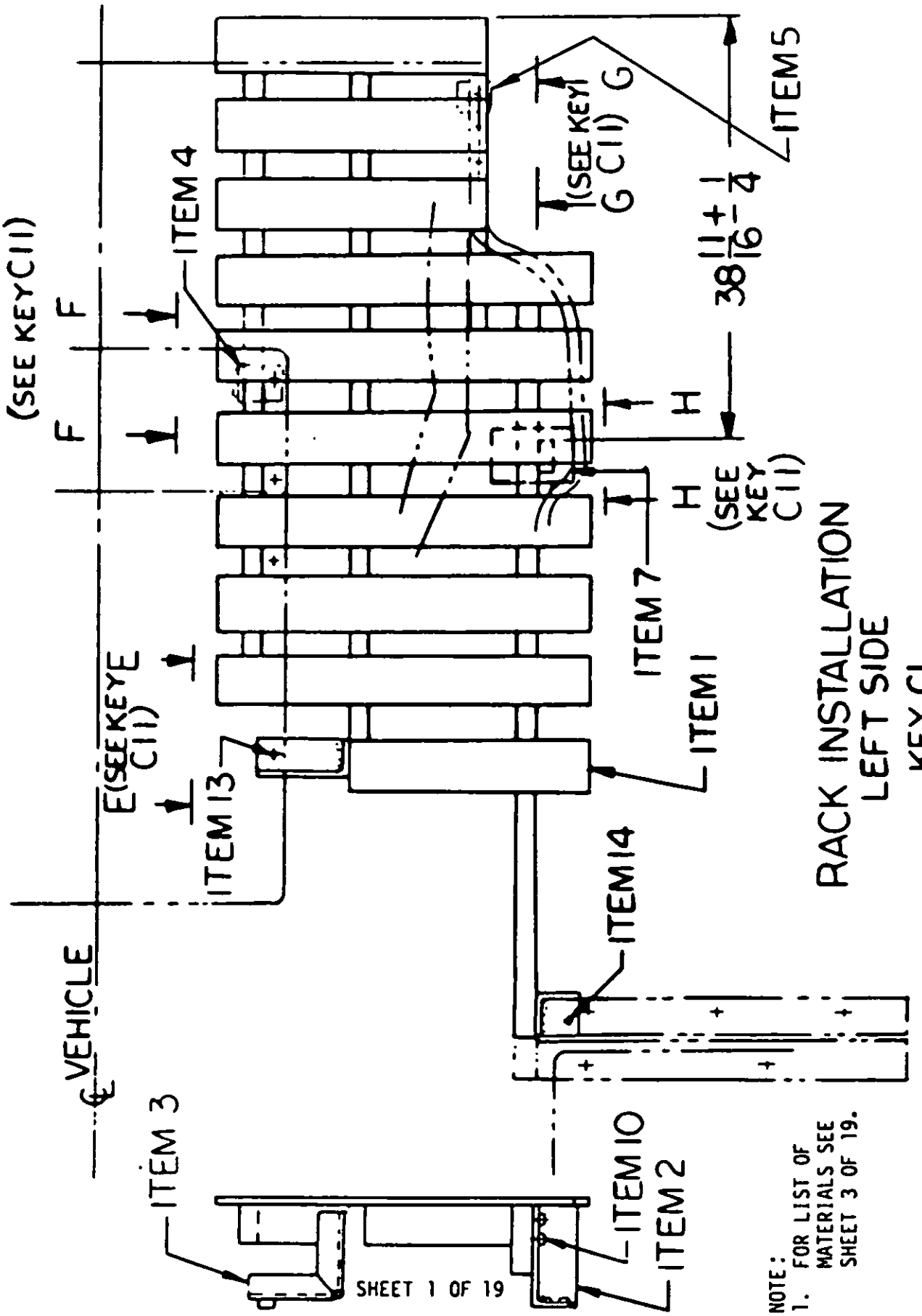
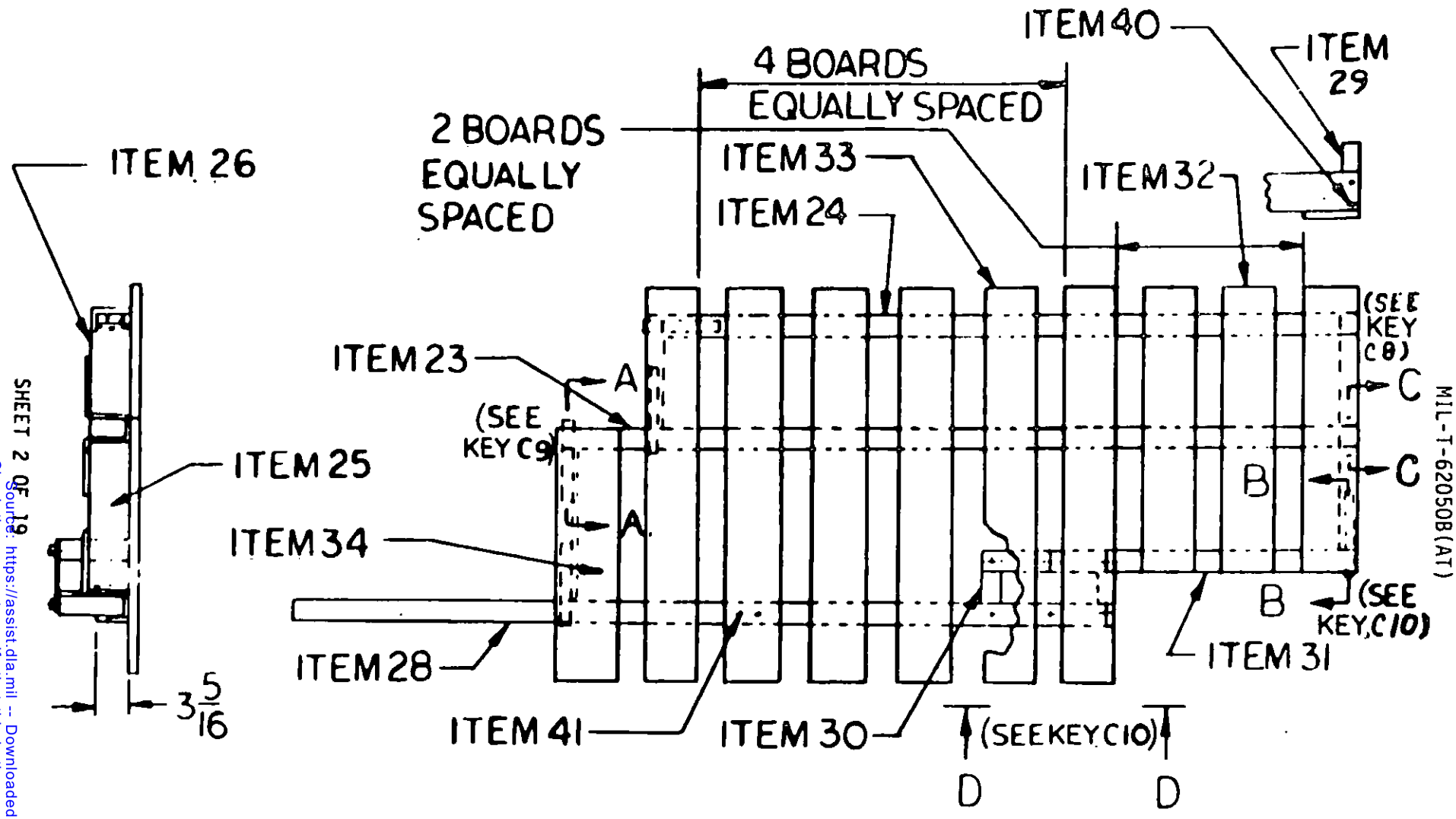


FIGURE 29. Rack assembly and installation.

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Check the source to verify that this is the current version before use.



NOTES:

1. FOR NAILING SPECIFICATIONS AND PATTERNS NOT SHOWN SEE KEY C21.
2. FOR LIST OF MATERIALS SEE SHEET 3 OF 19.

RACK ASSEMBLY
LEFT SIDE
ITEM 1
KEY C5

FIGURE 29. Rack assembly and installation - Continued.

MIL-T-62050B(AT)
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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MATERIAL LIST FOR RACK INSTALLATION (LEFT)						
ITEM NO.	KEY NO.	SHEET NO.	NO REQ'D	NAME	MAT'L	STOCK SIZE
1	C5	2	1	RACK ASSY	-----	-------
2	C19	17	1	BRACKET	ANGLE IRON	SEE DETAIL
3	C17	15	1	BRACKET	ANGLE IRON	SEE DETAIL
4	C12	10	1	BRACKET	STEEL	SEE DETAIL
5	C12	10	1	BRACKET	STEEL	SEE DETAIL
7	C14	12	1	SUPPORT ASSY	-----	SEE DETAIL
8	C14	12	1	SUPPORT	1/	SEE DETAIL
9	---		1	PLATE	1/	1 X 6 X 8
10	---		9	LAG BOLT	-----	1/2 X 2 LG.
11	---		2	LAG BOLT	-----	1/2 X 3 1/2 LG.
13	---		1	SCREW	-----	3/8-24 X 1 3/4 LG.
14	---		1	SCREW	-----	1/2-13 X 1 3/4 LG.
17	---		2	WASHER	-----	1/2 I.D. PLAIN
41	---		2	NAIL	-----	8D
46	---		1	SCREW	-----	3/8 - 24 X 3 LG.

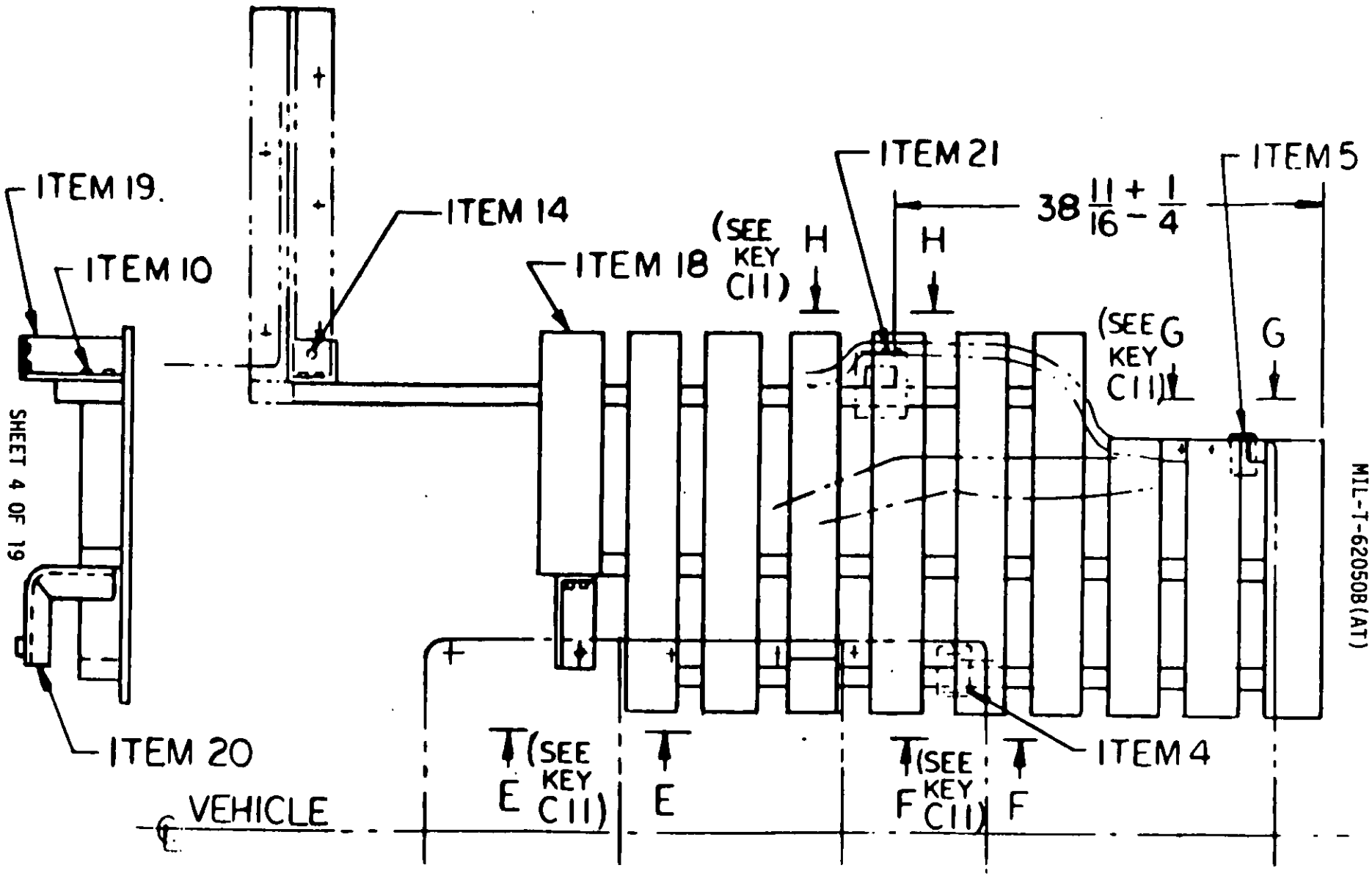
MATERIAL LIST FOR RACK ASSEMBLY (LEFT)						
ITEM NO.	KEY NO.	SHEET NO.	NO REQ'D	NAME	MAT'L	STOCK SIZE
10	---		6	LAG BOLT	-----	1/2 X 2 LG.
17	---		6	WASHER	-----	1/2 I.D. PLAIN
23	---		1	INTERMED. MEMBER	1/	2 X 4 X 85 1/8
24	C15	13	1	CENTER MEMBER	1/	SEE DETAIL
25	---		1	FRONT BRACE	1/	2 X 4 X 16 1/2
26	---		1	FRONT INTER. MEM.	1/	2 X 4 X 10 3/4
28	---		1	OUTSIDE MEMBER	1/	2 X 8 X 88
29	C13	11	1	REAR BRACE	1/	SEE DETAIL
30	---		2	BRACE	1/	2 X 8 X 4
31	---		1	REAR MEMBER	1/	2 X 8 X 38
32	---		3	FLOOR BOARD	1/	1 X 6 X 29 3/4
33	---		6	FLOOR BOARD	1/	1 X 6 X 41 1/2
34	---		1	FLOOR BOARD	1/	1 X 6 X 26
37	C15	13	1	BRACE	1/	SEE DETAIL
38	C16	14	1	STRAP	STEEL	SEE DETAIL
39	C16	14	4	STRAP	STEEL	SEE DETAIL
40	---		27	NAIL	-----	12D
41	---		62	NAIL	-----	8D
42	---		14	NAIL	-----	4D

1/ MATERIAL:
WOOD, GROUP II OR III, PER MIL-C-104.

SHEET 3 OF 19

FIGURE 29. Rack assembly and installation - Continued.

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NOTE:

1. FOR MATERIALS LIST SEE SHEET 6 OF 19.

KEY C3
RACK INSTALLATION
RIGHT SIDE

FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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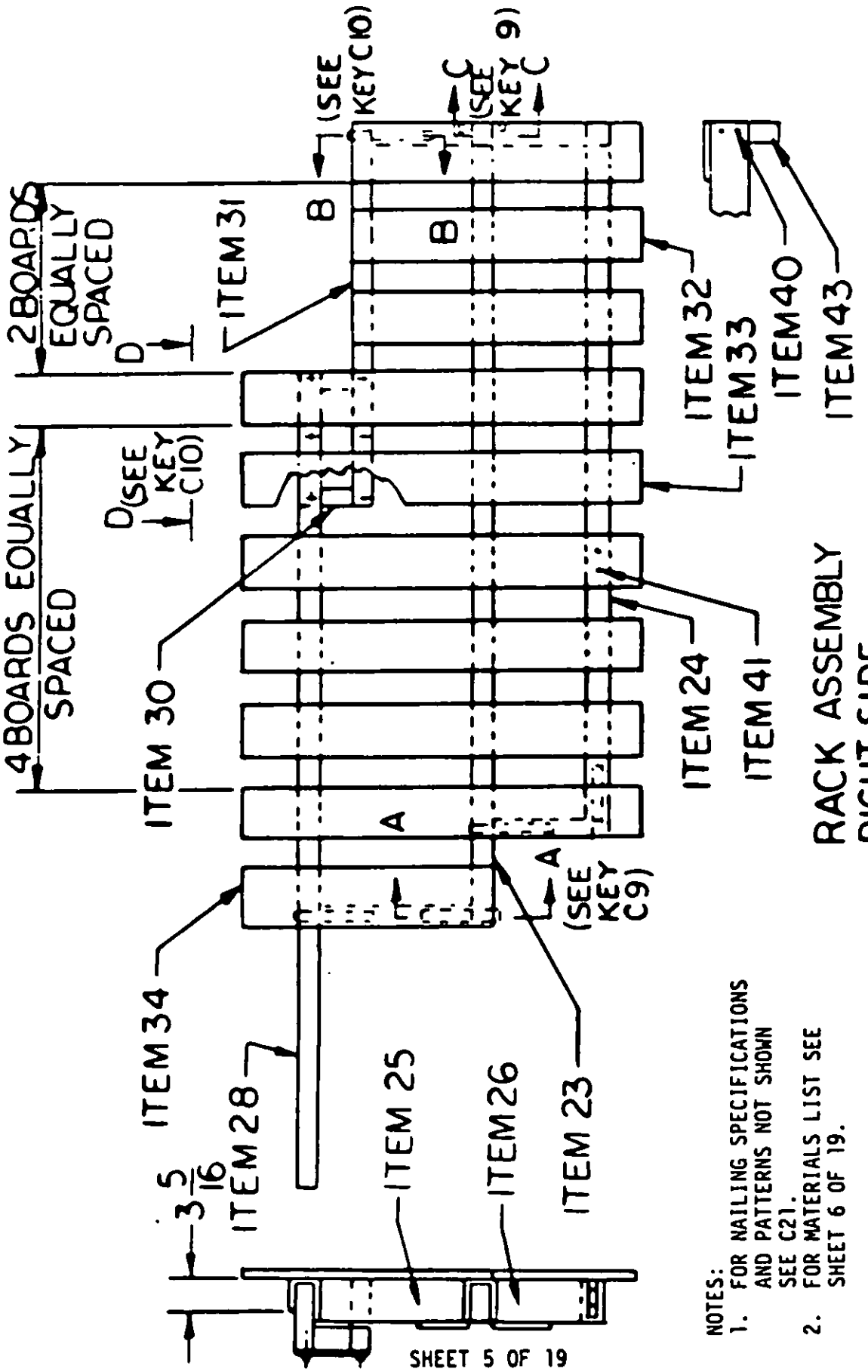


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

MATERIAL LIST FOR RACK INSTALLATION (RIGHT)						
ITEM NO.	KEY NO.	SHEET NO.	NO REQ'D	NAME	MAT'L	STOCK SIZE
4	C12	10	1	BRACKET	STEEL	SEE DETAIL
5	C12	10	1	BRACKET	STEEL	SEE DETAIL
9	---		1	PLATE	1/	1 X 6 X 8
10	---		9	LAG BOLT	-----	1/2 X 2 LG.
11	---		2	LAG BOLT	-----	1/2 X 3 1/2 LG.
13	---		1	SCREW	-----	3/8-24 X 1 3/4 LG.
14	---		1	SCREW	-----	1/2-13 X 1 3/4 LG.
17	---		2	WASHER	-----	1/2 I.D. PLAIN
18	C7	5	1	RACK ASSY	-----	-------
19	C19	17	1	BRACKET	ANGLE IRON	SEE DETAIL
20	C17	15	1	BRACKET	ANGLE IRON	SEE DETAIL
21	C14	12	1	SUPPORT ASSY	-----	SEE DETAIL
22	C14	12	1	SUPPORT	1/	SEE DETAIL
41	---		2	NAIL	-----	8D
46	---		1	SCREW	-----	3/8-24 X 3 LG.

MATERIAL LIST FOR RACK ASSEMBLY (RIGHT)						
ITEM NO.	KEY NO.	SHEET NO.	NO REQ'D	NAME	MAT'L	STOCK SIZE
10	---		6	LAG BOLT	-----	1/2 X 2 LG.
17	---		6	WASHER	-----	1/2 I.D. PLAIN
23	---		1	INTERMED. MEMBER	1/	2 X 4 X 85 1/8
24	C15	13	1	CENTER MEMBER	1/	SEE DETAIL
25	---		1	FRONT BRACE	1/	2 X 4 X 16 1/2
26	---		1	FRONT INTER. MEM.	1/	2 X 4 X 10 3/4
28	---		1	OUTSIDE MEMBER	1/	2 X 8 X 88
30	---		2	BRACE	1/	2 X 8 X 4
31	---		1	REAR MEMBER	1/	2 X 8 X 38
32	---		3	FLOOR BOARD	1/	1 X 6 X 29 3/4
33	---		6	FLOOR BOARD	1/	1 X 6 X 41 1/2
34	---		1	FLOOR BOARD	1/	1 X 6 X 26
37	C15	13	1	BRACE	1/	SEE DETAIL
38	C16	16	1	STRAP	STEEL	SEE DETAIL
39	C16	16	4	STRAP	STEEL	SEE DETAIL
40	---		27	NAIL	-----	12D
41	---		62	NAIL	-----	8D
42	---		14	NAIL	-----	4D
43	C18	16	1	REAR BRACE	1/	SEE DETAIL

1/ MATERIAL:
WOOD, GROUP II OR III, PER MIL-C-104.

SHEET 6 OF 19

FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

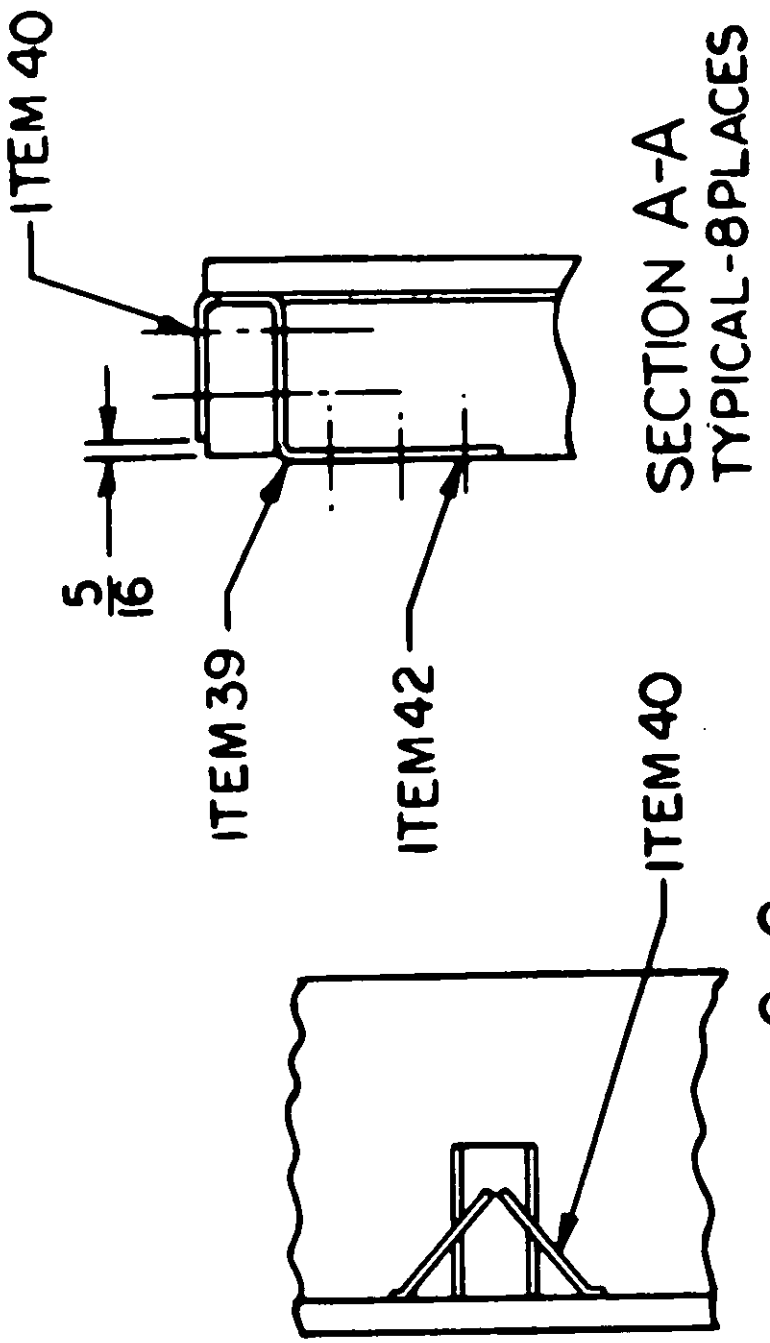


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

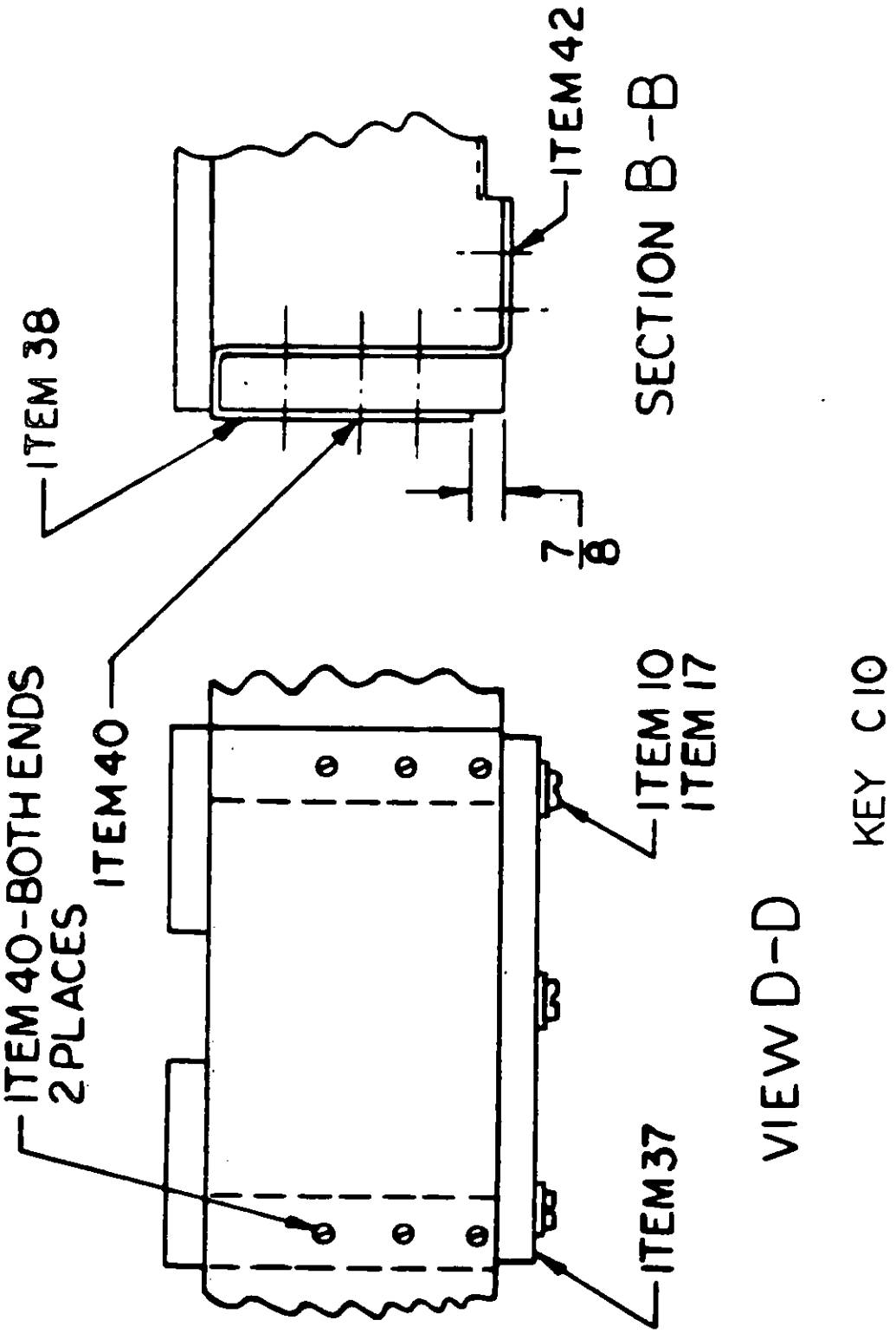


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

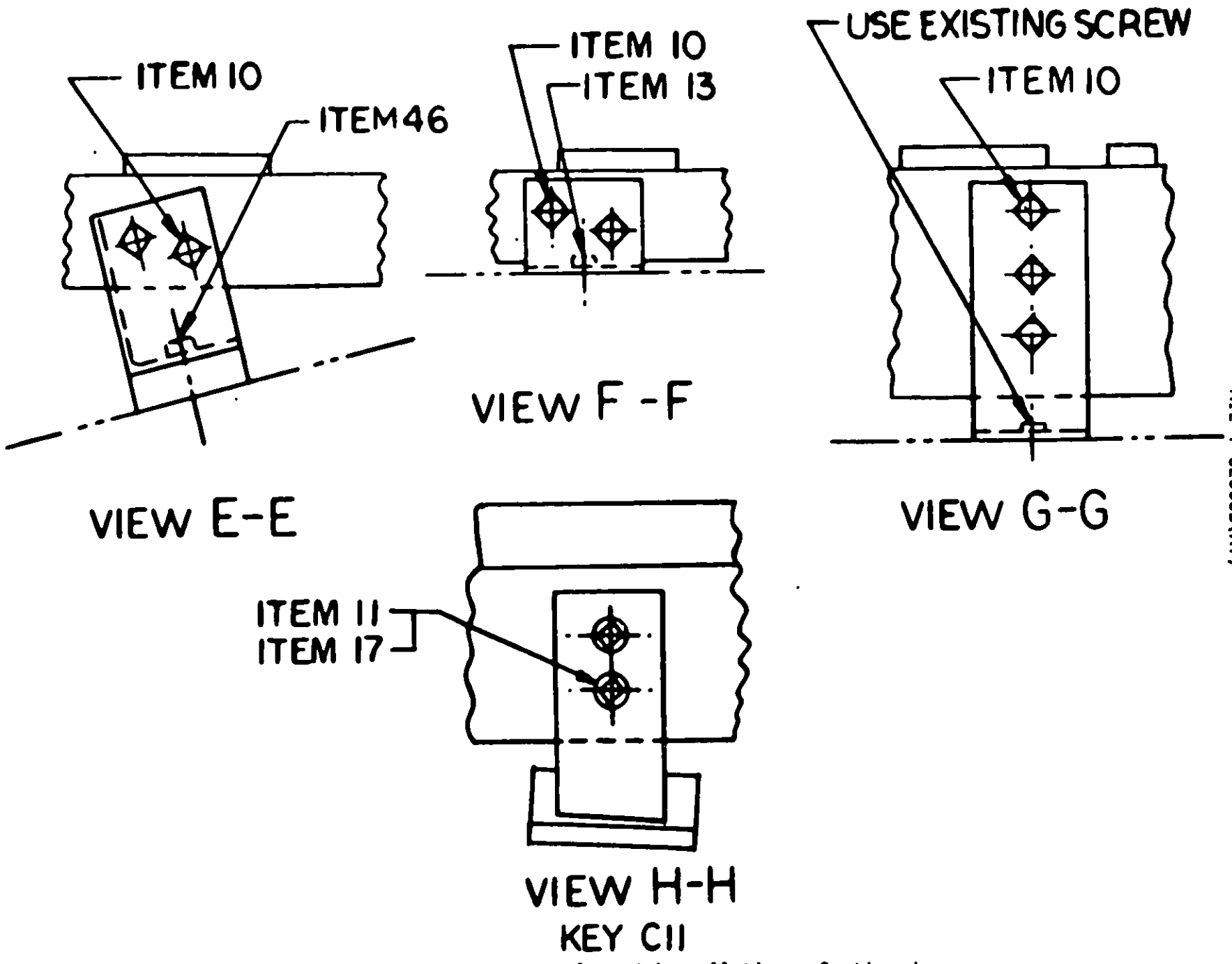


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

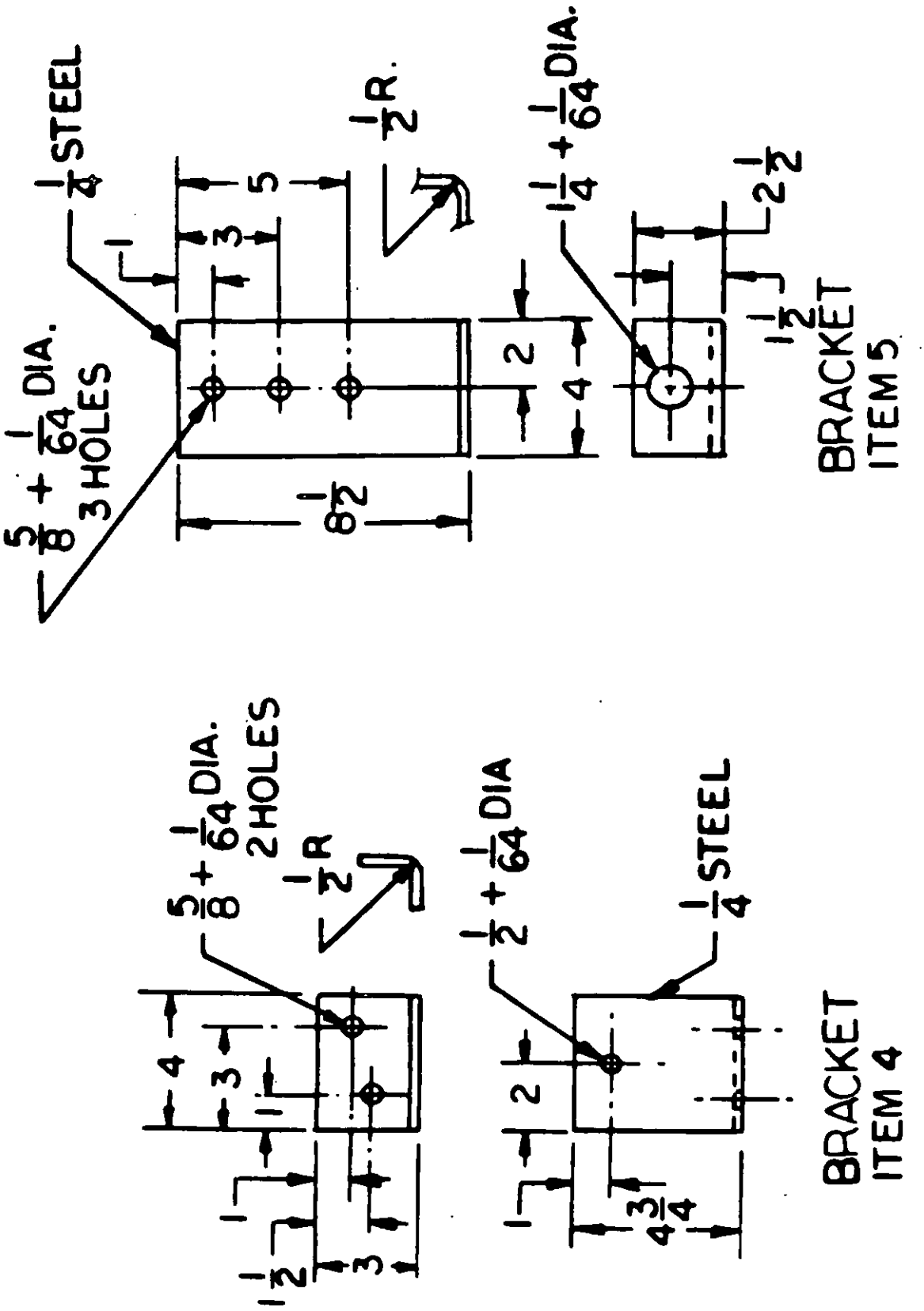


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

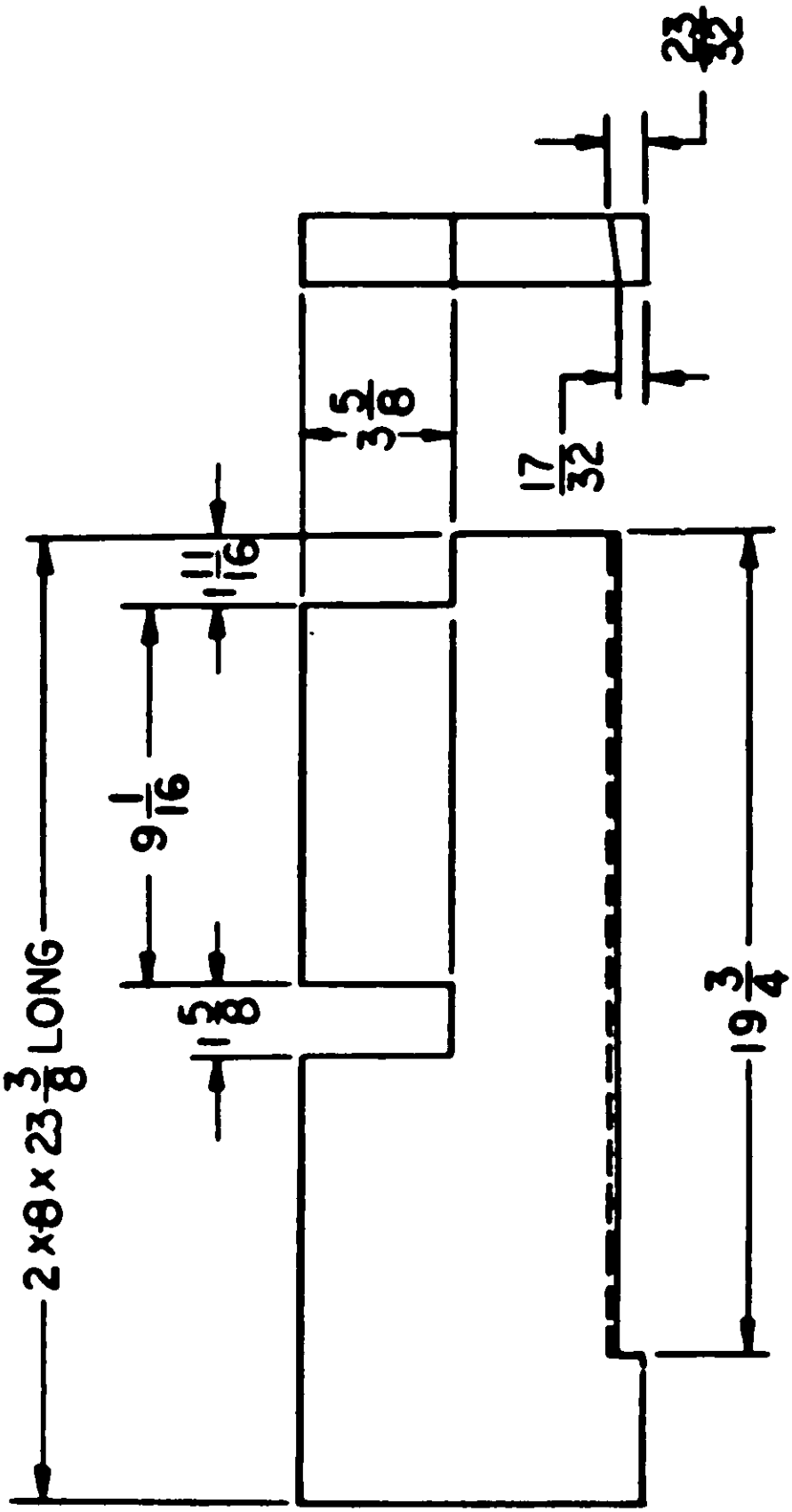
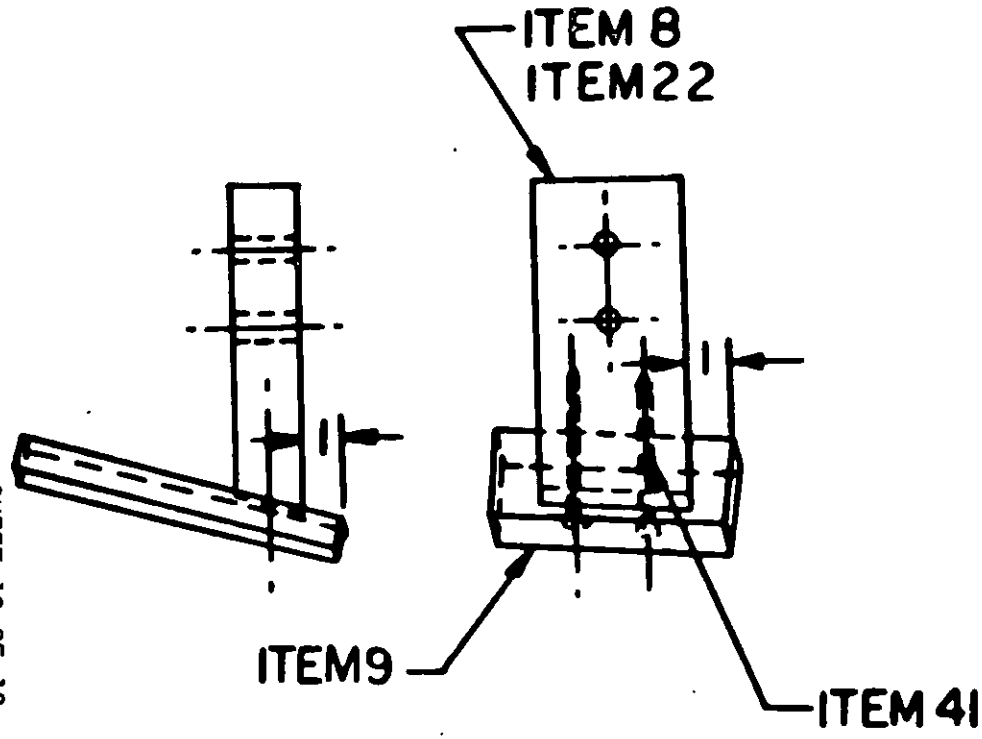


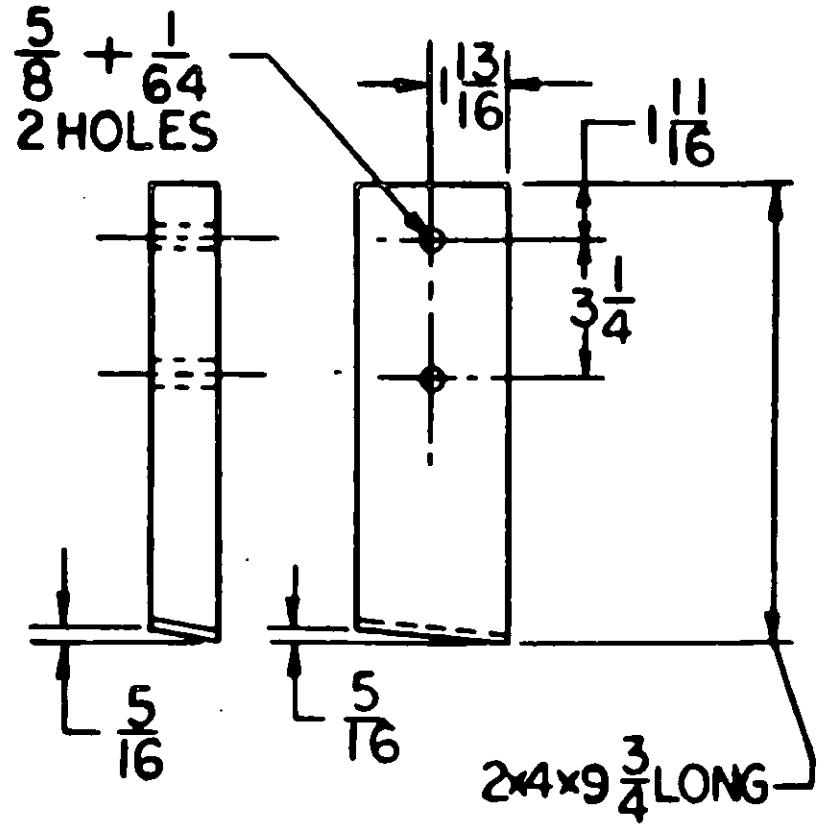
FIGURE 29. Rack assembly and installation - Continued.

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Check the source to verify that this is the current version before use.



SUPPORT ASSEMBLY

LEFT SHOWN-ITEM 7
RIGHT OPPOSITE-ITEM 21



SUPPORT

LEFT SHOWN - ITEM 8
RIGHT OPPOSITE - ITEM 22

KEY C14

FIGURE 29. Rack assembly and installation - Continued.

Source: MIL-T-62050B(AT) Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

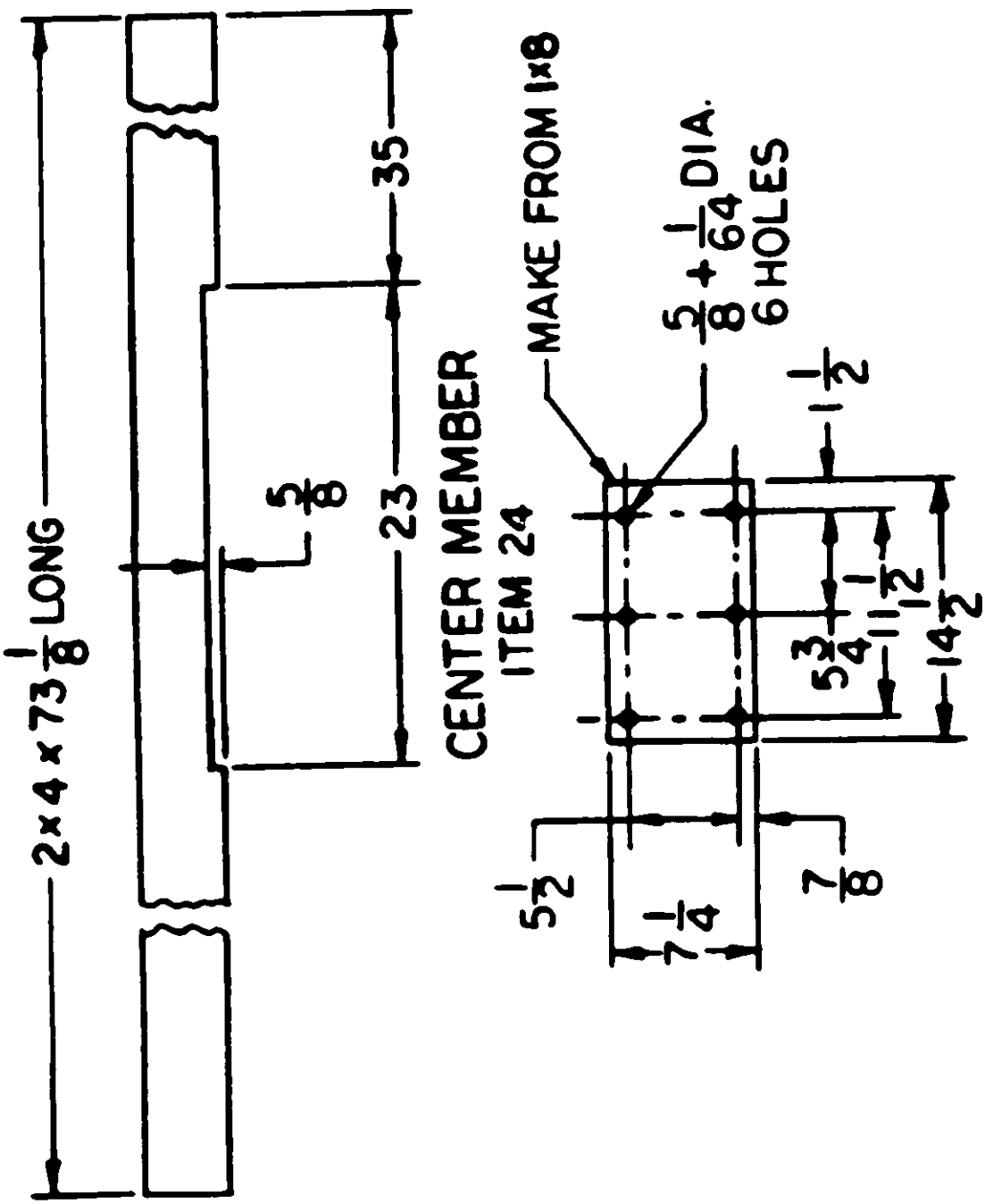


FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

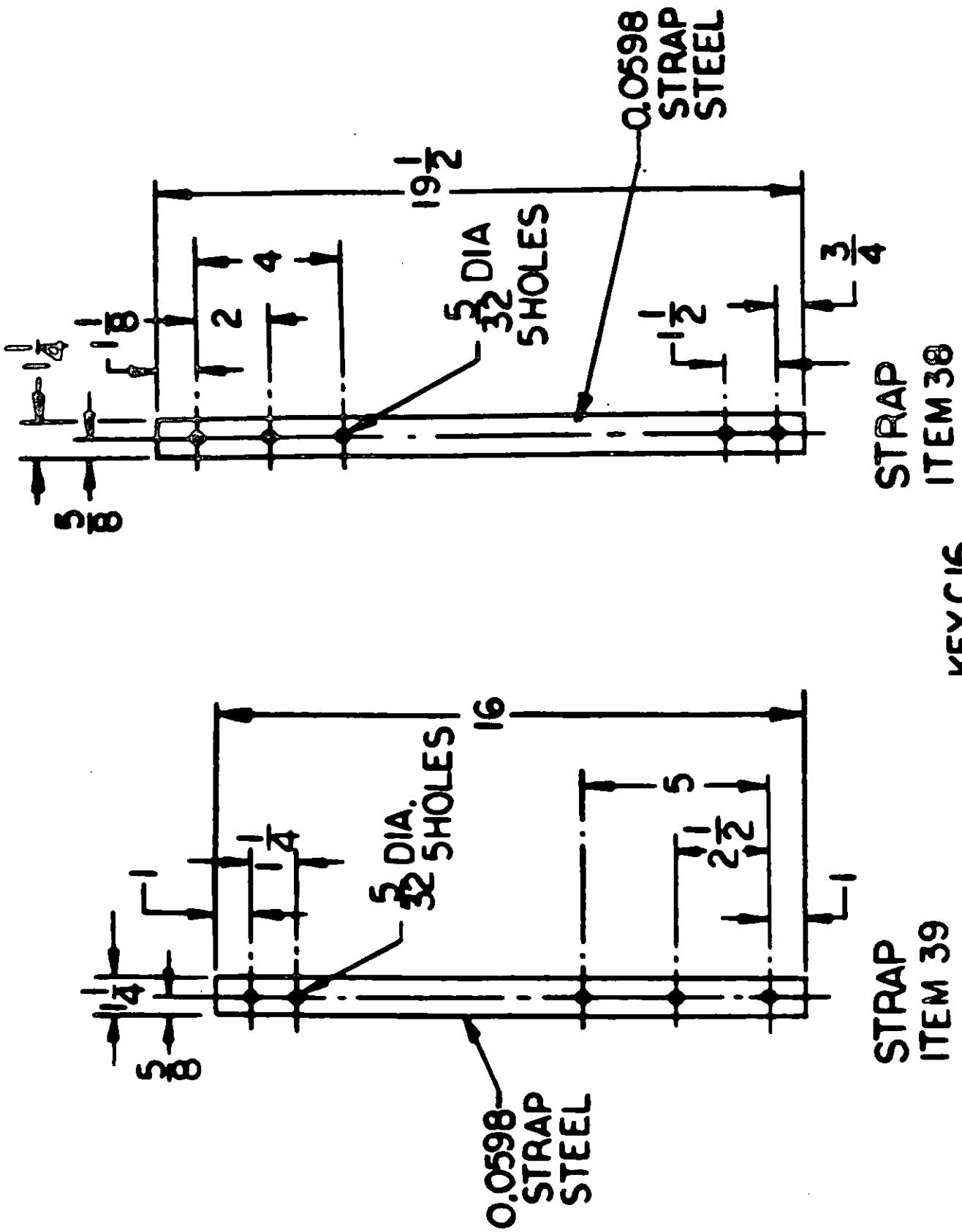
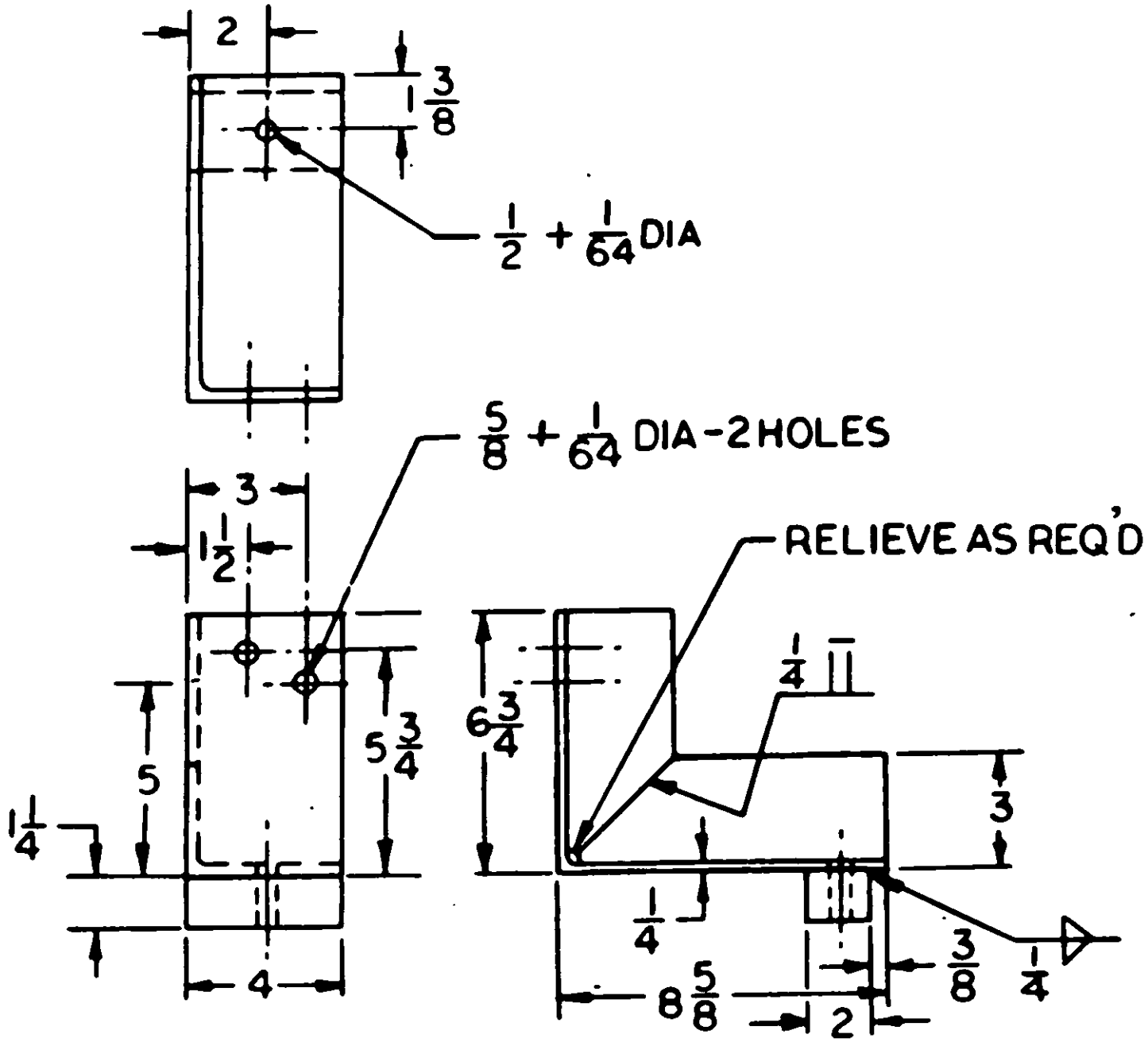


FIGURE 29. Rack assembly and installation - Continued.

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BRACKET ASSEMBLY
LEFT HAND SHOWN
ITEM-3
RIGHT HAND ITEM-20
KEY C17

FIGURE 29. Rack assembly and installation - Continued.

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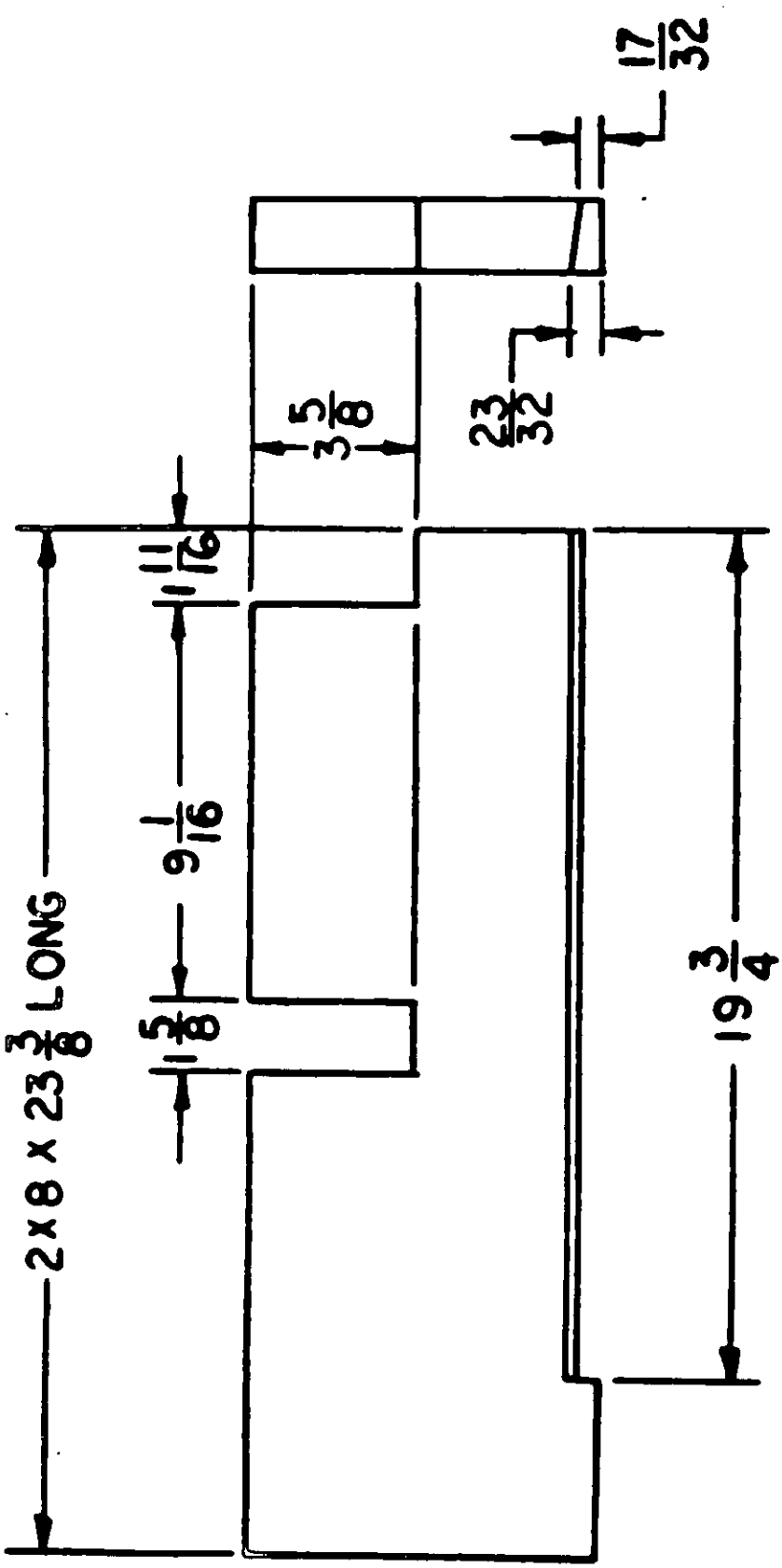
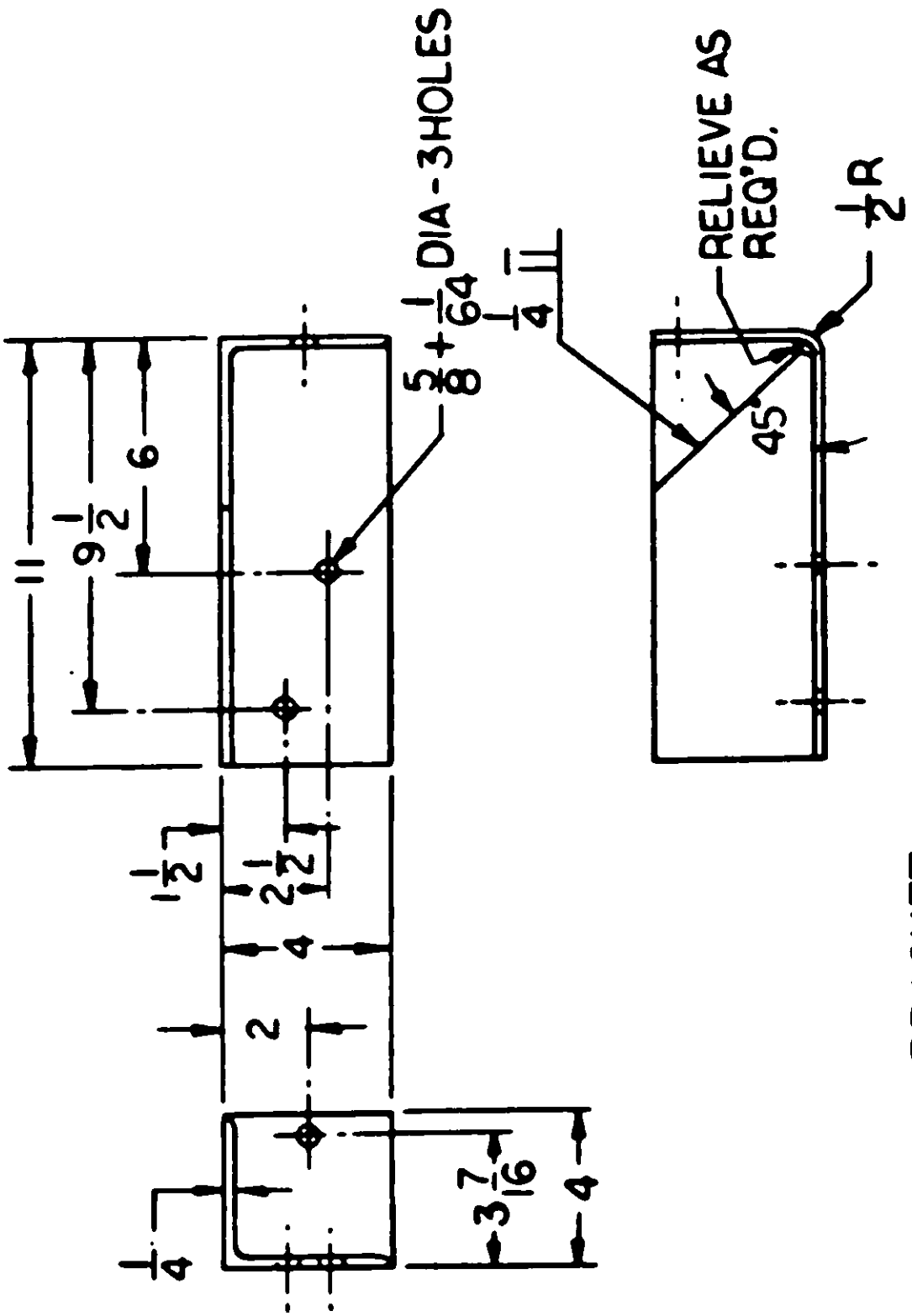


FIGURE 29. Rack assembly and installation - Continued.

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Check the source to verify that this is the current version before use.



BRACKET
LEFT SHOWN ITEM 2
RIGHT ITEM 19

KEY C19

FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

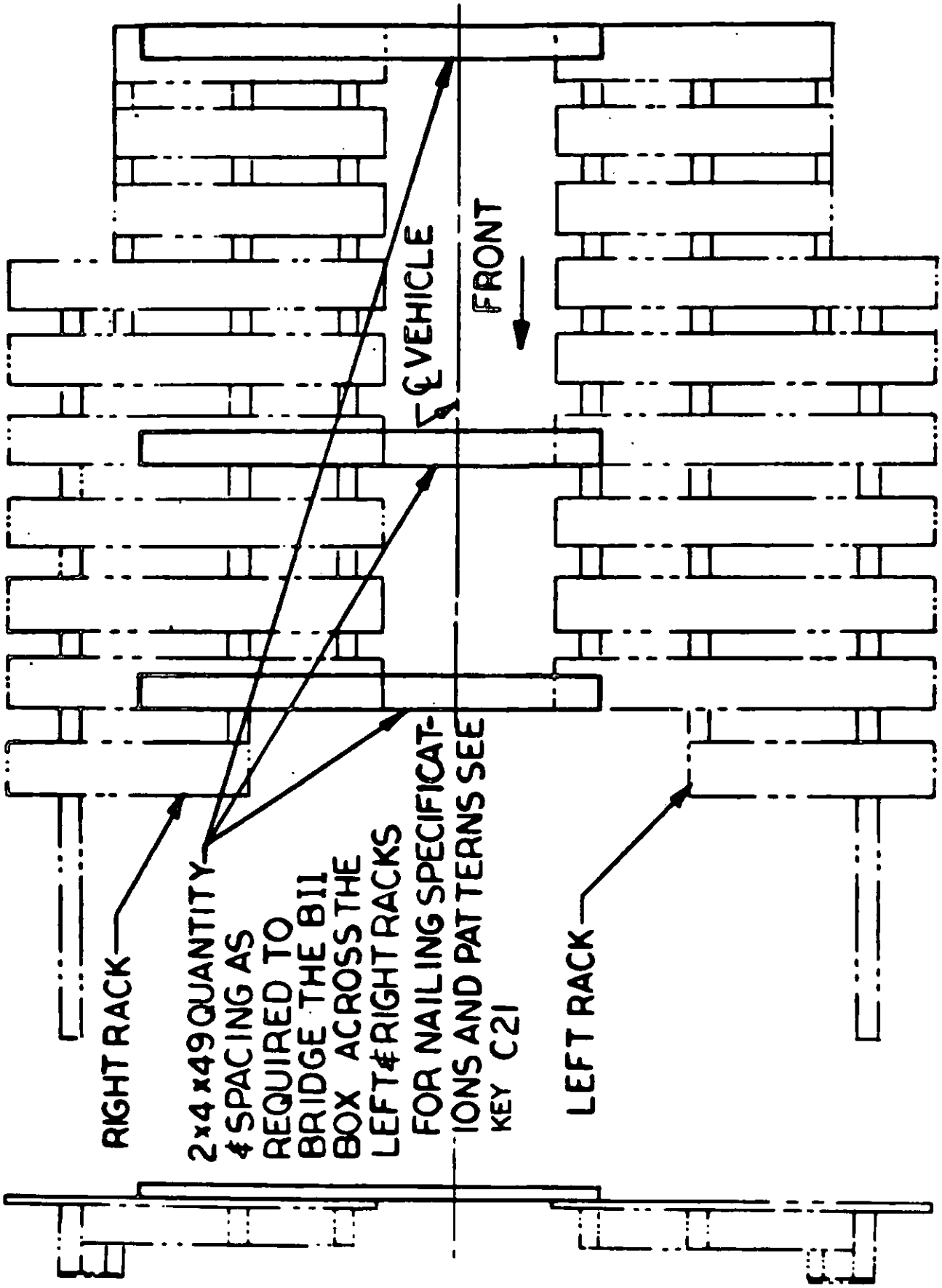
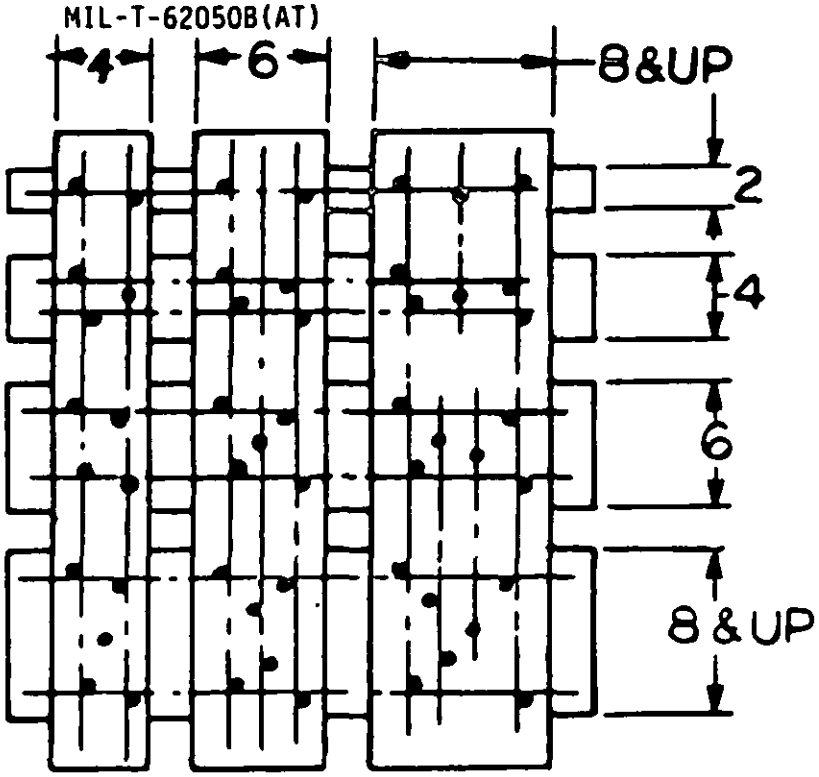


FIGURE 29. Rack assembly and installation - Continued.

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Check the source to verify that this is the current version before use.

WHEN PIECES ARE NOT AT RIGHT ANGLES A SIMILAR BUT SKEWED PATTERN SHALL BE USED



NAILING PATTERNS
(NOMINAL WOOD SIZES)

NAILING SPECIFICATIONS			
NOMINAL WOOD THICK,		NAIL SIZE	
NAILING	TO	CLINCHED	HIDDEN
1	1	5d	4dCC
1	2	8d	7dCC
1	4	——	8dCC
1	6	——	8dCC
1	8	——	8dCC
2	2	16d	12dCC
2	4	50d	20dCC
2	6	——	20dCC
2	8	——	20dCC
4 & UP	4 & UP	BOLTED CONSTRUCTION	
FOR GROUP I WOODS, INCREASE NAIL SIZE TO NEXT LARGER SIZE & REDUCE NAIL SPACING BY 1/4 INCH			
FOR GROUP IV WOODS, DECREASE NAIL SIZE TO NEXT SMALLER SIZE & NAIL SPACING MAY BE INCREASED BY 1/4 INCH. DRILL LEAD HOLES WHERE NECESSARY TO PREVENT SPLITTING			

SHEET 19 OF 19 KEY C21

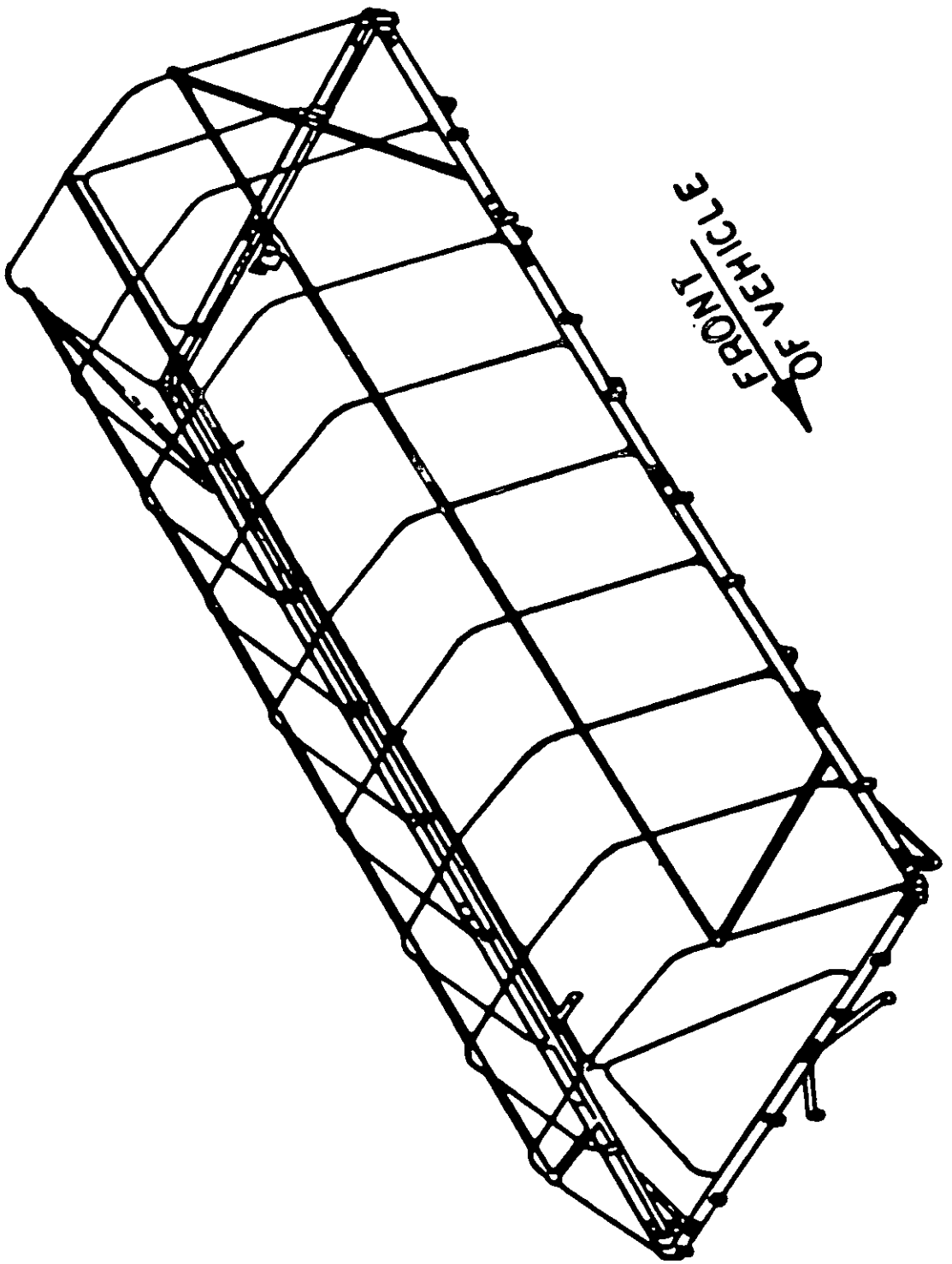
FIGURE 29. Rack assembly and installation - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

KEY D1					
KEY NO.	FIG. NO. - SHEET NO.	DESCRIPTION	KEY NO.	FIG. NO - SHEET NO.	DESCRIPTION
D2	31	FRAME ASSEMBLY	D41	40-8	TUBE, TOP FRAME
D3	32	PLAN VIEW OF THE FRAME	D42	40-9	LINK, FRAME
D4	33	RIGHT AND LEFT VIEW OF THE FRAME	D43	41-1	BRACKET BASE FRAME
D5	34	FRONT VIEW OF THE FRAME	D44	41-1	BRACKET BASE FRAME
D6	35	REAR VIEW OF THE FRAME	D45	41-2	FRAME TO FENDER INSTALLATION
D7	36-1	BASE FRAME	D46	41-2	FRAME TO FENDER INSTALLATION
D8	36-1	BASE FRAME	D47	41-3	FRAME TO FENDER INSTALLATION
D9	36-2	BASE FRAME ASSEMBLY	D48	41-3	FRAME TO FENDER INSTALLATION
D10	36-2	BASE FRAME ASSEMBLY	D49	41-4	FRAME TO FENDER INSTALLATION
D11	36-3	BASE FRAME	D50	41-4	FRAME TO FENDER INSTALLATION
D12	36-3	BASE FRAME	D51	41-5	FRAME TO FENDER INSTALLATION
D13	36-4	BASE FRAME ASSEMBLY	D52	41-5	FRAME TO FENDER INSTALLATION
D14	36-4	BASE FRAME ASSEMBLY	D53	41-6	FRAME TO FENDER
D15	36-5	BASE FRAME	D54	41-10	BRACE, FRAME
D16	36-5	BASE FRAME	D55	41-10	BRACE, FRAME
D17	36-6	BASE FRAME ASSEMBLY	D56	41-11	BRACKET BASE FRAME
D18	36-6	BASE FRAME ASSEMBLY	D57	42	BRACE, FRAME
D19	36-7	BASE FRAME	D58	42	BRACE, FRAME
D20	36-7	BASE FRAME	D59	41-8	TUBING, BASE
D21	36-8	BASE FRAME ASSEMBLY	D60	41-8	TUBING, BASE
D22	36-8	BASE FRAME ASSEMBLY	D61	41-9	SPACER BLOCK
D23	36-9	BASE FRAME	D62	41-9	SPACER BLOCK
D24	36-10	BASE FRAME	D63		INTENTIONALLY OMITTED
D25	36-11	BASE FRAME ASSEMBLY	D64	43	CLAMP
D26	36-12	BASE FRAME ASSEMBLY	D65	44	HARDWARE, BASE FRAME
D27	37	ENGAGEMENT PIPE	D66	44	SCREW, PAN HEAD
D28	38-1	SPLICE PLATE	D67	44	NUT, HEX
D29	38-2	TYPICAL BASE FRAME SPLICE	D68	44	WASHER, LOCK
D30	39-1	TYPICAL CORNER CONSTRUCTION	D69	44	WASHER, PLAIN
D31	39-2	TUBE, BASE FRAME CORNER	D70	44	SCREW, LOCK WASHER, HEX HEAD
D32	40-1	TUBE, TOP FRAME	D112	44	NUT, JAM HEX
D33	40-1	CONNECTOR	D113	44	NUT, HEX
D34	40-2	TUBE ASSEMBLY, TOP FRAME	D114	44	WASHER, LOCK
D35	40-3	TUBE, TOP FRAME	D115	44	SCREW, HEX HEAD
D36	40-4	TUBE ASSEMBLY, TOP FRAME	D116	44	WASHER, PLAIN
D37	40-5	CONNECTOR	D117	41-9	SPACER BLOCK
D38	40-6	TUBE, UPPER BRACE	D118	41-7	FRAME TO FENDER INSTALLATION
D39	40-7	TUBE ASSEMBLY	D119	41-7	FRAME TO FENDER INSTALLATION
D40	40-8	PLATE			

FIGURE 30. List of key numbers for the framework and base for closure on the M48A5.

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KEY D2

FIGURE 31. Frame assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

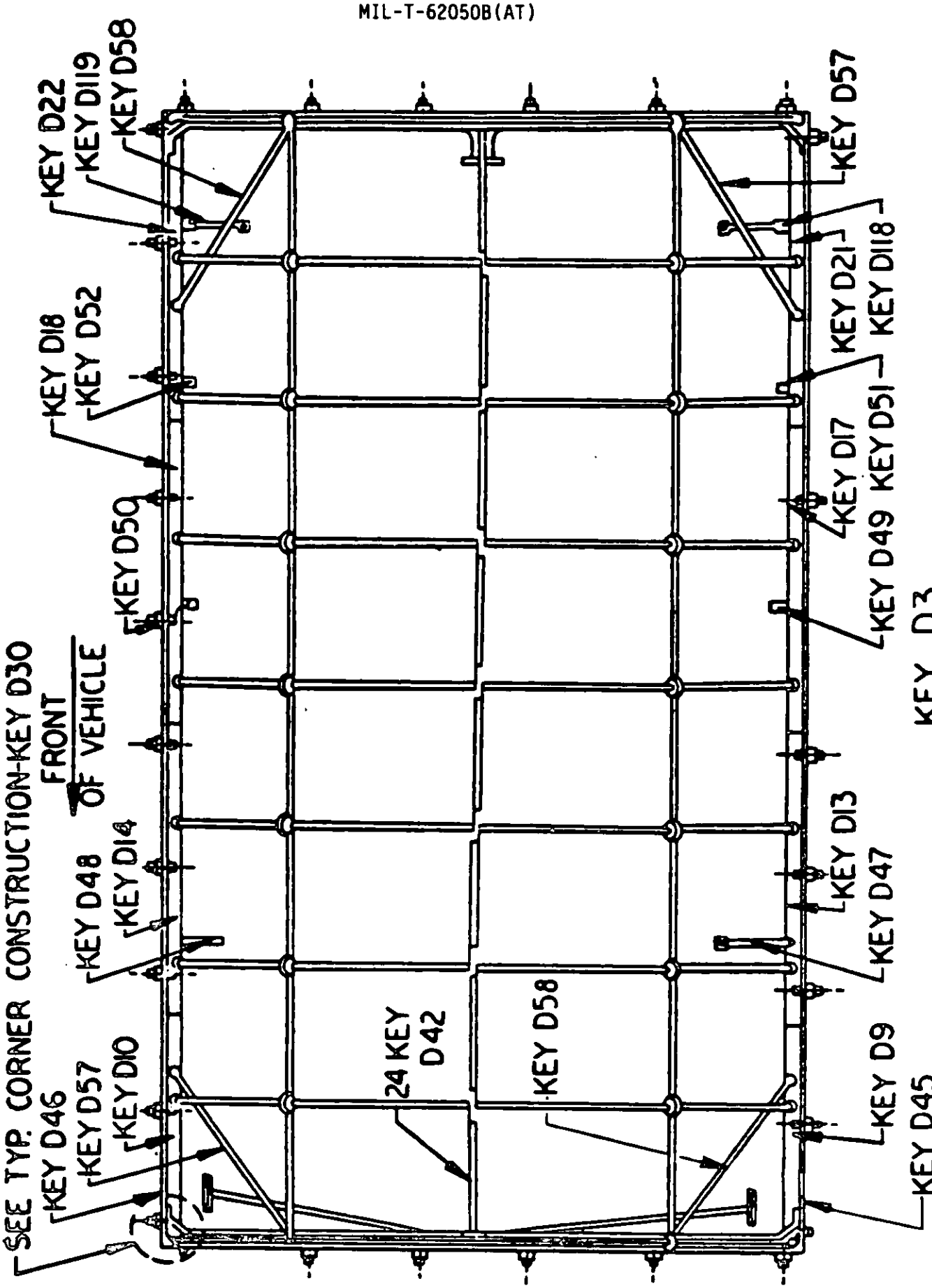


FIGURE 32. Plan view of frame.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

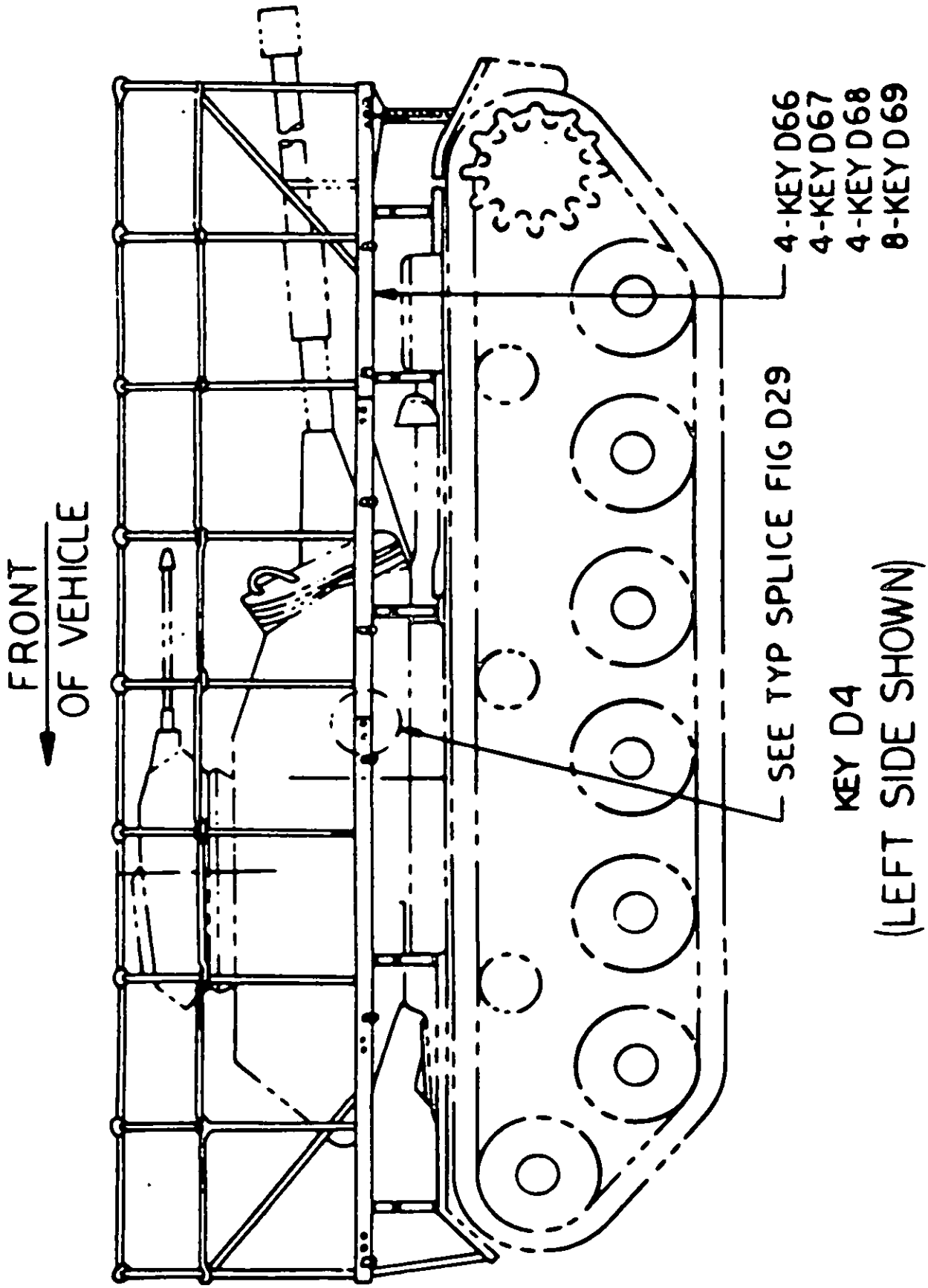


FIGURE 33. Right and left view of the frame.

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Check the source to verify that this is the current version before use.

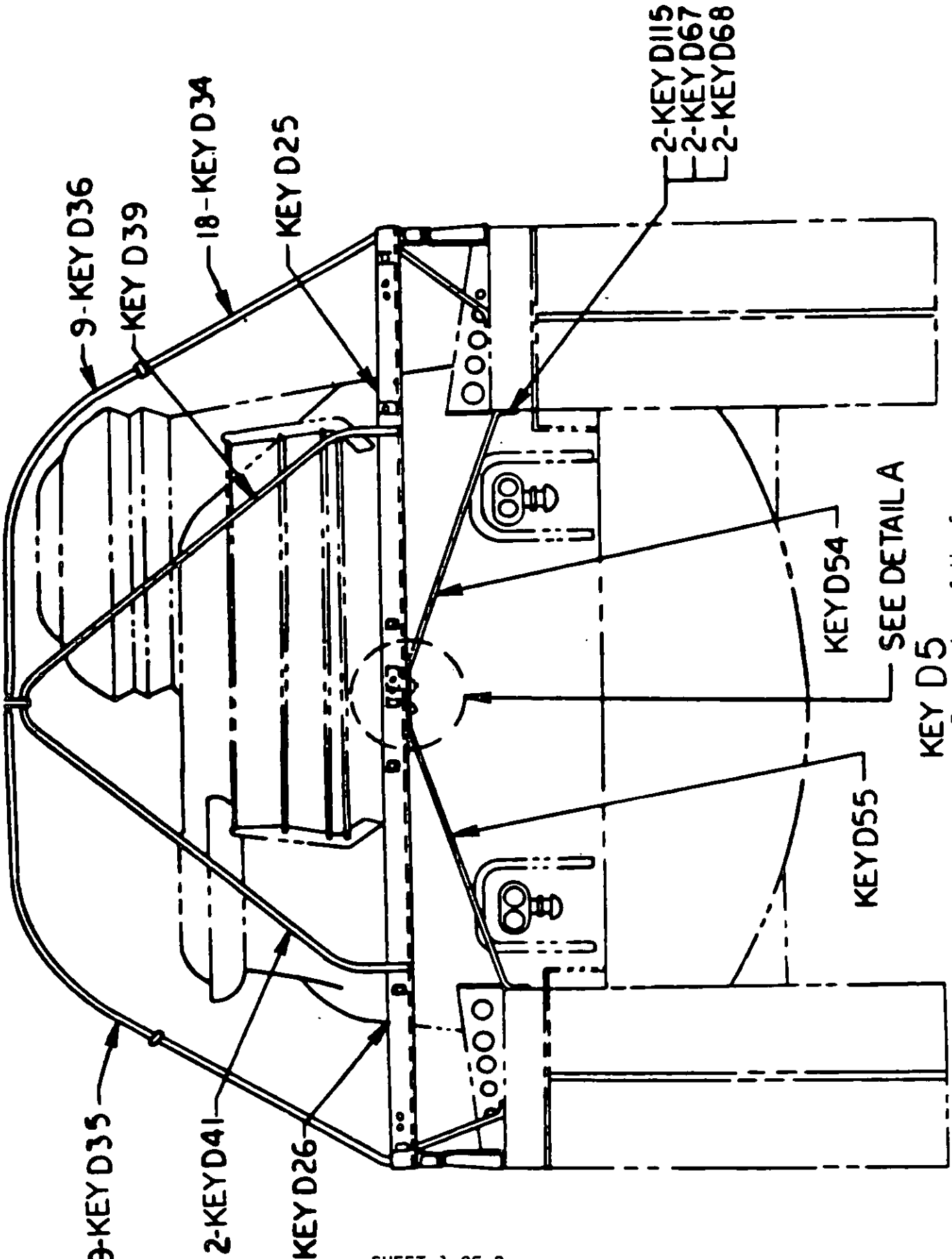
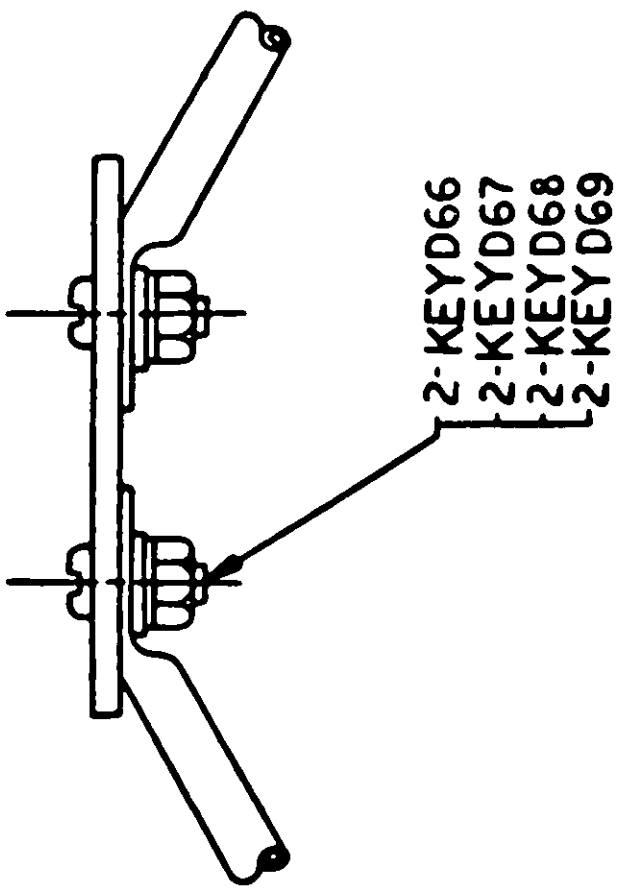


FIGURE 34. Front view of the frame.

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DETAIL A
KEY D5

FIGURE 34. Front view of the frame - Continued.

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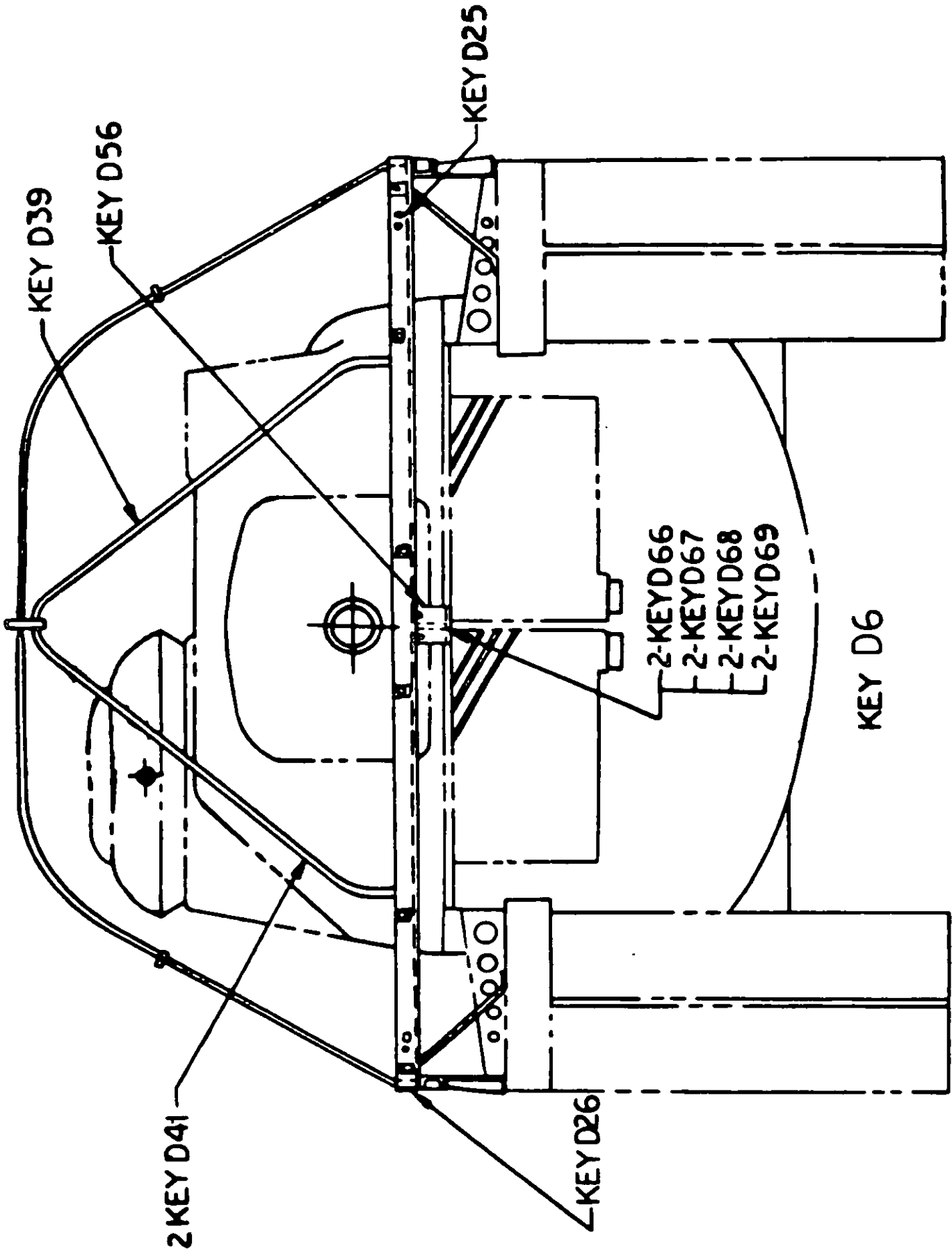
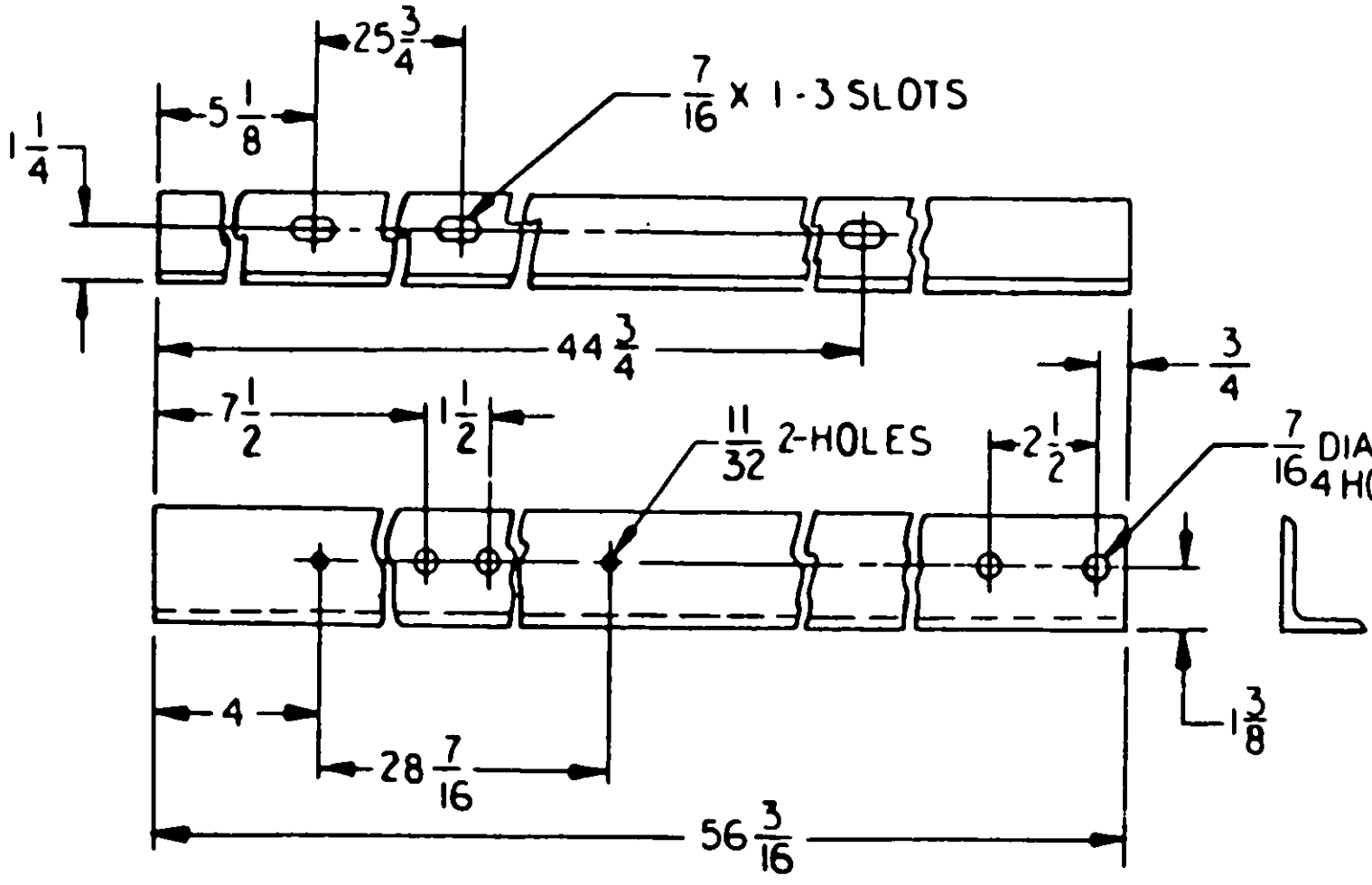


FIGURE 35. Rear view of the frame.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

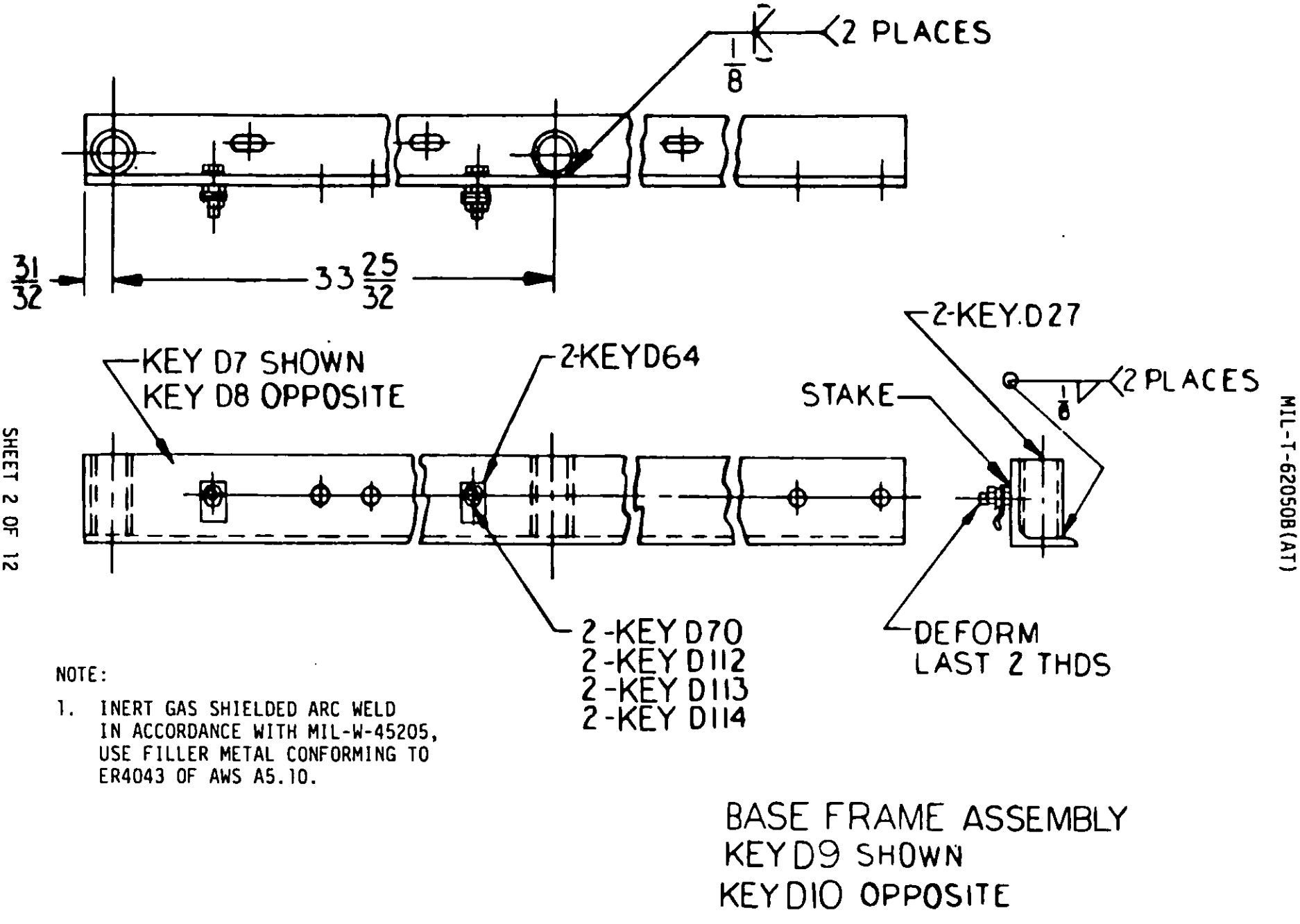


NOTE:
1. MATERIAL:
   ALUMINUM ALLOY
   T6-TEMPER PER
   QQ-A-200/8
   2-1/2 X 2 X 1/4 ANGLE.

BASE FRAME
KEY D7 SHOWN
KEY D8 OPPOSITE

FIGURE 36. Base frame and frame assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

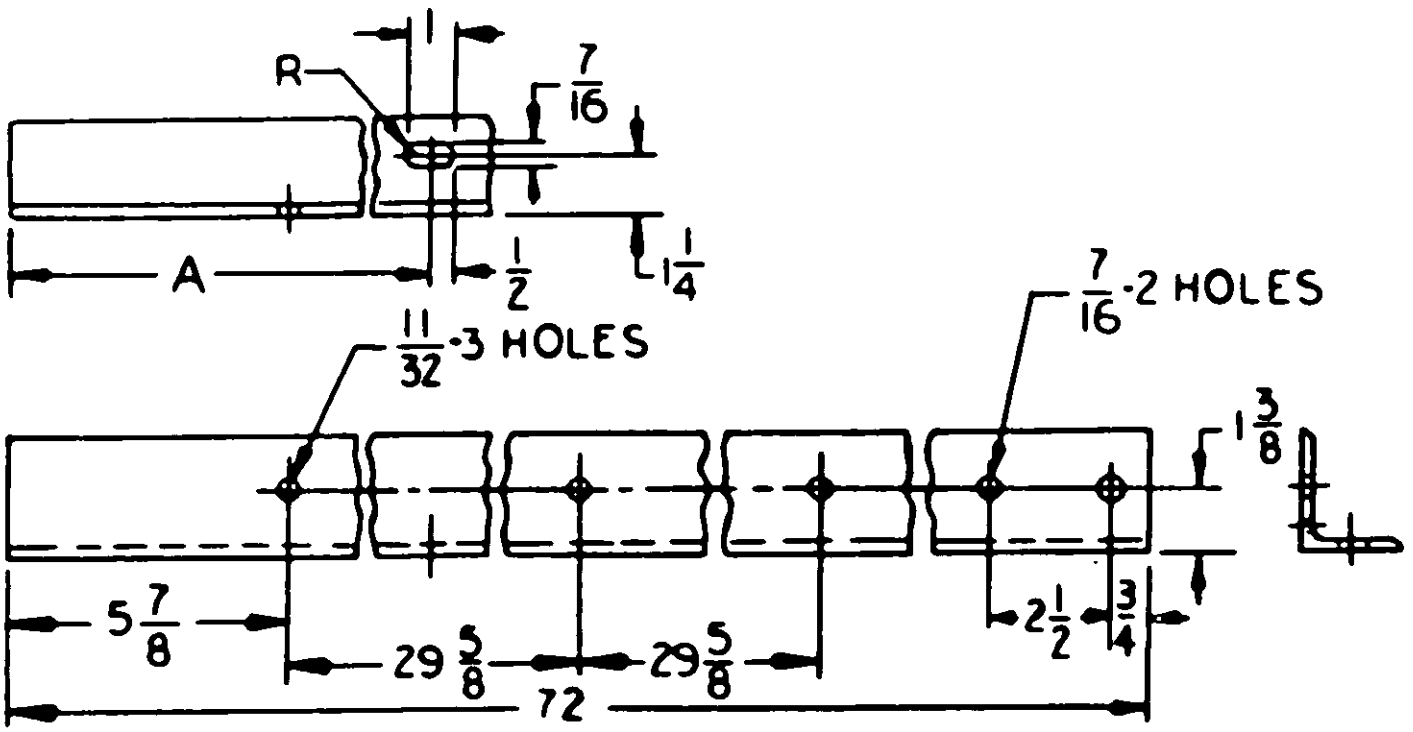


NOTE:

1. INERT GAS SHIELDED ARC WELD IN ACCORDANCE WITH MIL-W-45205, USE FILLER METAL CONFORMING TO ER4043 OF AWS A5.10.

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE:
1. MATERIAL:
   ALUMINUM ALLOY
   T6-TEMPER PER
   QQ-A-200/8
   2-1/2 X 2 X 1/4 ANGLE.

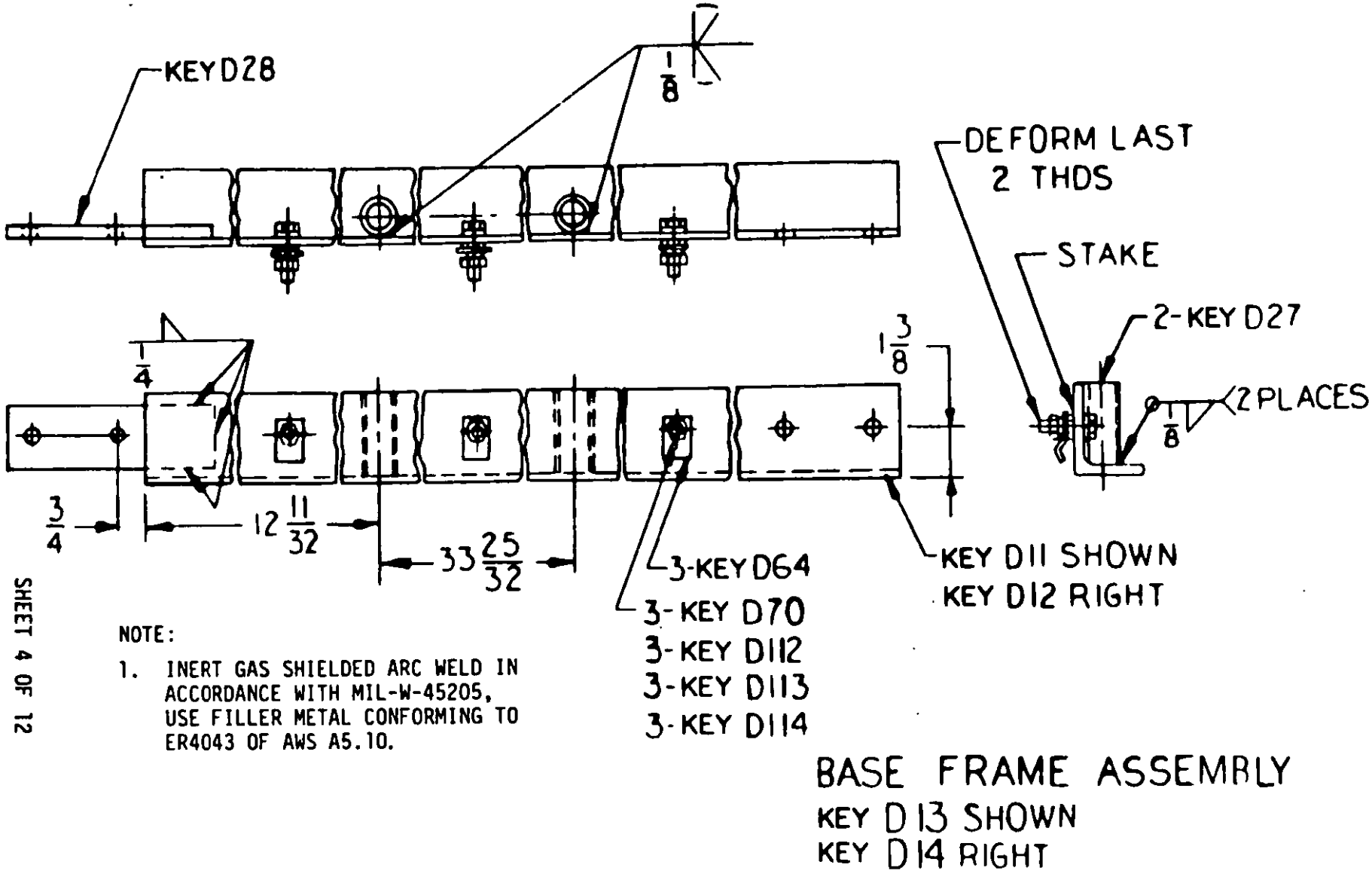
KEY	A
KEY D11	16
KEY D12	15 5/16

BASE FRAME
KEY D11 SHOWN
KEY D12 RIGHT

SHEET 3 OF 12

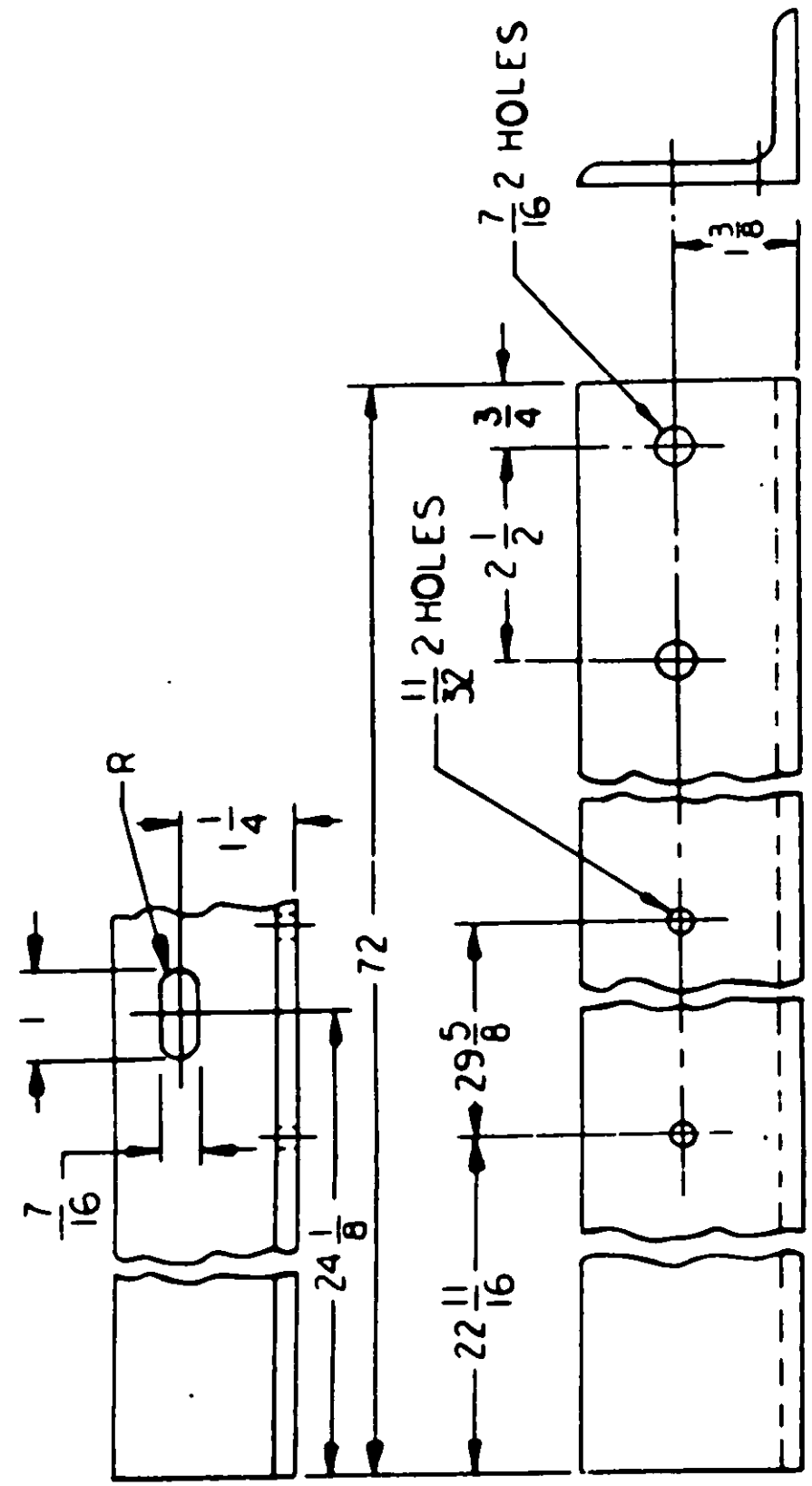
FIGURE 36. Base frame and frame assembly - Continued.

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Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FIGURE 36. Base frame and frame assembly - Continued.



NOTE:
1. MATERIAL:
ALUMINUM ALLOY
T6-TEMPER PER
QQ-A-200/8
2-1/2 X 2 X 1/4 ANGLE.

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

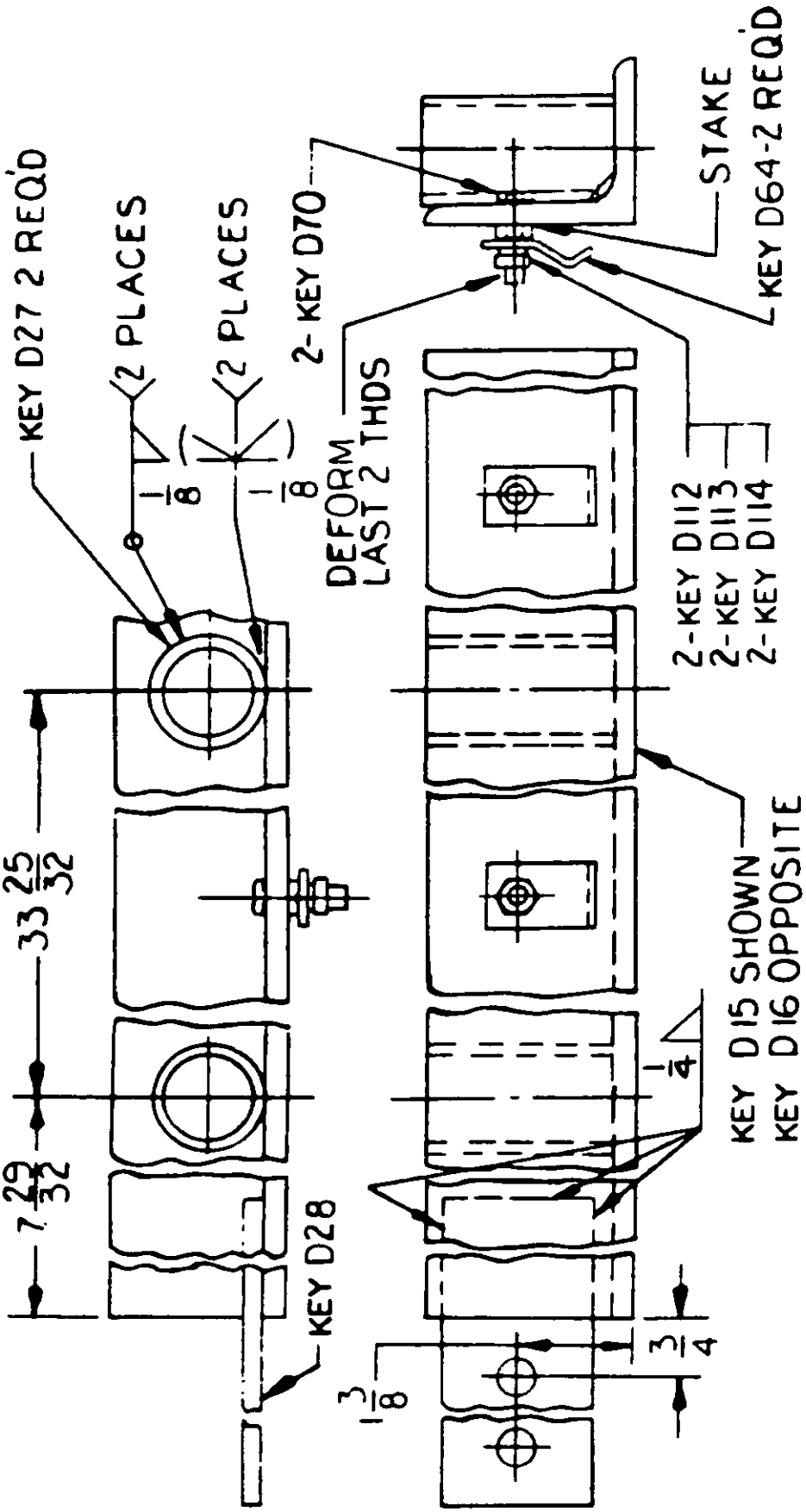
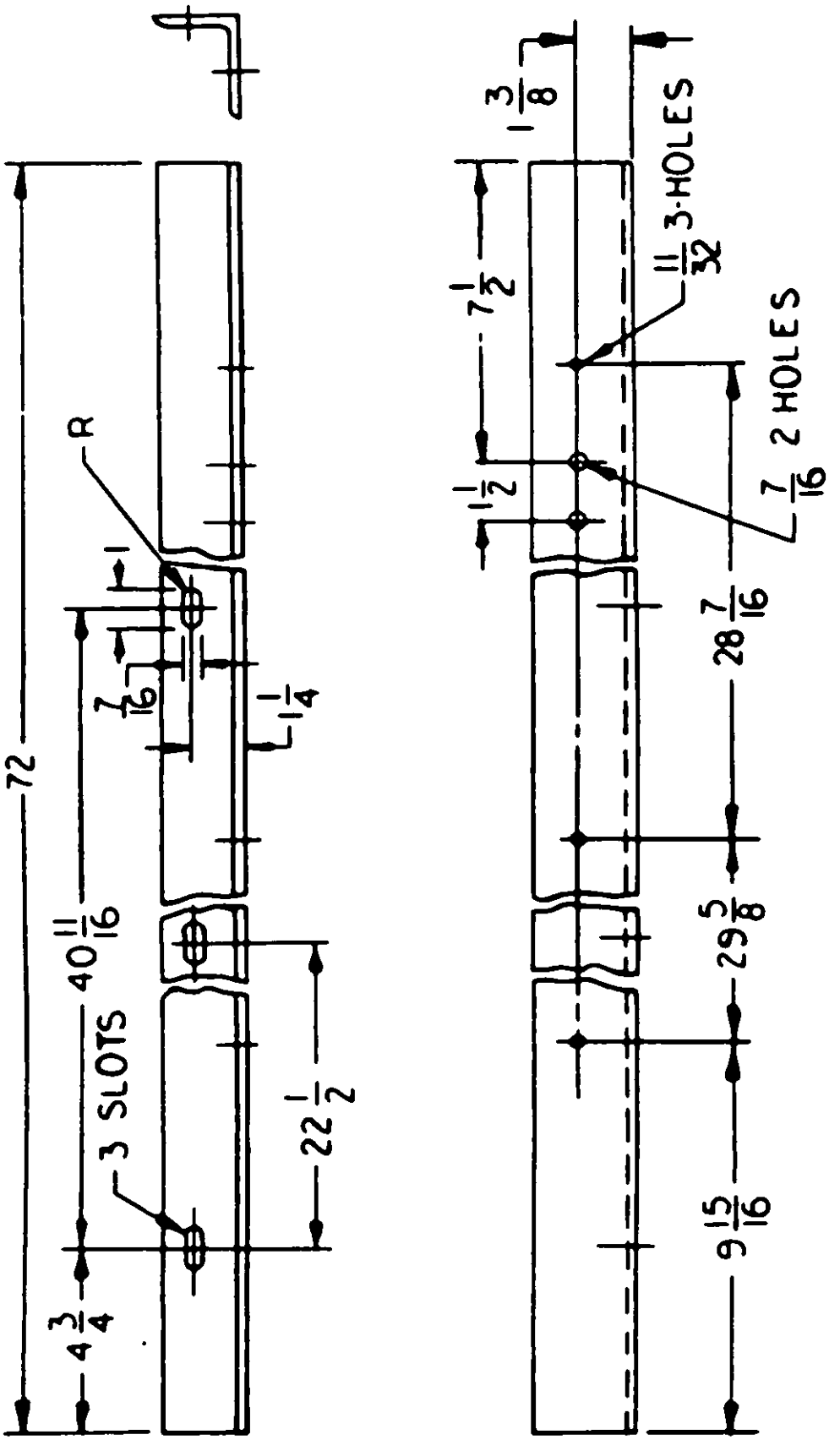


FIGURE 36. Base frame and frame assembly - Continued.

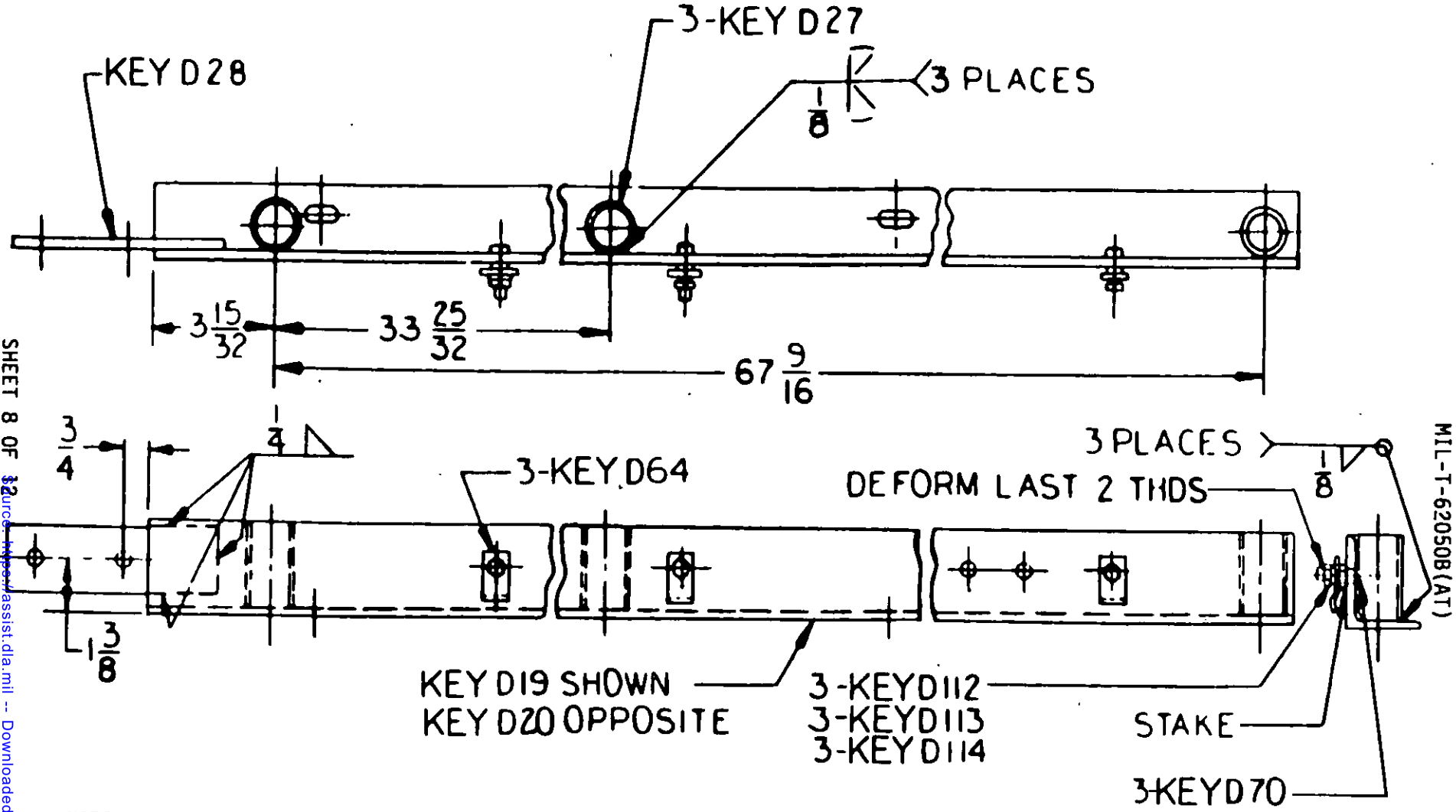
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE:
1. MATERIAL:
   ALUMINUM ALLOY
   T6-TEMPER PER
   QQ-A-200/8
   2-1/2 X 2 X 1/4 ANGLE.

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

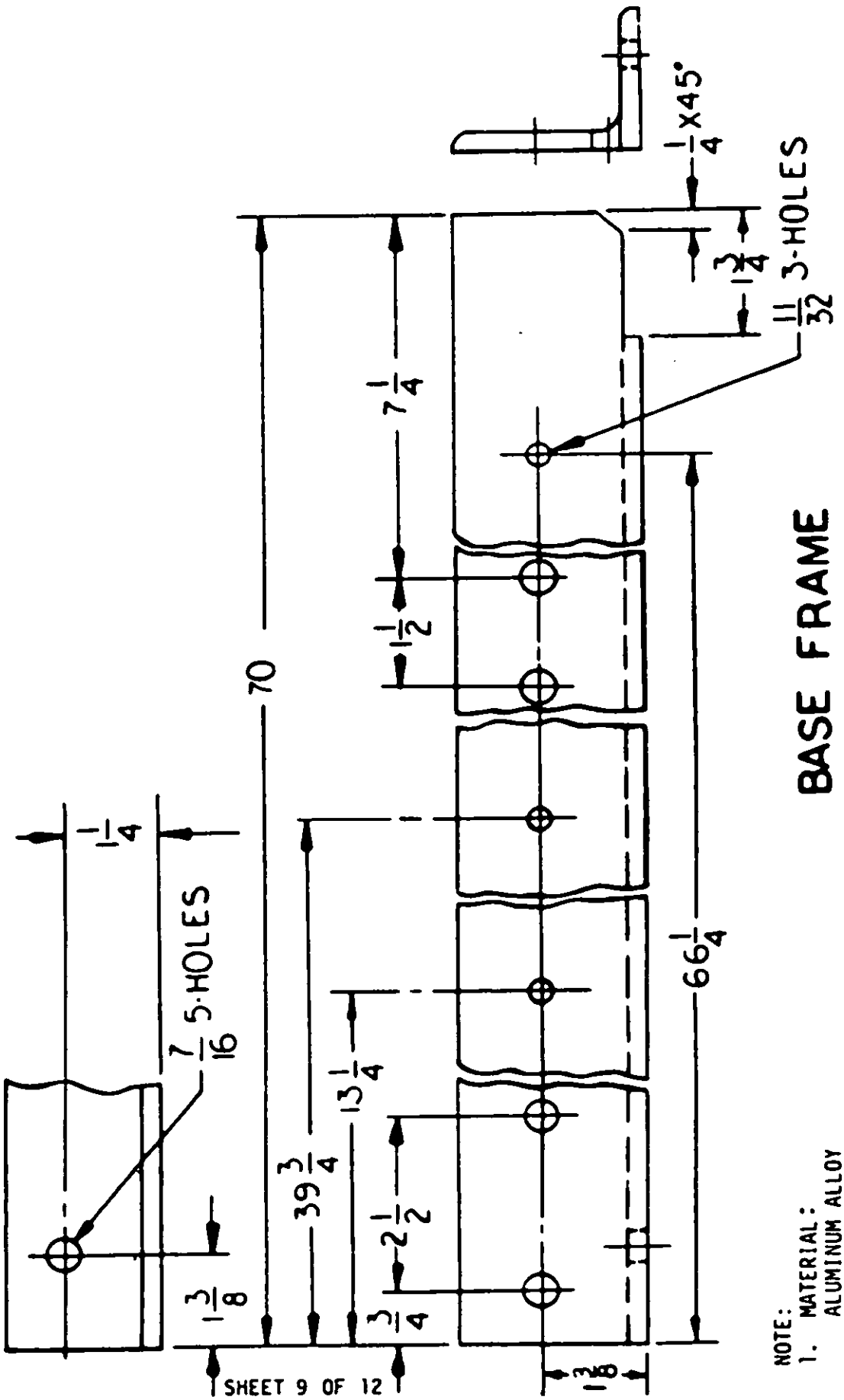


NOTE:
1. INERT GAS SHIELDED ARC WELD IN ACCORDANCE WITH MIL-W-45205, USE FILLER METAL CONFORMING TO ER4043 OF AWS A5.10.

BASE FRAME ASSEMBLY
KEY D21 SHOWN
KEY D22 OPPOSITE

FIGURE 36. Base frame and frame assembly - Continued.

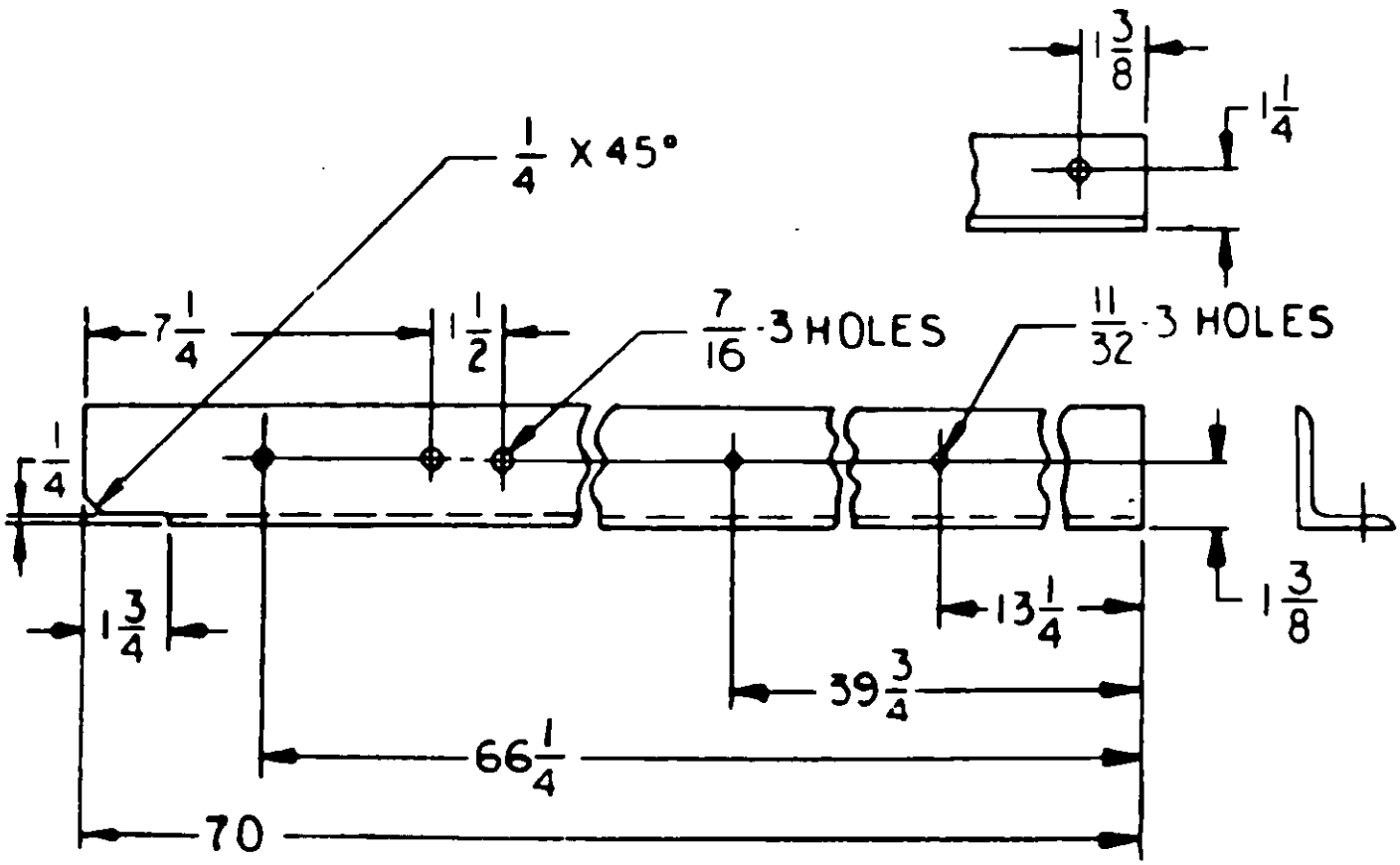
Source: http://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE:
1. MATERIAL:
ALUMINUM ALLOY
T6-TEMPER PER
QQ-A-200/8
2-1/2 X 2 X 1/4 ANGLE.

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

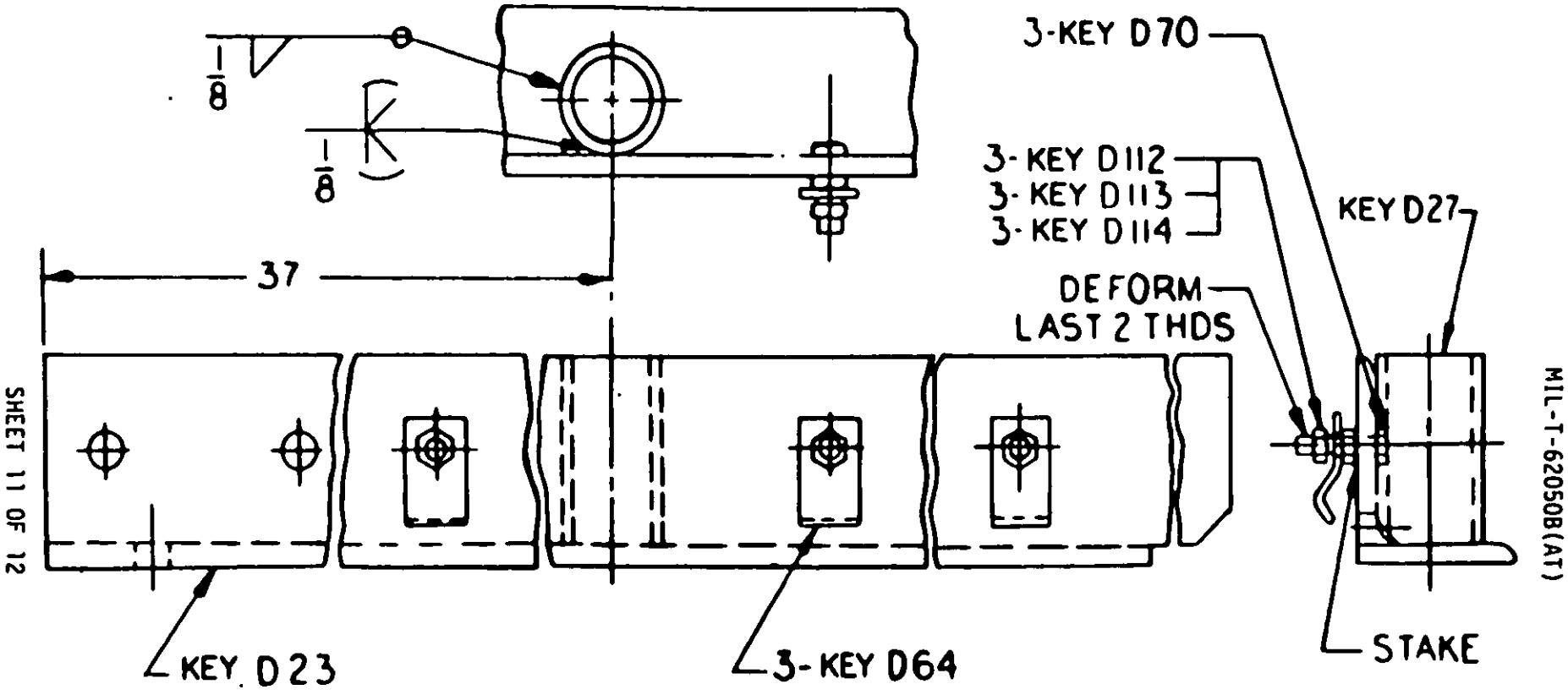


NOTE:
1. MATERIAL:
   ALUMINUM ALLOY
   T6-TEMPER PER
   QQ-A-200/8
   2-1/2 X 2 X 1/4 ANGLE.

BASE FRAME
2 REQ'D
KEY D 24

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

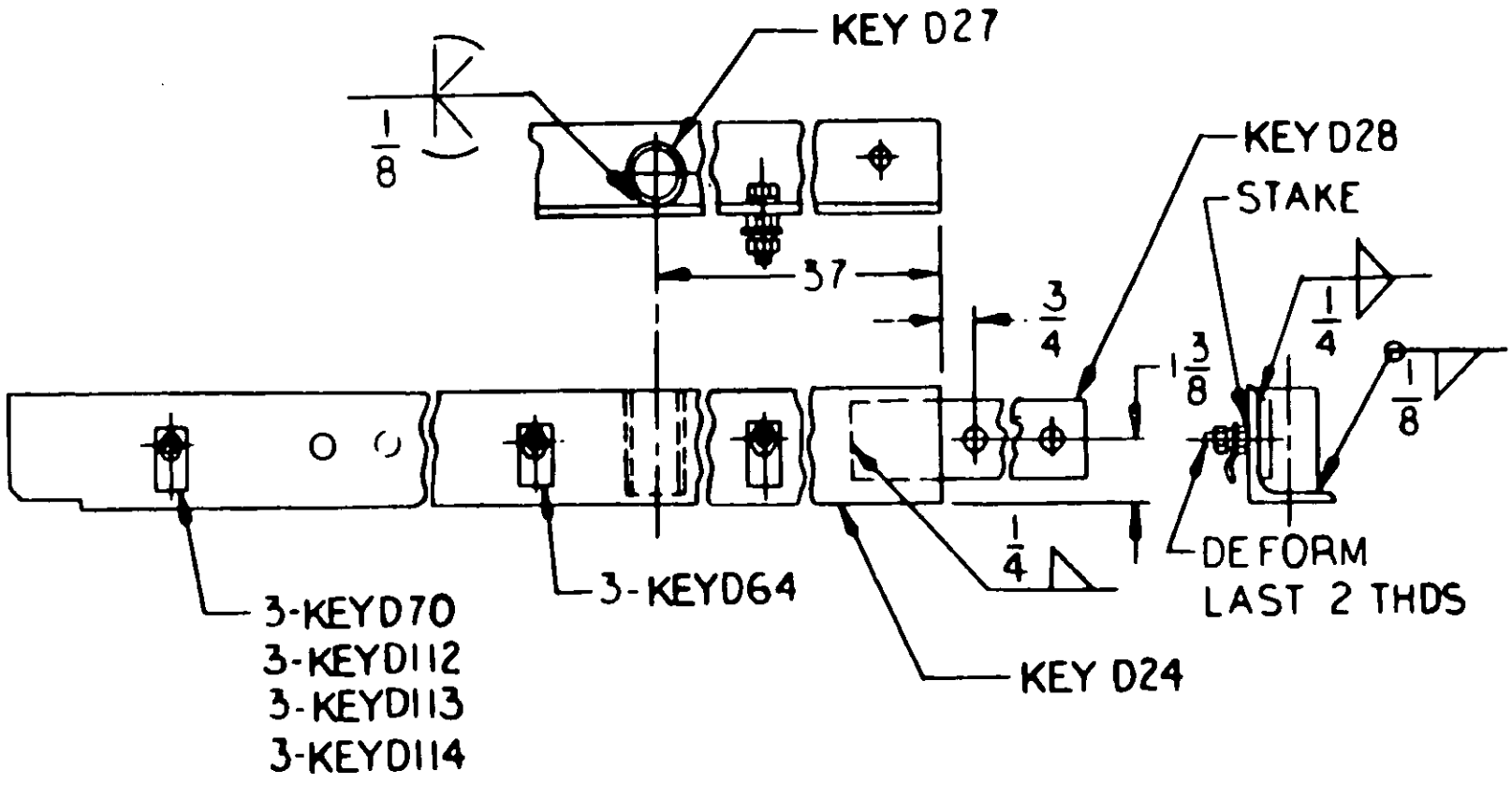


NOTE:
1. INERT GAS SHIELDED ARC WELD IN ACCORDANCE WITH MIL-W-45205, USE FILLER METAL CONFORMING TO ER4043 OF AWS A5.10.

BASE FRAME ASSEMBLY
2 REQ'D
KEY D25

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

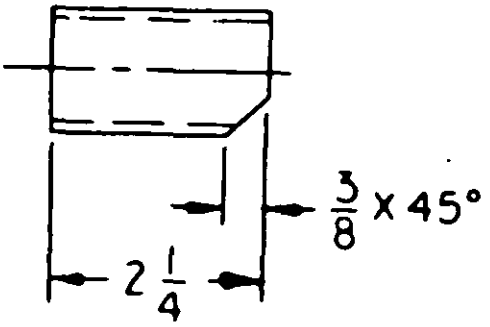


NOTE:
1. INERT GAS SHIELDED ARC WELD IN ACCORDANCE WITH MIL-W-45205, USE FILLER METAL CONFORMING TO ER4043 OF AWS 5.10.

BASE FRAME ASSEMBLY
2 REQ'D
KEY D26

FIGURE 36. Base frame and frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

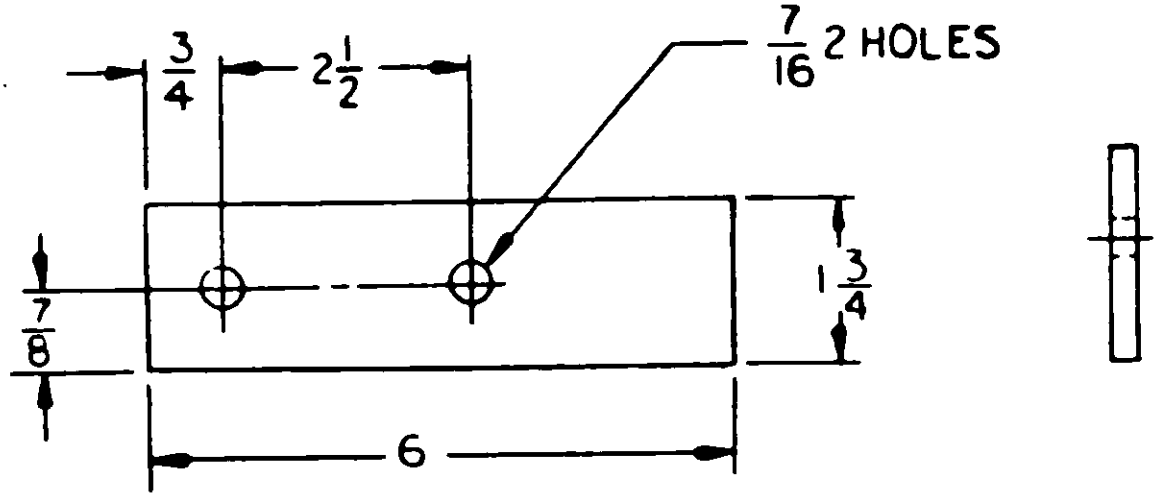


22 REQ'D
KEY D 27

NOTE:
1. MATERIAL: TUBING, ALUMINUM ALLOY, TYPE 1, T6-TEMPER, PER WW-T-700/6 1.315 O.D. x 0.133 WALL.

FIGURE 37. Engagement pipe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



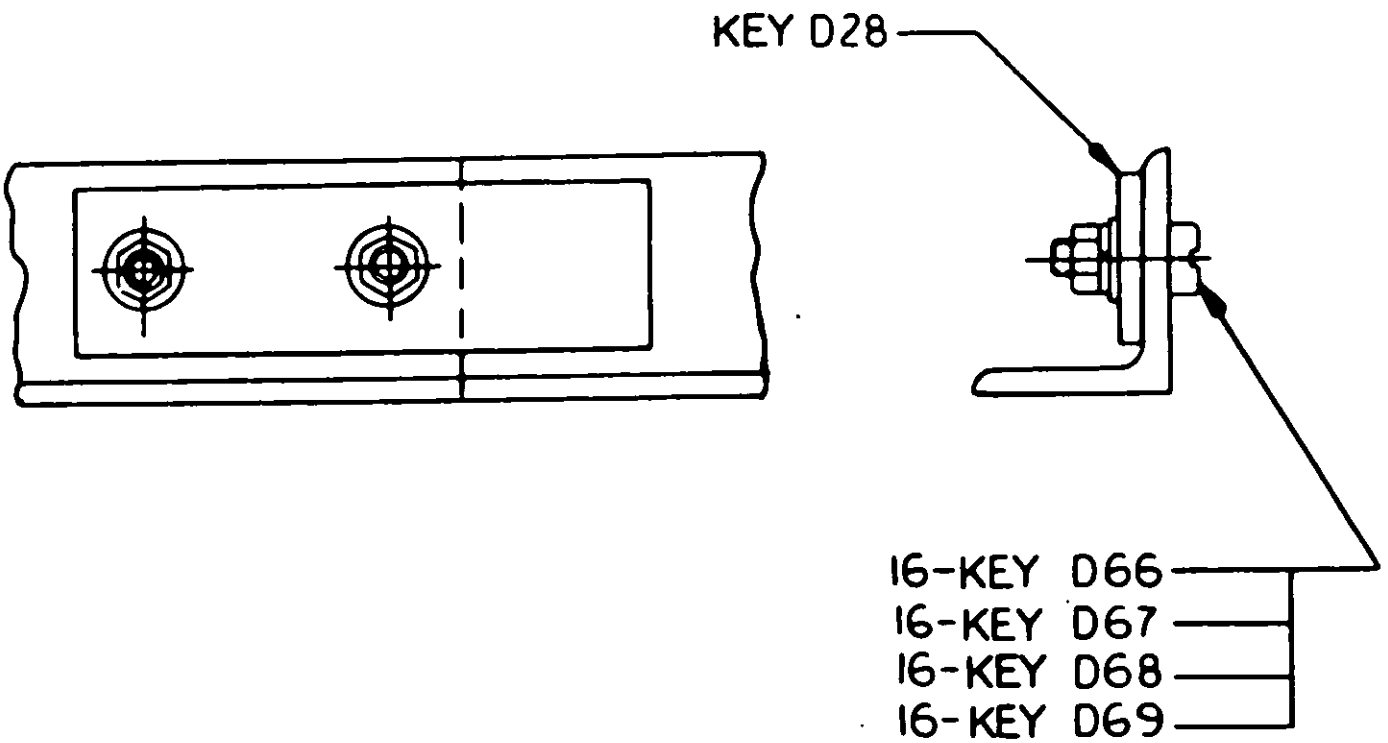
NOTE:
1. MATERIAL:
   ALUMINUM ALLOY, T6-TEMPER,
   PER QQ-A-250/11, 1/4 THICK.

SPLICE PLATE
8 REQ'D
KEY D28

SHEET 1 OF 2

FIGURE 38. Splice plate and typical base frame splice.

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Check the source to verify that this is the current version before use.

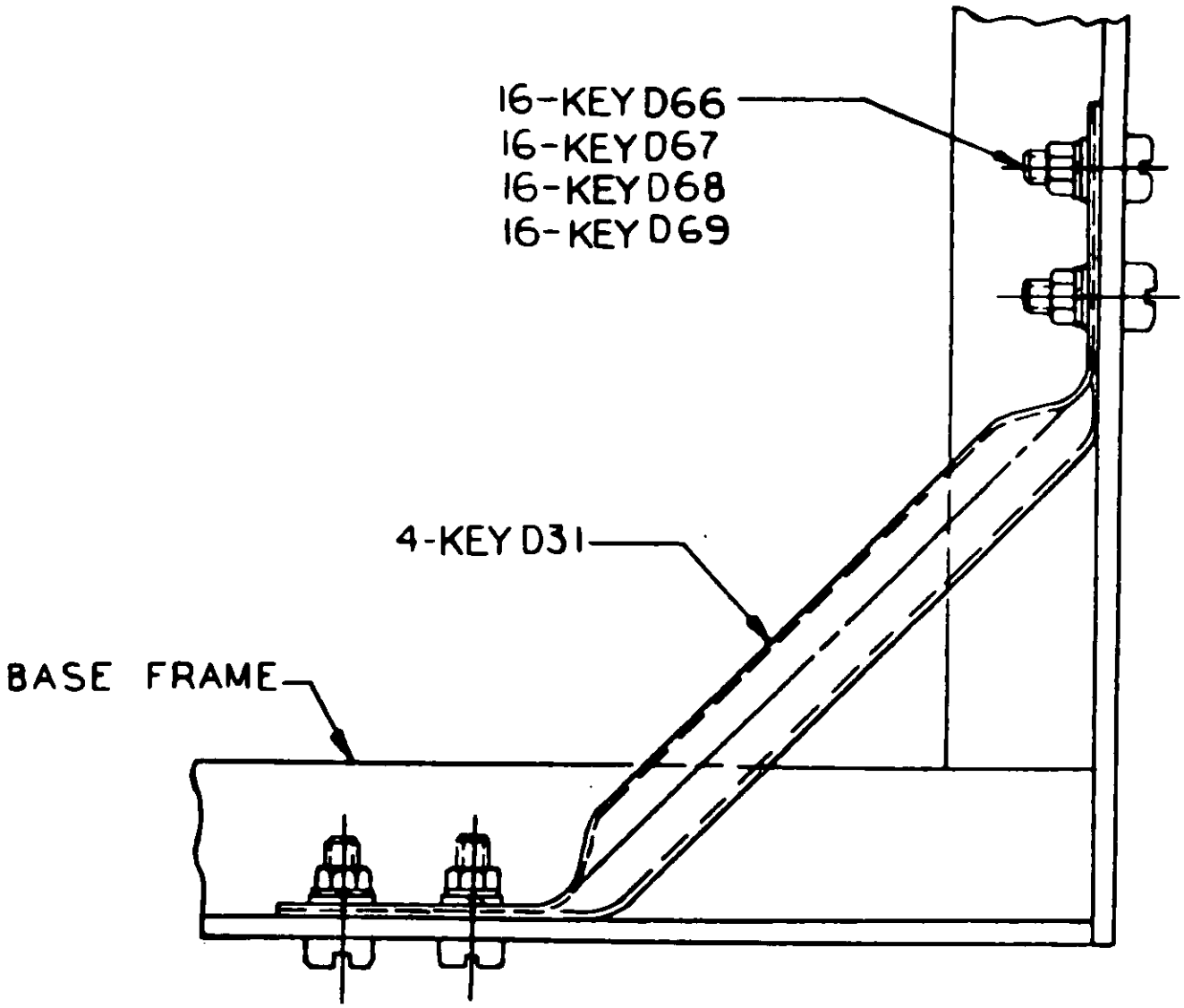


TYPICAL BASE FRAME SPLICE
KEY-D29

SHEET 2 OF 2

FIGURE 38. Splice plate and typical base frame splice - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

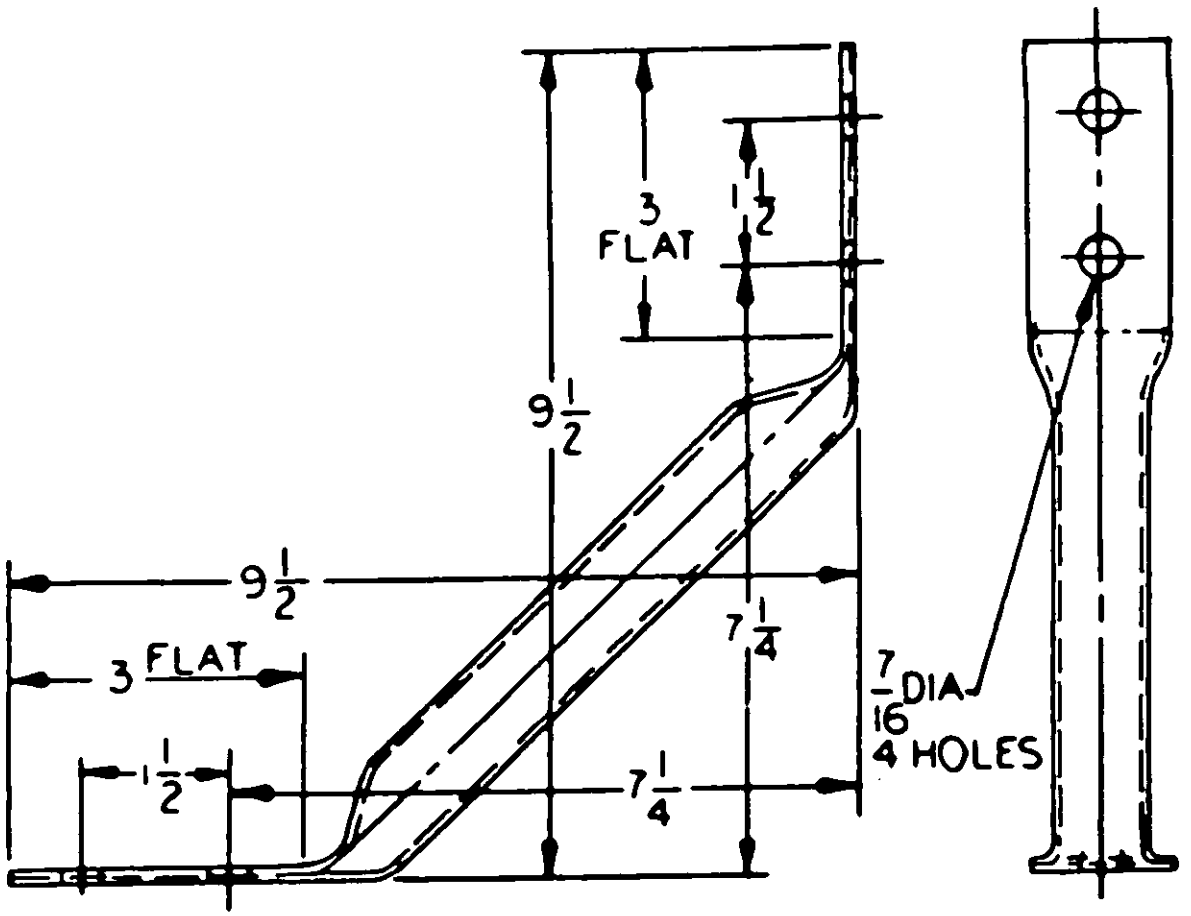


TYPICAL CORNER CONSTRUCTION
KEY D 30

SHEET 1 OF 2

FIGURE 39. Base frame corner and corner construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

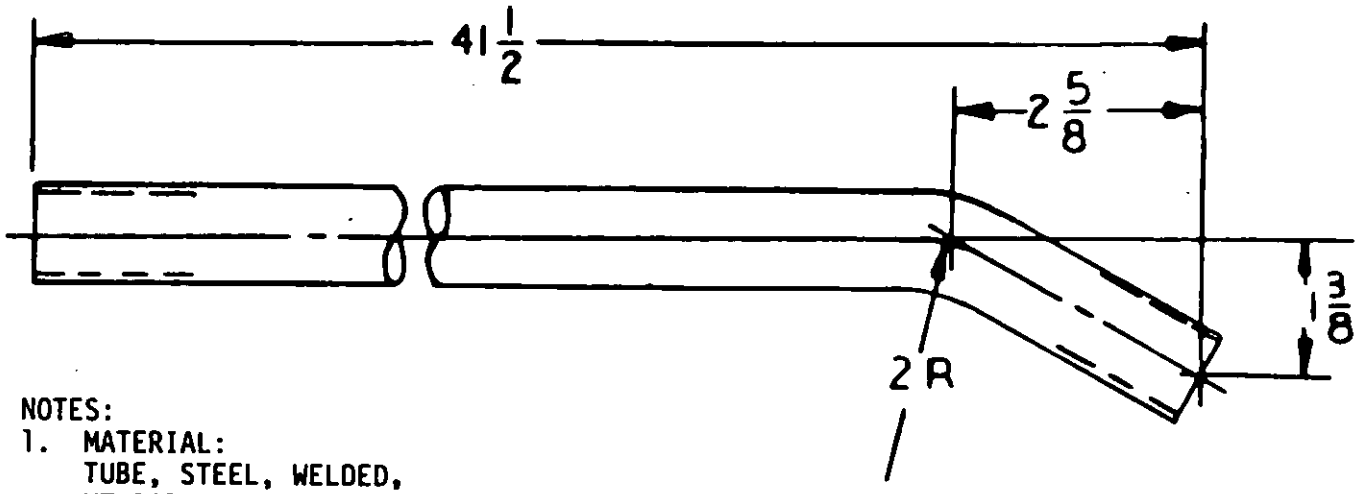


NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED, MT 1010 TO MT 1020,
   PER ASTM A512, A513, OR A519.
2. PRIME IN ACCORDANCE WITH TT-P-636
   ENAMEL SEMIGLOSS COLOR O.D. TYPE
   OPTIONAL PER TT-E-485 OR TT-E-529.

BASE FRAME CORNER TUBE
4 REQ'D
KEY D31

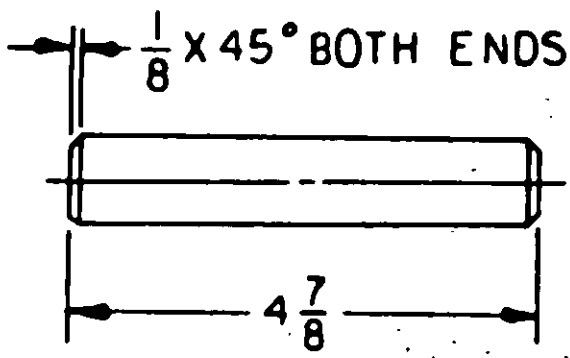
FIGURE 39. Base frame corner and corner construction - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED,
   MT 1010 TO MT 1020
   PER ASTM A512, A513,
   OR A519.
2. I.D. OF TUBE MUST BE
   FREE OF BURRS.

TOP FRAME TUBE
18 REQ'D
KEY D 32



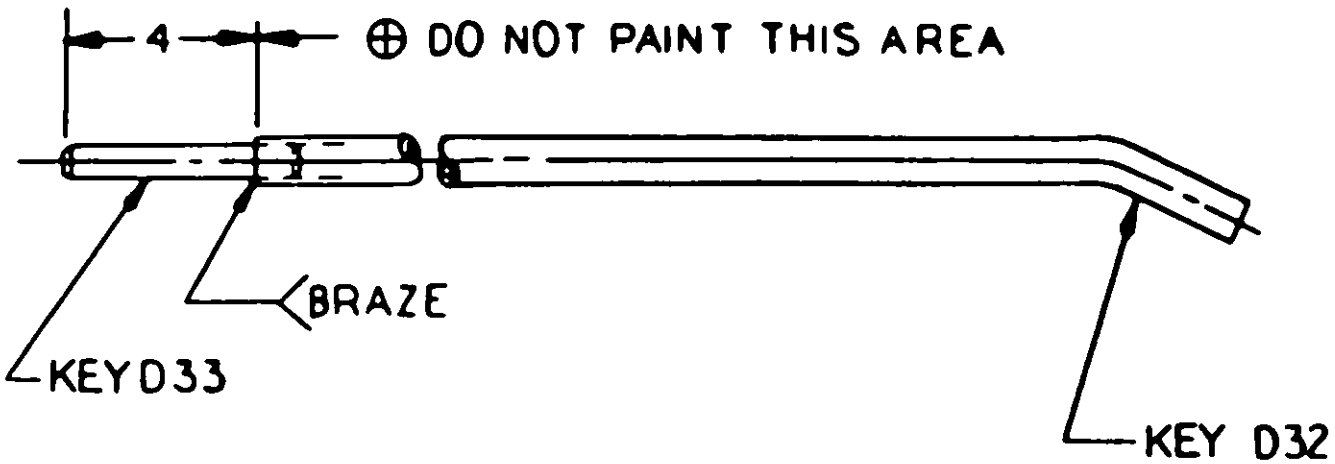
CONNECTOR
22 REQ'D.
KEY D33

NOTES:
1. MATERIAL:
   STEEL 1015 TO 1025,
   PER ASTM A108,
   7/8 DIA.

SHEET 1 OF 9

FIGURE 40. Top frame and top frame assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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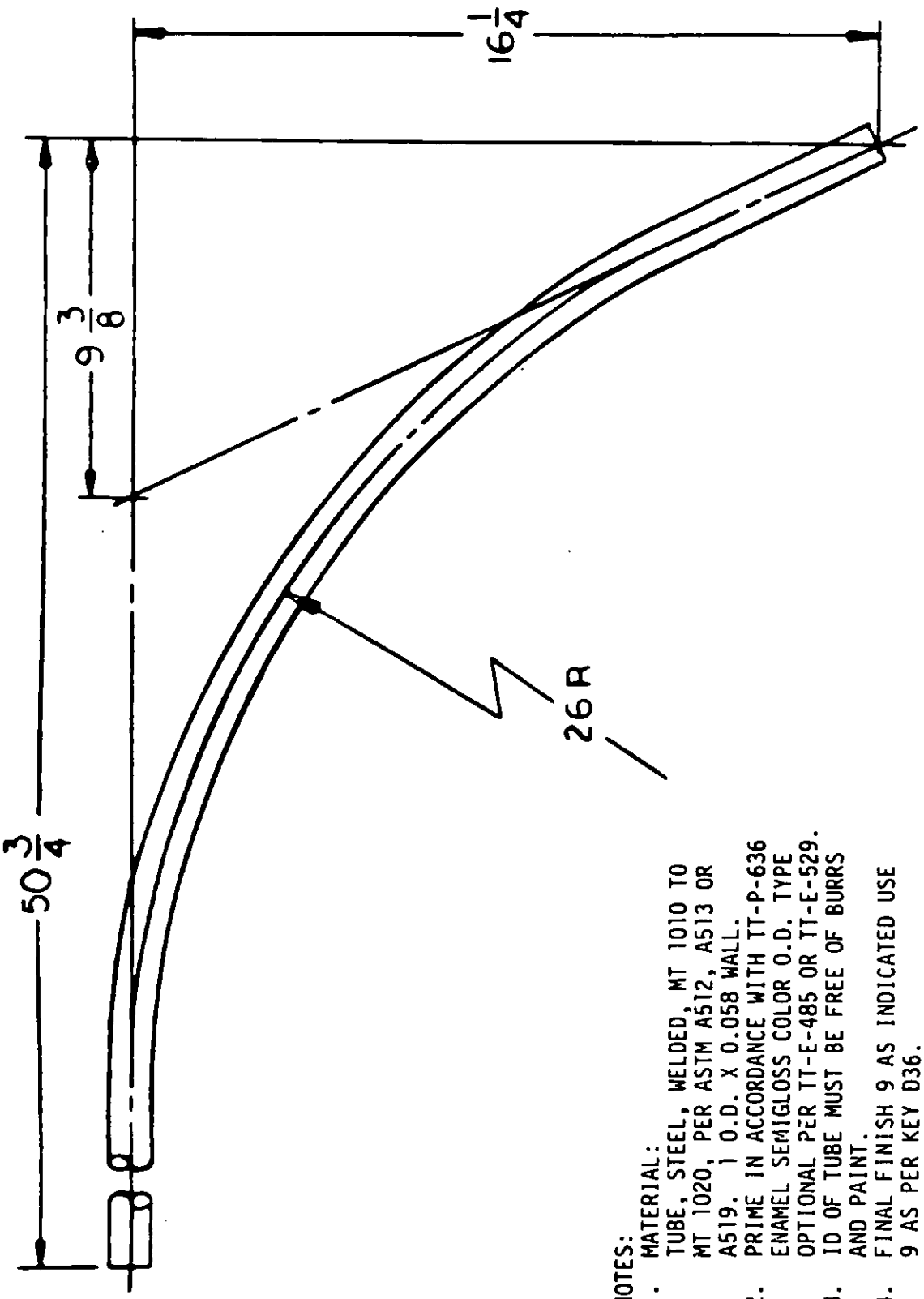


TOP FRAME TUBE ASSEMBLY
18 REQD
KEY D34

NOTES:
1. FURNACE BRAZE IN ACCORDANCE WITH MIL-B-7883, USE FILLER METAL, CLASS FS BCU PER QQ-B-650.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
3. UNPAINTED AREA TO BE COATED WITH MIL-C-16173 CORROSION PREVENTIVE OR TT-C-490 CLASS C, GRADE I, PHOSPHATE PROTECTIVE COATING.

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

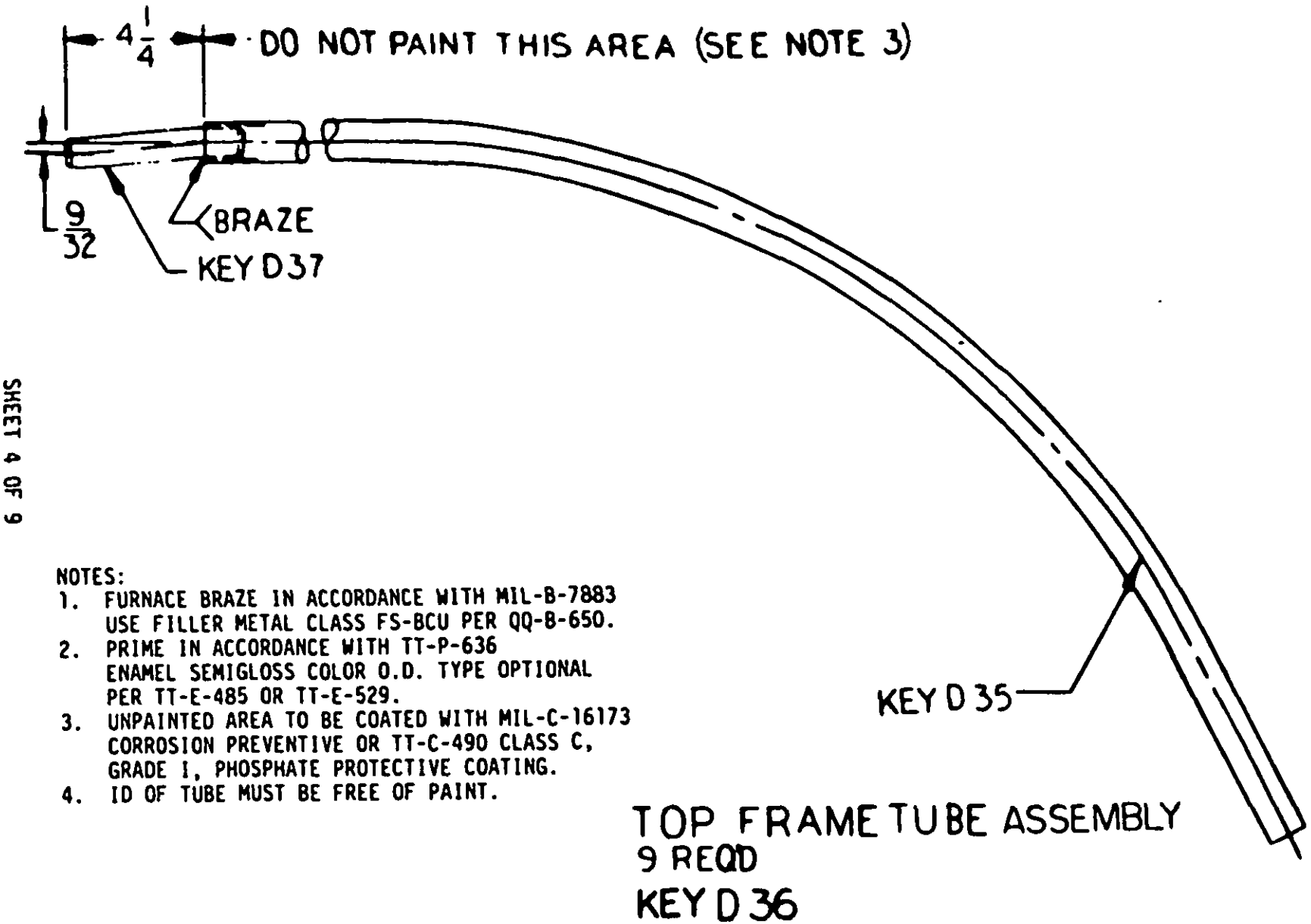


TOP FRAME TUBE
18 REQD (SEE NOTES)
KEY D35

NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED, MT 1010 TO MT 1020, PER ASTM A512, A513 OR A519. 1 O.D. X 0.058 WALL.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
3. ID OF TUBE MUST BE FREE OF BURRS AND PAINT.
4. FINAL FINISH 9 AS INDICATED USE 9 AS PER KEY D36.

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

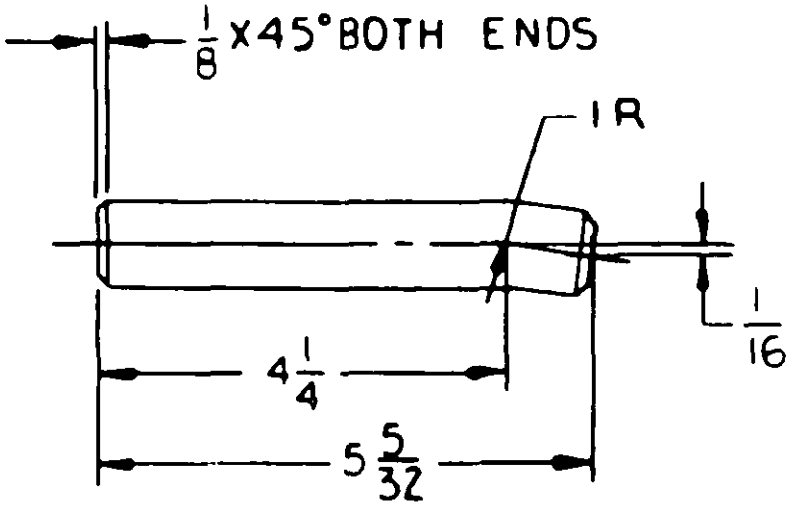


NOTES:
1. FURNACE BRAZE IN ACCORDANCE WITH MIL-B-7883 USE FILLER METAL CLASS FS-BCU PER QQ-B-650.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
3. UNPAINTED AREA TO BE COATED WITH MIL-C-16173 CORROSION PREVENTIVE OR TT-C-490 CLASS C, GRADE I, PHOSPHATE PROTECTIVE COATING.
4. ID OF TUBE MUST BE FREE OF PAINT.

SHEET 4 OF 9

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:22Z
Check the source to verify that this is the current version before use.

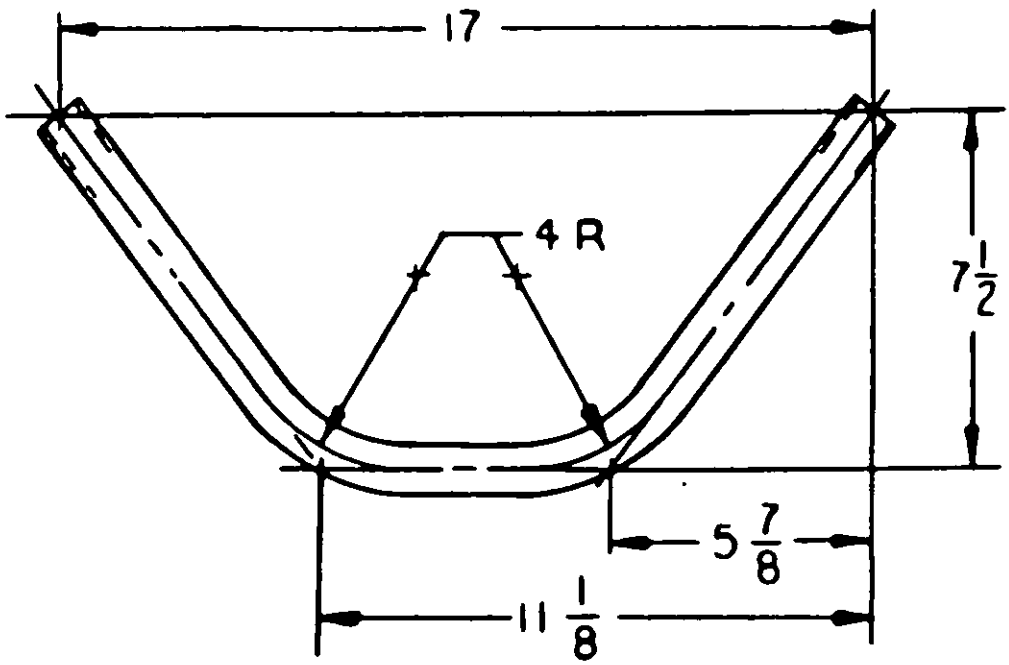


NOTE:
1. MATERIAL:
   STEEL, 1015 TO 1025,
   PER ASTM A108,
   7/8 DIA.

CONNECTOR
9 REQ'D
KEY D37

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

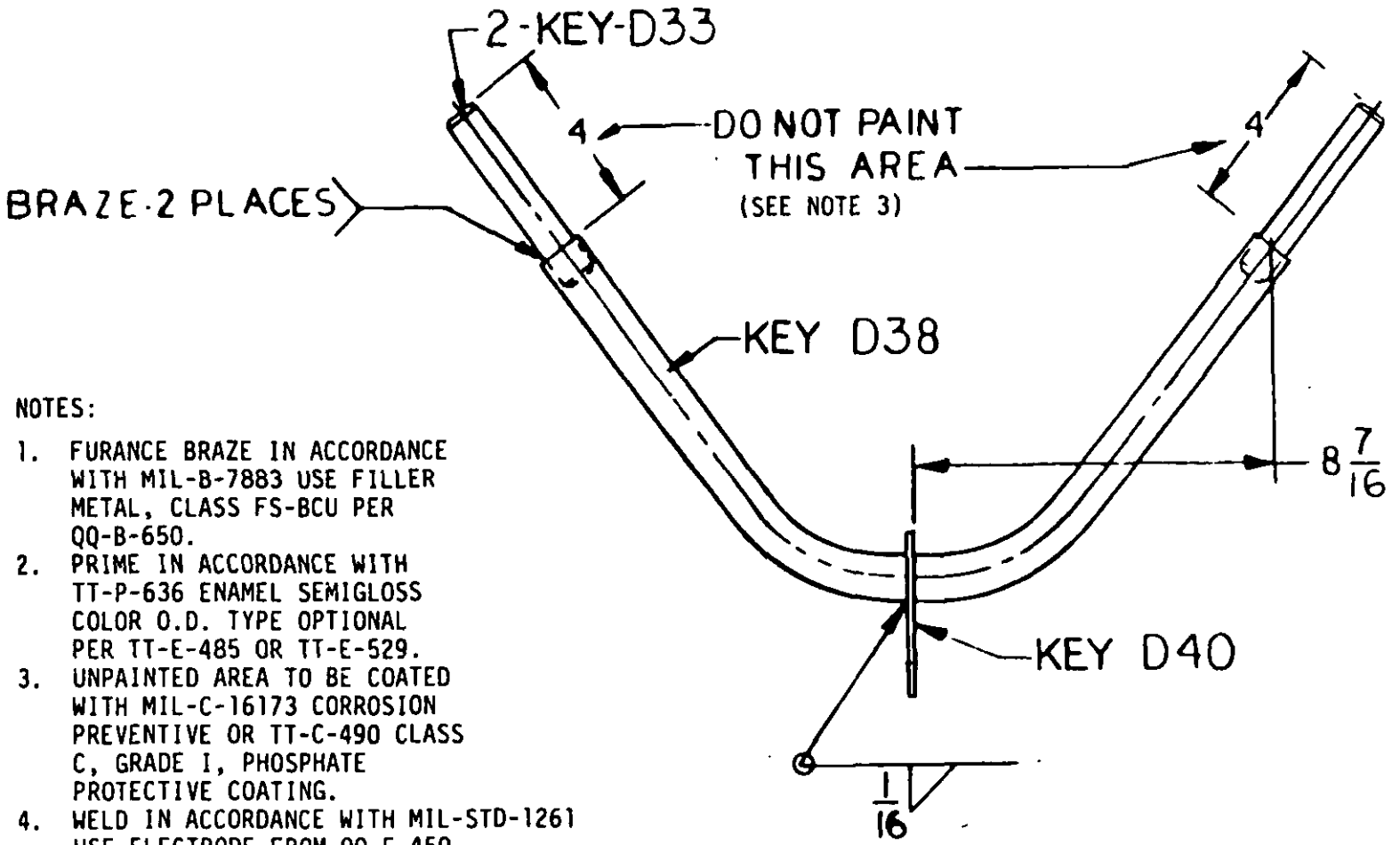


NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED,
   MT 1010 TO 1020, PER
   ASTM A512, A513 OR A519
   1 O.D. X 0.058 WALL.
2. I.D. OF TUBE MUST BE
   FREE OF BURRS.

UPPER BRACE TUBE
2 REQ'D
KEY D 38

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



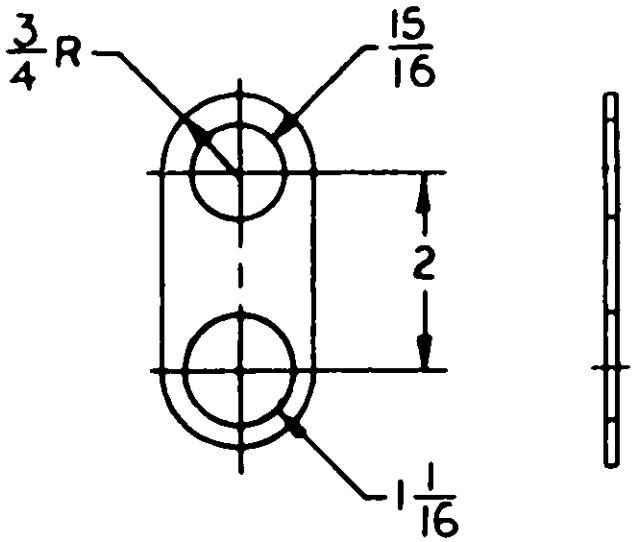
NOTES:

1. FURANCE BRAZE IN ACCORDANCE WITH MIL-B-7883 USE FILLER METAL, CLASS FS-BCU PER QQ-B-650.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
3. UNPAINTED AREA TO BE COATED WITH MIL-C-16173 CORROSION PREVENTIVE OR TT-C-490 CLASS C, GRADE I, PHOSPHATE PROTECTIVE COATING.
4. WELD IN ACCORDANCE WITH MIL-STD-1261 USE ELECTRODE FROM QQ-E-450.

TUBE ASSEMBLY
2 REQ'D
KEY D39

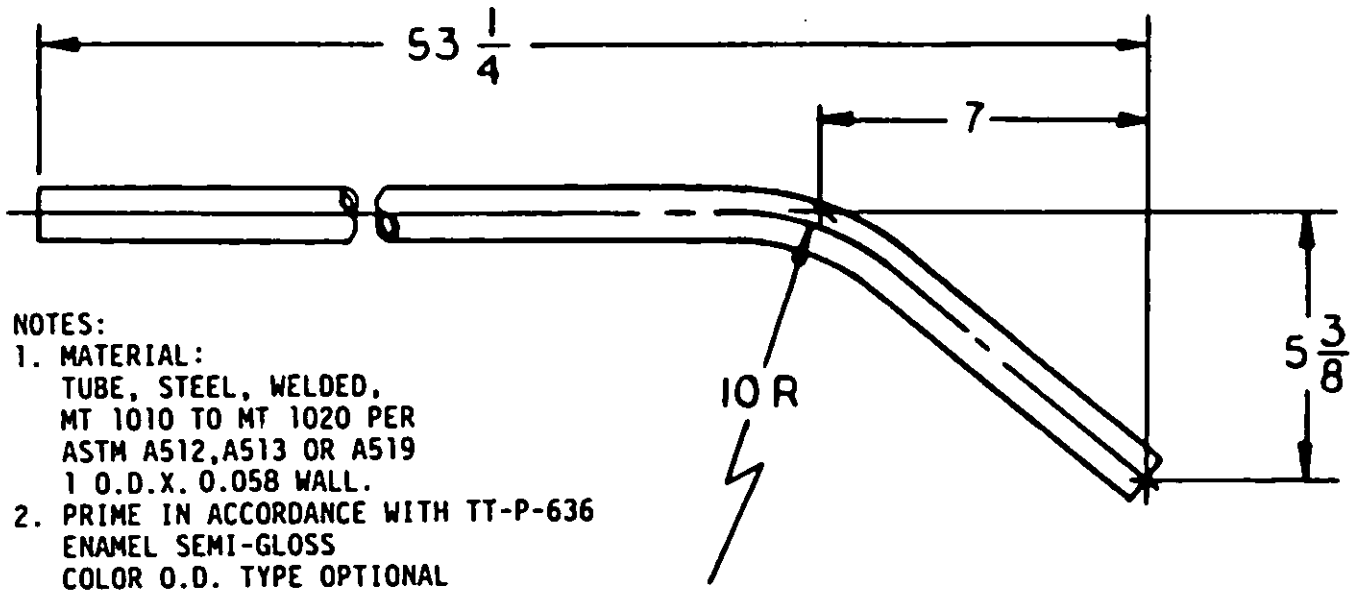
FIGURE 40. Top frame and top frame assembly - Continued.

Source: http://assist.dla.mil -- Downloaded: 2026/0-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE:
1. MATERIAL:
   STEEL,FS1009 TO FS1020
   CONDITION OPTIONAL PER
   QQ-S-698
   0.125 THICK.

PLATE
2 REQ'D
KEY D40

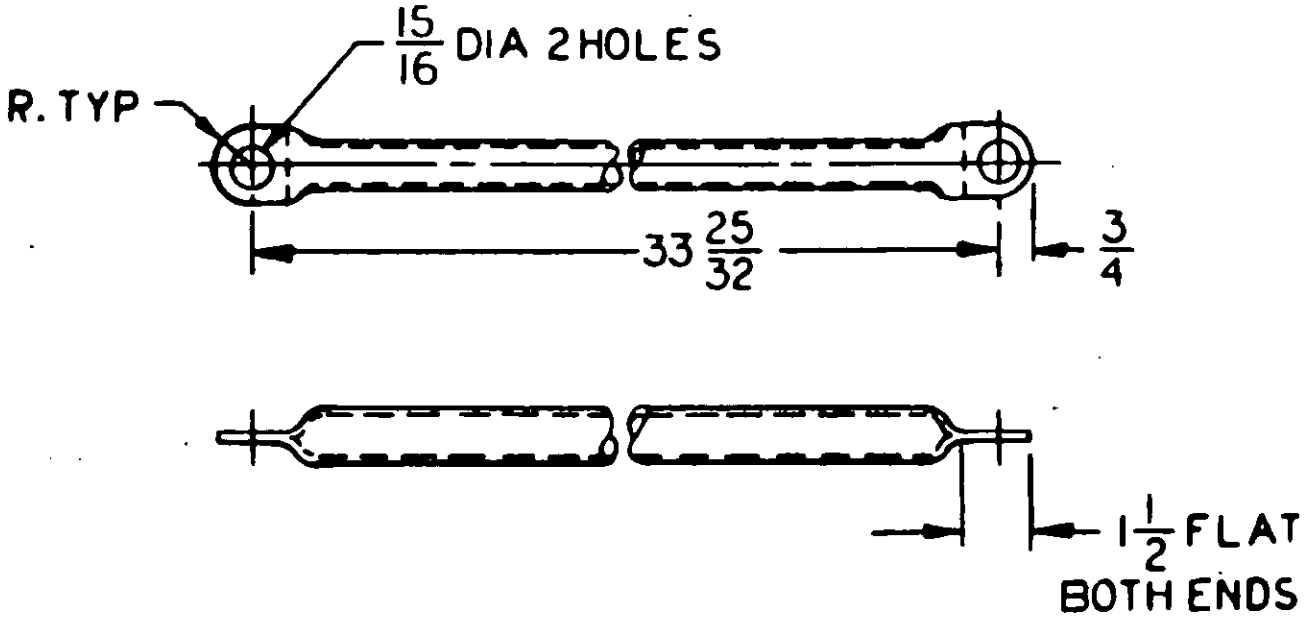


NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED,
   MT 1010 TO MT 1020 PER
   ASTM A512,A513 OR A519
   1 O.D.X. 0.058 WALL.
2. PRIME IN ACCORDANCE WITH TT-P-636
   ENAMEL SEMI-GLOSS
   COLOR O.D. TYPE OPTIONAL
   PER TT-E-485 OR TT-E-529.
3. I.D. OF TUBE MUST BE FREE
   OF BURR AND PAINT.

TOP FRAME TUBE
4 REQ'D
KEY D41

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

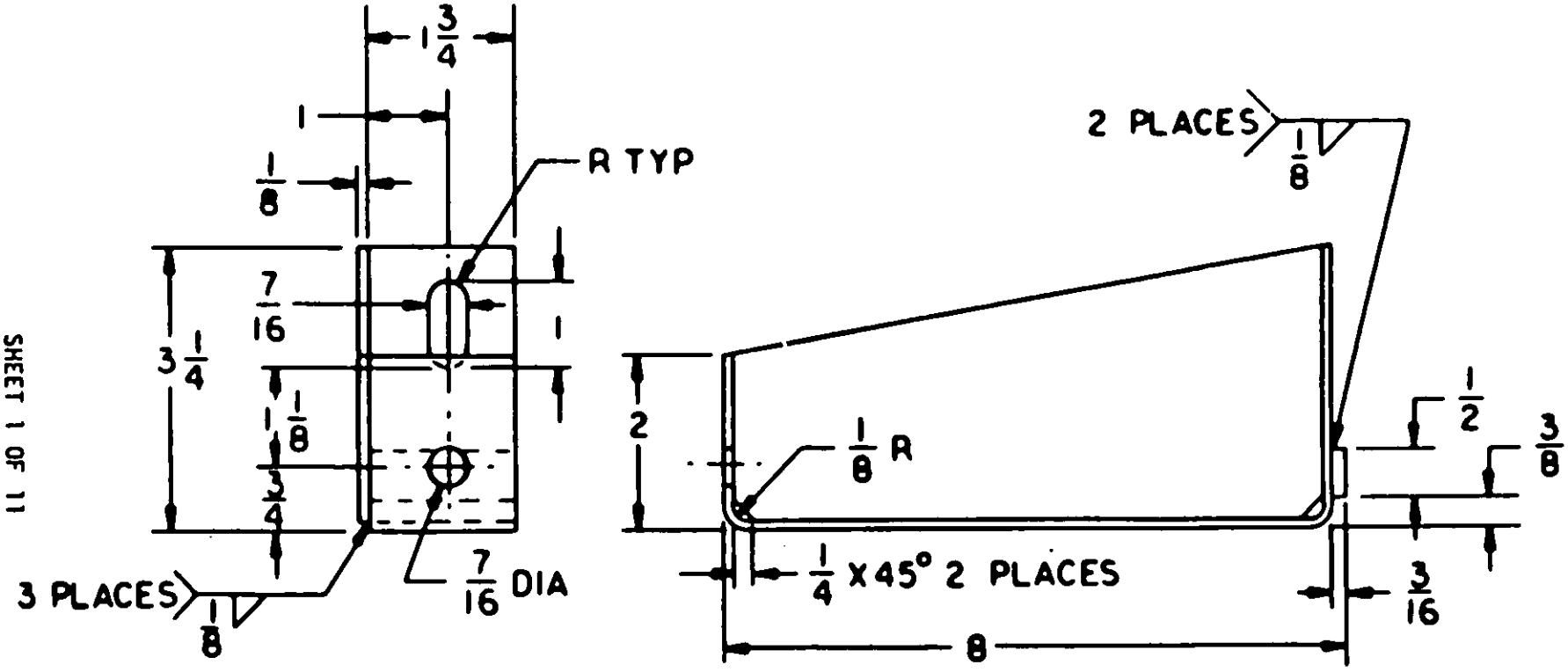


NOTES:
1. MATERIAL:
   TUBE, STEEL, WELDED,
   MT 1010 TO MT 1020, PER
   ASTM A512, A513 OR A519
   1.00 O.D. X 0.058 WALL.
2. PRIME IN ACCORDANCE WITH
   TT-P-636 ENAMEL SEMI-GLOSS
   COLOR O.D. TYPE OPTIONAL
   PER TT-E-485 OR TT-E-529.
3. BREAK ALL SHARP EDGES.

FRAME LINK
REQ'D 24
KEY D42

FIGURE 40. Top frame and top frame assembly - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



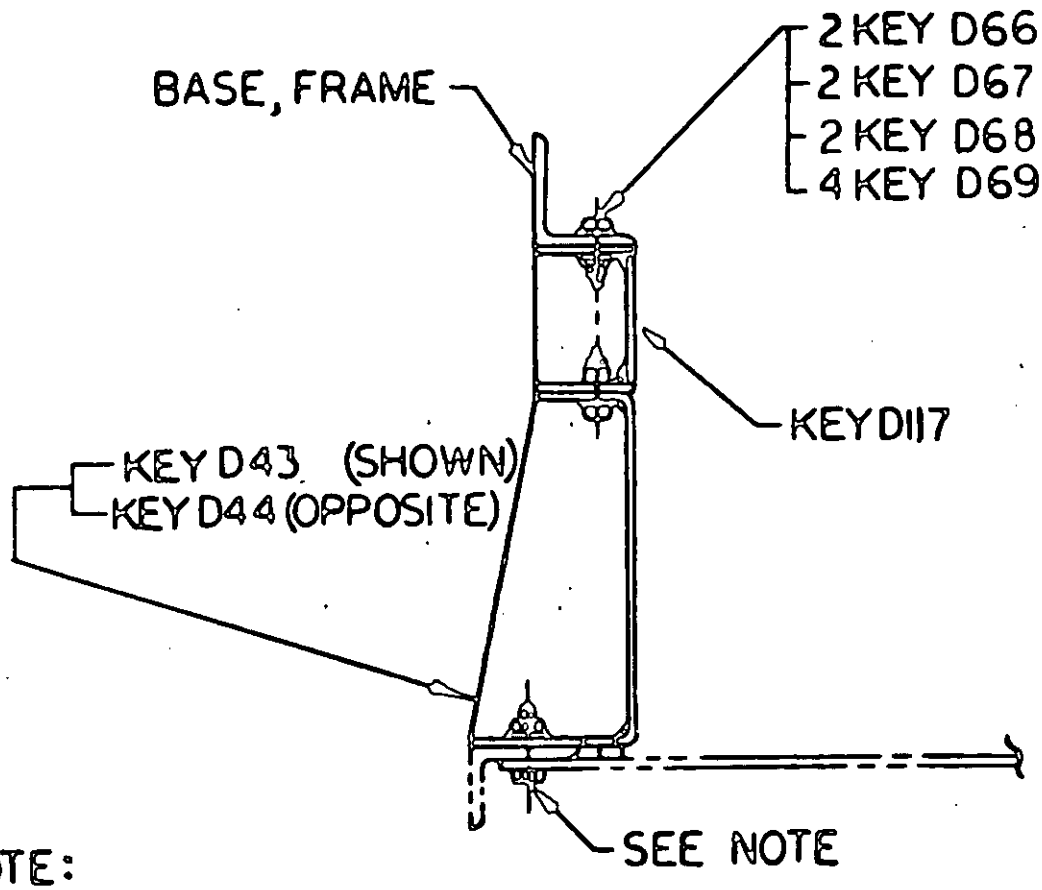
NOTES:
1. MATERIAL: STEEL PLATE FS 1020, PER QQ-S-635.
2. PRIME IN ACCORDANCE WITH TT-P-636, ENAMEL SEMI-GLOSS COLOR O.D., TYPE OPTIONAL, PER TT-E-485 OR TT-E-529.
3. STEEL WELD IN ACCORDANCE WITH MIL-STD-1261, ELECTRODE QQ-E-450.
4. REMOVE ALL BURRS AND SHARP EDGES.

BASE FRAME BRACKET

KEY D43 SHOWN 5 REQ'D
KEY D44 OPPOSITE 5 REQ'D

FIGURE 41. Installation of frame on fender.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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NOTE:

1. USE EXISTING SCREWS NUTS AND WASHERS.

FRAME TO FENDER INSTALLATION

KEY D45 (SHOWN)
KEY D46 (OPPOSITE)

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

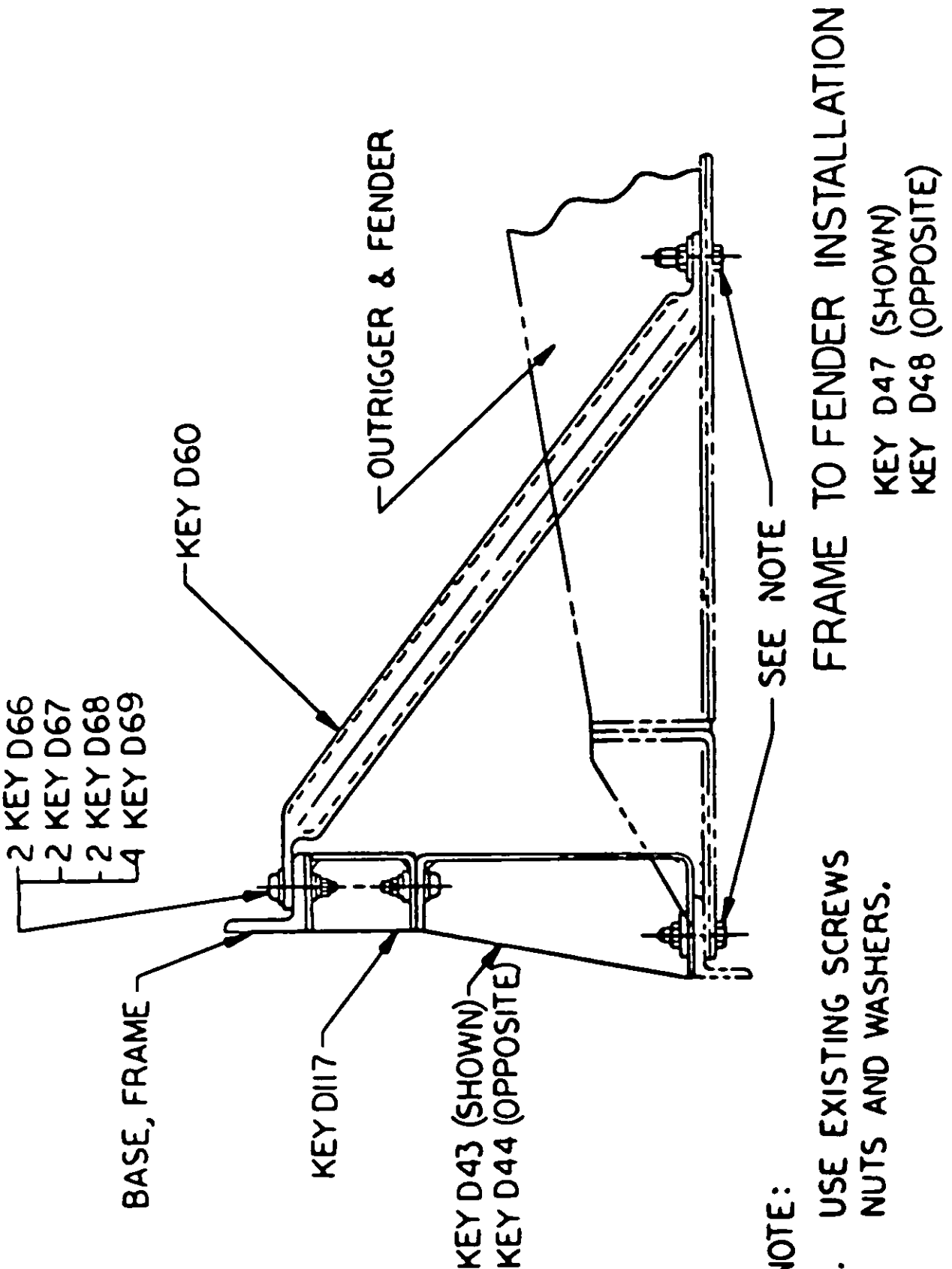
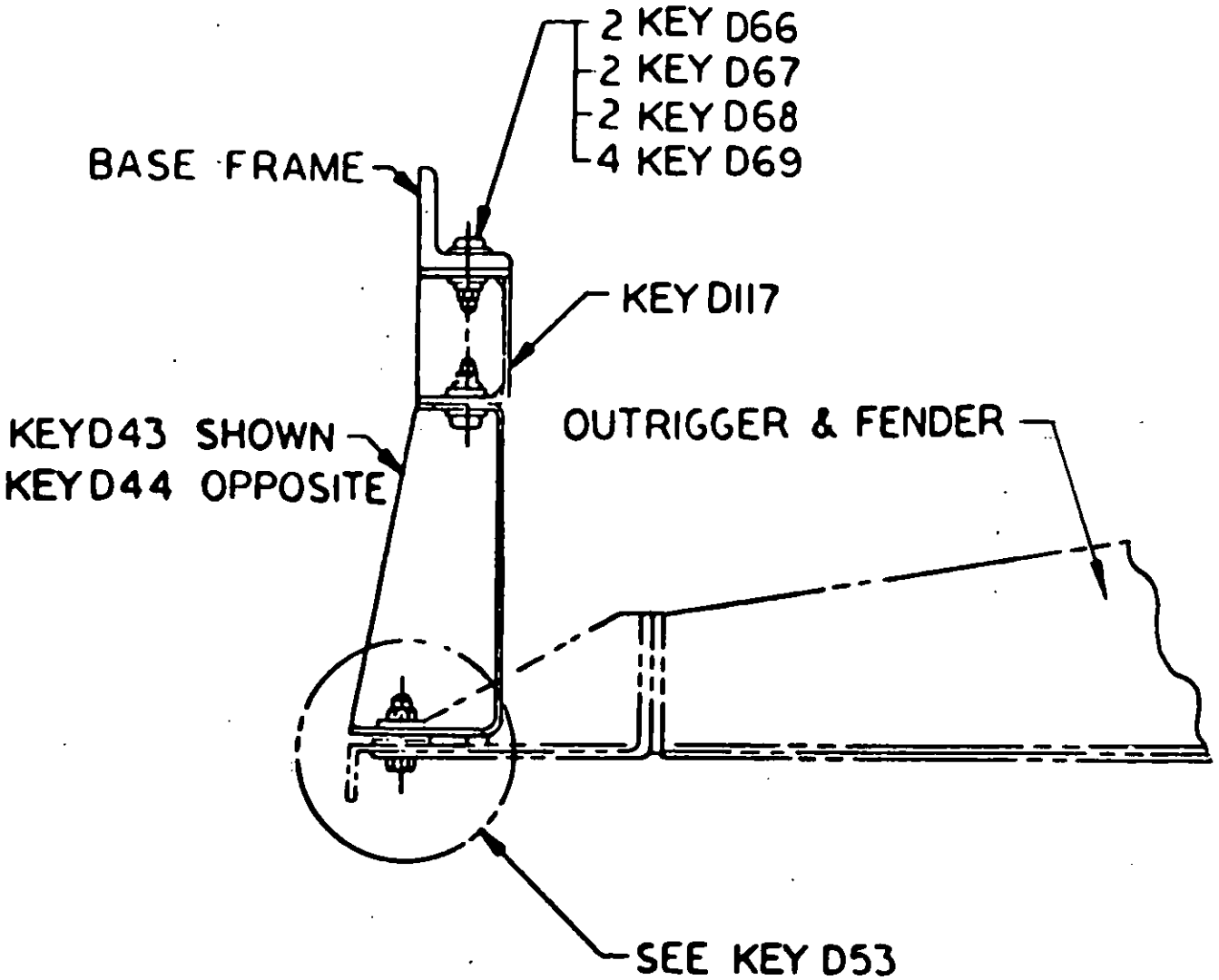


FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

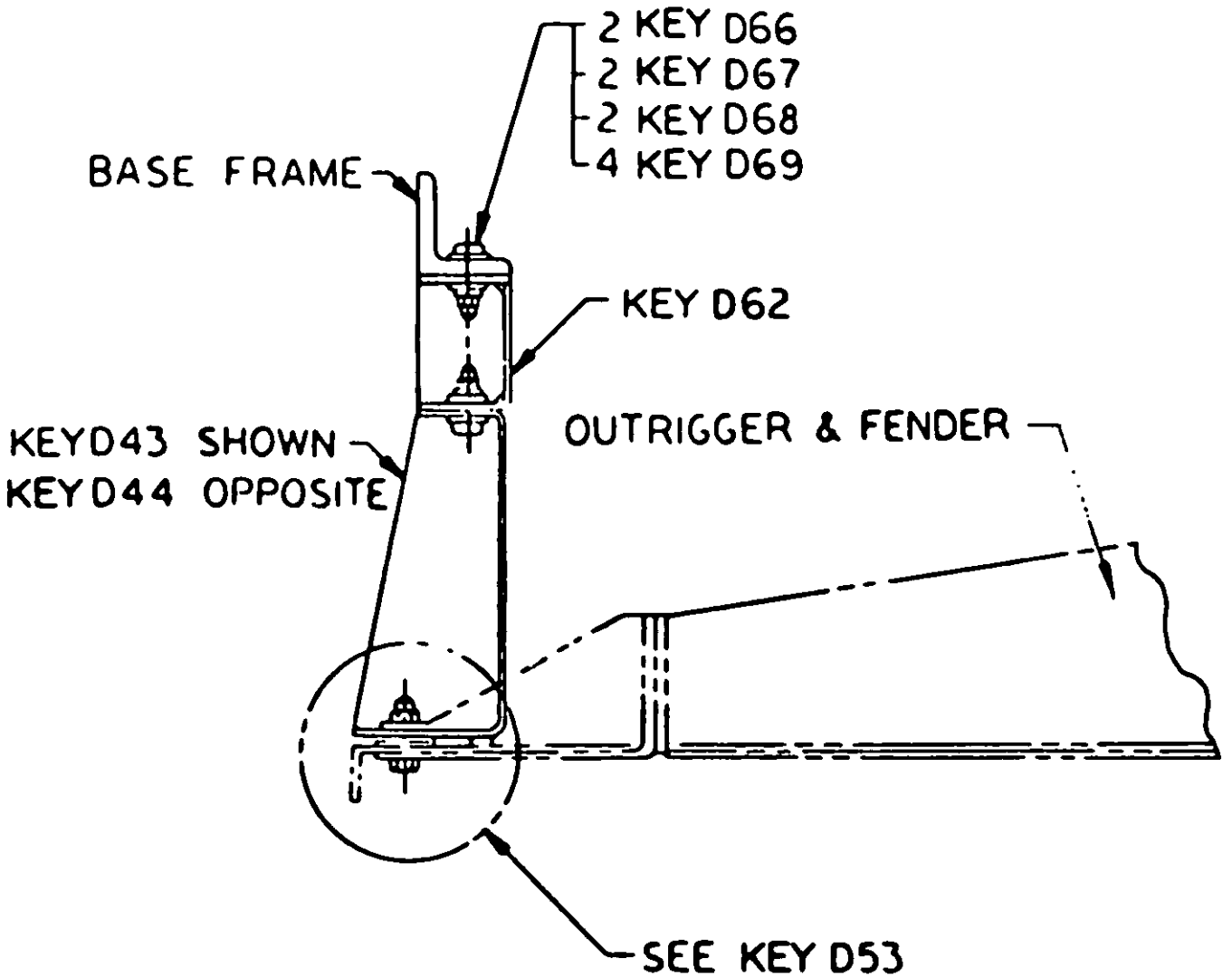


FRAME TO FENDER
INSTALLATION
KEY D49 SHOWN
KEY D50 OPPOSITE

SHEET 4 OF 11

FIGURE 41. Installation of frame on fender - Continued.

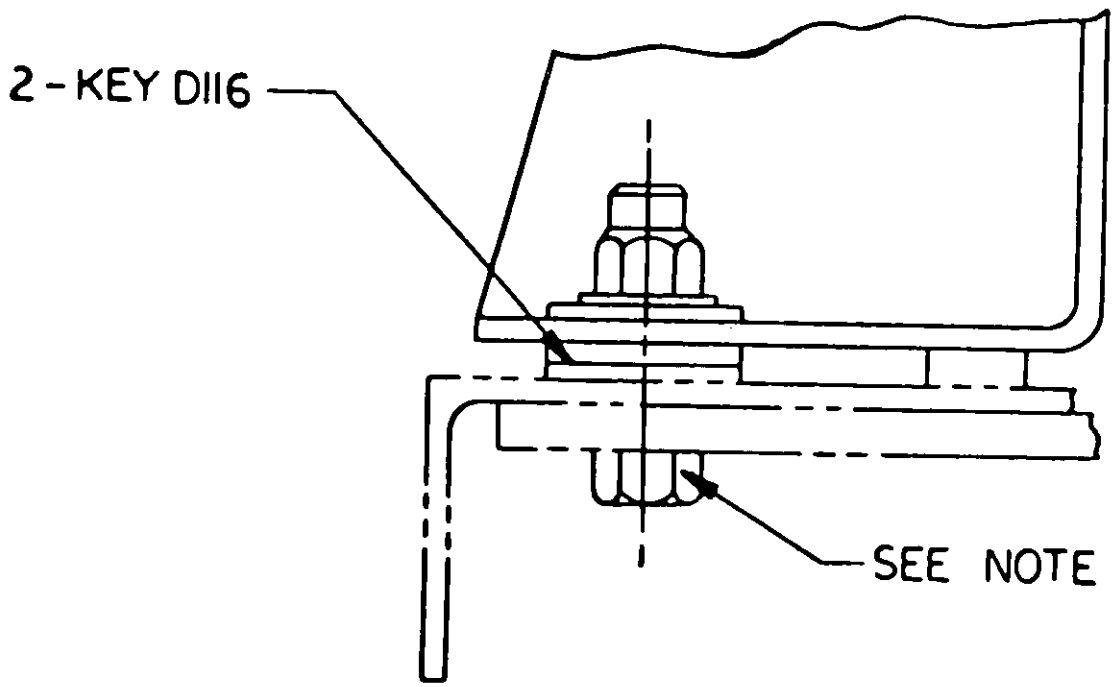
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



FRAME TO FENDER
INSTALLATION
KEY D51 SHOWN
KEY D52 OPPOSITE

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE:

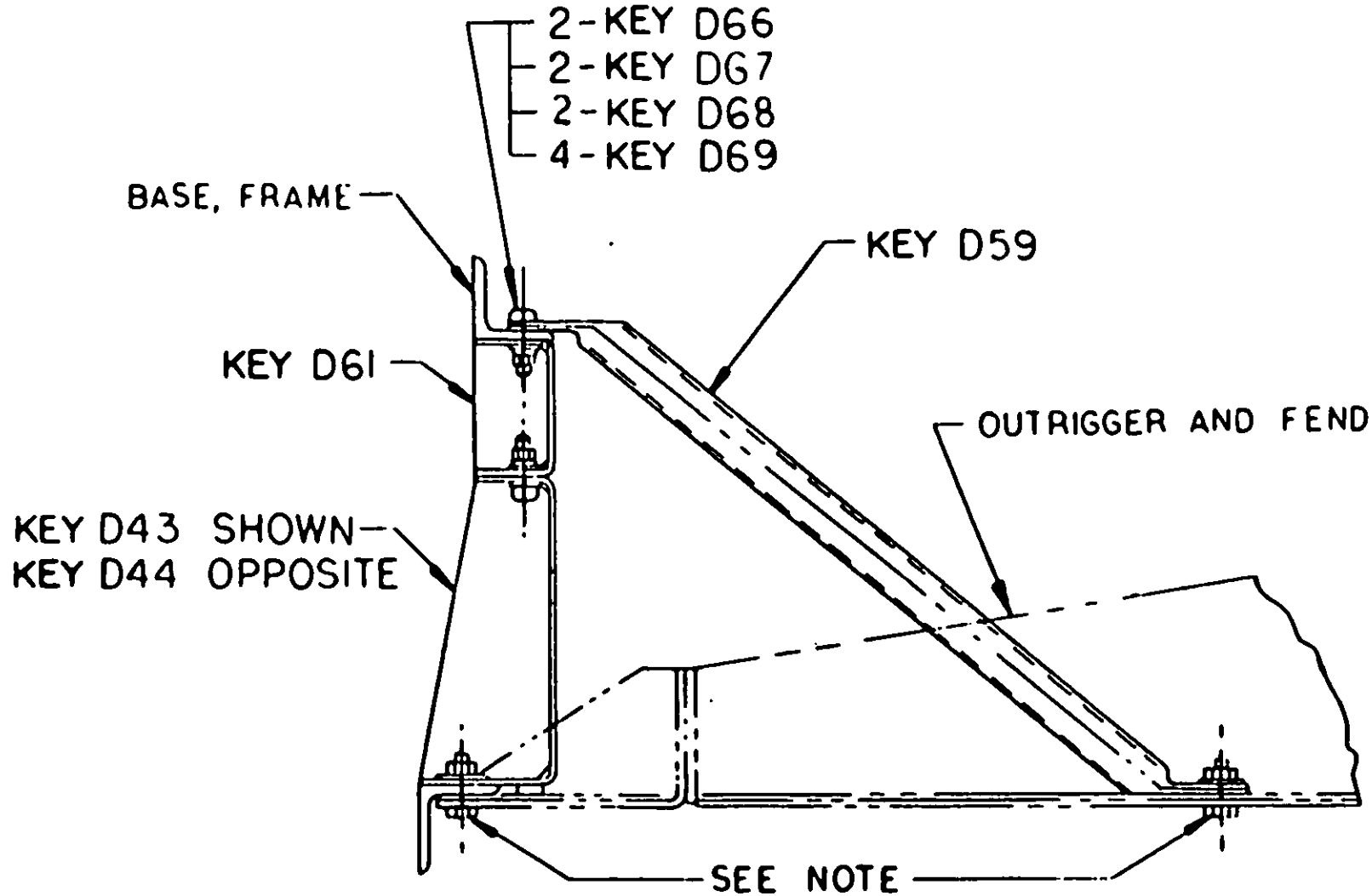
1. USE EXISTING SCREWS NUTS AND WASHERS.

FRAME TO FENDER
KEY D53

SHEET 6 OF 11

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



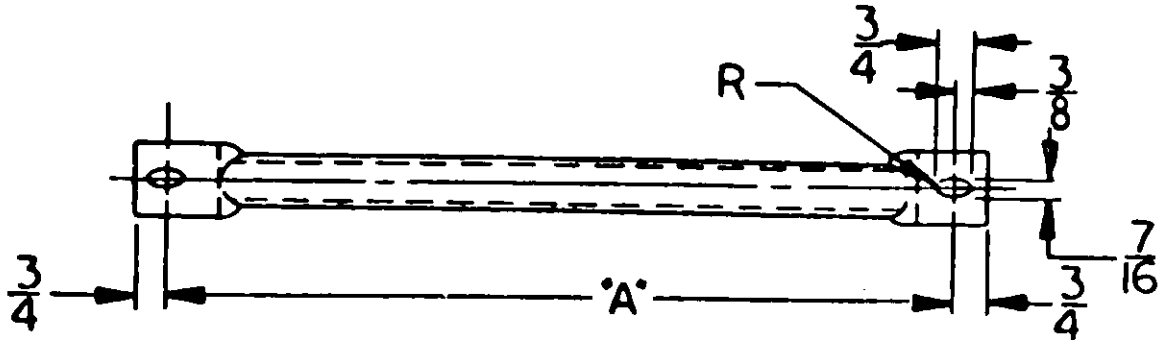
NOTE:

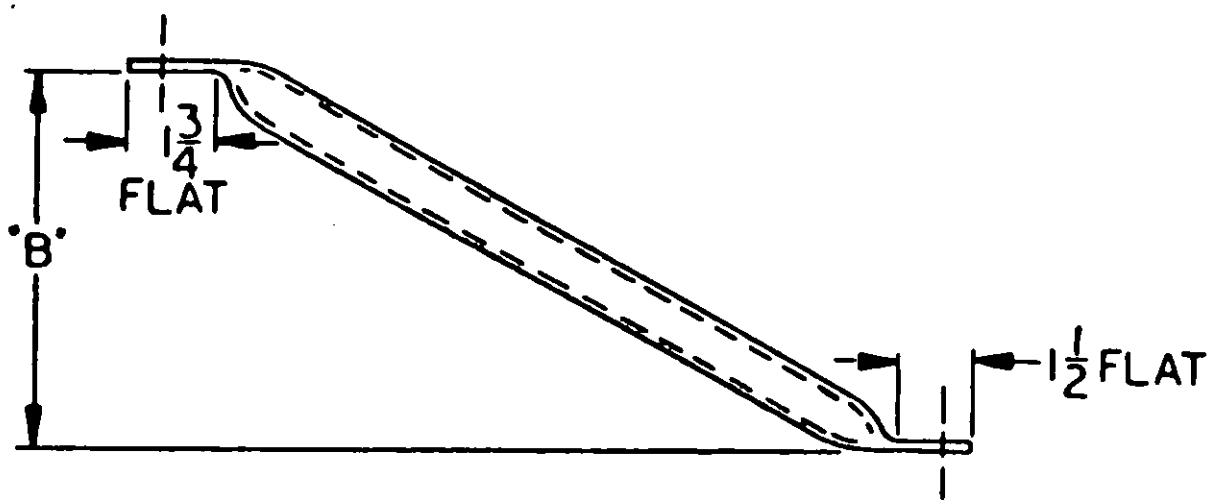
1. USE EXISTING SCREWS NUTS AND WASHERS.

FRAME TO FENDER INSTALLATION
KEY D118 SHOWN
KEY D119 OPPOSITE

FIGURE 41. Installation of frame on fender - Continued.

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KEY NO.	REQ'D	"A"	"B"
KEY D59	2	17 7/16	11 1/2
KEY D60	2	14 1/8	11 1/4

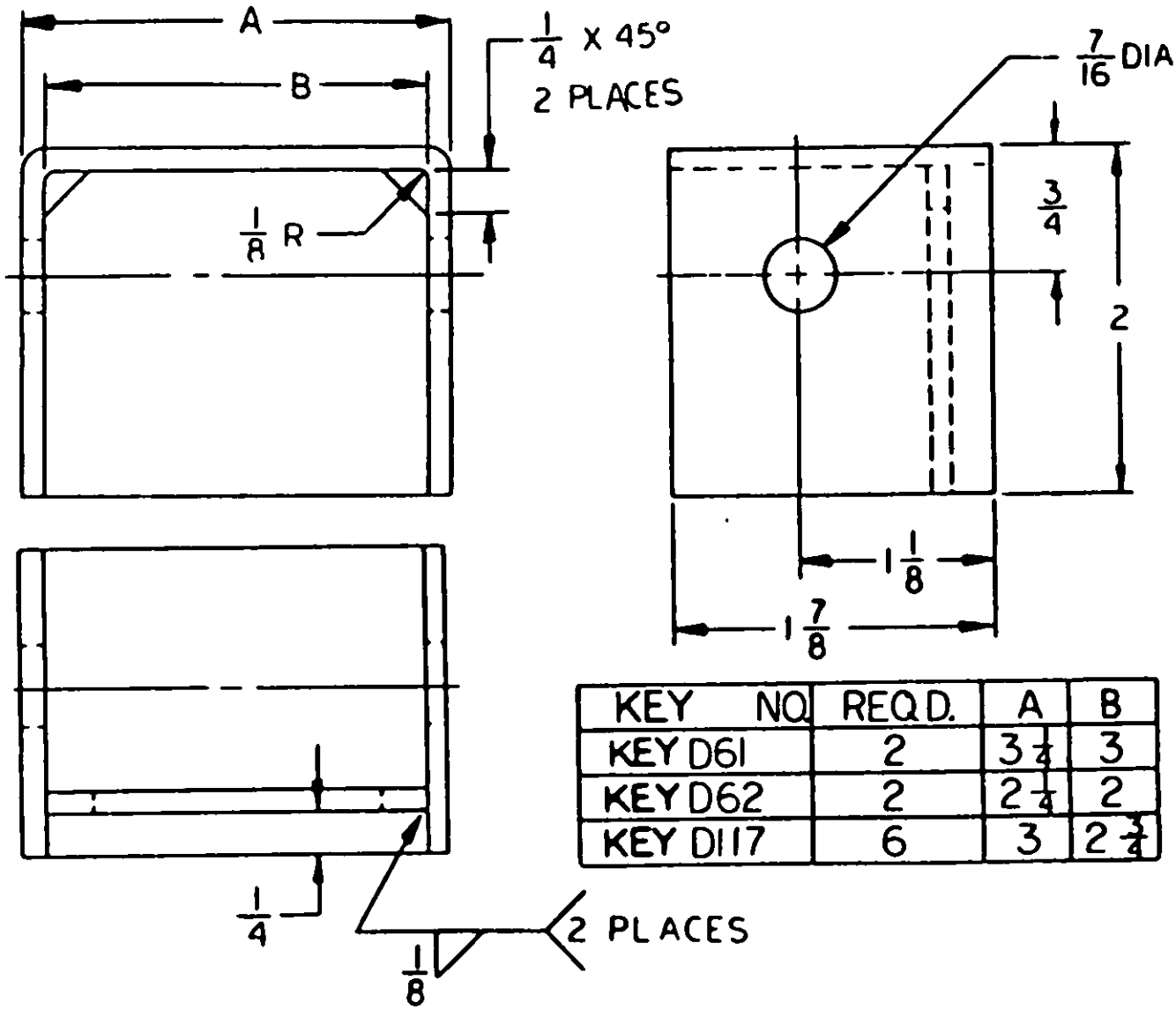
NOTES:
1. MATERIAL: TUBE, STEEL, WELDED, MT 1010 TO MTX 1020, PER ASTM A512, A513 OR A519 1.00 O.D. X 0.058 WALL.
2. TREAT IN ACCORDANCE WITH TYPE I OR TYPE III OF TT-C-490.
3. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.

BASE FRAME SUPPORT TUBING
KEY D59 AND D60

SHEET 8 OF 11

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



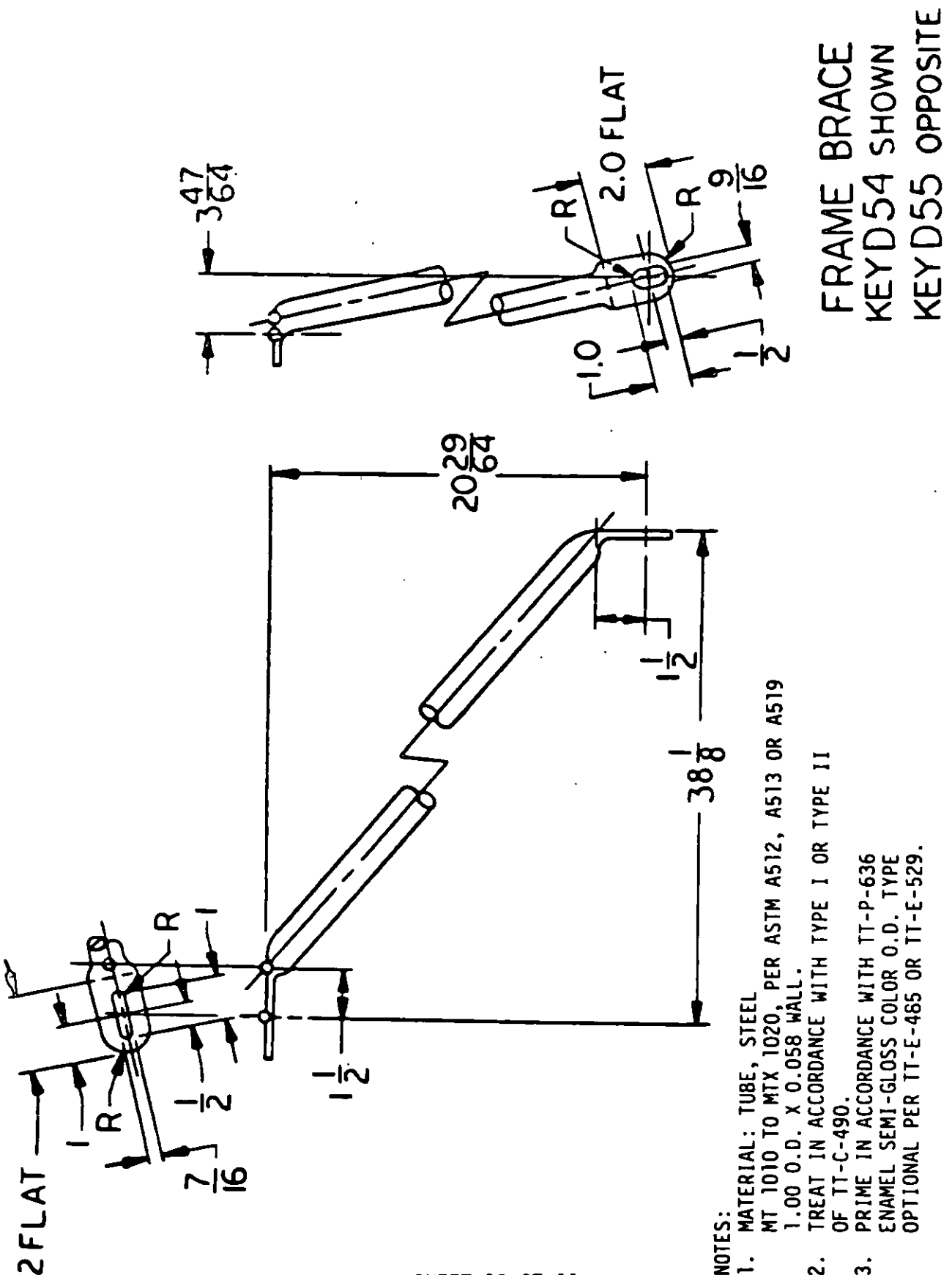
KEY NO.	REQ'D.	A	B
KEY D61	2	3 3/4	3
KEY D62	2	2 3/4	2
KEY D117	6	3	2 3/4

NOTES:
1. MATERIAL: STEEL, COMMERCIAL QUALITY, PER QQ-S-698, 0.125 THICK.
2. WELD IN ACCORDANCE WITH MIL-STD-1261, CLASS 1.
3. FINISH: PRIME IN ACCORDANCE WITH TT-P-636, ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
4. REMOVE ALL BURRS AND SHARP EDGES.

SPACER BLOCK
KEY D61 D62 AND D117

FIGURE 41. Installation of frame on fender - Continued.

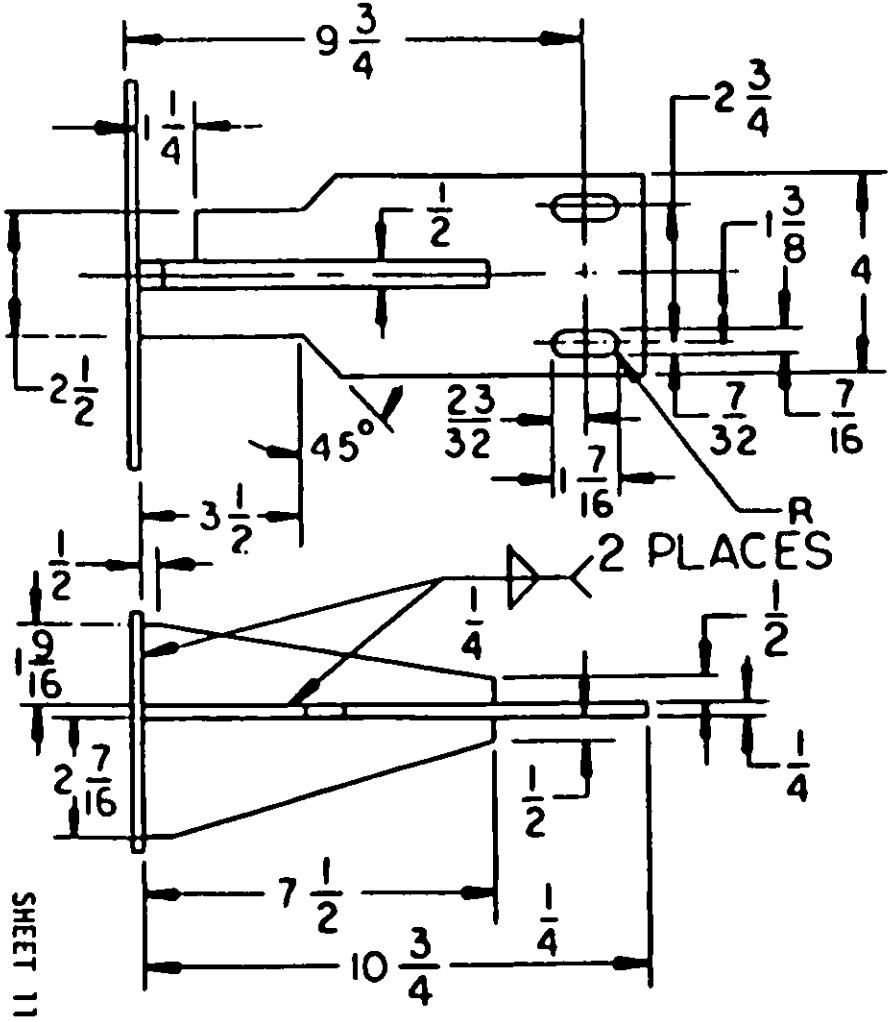
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



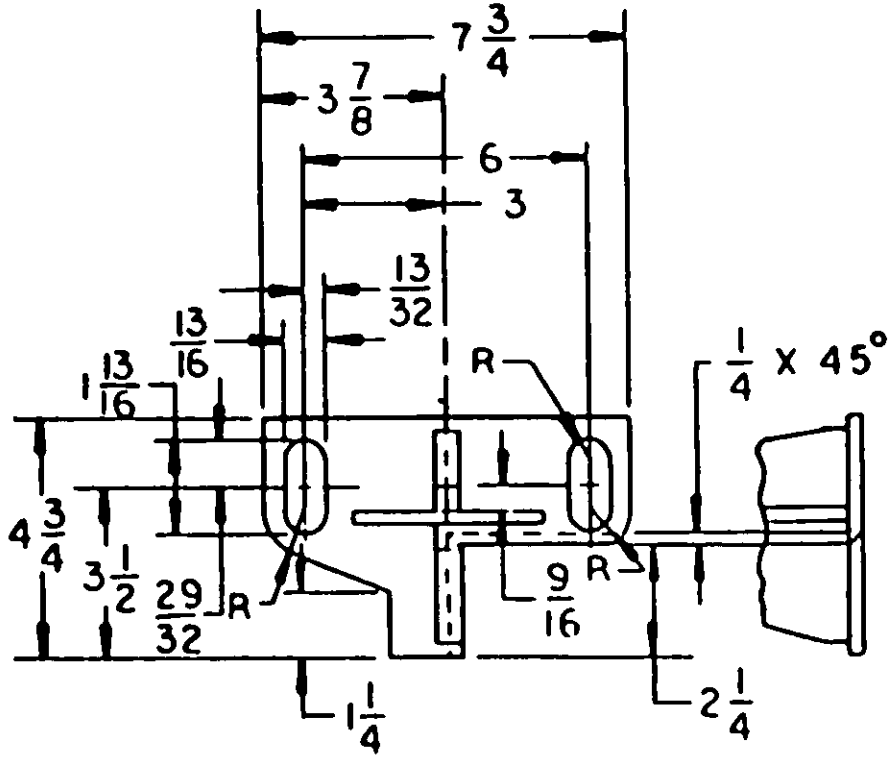
NOTES:
1. MATERIAL: TUBE, STEEL
   MT 1010 TO MTX 1020, PER ASTM A512, A513 OR A519
   1.00 O.D. X 0.058 WALL.
2. TREAT IN ACCORDANCE WITH TYPE I OR TYPE II
   OF TT-C-490.
3. PRIME IN ACCORDANCE WITH TT-P-636
   ENAMEL SEMI-GLOSS COLOR O.D. TYPE
   OPTIONAL PER TT-E-465 OR TT-E-529.

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



BASE FRAME BRACKET
KEY D56

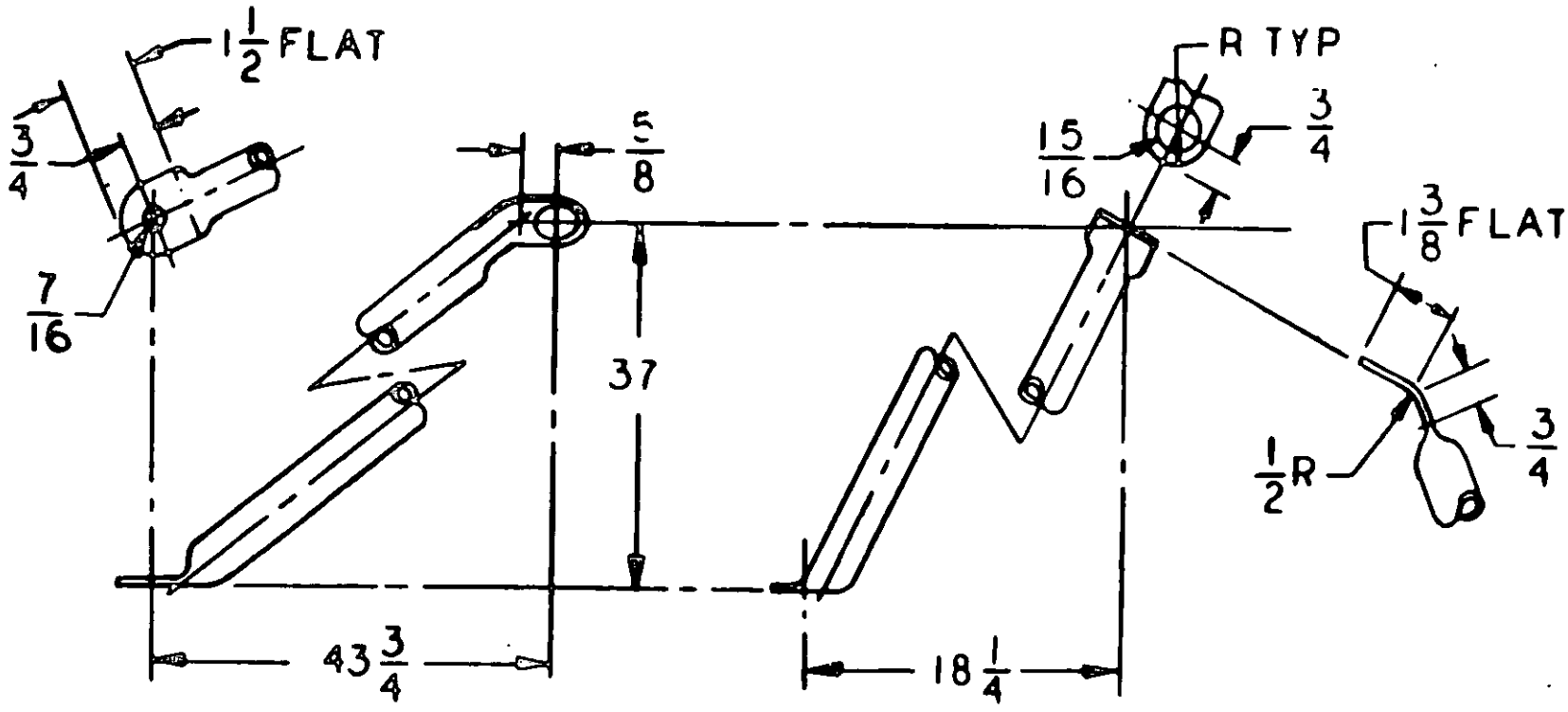


NOTES:

1. MATERIAL: STEEL, 1015 TO 1025, PER ASTM A108, A576, OR A675.
2. FINISH: PRIME IN ACCORDANCE WITH TT-P-636, ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL, TT-E-485 OR TT-E-529.
3. WELD IN ACCORDANCE WITH MIL-STD-1261, CLASS 1.
4. REMOVE ALL BURRS AND SHARP EDGES.

FIGURE 41. Installation of frame on fender - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T04:22Z
Check the source to verify that this is the current version before use.

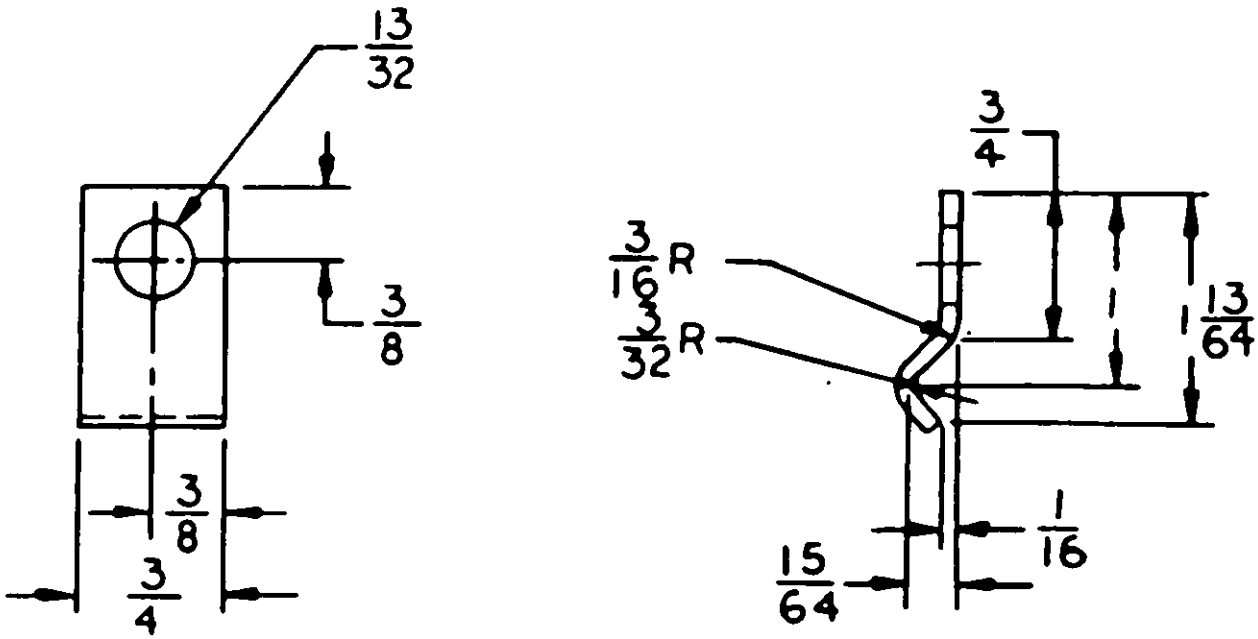


NOTES:
1. MATERIAL: TUBE, STEEL, WELDED, MT 1010 TO MT 1020, PER ASTM A512, A513 OR A519 1.00 O.D. X 0.058 WALL.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.
3. BREAK ALL SHARP EDGES.

KEY D57 SHOWN 2REQ'D
KEY D58 OPPOSITE 2REQ'D

FIGURE 42. Frame brace.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



KEY D64
32 REQ'D

NOTES:
1. MATERIAL:
   STEEL, FS1009 TO FS1020 CONDITION OPTIONAL PER QQ-S-698, 0.090 THICK.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.

FIGURE 43. Clamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

HARDWARE, BASE, FRAME KEY D65 KEY D66 THRU D70 & D112 THRU D116				
KEY NUMBER	REQ'D	ORDNANCE PART NUMBER	DESCRIPTION	MATERIAL
D66	62	MS35206-312	SCREW, PAN HD. 3/8-16 UNC-2A X 1 1/4 LONG	CARBON STEEL CADMIUM PLATE
D67	62	MS51967-8	NUT, HEX 3/8-16 UNC-2B	STEEL CADMIUM PLATE
D68	62	MS35338-46	WASHER, LOCK 3/8	CARBON STEEL CADMIUM PLATE
D69	82	MS27183-14	WASHER, PLAIN 3/8	STEEL CADMIUM PLATE
D70	32	425594 1/	SCREW LOCK WASHER, HEX HD. 5/16-24 UNF-2A 1 1/4 LONG, TYPE I	STEEL CADMIUM PLATE
D112	32	MS35691-522	NUT, JAM HEX 5/16-24 UNF-2B	STEEL CADMIUM PLATE
D113	32	MS51968-5	NUT, HEX 5/16-24 UNF 2B	STEEL CADMIUM PLATE
D114	32	MS35338-45	WASHER, LOCK 5/16	CARBON STEEL CADMIUM PLATE
D115	2	MS90725-11	SCREW, HEX HD. 3/8-16 UNC-2A X 1 3/8 LONG	CARBON STEEL CADMIUM PLATE
D116	8	MS27183-15	WASHER, PLAIN 3/8	STEEL CADMIUM PLATE

1/ SEE ARMY DRAWING BCYX6.

FIGURE 44. Base frame hardware.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

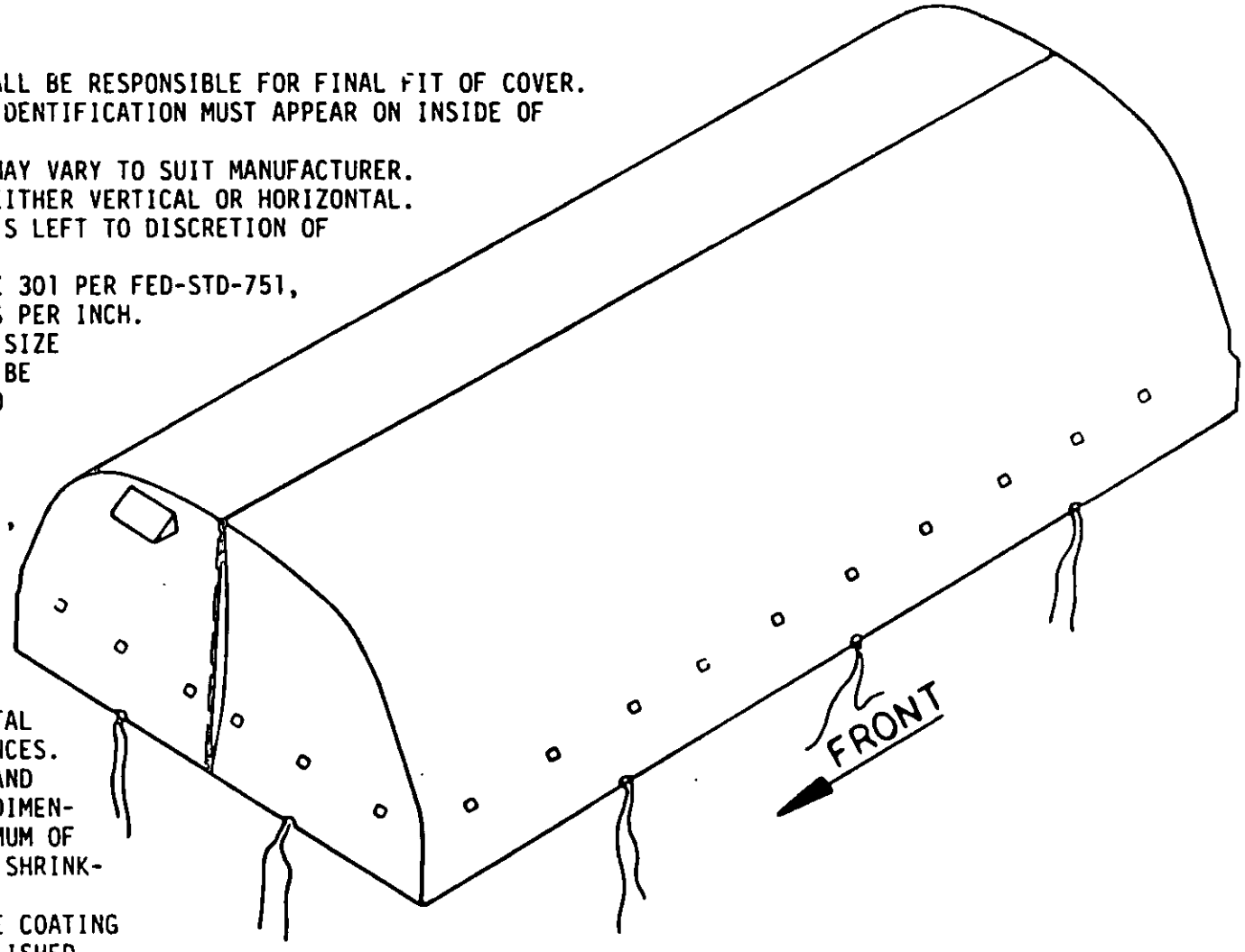
KEY D71		
KEY NO.	FIG. NO. - SHEET NO.	DESCRIPTION
D72	46-1	COVER ASSEMBLY
D73	46-2	PLAN VIEW, COVER
D74	46-3	LEFT SIDE ELEVATION
D75	46-4	SECTION VIEWS FOR KEY D74
D76	46-5	FRONT ELEVATICN
D77	46-6	REAR ELEVATION
D78	46-7	SECTION VIEWS FOR KEYS D76 AND D77
D79	46-8	LEFT SIDE AND REAR ELEVATION
D80	46-9	DETA AND CROSS SECTION ON DEY D79
D81	46-10	VENT, COVER
D82	46-11	VENT SIDE
D83	46-11	VENT SIDE
D84	46-12	VENT REINFORCEMENT
D85		(NOT USED)
D86		(NOT USED)
D87	46-13	BINDING
D88	46-14	WEBBING
D89	46-14	WEBBING
D90	46-15	PULLEY ASSEMBLY
D91	46-16	PULLEY INSTALLATION FRONT, AND REAR
D92	46-17	SECTION AND DETAIL FOR KEY D74
D93	46-18	ROD
D94	46-18	ROD
D95	46-19	ROD TIE DOWN
D96	46-19	ROD TIE DOWN
D97	46-20	STRIP TIE DOWN
D98	46-20	STRIP TIE DOWN
D99	46-20	STRIP TIE DOWN
D100	46-21	REINFORCEMENT TIE DOWN
D101	46-22	REINFORCEMENT
D102	46-23	HARDWARE CLOSURE ATTACHMENT
D103	46-23	TIP, RUBBER
D104	46-23	CHAPE, WEB
D105		(NOT USED)
D106	46-24	PARTS, CLOSURE ATTACHMENT
D107	46-24	DOUBLE FASTENER
D108	46-24	FASTENER
D109	46-24	FASTENER
D110	46-24	ROPE, JUTE
D111	46-24	ROPE, JUTE

FIGURE 45. List of key numbers for cover assembly, installation and attachment.

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NOTES:

1. MANUFACTURER SHALL BE RESPONSIBLE FOR FINAL FIT OF COVER.
2. MANUFACTURER'S IDENTIFICATION MUST APPEAR ON INSIDE OF COVER.
3. SEAM ALLOWANCE MAY VARY TO SUIT MANUFACTURER. SEAMS SHALL BE EITHER VERTICAL OR HORIZONTAL. MATERIAL WIDTH IS LEFT TO DISCRETION OF MANUFACTURER.
4. STITCHING: TYPE 301 PER FED-STD-751, 8 TO 10 STITCHES PER INCH. SMALLEST NEEDLE SIZE FEASIBLE SHOULD BE USED IN ORDER TO ASSURE WEATHER-PROOF SEAMS.
5. THREAD: NYLON, TYPE II, CLASS 1, SIZE F, COLOR O.D., PER V-T-295.
6. MATERIAL: CLOTH, TYPE II, CLASS 3, PER MIL-C-20696, TOTAL WEIGHT OF 14 OUNCES.
7. INCREASE COVER AND SKIRT MATERIAL DIMENSIONS BY A MINIMUM OF 1% TO ALLOW FOR SHRINKAGE.
8. BALANCED SURFACE COATING SHALL BE ACCOMPLISHED.
9. COLOR: GREEN NO. 34087 PER FED-STD-595.



COVER ASSEMBLY
KEY D72

FIGURE 46. Cover assembly, installation, and attachment.

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Check the source to verify that this is the current version before use.

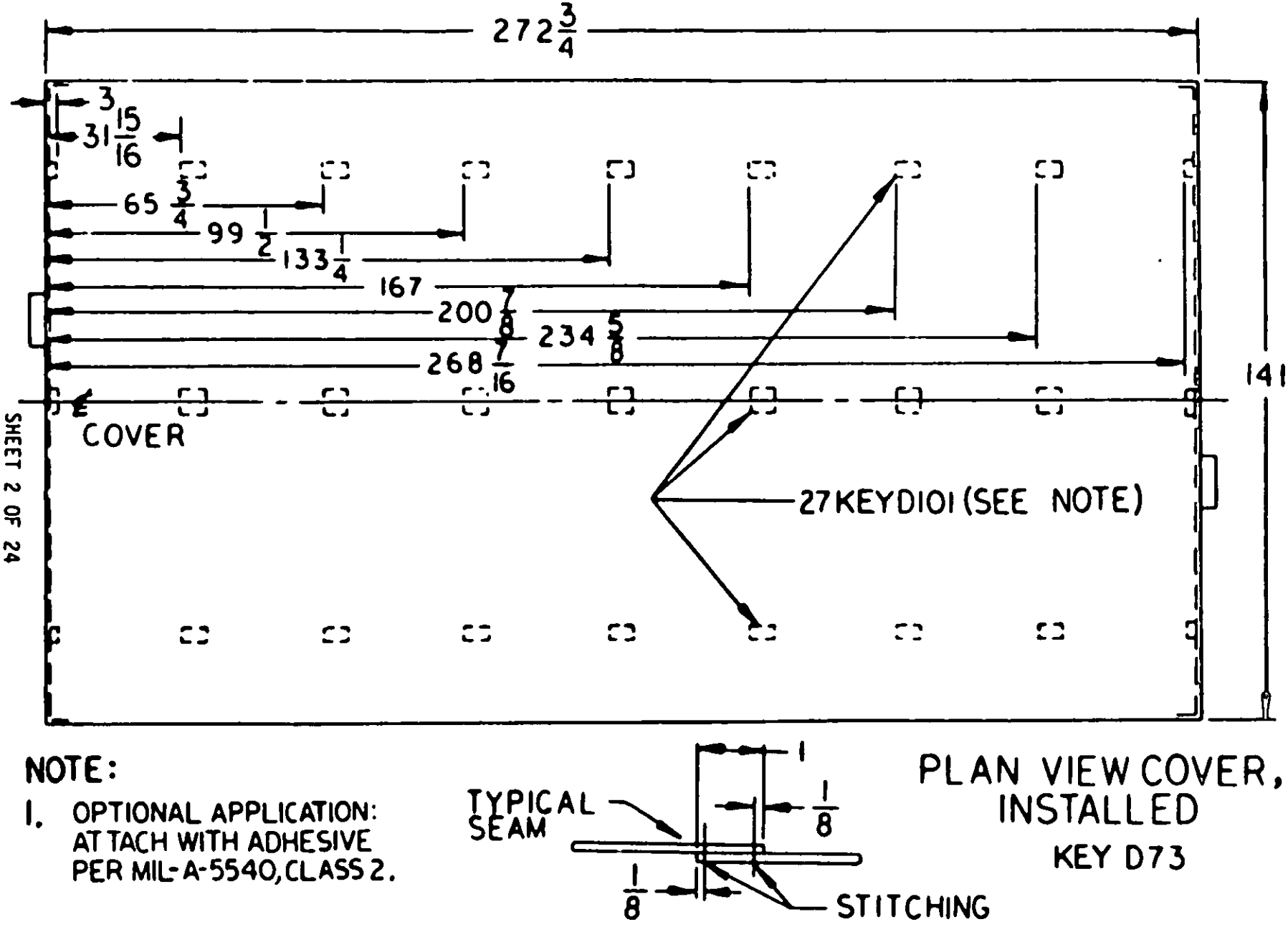
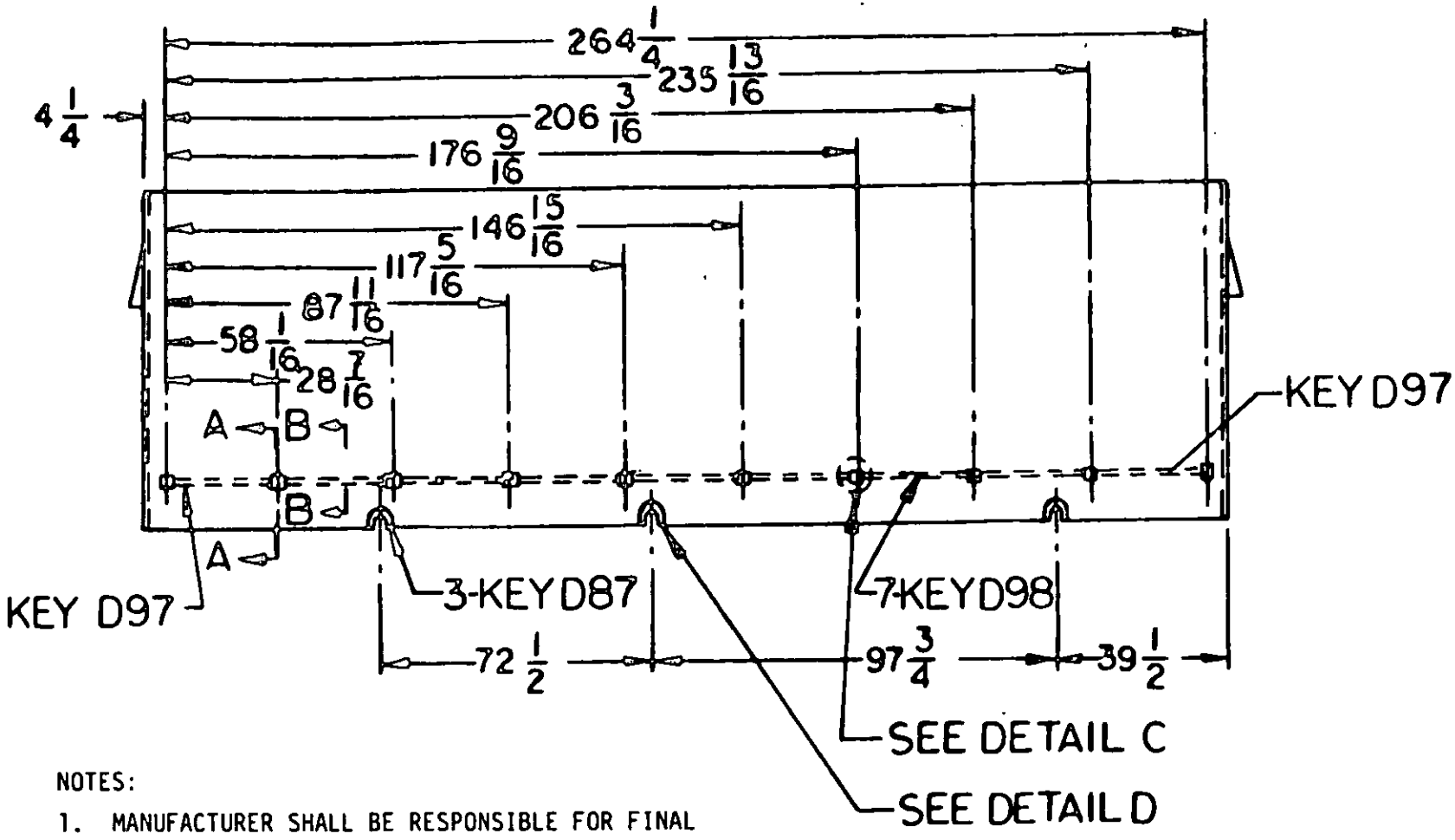


FIGURE 46. Cover assembly, installation, and attachment - Continued.

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Check the source to verify that this is the current version before use.



NOTES:

1. MANUFACTURER SHALL BE RESPONSIBLE FOR FINAL FIT OF COVER.
2. MANUFACTURERS IDENTIFICATION MUST APPEAR ON INSIDE OF COVER.
3. SEAM ALLOWANCE MAY VARY TO SUIT MANUFACTURER.
4. STITCHING TYPE 301 PER FED-STD-751, 8-10 STITCHES PER INCH SMALLEST NEEDLE SIZE FEASIBLE SHOULD BE USED IN ORDER TO ASSURE WEATHER PROOF SEAMS.
5. THREAD NYLON TYPE II, CLASS I, SIZE F, COLOR O.D. PER V-T-295.
6. FOR SECTIONS AND DETAIL SEE KEY D75 & D92.
7. QUANTITY OF KEY D87, D97 AND D98 SYMMETRICAL FOR RIGHT SIDE.

COVER INSTALLATION
LEFT SIDE ELEVATION
KEY D74

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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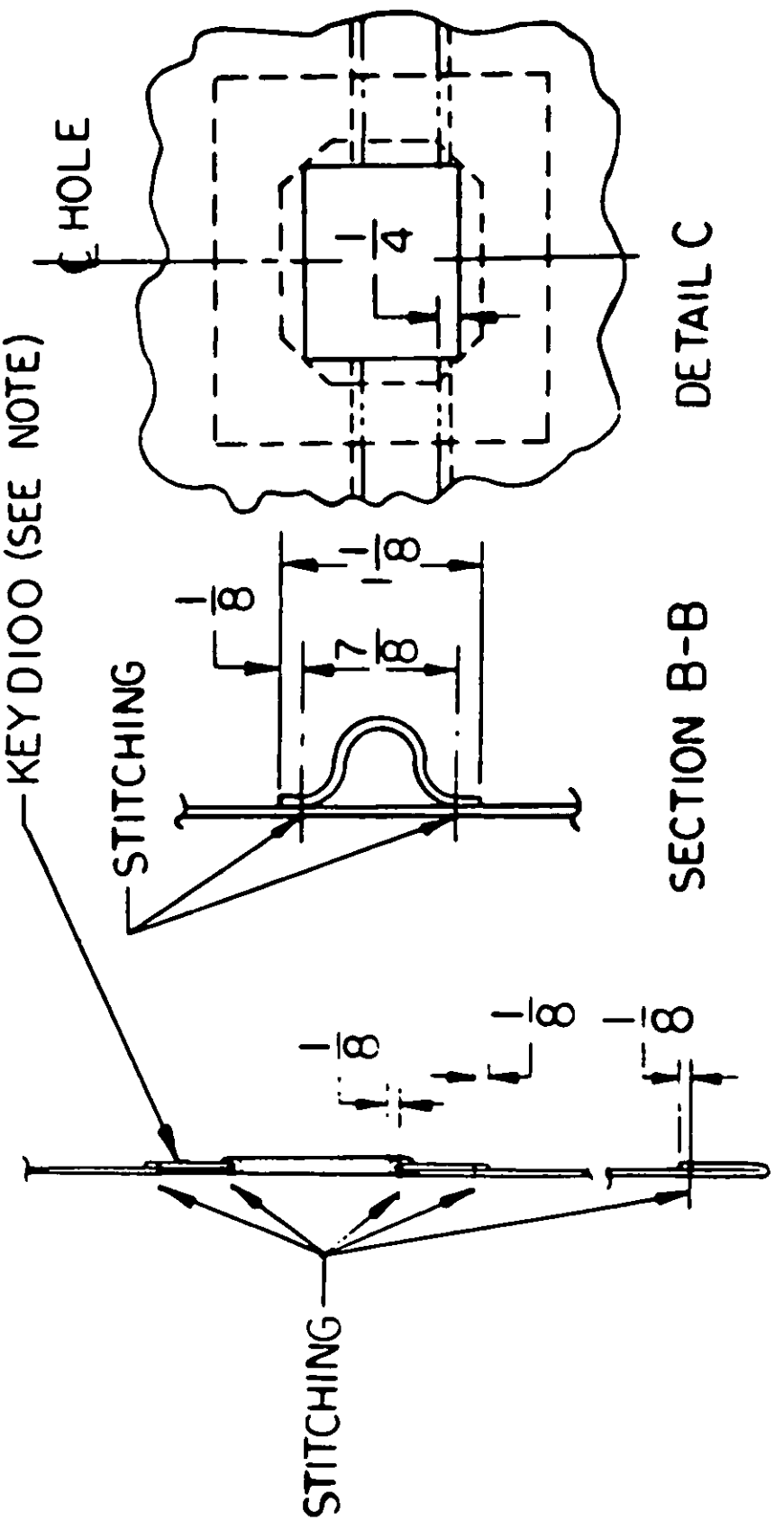
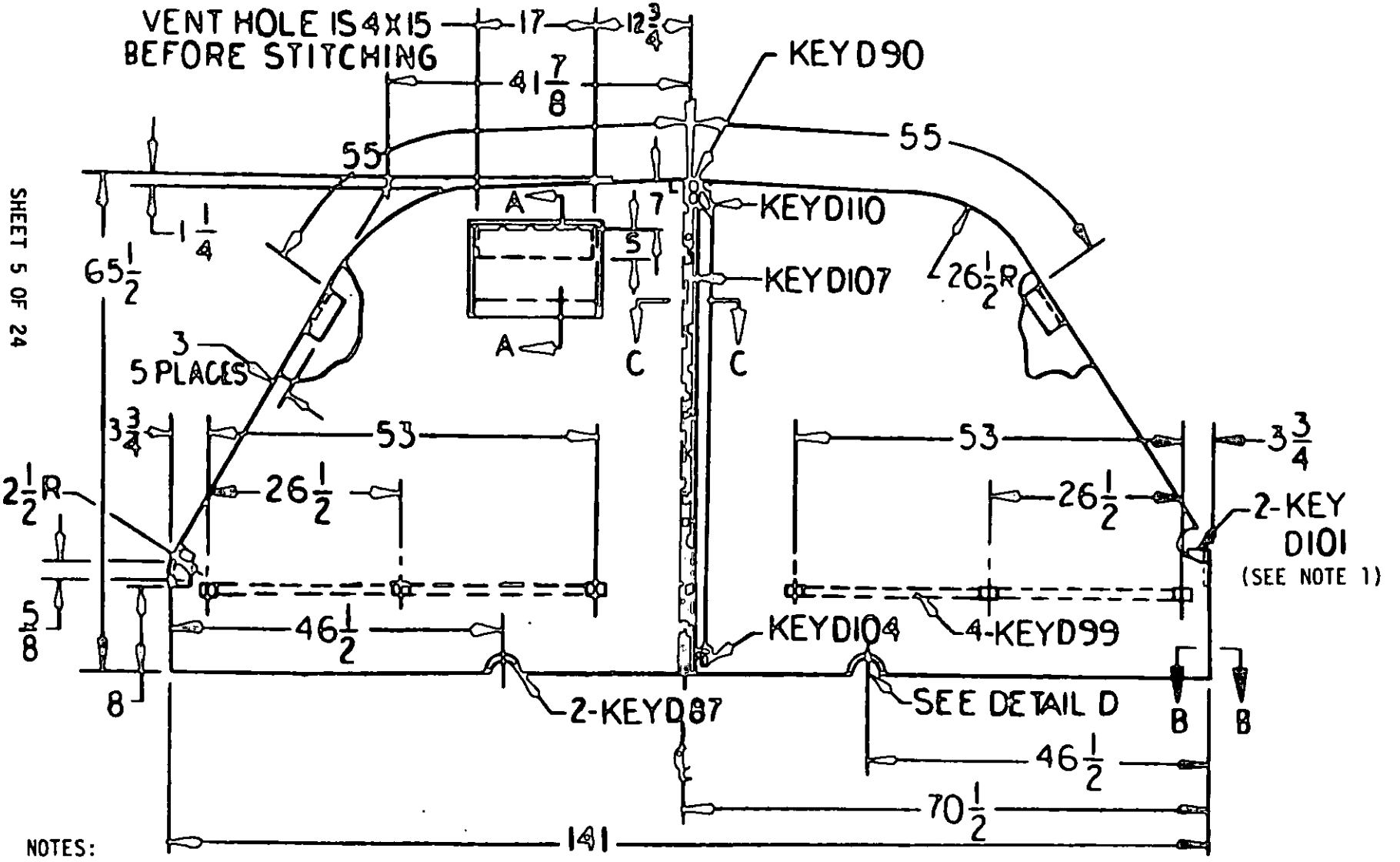


FIGURE 46. Cover assembly, installation, and attachment - Continued.

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Check the source to verify that this is the current version before use.

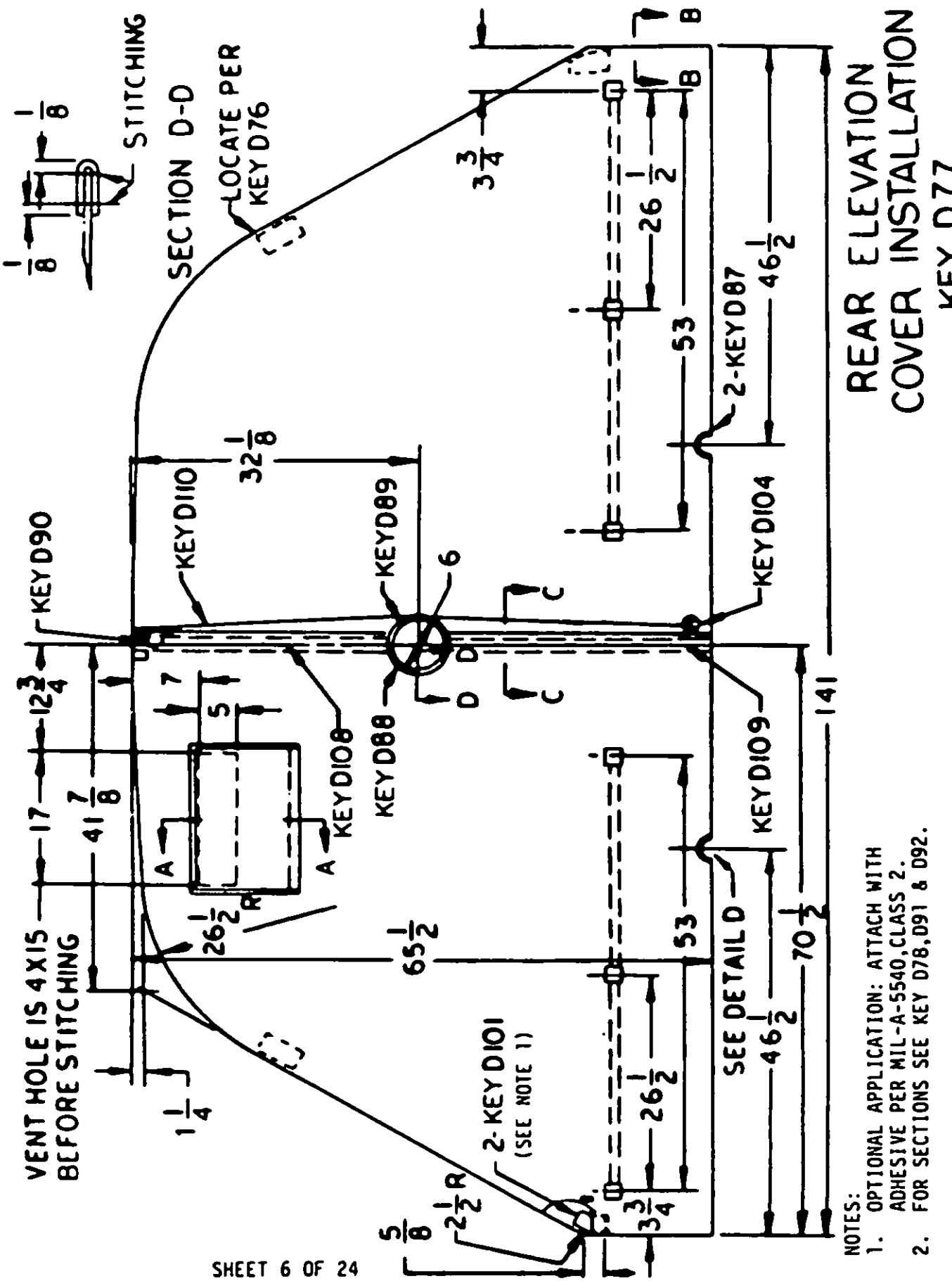
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTES:
1. OPTIONAL APPLICATION:
   ATTACH WITH ADHESIVE
   PER MIL-A-5540, CLASS 2.
2. FOR SECTIONS
   SEE KEY: D78, D91 & D92.

FRONT ELEVATION
COVER INSTALLATION
KEY D76

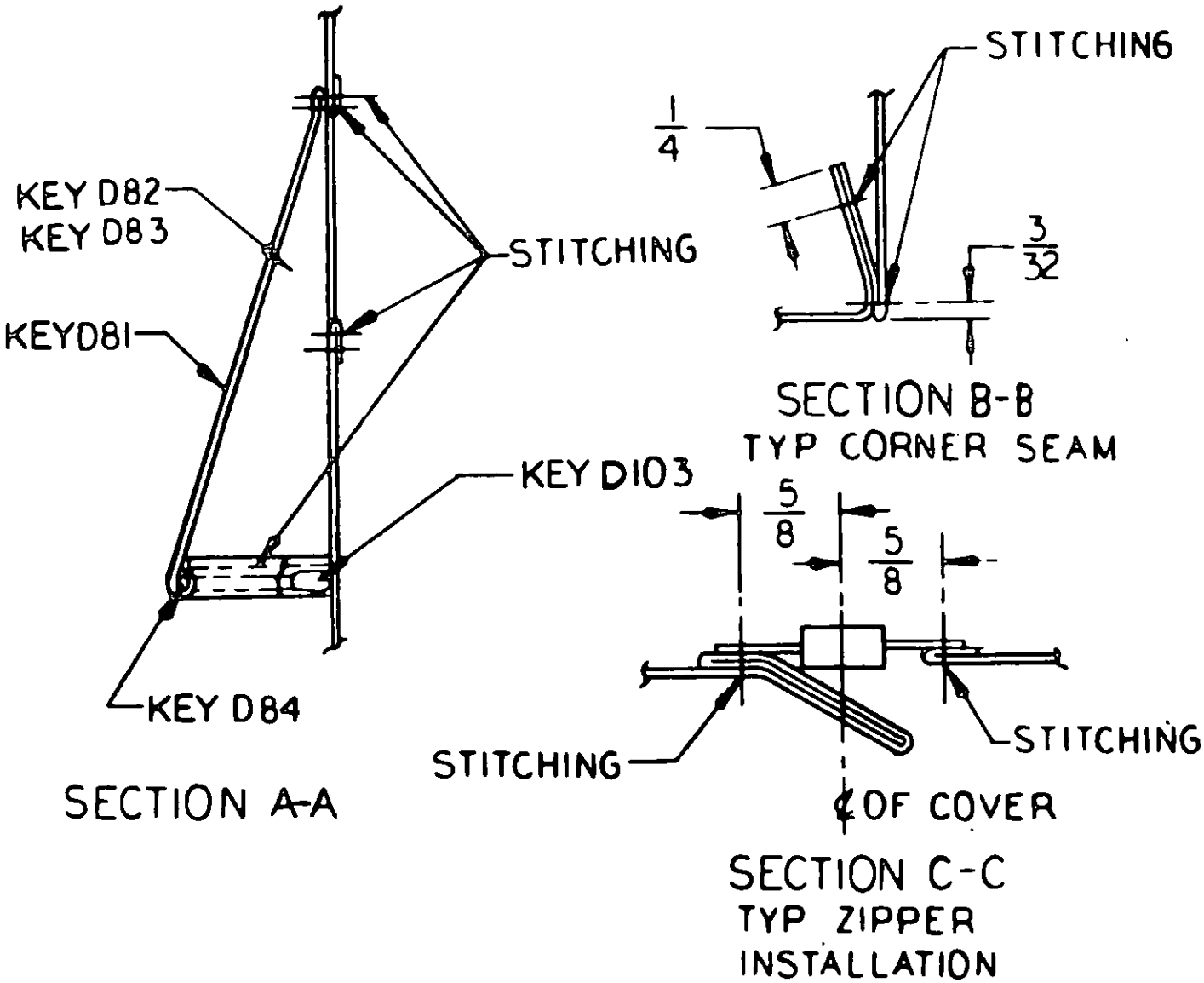
FIGURE 46. Cover assembly, installation, and attachment - Continued.



NOTES:
1. OPTIONAL APPLICATION: ATTACH WITH ADHESIVE PER MIL-A-5540,CLASS 2.
2. FOR SECTIONS SEE KEY D78,D91 & D92.

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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SECTION VIEW, FOR KEY D76 & D77
KEY D78

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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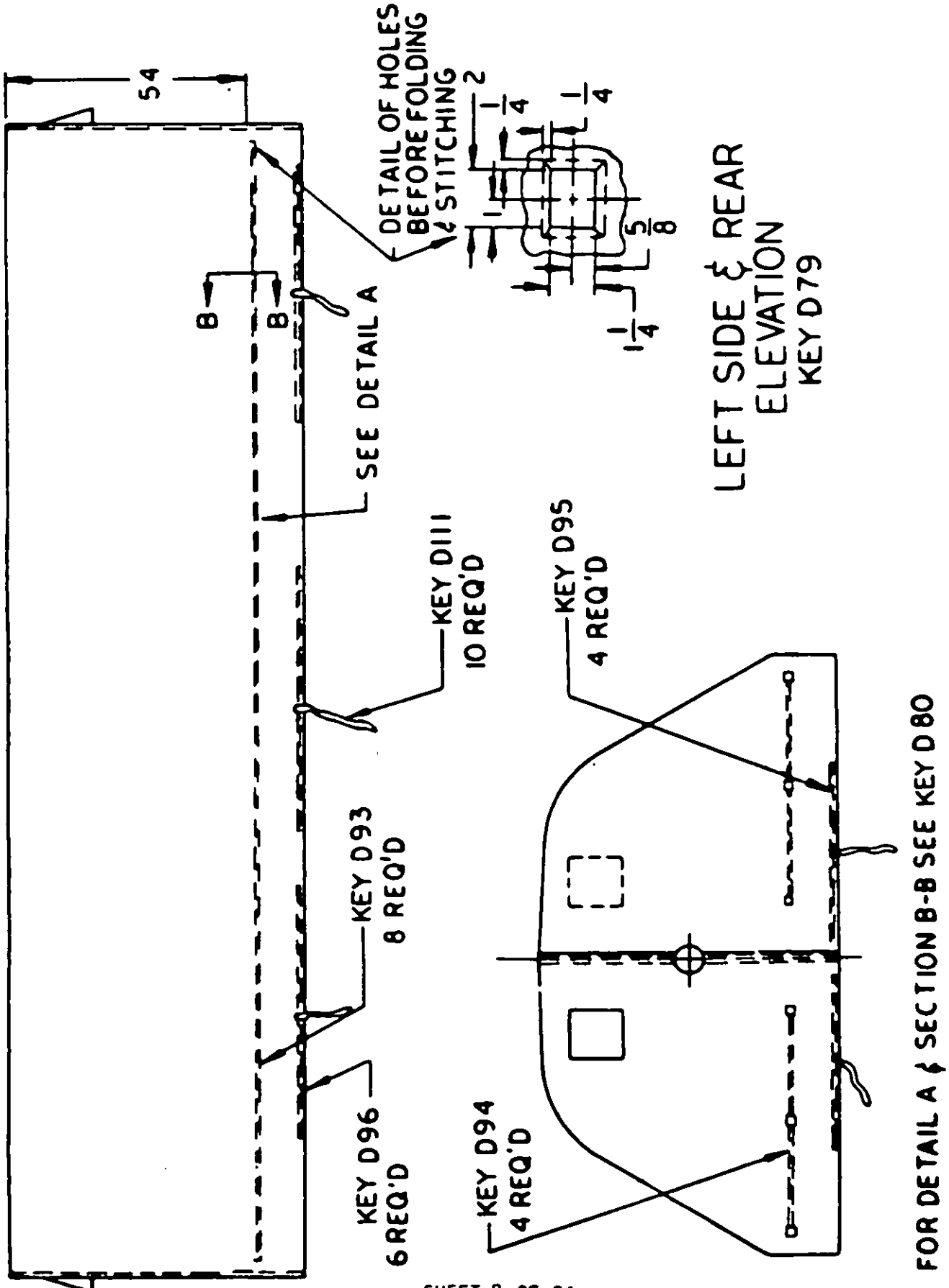
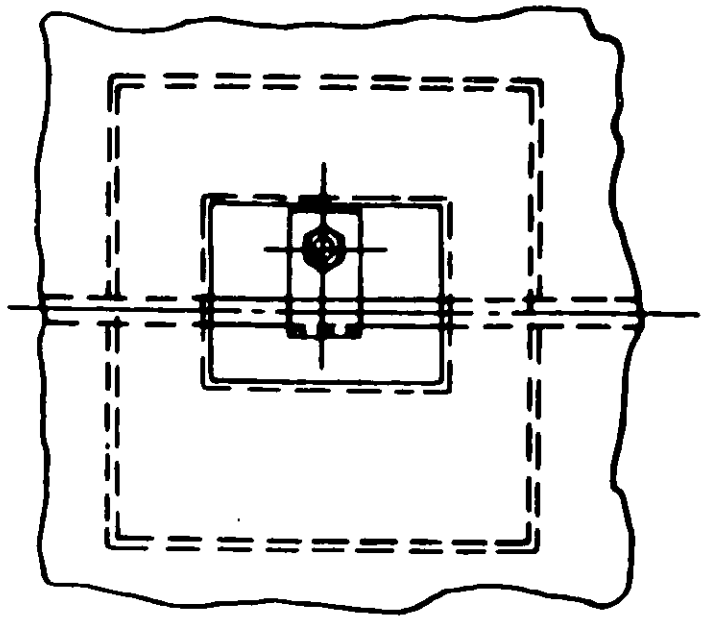
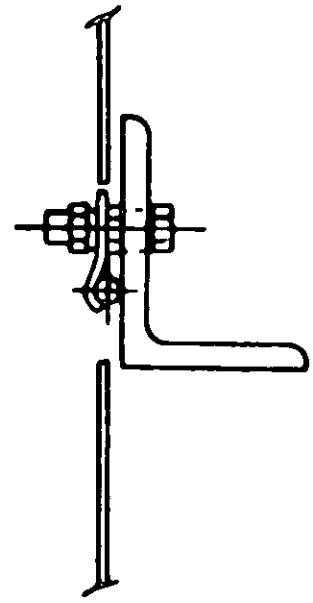


FIGURE 46. Cover assembly, installation, and attachment - Continued.

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DETAIL-A
TYPICAL 32 PLACES

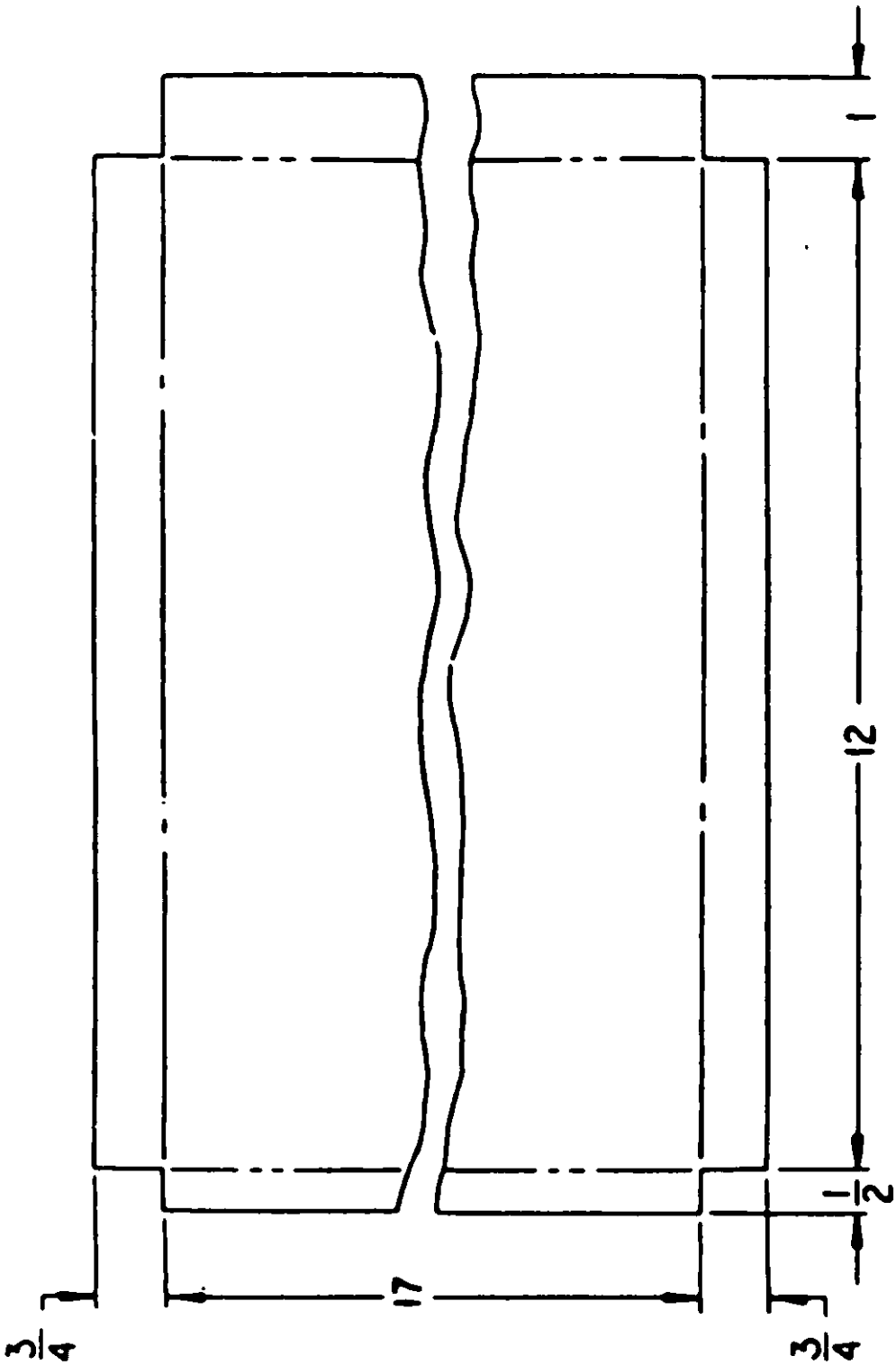


SECTION B-B
TYPICAL 32 PLACES

DETAIL-A-AND CROSS SECTION ON KEY D79
KEY D80

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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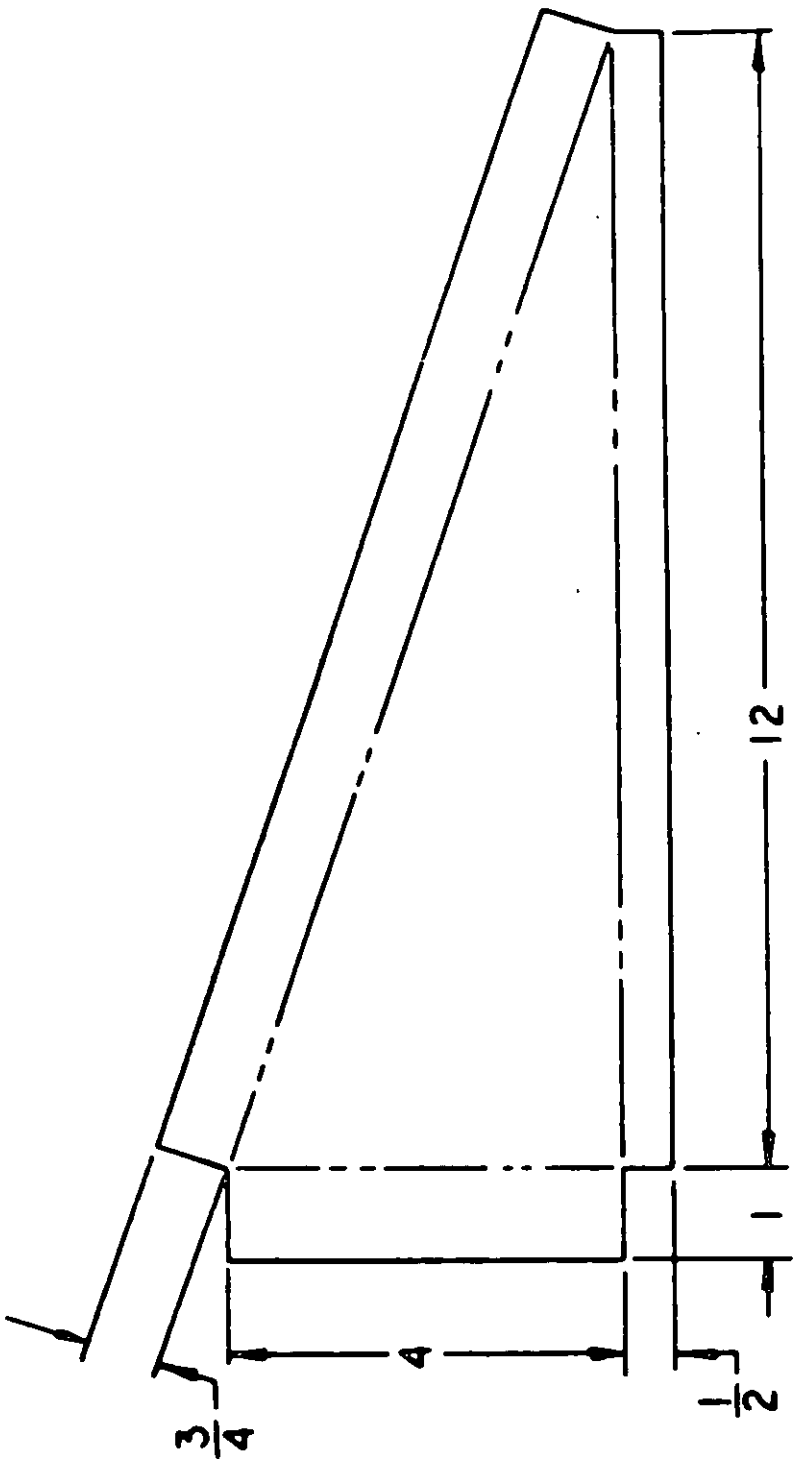
VENT, COVER
KEY D81
2 REQ'D

FOR MATERIAL NOTE
SEE KEY D72

——·——·—— FOLD LINE

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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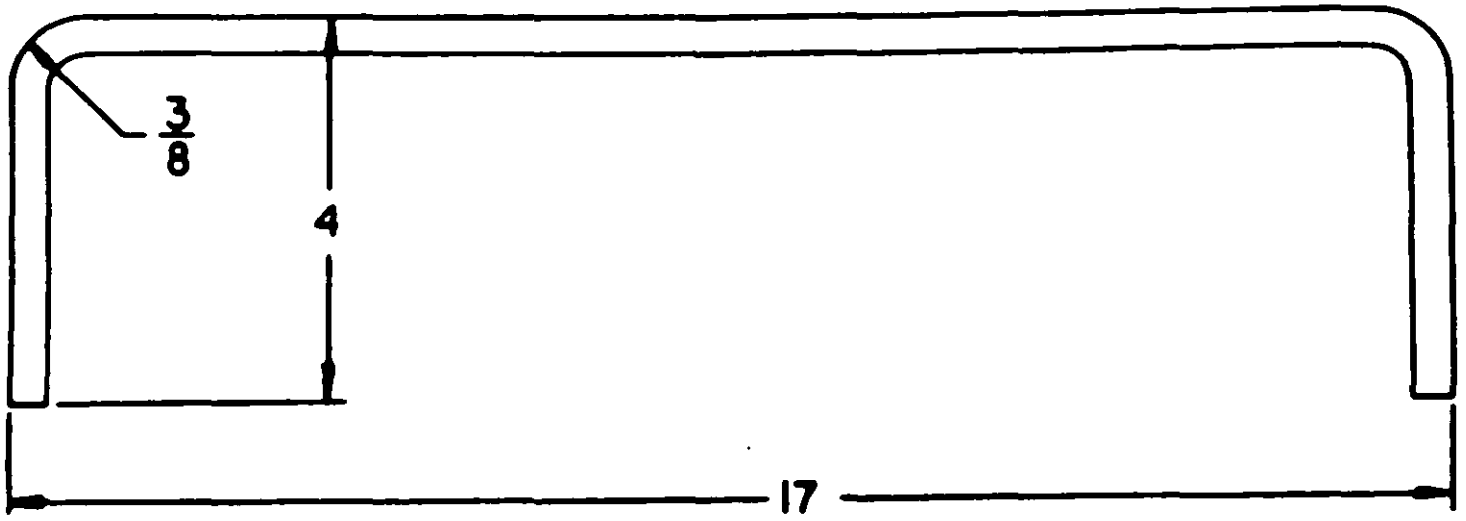


VENT, SIDE
KEY D82 SHOWN 2REQ'D
KEY D83 OPPOSITE 2REQ'D

FOR MATERIAL NOTE
SEE KEY D72
----FOLD LINE

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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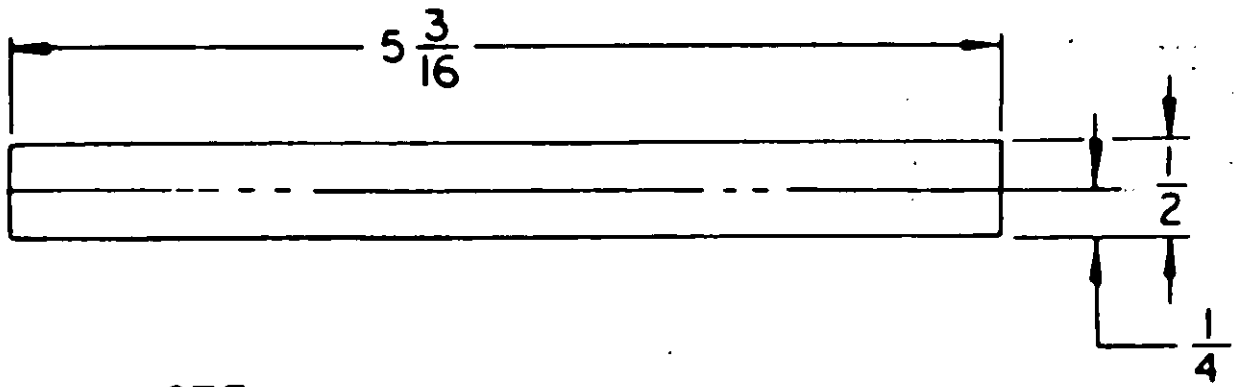


VENT REINFORCEMENT
KEY D84
2 REQ'D

NOTES:
1. STEEL BAR, CARBON 1010 TO 1020, PER ASTM A108, 3/8 DIA.
2. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMIGLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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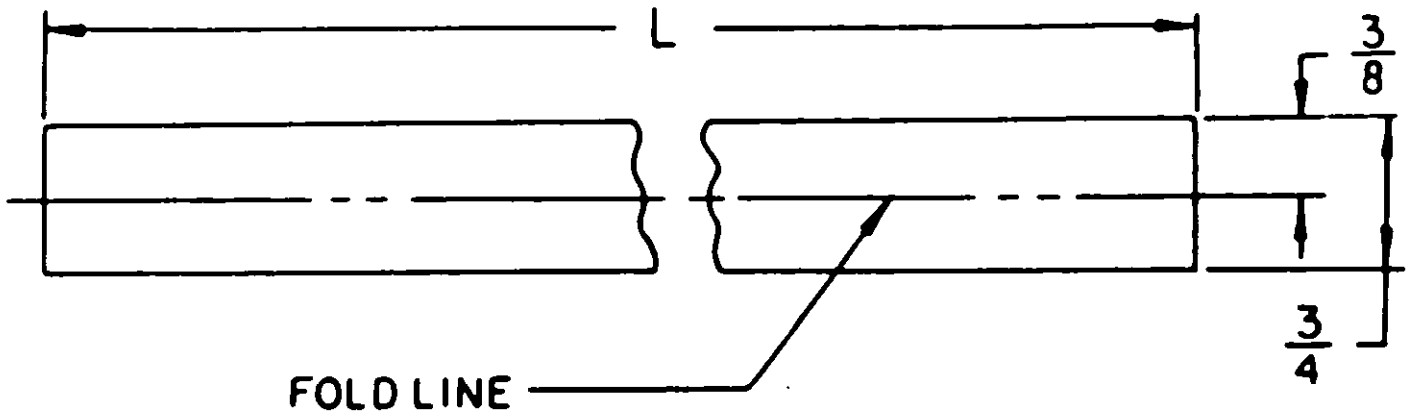
FOR MATERIAL NOTE
SEE KEY D72

—··—FOLD LINE

BINDING
10 REQ'D
KEY D87

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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KEY NO.	REQ'D	L
KEY D88	1	11 3/4
KEY D89	1	8 1/4

NOTE:
1. MATERIAL:
   COTTON WEBBING TYPE II, COLOR O.D.
   WATER REPELLENT AND MILDEW RESISTANT
   PER MIL-W-530.

WEBBING

KEY D88 & 89

—·-·— FOLD LINE

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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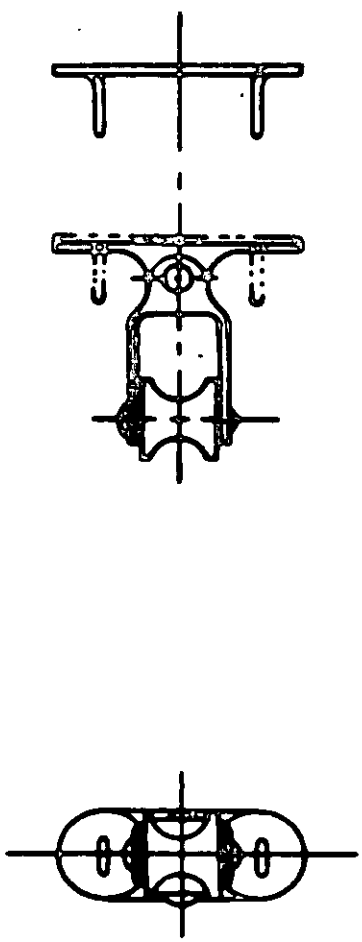
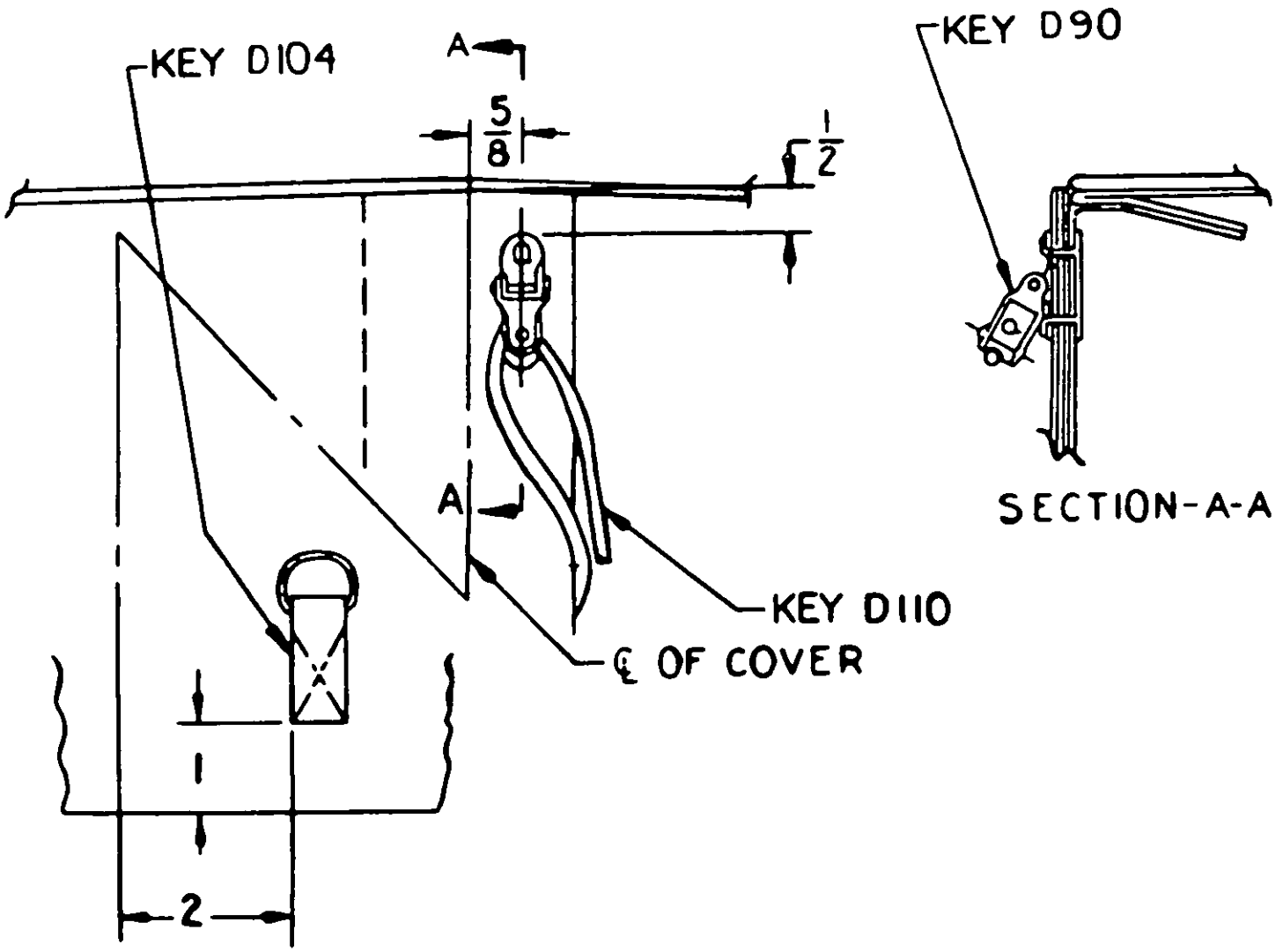


FIGURE 46. Cover assembly, installation, and attachment - Continued.

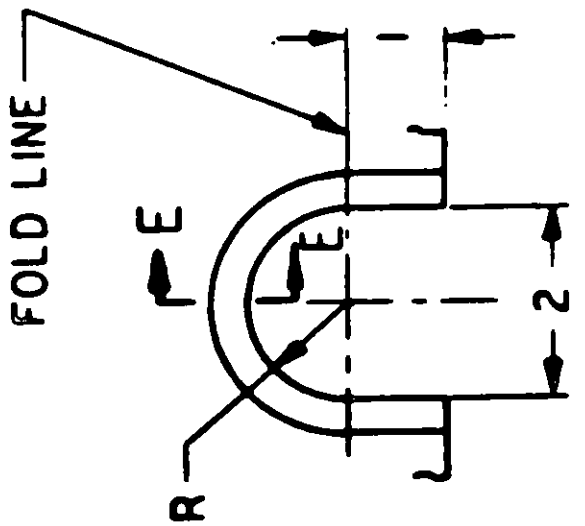
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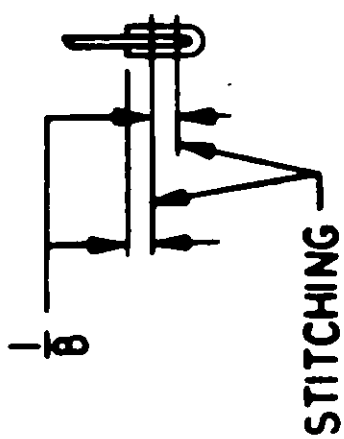
PULLEY INSTALLATION FRONT & REAR
KEY D91

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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DETAIL D
(BEFORE HEMMING)

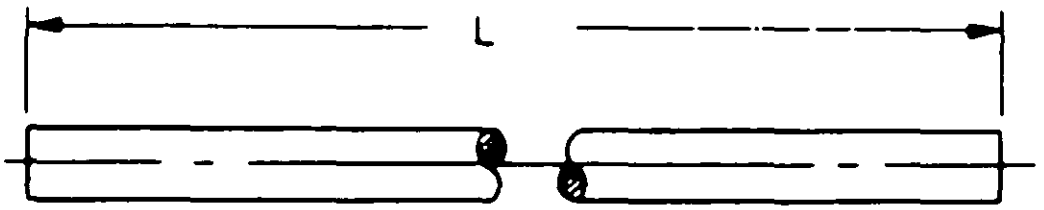


SECTION E-E
FULL SCALE

SECTION ◖ DETAIL FOR KEY D74
KEY D92

FIGURE 46. Cover assembly, installation and attachment - Continued.

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KEY NO.	REQD	L
KEY D93	8	67
KEY D94	4	60

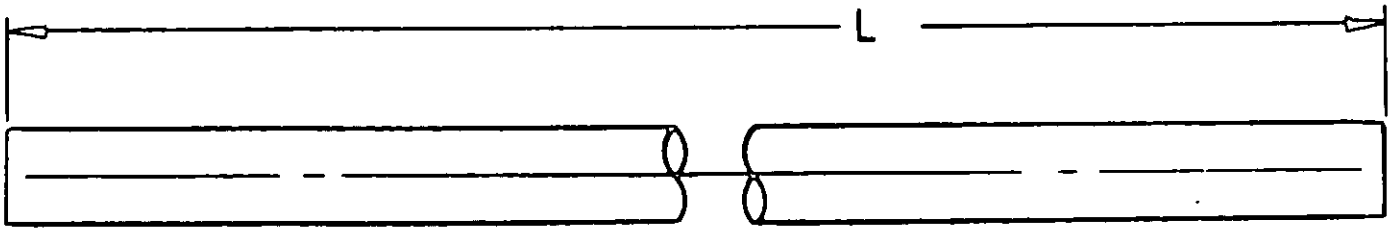
NOTES:
1. MATERIAL: STEEL, BAR, CARBON 1010 TO 1020 PER ASTM A108, $\frac{3}{8}$ DIA.
2. MUST BE FREE OF BURRS.
3. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.

ROD
KEY D93 & D94

SHEET 18 OF 24

FIGURE 46. Cover assembly, installation, and attachment - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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KEY NO.	REQD	L
KEY D93	4	60
KEY D94	6	76

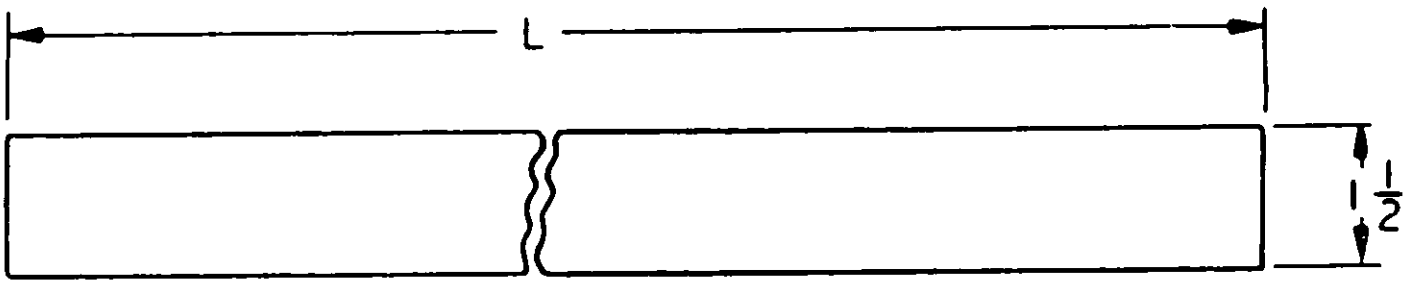
NOTES:
1. MATERIAL: STEEL, BAR, CARBON 1010 TO 1020 PER ASTM A108, $\frac{1}{4}$ DIA.
2. MUST BE FREE OF BURRS.
3. PRIME IN ACCORDANCE WITH TT-P-636 ENAMEL SEMI-GLOSS COLOR O.D. TYPE OPTIONAL PER TT-E-485 OR TT-E-529.

ROD TIE DOWN
D95 & D96

SHEET 19 OF 24

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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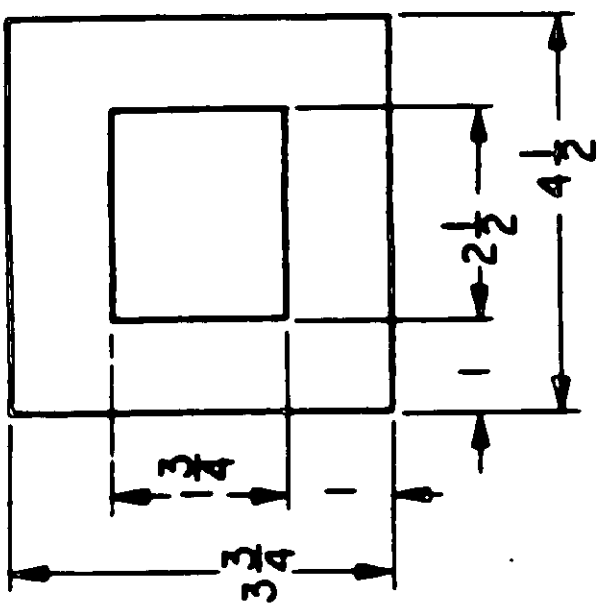
KEY NO.	REQD	L
KEY D97	4	25 3/4
KEY D98	14	27
KEY D99	8	24

FOR MATERIAL NOTE
SEE KEY D72

STRIP, TIE DOWN
KEY D97 D98 & D99

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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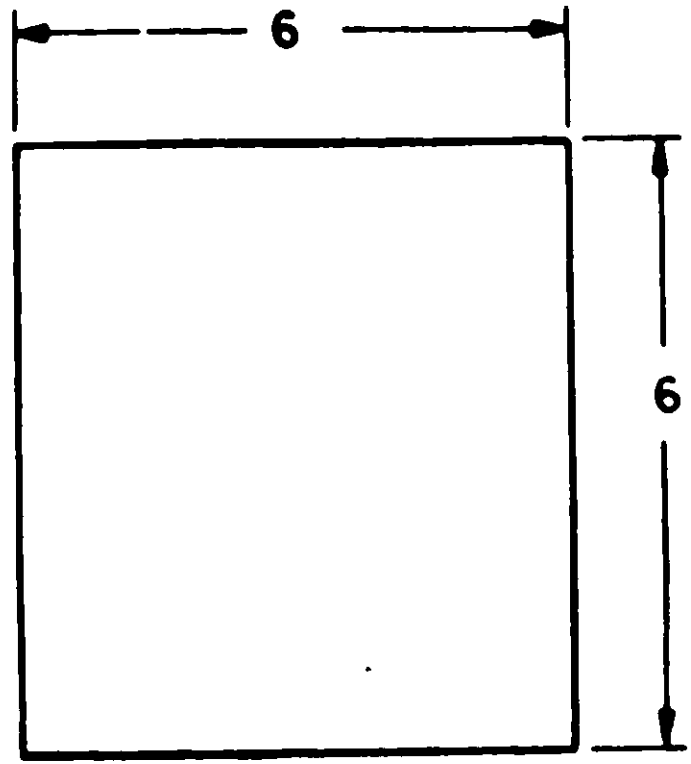


REINFORCEMENT, TIE DOWN
KEY D100
32 REQD

FOR MATERIAL NOTE
SEE KEY D72

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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NOTE:
1. MATERIAL: TYPE II, CLASS 3, PER MIL-C-20696. TOTAL WT OF 14 OUNCES. BALANCED SURFACE COATING SHALL BE ACCOMPLISHED. 2 THICKNESSES OF MATERIAL.

REINFORCEMENT
31 REQD
KEY D 101

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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KEY NUMBER	REQ'D	ORD. PART NO.	DESCRIPTION
D103	4		TIP, RUBBER SIZE 15 PER ZZ-T-351
D104	2	547557 1/	CHAPE, WEB, TYPE 3 WITH O RING

1/ SEE ARMY DRAWING QACX1.

HARDWARE, CLOSURE ATTACHMENT
KEY D102

SHEET 23 OF 24

FIGURE 46. Cover assembly, installation, and attachment - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

KEY NUMBER	REQ'D	DESCRIPTION	SPEC.
D107	1	DOUBLE FASTENER, SLIDE, INTERLOCKING, NON-SEPARATING, HEAVY DUTY (SINGLE ACTION, OPEN TOP CLOSED BOTTOM WITH NON-LOCK DOUBLE PULL REVERSIBLE SLIDER) 62 IN. LONG.	V-F-106 OR CLASS B SIZE H
D108	1	FASTENER, SLIDE, INTERLOCKING, NON-SEPARATING HEAVY DUTY (SINGLE ACTION, OPEN TOP, CLOSED BOTTOM WITH NON-LOCK DOUBLE PULL REVERSIBLE SLIDER) 28 IN. LONG.	V-F-106 OR CLASS B SIZE H
D109	1	FASTENER, SLIDE, INTERLOCKING, SEPARATING, HEAVY DUTY (REGULAR, LEFT HAND, WITH NON-LOCK DOUBLE PULL REVERSIBLE SLIDER) 29 IN. LONG.	V-F-106 OR CLASS B SIZE H
D110	2	ROPE, JUTE, TYPE I, MILDEW RESISTANT, 1/4 DIA 150 IN. LONG.	T-R-650 1/
D111	10	ROPE, JUTE, TYPE I, MILDEW RESISTANT, 1/4 DIA 72 IN. LONG.	T-R-650 1/

1/ OPTIONAL SPECIFICATIONS - MIL-R-17343 (NYLON), MIL-R-24049 (POLYPROPYLENE).

PARTS, CLOSURE ATTACHMENT
KEY D106

SHEET 24 OF 24

FIGURE 46. Cover assembly, installation, and attachment - Continued.

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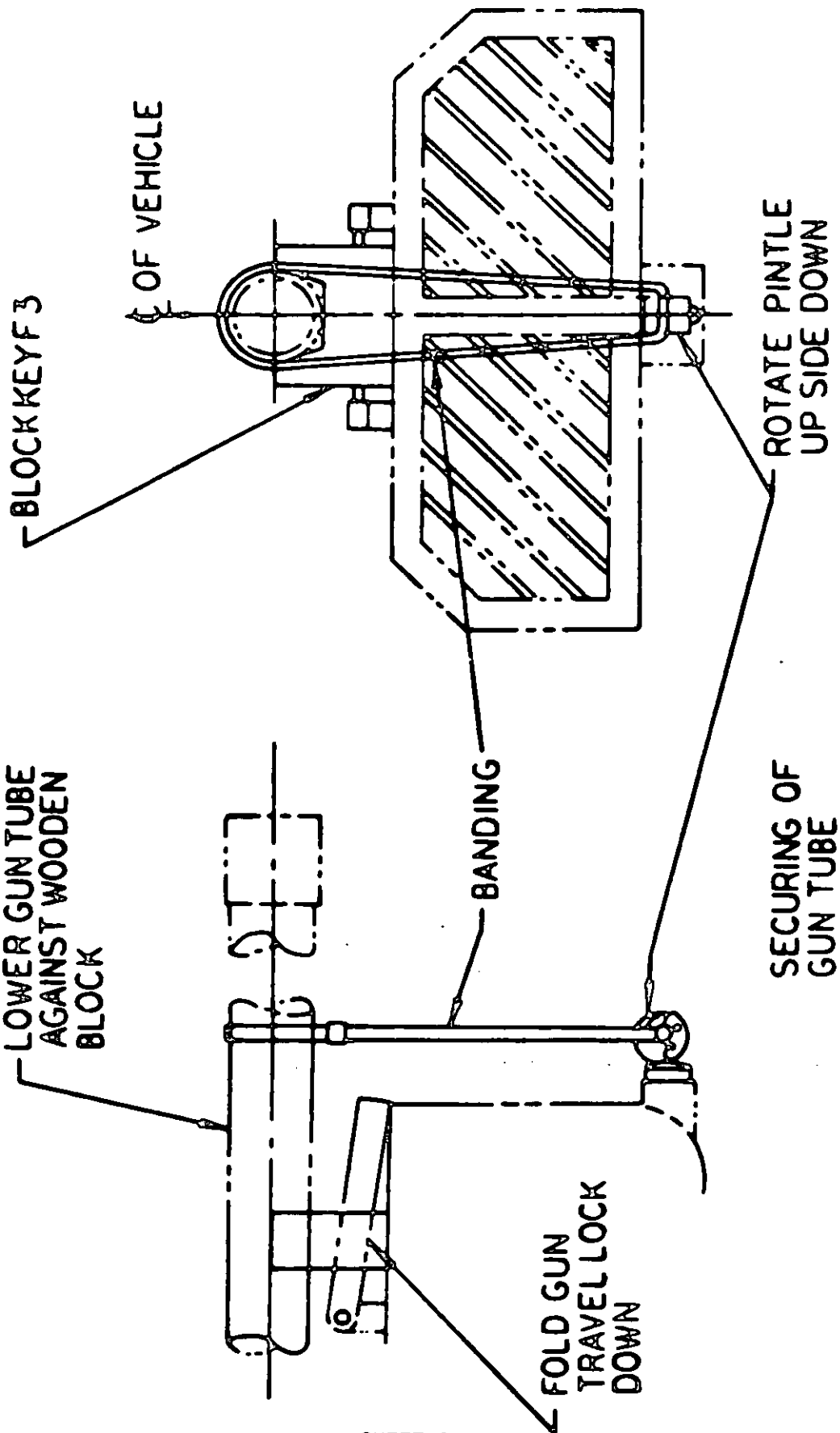
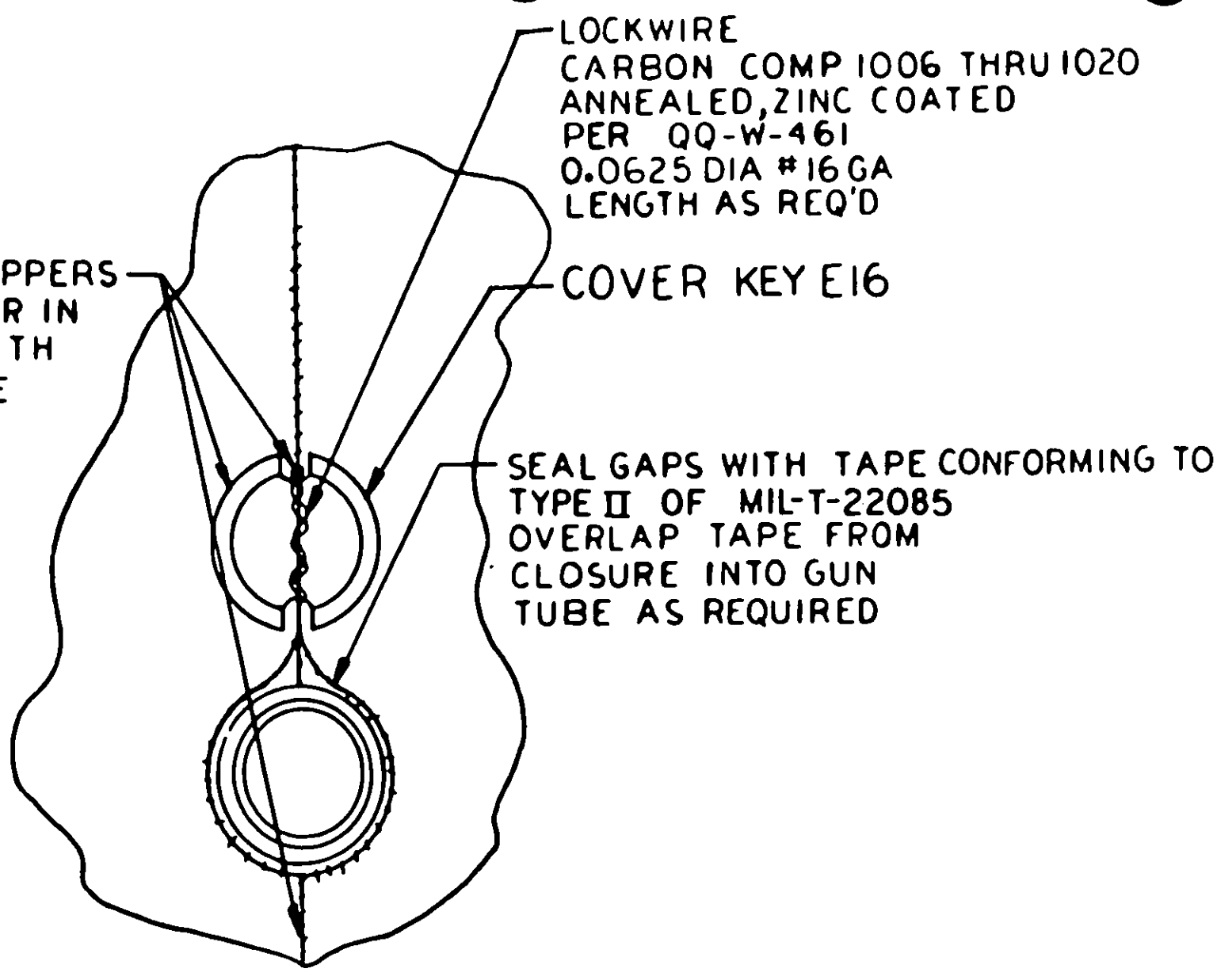


FIGURE 47. Shipment of two vehicles on one flat car.

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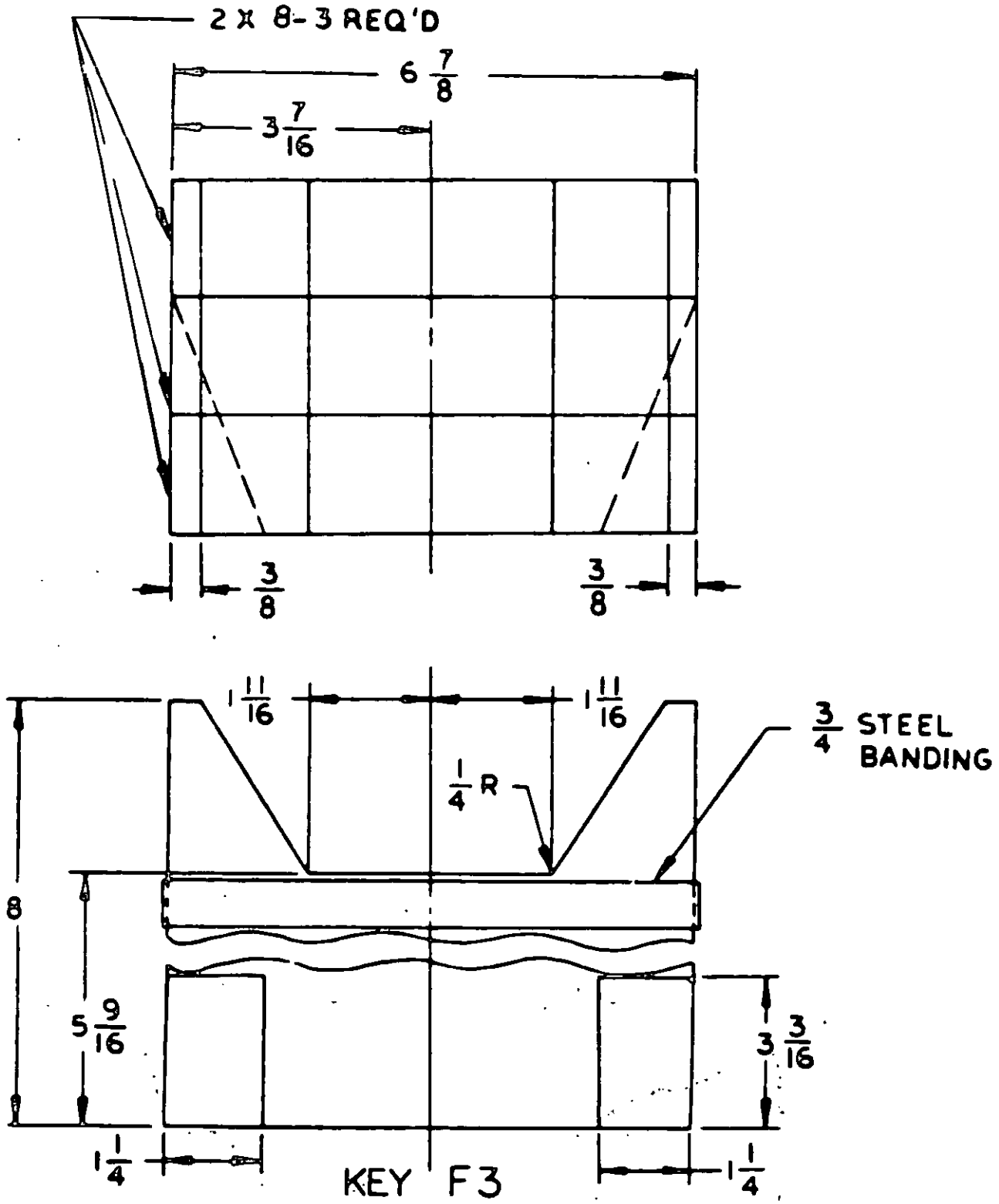


SEALING OF CLOSURE COVER

KEY F2

FIGURE 47. Shipment of two vehicles on one flat car - Continued.

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Check the source to verify that this is the current version before use.



NOTES:
1. MATERIAL: WOOD, GROUP II OR III, MIL-C-104.
2. NAIL 2 x 8 TOGETHER AT RANDOM USING 12d CC NAILS.

SHEET 3 OF 3

FIGURE 47. Shipment of two vehicles on one flat car - Continued.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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